Integrals for fun

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Abstract

This document compiles over a year and a half's worth of (mostly) integrals that I have written out the solutions to. Many of the problems were solved completely by me, many were solved with hints (see acknowledgements), and many were solved completely by someone else found online. The document begins with a chronological list of the problems, and each of the following sections are titled by the month in which I did each particular problem.

Contents

1 List of integrals

A list of integrals that I have done (with differing levels of assistance from none to all of it).

$$\int x^x \, dx = \sum_{n=1}^{\infty} \sum_{k=0}^{n-1} \frac{-x^n \ln^k x}{(-n)^{n-k} k!} + C$$

$$\int_0^1 x^x \, dx = -\sum_{n=1}^{\infty} (-n)^{-n}$$

$$\int_1^e x^x \, dx = \sum_{n=1}^{\infty} \sum_{k=0}^{\infty} \frac{n^{k+1}}{k! n! (n+k)}$$

$$\int_{-\infty}^{\infty} \frac{\cos x}{1+x^2} \, dx = \frac{\pi}{e}$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x}{\tan x} \, dx = \pi \ln 2$$

$$\int_{-\infty}^{\infty} \frac{\sin x}{x} \, dx = \pi$$

$$\int_0^1 \frac{\ln(x+1)}{x^2+1} \, dx = \frac{\pi}{8} \ln 2$$

$$\int_0^{\infty} \frac{\ln(x^2+1)}{x^2+1} \, dx = \pi \ln 2$$

$$\int_2^{\infty} \frac{(-1)^{\lfloor x \rfloor}}{\lfloor x-2 \rfloor ! \lceil x-2 \rceil x^2} \, dx = \frac{1}{2} - \frac{1}{e}$$

$$\int_0^1 \left\lfloor \frac{1}{x} \right\rfloor^{-1} \, dx = \frac{\pi^2}{6} - 1$$

$$\int_0^{\infty} \frac{\cosh x}{\lfloor e^x \rfloor !} \, dx = \frac{1}{2} \big(\mathrm{Ei}(1) - \gamma + 1 \big)$$

$$\int_0^{\infty} \frac{\cosh x}{\lceil e^x \rceil !} \, dx = \frac{1}{2}$$

$$\int_0^{\infty} \frac{\sinh x}{\lfloor e^x \rfloor !} \, dx = \frac{1}{2} \big(2e + \gamma - 3 - \mathrm{Ei}(1) \big)$$

$$\int_0^{\infty} \frac{\sinh x}{\lfloor e^x - 1 \rfloor !} \, dx = 1$$

$$\lim_{n \to \infty} \int_0^1 x \sqrt{x \sqrt[3]{x \sqrt[4]{x \cdots \sqrt[n]{x}}}} \, dx = \frac{1}{e}$$

$$\int \frac{1}{4 + \sin x} \, dx = \frac{2}{\sqrt{15}} \arctan\left(\frac{4 \tan(\frac{x}{2}) + 1}{\sqrt{15}} \right) + C$$

$$\int_0^{\frac{\pi}{2}} \frac{\cos x}{\left(1 + \sqrt{\sin 2x}\right)^2} \, dx = \frac{1}{3}$$

$$\sum_{n=0}^{\infty} \frac{(k!)^2}{(2k+1)!} = \frac{2\pi}{3\sqrt{3}}$$

$$\int_{-\infty}^{\infty} \frac{\cos x}{\cosh x}\, dx = \pi \operatorname{sech} \frac{\pi}{2}$$

$$\int_{0}^{\infty} \frac{\cos x}{\cosh\left(\frac{\pi x}{2}\right)}\, dx = \operatorname{sech} 1$$

$$\int_{-\infty}^{\infty} \frac{\cos^2 x}{\cosh x}\, dx = \frac{\pi}{2}(\operatorname{sech}(\pi) + 1)$$

$$\int_{-\infty}^{\infty} \frac{\cos x}{\cosh^2 x}\, dx = \pi \operatorname{csch} \frac{\pi}{2}$$

$$\int_{-\infty}^{\infty} \frac{\cos^2 x}{\cosh^2 x}\, dx = \pi \operatorname{csch}(\pi) + 1$$

$$\int_{0}^{\frac{\pi}{2}} \cos(\ln(\tan x))\, dx = \frac{\pi}{2} \operatorname{sech} \frac{\pi}{2}$$

$$\int_{0}^{\frac{\pi}{2}} x\cos(\ln(\tan x))\, dx = \frac{\pi^2}{8} \operatorname{sech} \frac{\pi}{2}$$

$$\int_{0}^{\infty} \frac{1}{x^{\pi}+1}\, dx = \csc 1$$

$$\int_{0}^{\infty} \frac{1}{x^{10}+1}\, dx = \frac{\phi\pi}{5}$$

$$\int_{0}^{\infty} \frac{\ln x}{x^{\pi}+1}\, dx = -\cot(1)\csc(1)$$

$$\int_{0}^{\infty} \frac{\ln x}{x^{2}+1}\, dx = 0$$

$$\int_{0}^{\infty} \frac{\ln x}{x^{3}+1}\, dx = -\frac{2\pi^2}{27}$$

$$\int_{0}^{\infty} \frac{\ln x}{x^{4}+1}\, dx = -\frac{\pi^2}{8\sqrt{2}}$$

$$\int_{0}^{\infty} \frac{\ln x}{x^{6}+1}\, dx = -\frac{\pi^2}{6\sqrt{3}}$$

$$\int_{-\infty}^{\infty} \cos(x^2)\, dx = \sqrt{\frac{\pi}{2}}$$

$$\int_{-\infty}^{\infty} \sin(x^2)\, dx = \sqrt{\frac{\pi}{2}}$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} \cos(kx^2)\, dx = \sqrt{\frac{\pi\left(\sqrt{\alpha^2+k^2}+\alpha\right)}{2\left(\alpha^2+k^2\right)}}$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} \sin(kx^2)\, dx = \sqrt{\frac{\pi\left(\sqrt{\alpha^2+k^2}-\alpha\right)}{2\left(\alpha^2+k^2\right)}}$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} \cos^n(kx^2)\, dx = \frac{1}{2^n}\sqrt{\frac{\pi}{2}} \sum_{h=0}^{n} \binom{n}{h} \sqrt{\frac{\alpha + \sqrt{\alpha^2 + (k(2h-n))^2}}{\alpha^2 + (k(2h-n))^2}}$$

$$k \in \mathbb{R}, \alpha \in \mathbb{R}^+, n \in \mathbb{N}.$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} \sin^n(kx^2)\, dx = \frac{1}{2^n}\sqrt{\frac{\pi}{2}} \sum_{h=0}^{n} i^{2h-n} \binom{n}{h} \sqrt{\frac{\alpha + \sqrt{\alpha^2 + (k(2h-n))^2}}{\alpha^2 + (k(2h-n))^2}}$$

$$k \in \mathbb{R}, \alpha \in \mathbb{R}^+, n \in 2\mathbb{N}.$$

$$\int_{-\infty}^{\infty} e^{-x^2} \cos(2x^2)\, dx = \sqrt{\frac{\phi\pi}{5}}$$

$$\int_{-\infty}^{\infty} e^{-x^2} \cos\left(\sqrt{\phi}\, x^2\right) dx = \sqrt{\frac{\pi}{2}}$$

$$\int_{-\infty}^{\infty} e^{-x^4} \cos(x^4)\, dx = \left(\frac{1}{4}\right)! \sqrt{\frac{2 + \sqrt{2 + \sqrt{2}}}{\sqrt[4]{2}}}$$

$$\int_{-\infty}^{\infty} x^2 e^{-2x^2} \cos^2(2x^2)\, dx = \frac{\sqrt{\pi}}{16}\left(\sqrt{2} - \frac{1}{5}\sqrt{\sqrt{5} - \frac{11}{5}}\right)$$

$$\int_{0}^{\infty} \frac{W(x)}{x\sqrt{x}}\, dx = 2\sqrt{2\pi}$$

$$\int_{0}^{1} \frac{\sin^2(\ln\sqrt{x})}{\ln^2 x}\, dx = \frac{1}{4}\left(\frac{\pi}{2} - \ln 2\right)$$

$$\int_{0}^{1} \frac{\sin^2(\ln x)}{\ln x}\, dx = -\frac{1}{4}\ln 5$$

$$\int_{0}^{\infty} \frac{\sin x}{e^x - 1}\, dx = \frac{\pi}{2}\coth(\pi) - \frac{1}{2}$$

$$\int_{0}^{1} \frac{\ln x}{x^2+1}\, dx = -G$$

$$\int_{1}^{\infty} \operatorname{arccsc}^2 x\, dx = 4G - \frac{\pi^2}{4}$$

$$\int_{0}^{\pi} \ln(\sin x)\, dx = -\pi \ln 2$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \ln(\cos x)\, dx = -\pi \ln 2$$

$$\int_{0}^{1} \ln\Gamma(x)\, dx = \frac{1}{2}\ln(2\pi)$$

$$\int_{a}^{a+1} \ln\Gamma(x)\, dx = \frac{1}{2}\ln(2\pi) + a\ln(|a|) - a \quad (a \geq 0, a \in 2\mathbb{Z}^-)$$

$$\int_0^\infty \operatorname{arccot}(x^\pi)\,dx = \frac{\pi}{2}\sec\frac{1}{2}$$

$$\int_0^\infty \operatorname{arccot}(x^{5/2})\,dx = \frac{\pi}{\phi}$$

$$\int_0^\infty \operatorname{arccot}(x^2)\,dx = \frac{\pi}{\sqrt{2}}$$

$$\int_0^\infty \operatorname{arccot}(x^3)\,dx = \frac{\pi}{\sqrt{3}}$$

$$\int_0^\infty \operatorname{arccot}(x^4)\,dx = \frac{\pi}{\sqrt{2+\sqrt{2}}}$$

$$\int_{-\infty}^\infty \arctan(e^x)\arctan(e^{-x})\,dx = \frac{7}{4}\zeta(3)$$

$$\int_0^{\frac{\pi}{2}} x\ln(\sin x)\,dx = \frac{7}{16}\zeta(3) - \frac{\pi^2}{8}\ln 2$$

$$\int_0^{\frac{\pi}{2}} x\ln(\cos x)\,dx = -\frac{7}{16}\zeta(3) - \frac{\pi^2}{8}\ln 2$$

$$\int_0^{\frac{\pi}{2}} x\ln(\tan x)\,dx = \frac{7}{8}\zeta(3)$$

$$\int_0^1 \frac{1}{\sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}}\,dx = \frac{2}{\phi} - \ln\phi$$

$$\int_0^{\frac{\pi}{2}} \frac{1}{\sin^2 x}\ln\left(\frac{1+\sin^2 x}{1-\sin^2 x}\right)dx = \pi\sqrt{2}$$

$$\int_0^{\frac{\pi}{2}} \sqrt{\tan x}\,dx = \frac{\pi}{\sqrt{2}}$$

$$\int_0^1 (x^x)^{(x^x)^{(x^x)^{\cdots}}}\,dx = \frac{\pi^2}{12}$$

$$\int_0^k (x^{-x})^{(x^{-x})^{(x^{-x})^{\cdots}}}\,dx = \frac{\pi^2}{6} + \ln k \quad (1/e \le k \le e)$$

$$\int_0^1 (x^{-x/2})^{(x^{-x/2})^{(x^{-x/2})^{\cdots}}}\,dx = \frac{\pi^2}{6} - \ln^2 2$$

$$\int_0^{\frac{1}{e}} (e^x)^{(e^x)^{(e^x)^{\cdots}}}\,dx = \frac{1}{2}$$

$$\int_0^e (e^{-x})^{(e^{-x})^{(e^{-x})^{\cdots}}}\,dx = \frac{3}{2}$$

$$\int_{-1}^\infty \left(e^{-xe^x}\right)^{\left(e^{-xe^x}\right)^{\cdots}}\,dx = e$$

$$\int_0^\infty (a^{-x})^{(a^x)^{(a^{-x})^{(a^x)^{\cdots}}}}\,dx = \frac{\pi^2}{12\ln a} \quad (a > 1)$$

$$\int_0^\infty (e^{-x})^{(e^{\phi x})^{(e^{-x})^{(e^{\phi x})^{\cdots}}}}\,dx = \frac{1}{\phi}\left(\frac{\pi^2}{10} + \ln^2\phi\right)$$

$$\int_0^1 \frac{\ln(x)\ln(-\ln x)}{1-x}\,dx = \frac{\pi^2}{6}(12\ln(A) - \ln(2\pi) - 1)$$

$$\int_{-\infty}^{\infty} \frac{\sin x\cos x}{\sinh x\cosh x}\,dx = \frac{\pi}{2}\tanh\frac{\pi}{2}$$

$$\int_0^{\infty} \frac{\ln x}{\cosh x + 1}\,dx = \ln\left(\frac{\pi}{2}\right) - \gamma$$

$$\int_0^{\infty} \frac{x\ln x}{\cosh x + 1}\,dx = 2\ln 2 - \ln^2 2$$

$$\int_0^{\infty} \frac{x^2\ln x}{\cosh x + 1}\,dx = \frac{\pi^2}{6}\big(3 + 2\ln(4\pi) - 24\ln A\big)$$

$$\int_0^1 \sin(\sqrt{-\ln x})\,dx = \frac{1}{2}\sqrt{\frac{\pi}{\sqrt{e}}}$$

$$\int_0^1 \sinh(\sqrt{-\ln x})\,dx = \frac{1}{2}\sqrt{\pi\sqrt{e}}$$

$$\int_0^{\infty} \frac{\sin x\ln x}{x}\,dx = -\frac{\gamma\pi}{2}$$

$$\int_0^{\infty} \frac{e^{-x}\sin x\ln x}{x}\,dx = -\frac{\pi}{4}\left(\gamma + \frac{1}{2}\ln 2\right)$$

$$\int_0^{\infty} \frac{e^{-2x}\sinh x\ln x}{x}\,dx = -\frac{1}{4}\ln^2 3 - \frac{\gamma}{2}\ln 3$$

$$\int_0^{\infty} \frac{e^{-\sqrt{2}x}\sinh x\ln x}{x}\,dx = \frac{\gamma}{2}\ln\left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)$$

$$\int_0^{\infty} \frac{1}{\sqrt{\cosh x}}\,dx = \varpi$$

$$\int_0^{\infty} \frac{1}{\sqrt{\sinh x}}\,dx = \varpi\sqrt{2}$$

$$\int_0^{\infty} \ln(x)\sin(x^2)\,dx = \frac{1}{8}\sqrt{\frac{\pi}{2}}(\pi - 2\gamma - 4\ln 2)$$

$$\int_0^{\infty} \ln^2(x)\sin(x^2)\,dx = \frac{1}{32}\sqrt{\frac{\pi}{2}}(2\gamma - \pi + 4\ln 2)^2$$

$$\int_0^{\infty} \frac{\ln^2(x)\sin(x^2)}{x^2}\,dx = \frac{1}{16}\sqrt{\frac{\pi}{2}}\big((4 - 2\gamma - \pi - 4\ln 2)^2 + 16\big)$$

$$\int_0^{\infty} \sqrt{x}\ln^2(x)\sin(x^2)\,dx = \frac{\pi}{64}\sqrt{\frac{2+\sqrt{2}}{\varpi\sqrt{2\pi}}}\big((\pi - 2\gamma - 6\ln 2)^2 + 2\pi(\sqrt{2}-1)(\pi - 2\gamma - 6\ln 2) + 3\pi^2 - 32G\big)$$

$$\int_0^{\infty} \frac{\ln^2(x)\sin(x^2)}{\sqrt{x}}\,dx = \frac{1}{16}\sqrt[4]{\frac{\pi}{2}}\sqrt{\varpi(2-\sqrt{2})}\big(2(\gamma + 3\ln 2)^2 - 2\pi\sqrt{2}(\gamma + 3\ln 2) + (1-\sqrt{2})\pi^2 + 16G\big)$$

$$\int_0^{\pi} e^{\frac{3(1+\cos x)}{5+4\cos x}} \cos\left(\frac{\sin x}{5+4\cos x}\right) dx = \pi\sqrt{e}$$

$$\int_0^{2\pi} \arctan\left(\frac{1}{1+2\cos x}+1\right) dx = \frac{\pi^2}{3}$$

$$\int_0^{\frac{\pi}{2}} \ln\big((\ln(2\cos x)-2)^2+x^2\big)\, dx = \pi\ln 2$$

$$\int_0^{2\pi} \cosh\big(e^{-\sin x}\cos(\cos x)\big)\cos\big(e^{-\sin x}\sin(\cos x)\big)\, dx = \pi e + \frac{\pi}{e}$$

$$\int_{-\pi}^{\pi} e^{\frac{1}{2}\cos(x)\ln(2(1+\cos x))-\frac{x}{2}\sin x} \cos\left(\frac{x}{2}\cos(x)+\frac{1}{2}\sin(x)\ln\big(2(1+\cos x)\big)\right) dx = \pi$$

$$\int_0^{\pi} \ln\left(\sin^2\left(\cos(x)+\frac{\pi}{3}\right)\cosh^2(\sin x)+\cos^2\left(\cos(x)+\frac{\pi}{3}\right)\sinh^2(\sin x)\right) dx = \pi\ln\frac{3}{4}$$

$$\int_0^{\pi} \frac{\sinh\left(\arctan\left(\frac{1+\cos x}{1+\sin x}\right)+\arctan\left(\frac{1+\cos x}{1-\sin x}\right)\right)}{\cosh\left(\arctan\left(\frac{1+\cos x}{1+\sin x}\right)+\arctan\left(\frac{1+\cos x}{1-\sin x}\right)\right)+\cos\left(\frac{1}{2}\ln\left(\frac{3+2(\cos x+\sin x)}{3+2(\cos x-\sin x)}\right)\right)} dx = \pi\tanh\frac{\pi}{4}$$

$$\int_0^{a} \ln\Gamma(x)\, dx = \frac{a}{2}\ln(2\pi) + a\ln(\Gamma(a)) - \ln(G(1+a)) + \frac{a(1-a)}{2}$$

$$\int_0^{\frac{1}{2}} \ln\Gamma(x)\, dx = \frac{3}{2}\ln(A) + \frac{1}{4}\ln(\pi) + \frac{5}{24}\ln(2)$$

$$\int_0^{\frac{1}{3}} \ln\Gamma(x)\, dx = \frac{4}{3}\ln(A) + \frac{1}{6}\ln(2\pi) - \frac{1}{72}\ln(3) + \frac{V}{6\pi}$$

$$\int_0^{\frac{1}{4}} \ln\Gamma(x)\, dx = \frac{9}{8}\ln(A) + \frac{1}{8}\ln(2\pi) + \frac{G}{4\pi}$$

$$\int_0^{\frac{1}{6}} \ln\Gamma(x)\, dx = \frac{5}{6}\ln(A) + \frac{1}{12}\ln(2\pi) + \frac{1}{144}\ln(3) + \frac{1}{72}\ln(2) + \frac{V}{4\pi}$$

$$\Re\int_{-a}^{a} \ln\Gamma(x)\, dx = a\ln\left(\frac{2\pi}{|a|}\right) + \frac{1}{2\pi}\mathrm{Cl}_2(2\pi a) + a$$

$$\Re\int_{-\frac{1}{2}}^{\frac{1}{2}} \ln\Gamma(x)\, dx = \frac{1}{2}\ln(4\pi) + \frac{1}{2}$$

$$\Re\int_{-\frac{1}{3}}^{\frac{1}{3}} \ln\Gamma(x)\, dx = \frac{1}{3}\ln(6\pi) + \frac{V}{3\pi} + \frac{1}{3}$$

$$\Re\int_{-\frac{1}{4}}^{\frac{1}{4}} \ln\Gamma(x)\, dx = \frac{1}{4}\ln(8\pi) + \frac{G}{2\pi} + \frac{1}{4}$$

$$\Re\int_{-\frac{1}{6}}^{\frac{1}{6}} \ln\Gamma(x)\, dx = \frac{1}{6}\ln(12\pi) + \frac{V}{2\pi} + \frac{1}{6}$$

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\tan^2 x\cosh x}{\cos x}\, dx = \frac{\sqrt{2}}{e^{\frac{\pi}{4}}}$$

$$\int_0^{\frac{\pi}{4}} \frac{\tan^2 x\sinh x}{\cos x}\, dx = \frac{1}{2} - \frac{\sqrt{2}}{2e^{\frac{\pi}{4}}}$$

$$\int_0^\infty \frac{1-\cos x}{x(1-e^x)}\,dx = \frac{1}{2}\ln\left(\frac{\pi}{\sinh \pi}\right)$$

$$\int_0^\pi \int_0^\pi \frac{\arctan(\sin x \sin y)}{\sin x}\,dx\,dy = 2G\pi$$

$$\int_0^{2\pi} \zeta(e^{ix}-1)\,dx = -\frac{\pi}{6}$$

$$\int_0^{2\pi} \zeta(e^{ix})\,dx = -\pi$$

$$\int_0^{2\pi} \zeta(e^{ix}+1)\,dx = 2\pi\gamma$$

$$\int_0^{2\pi} \zeta(e^{ix}+2)\,dx = \frac{\pi^3}{3}$$

$$\int_0^{2\pi} \Gamma(e^{ix})\,dx = -2\pi\gamma$$

$$\int_0^{2\pi} \Gamma\left(e^{ix}+\frac{1}{2}\right)dx = 2\pi(\sqrt{\pi}-2)$$

$$\int_0^{2\pi} \Gamma(e^{ix}+1)\,dx = 2\pi$$

$$\int_0^{2\pi} \Gamma\left(e^{ix}+\frac{3}{2}\right)dx = \pi^{\frac{3}{2}}$$

$$\int_0^{2\pi} \psi_0(e^{ix}+1)\,dx = -2\pi\gamma$$

$$\int_0^{2\pi} \beta(e^{ix}+2)\,dx = 2\pi G$$

$$\int_0^{2\pi} \mathrm{Li}_2\left(e^{ix}+\frac{1}{2}\right)dx = \frac{\pi^3}{6} - \pi\ln^2 2$$

$$\int_0^{2\pi} \mathrm{Li}_3\left(e^{ix}+\frac{1}{2}\right)dx = \frac{\pi}{12}\big(21\zeta(3)+4\ln^3 2 - 2\pi^2\ln 2\big)$$

$$\int_0^{2\pi} G\left(e^{ix}+\frac{1}{2}\right)dx = \frac{2^{\frac{25}{24}}\pi^{\frac{3}{4}}e^{\frac{1}{8}}}{A^{\frac{3}{2}}} = \sqrt{\frac{\sqrt{\pi^3\sqrt{2^{\frac{25}{3}}e}}}{A^3}}$$

$$\int_0^{\pi} e^{\cos x}\,dx = \pi I_0(1)$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{\sin x}\,dx = \pi I_0(1)$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(x)\cos(\tan x)\,dx = \frac{\pi}{e}$$

$$\int_{-1}^{1} \frac{1}{x}\sqrt{\frac{1+x}{1-x}} \ln\left(\frac{2x^2+2x+1}{2x^2-2x+1}\right) dx = 4\pi \operatorname{arccot}\big(\sqrt{\phi}\big)$$

$$\int_0^{\pi} \arctan(1+\cos x)\, dx = \pi \operatorname{arccot}\big(\sqrt{\phi}\big)$$

$$\int_0^{\pi} \operatorname{arccot}(\cos x - 1)\, dx = \pi \operatorname{arccot}\big(\sqrt{\phi}\big) + \frac{\pi^2}{2}$$

$$\int_0^{\infty} W\left(\frac{1}{x^2}\right) dx = \sqrt{2\pi}$$

$$\int_{-\infty}^{\infty} e^{-e^x} x^2 e^{2x}\, dx = \frac{\pi^2}{6} + \gamma^2 - 2\gamma$$

$$\int_0^{\infty} \ln\left(\frac{\sinh x}{x}\right) \frac{1}{x^2(x^2+1)}\, dx = \pi \ln\Gamma\left(\frac{1}{\pi}+1\right) + \gamma$$

$$\int_0^{\infty} \ln(\cosh x) \frac{1}{x^2(x^2+1)}\, dx = \pi \ln\Gamma\left(\frac{1}{\pi}+\frac{1}{2}\right) - \frac{\pi}{2}\ln(\pi) + \gamma + 2\ln 2$$

$$\int_0^{\infty} \ln\left(\frac{\tanh x}{x}\right) \frac{1}{x^2(x^2+1)}\, dx = \pi \ln\left(\frac{\Gamma\big(\frac{1}{\pi}+1\big)}{\Gamma\big(\frac{1}{\pi}+\frac{1}{2}\big)}\right) + \frac{\pi}{2}\ln(\pi) - 2\ln 2$$

$$\int_0^{\infty} x^p e^{-kx^q}\, dx = \frac{1}{q} k^{-\frac{1+p}{q}} \Gamma\left(\frac{1+p}{q}\right) \quad (k, q > 0, p \geq 0)$$

$$\int_0^{\infty} \frac{1}{(cx^p + kx^q)^s}\, dx = \frac{1}{(q-p)\Gamma(s)} c^{\frac{sq-1}{p-q}} k^{\frac{1-sp}{p-q}} \Gamma\left(\frac{sq-1}{q-p}\right) \Gamma\left(\frac{1-sp}{q-p}\right)$$

$$\int_0^{\infty} \frac{1}{\sqrt{x^3+x^2+x}}\, dx = \frac{1}{2\sqrt{\pi}} \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \Gamma^2\left(\frac{2m+1}{4}\right)$$

$$\int_0^{\infty} \frac{1}{\sqrt{x^5+x^4+1}}\, dx = \frac{1}{5\sqrt{\pi}} \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \Gamma\left(\frac{4m+1}{5}\right) \Gamma\left(\frac{2m+3}{10}\right)$$

$$\sum_{m=0}^{\infty} \frac{\alpha^m}{m!} \Gamma^2\left(\frac{2m+1}{4}\right) = 4\sqrt{\pi} K\left(\frac{\sqrt{2+\alpha}}{2}\right) \quad (|\alpha| \leq 2)$$

$$\sum_{m=0}^{\infty} \frac{\big(2\big(5-4\sqrt{2}\big)\big)^m}{m!} \Gamma^2\left(\frac{2m+1}{4}\right) = \frac{\sqrt{\sqrt{2}+1}}{2^{\frac{5}{4}}} \Gamma\left(\frac{1}{8}\right) \Gamma\left(\frac{3}{8}\right)$$

$$\sum_{m=0}^{\infty} \frac{\big(-\sqrt{3}\big)^m}{m!} \Gamma^2\left(\frac{2m+1}{4}\right) = \frac{3^{\frac{1}{4}}}{2^{\frac{1}{3}}\sqrt{\pi}} \Gamma^3\left(\frac{1}{3}\right)$$

$$\sum_{m=0}^{\infty} \frac{\big(6\big(11-8\sqrt{2}\big)\big)^m}{m!} \Gamma^2\left(\frac{2m+1}{4}\right) = \frac{\sqrt{2}+1}{2\sqrt{2}} \Gamma^2\left(\frac{1}{4}\right)$$

$$\sum_{m=0}^{\infty} \frac{1}{m!} \left(-\frac{3\sqrt{7}}{4}\right)^m \Gamma^2\left(\frac{2m+1}{4}\right) = \frac{1}{\sqrt{\pi\sqrt{7}}} \Gamma\left(\frac{1}{7}\right) \Gamma\left(\frac{2}{7}\right) \Gamma\left(\frac{4}{7}\right)$$

$$\sum_{m=0}^{\infty} \frac{(-2)^m}{m!} \Gamma^2\left(\frac{2m+1}{4}\right) = 2\pi^{\frac{3}{2}}$$

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\beta)^2+\alpha^2} = \frac{\pi\sinh(2\pi\alpha)}{\alpha(\cosh(2\pi\alpha)-\cos(2\pi\beta))}$$

$$\sum_{n=-\infty}^{\infty} \frac{1}{an^2+bn+c} = \frac{2\pi}{\sqrt{\Delta}} \frac{\sin\left(\frac{\pi}{a}\sqrt{\Delta}\right)}{\cos\left(\frac{\pi}{a}\sqrt{\Delta}\right)-\cos\left(\frac{b\pi}{a}\right)}$$

$$\int_0^1 \prod_{n=1}^{\infty}(1-x^n)\,dx = \frac{4\pi\sqrt{\frac{3}{23}}\sinh\left(\frac{\pi\sqrt{23}}{3}\right)}{\cosh\left(\frac{\pi\sqrt{23}}{2}\right)}$$

$$\int_0^1 \prod_{n=1}^{\infty}(1-x^{24n})\,dx = \frac{\pi^2}{6\sqrt{3}}$$

$$\int_0^1 G(x)\phi(x)\,dx = \sqrt{\frac{10+2\sqrt{5}}{39}}\frac{8\pi\sinh\left(\frac{\pi\sqrt{39}}{10}\right)}{4\cosh\left(\frac{\pi\sqrt{39}}{5}\right)-1-\sqrt{5}}$$

$$\int_0^1 H(x)\phi(x)\,dx = \sqrt{\frac{10-2\sqrt{5}}{31}}\frac{8\pi\sinh\left(\frac{\pi\sqrt{31}}{10}\right)}{4\cosh\left(\frac{\pi\sqrt{31}}{5}\right)-1+\sqrt{5}}$$

$$\int_1^{\infty} \frac{\ln(\ln x)}{x^2+2x\cos a+1}\,dx = \frac{1}{2\sin a}\left(\pi\ln\left(\frac{\Gamma\left(\frac{1}{2}+\frac{a}{2\pi}\right)}{\Gamma\left(\frac{1}{2}-\frac{a}{2\pi}\right)}\right)+a\ln(2\pi)\right)$$

$$\int_1^{\infty} \frac{\ln(\ln x)}{x^2+1}\,dx = \pi\ln\left(\frac{\sqrt{2\sqrt{\pi^3}}}{\Gamma\left(\frac{1}{4}\right)}\right)$$

$$\int_1^{\infty} \frac{\ln(\ln x)}{x^2+x+1}\,dx = \frac{2\pi}{\sqrt{3}}\ln\left(\frac{(2\pi)^{\frac{2}{3}}}{3^{\frac{1}{4}}\Gamma\left(\frac{1}{3}\right)}\right)$$

$$\int_1^{\infty} \frac{\ln(\ln x)}{x^2-x+1}\,dx = \frac{2\pi}{\sqrt{3}}\ln\left(\frac{(2\pi)^{\frac{5}{6}}}{\Gamma\left(\frac{1}{6}\right)}\right)$$

$$\int_1^{\infty} \frac{\ln(\ln x)}{(x+1)^2}\,dx = \frac{1}{2}\left(\ln\left(\frac{\pi}{2}\right)-\gamma\right)$$

$$\int_0^{\infty} x^{-s}\psi_k(x+1)\,dx = (-1)^{k+1}\zeta(k+s)\Gamma(1-s)\Gamma(k+s) \quad (s<1, k\in\mathbb{N})$$

$$\int_0^{\infty} x\psi_3(x+1)\,dx = \frac{\pi^2}{6}$$

$$\int_0^{\infty} x^2\psi_4(x+1)\,dx = -\frac{\pi^2}{3}$$

$$\int_0^{\infty} x\psi_5(x+1)\,dx = \frac{\pi^4}{15}$$

$$\int_0^{\infty} x^2\psi_6(x+1)\,dx = -\frac{2\pi^4}{15}$$

$$\sum_{n=0}^{\infty}\left(\frac{1}{3n+1}-\frac{1}{3n+2}\right) = \frac{\pi}{3\sqrt{3}}$$

$$\int_0^1 x^{\alpha-1}\ln(x)\sin(\ln x)\,dx = \frac{2\alpha}{(\alpha^2+1)^2} \quad (\alpha>0)$$

$$\int_0^{\frac{1}{2}} \frac{(1-x)^{x-1}}{x^x}\,dx = \sum_{n=1}^{\infty} \frac{(1-n)^{n-1}}{n^n}$$

$$\int_0^1 \sin(\pi x) x^x (1-x)^{1-x}\,dx = \frac{\pi e}{24}$$

$$\int_0^1 \frac{\sin(\pi x)}{x^x(1-x)^{1-x}}\,dx = \frac{\pi}{e}$$

$$\int_0^\infty \frac{\sin(x)\ln(1+\cos x)}{x\cos x}\,dx = \frac{\pi}{2}\left(\frac{\pi}{2} - \ln 2\right)$$

$$\int_0^\infty \frac{\{e^x\}}{e^x}\,dx = 1-\gamma$$

$$\int_{-1}^0 \sqrt{x-\ln(x+1)}\,dx = \frac{\sqrt{\pi}}{2}\sum_{n=1}^{\infty}\frac{n^{n-\frac{3}{2}}}{n!e^n}$$

$$\int_0^1 \sin\left\lfloor\frac{1}{x}\right\rfloor dx = \frac{\pi-1}{2}(1-\cos 1) - \sin(1)\ln\left(2\sin\left(\frac{1}{2}\right)\right)$$

$$\int_0^\infty \frac{1}{\mathrm{Ai}^2(x)+\mathrm{Bi}^2(x)}\,dx = \frac{\pi^2}{6}$$

$$\int_0^1 \frac{\log_2(\Gamma(2+x)\Gamma(2-x))}{x(1+x)}\,dx = \ln\left(\frac{K_0}{2}\right)$$

$$\int_0^1 \ln(1-x)\ln(x)\ln(1+x)\,dx = \frac{21}{8}\zeta(3) + \frac{5\pi^2}{12} - \ln^2 2 - \left(\frac{\pi^2}{2}-4\right)\ln 2 - 6$$

$$\int_0^\infty \frac{1}{(x^4-x^2+1)^3}\,dx = \frac{9\pi}{16}$$

$$\prod_{n=1}^{\infty}\left(1-\frac{1}{(2n+1)^2}\right)^{(-1)^n(2n+1)} = \frac{\pi}{8}e^{\frac{4G}{\pi}}$$

$$\prod_{n=2}^{\infty}\left(1-\frac{1}{n^2}\right)^{(-1)^n} = \frac{8}{\pi^2}$$

$$\prod_{n=1}^{\infty}\left(1-\frac{1}{1+n^2}\right)^{(-1)^n} = \frac{\pi}{2}\coth\left(\frac{\pi}{2}\right)$$

$$\int_0^\pi i^{\tan x}\,dx = \pi e^{-\frac{\pi}{2}}$$

$$\int_{-\infty}^\infty e^{-x^2}\cos x\,dx = \sqrt{\frac{\pi}{\sqrt{e}}}$$

$$\int_{-\infty}^\infty e^{-x^2}\cosh x\,dx = \sqrt{\pi\sqrt{e}}$$

$$\int_1^\infty \frac{1}{(x-\ln x)^2}\,dx = \sum_{n=1}^{\infty}\frac{n!}{n^n}$$

$$\int_0^1 \Gamma\left(1+\frac{x}{2}\right)\Gamma\left(1-\frac{x}{2}\right)dx = \frac{4G}{\pi}$$

$$\int_0^1 \Gamma\left(1+\frac{x}{3}\right)\Gamma\left(1-\frac{x}{3}\right)dx = \frac{5V}{\pi} - \frac{1}{2}\ln 3$$

$$\int_0^1 \Gamma\left(1+\frac{2x}{3}\right)\Gamma\left(1-\frac{2x}{3}\right)dx = \frac{5V}{2\pi} + \frac{1}{2}\ln 3$$

$$\zeta(2k) = \frac{(2\pi)^{2k}|B_{2k}|}{2(2k)!} \quad (k \in \mathbb{Z}^+)$$

$$\beta(2k+1) = \frac{\pi^{2k+1}|E_{2k}|}{2^{2k+2}(2k)!} \quad (k \in \mathbb{N})$$

$$\int_0^1 x \ln\Gamma(x)\, dx = \frac{1}{4}\ln(2\pi) - \ln A$$

$$\int_0^1 \ln G(x)\, dx = \frac{1}{12} - \frac{1}{4}\ln(2\pi) - 2\ln A$$

$$\int_0^1 \ln K(x)\, dx = \ln A - \frac{1}{12}$$

$$\sum_{n=0}^{\infty} \frac{(-2)^n}{n!}\Gamma\left(\frac{n}{2}+\alpha\right)\Gamma\left(\frac{n+1}{2}-\alpha\right) = 2\pi^{\frac{3}{2}}\csc(2\pi\alpha)$$

$$\sum_{n=0}^{\infty} \frac{(-2)^n}{n!}\Gamma^2\left(\frac{2n+1}{4}\right)\psi\left(\frac{2n+1}{4}\right) = -2\pi^{\frac{3}{2}}(6\ln 2+\gamma)$$

$$\sum_{n=0}^{\infty} \frac{(-2)^n}{n!}\Gamma^2\left(\frac{4n+1}{8}\right)\psi\left(\frac{4n+1}{8}\right) = -\frac{2}{\sqrt{\pi}}\Gamma^3\left(\frac{1}{4}\right)\left(\frac{3\pi}{2}+5\ln 2+\gamma\right)$$

$$\int_{-\infty}^{\infty} \frac{\sin x}{x(x^2+1)}\, dx = \pi\left(1-\frac{1}{e}\right)$$

$$\int_{-\infty}^{\infty} \frac{\sin x}{x(x^4+5x^2+4)}\, dx = \frac{\pi}{3e}\left(\frac{1}{4e}-1\right) + \frac{\pi}{4}$$

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^4+5x^2+4}\, dx = \frac{\pi}{3e}\left(1-\frac{1}{2e}\right)$$

$$\int_0^{\infty} \frac{\ln x}{x^4-1}\, dx = \frac{\pi^2}{8}$$

$$\int_0^{\infty} \frac{x^n}{e^x+1}\, dx = \Gamma(n+1)\zeta(n+1)(1-2^{-n})$$

$$\int_0^{\infty} \frac{x^n}{e^x-1}\, dx = \Gamma(n+1)\zeta(n+1)$$

$$\int_{-\infty}^{\infty} \prod_{n=1}^{\alpha} \frac{1}{x^2+n^2}\,dx = \pi \frac{(\alpha-1)!}{(2\alpha-1)!}$$

$$\int_0^{\infty} \ln(x) \prod_{n=1}^{\alpha} \frac{1}{x^2+n^2}\,dx = \pi \sum_{k=1}^{\alpha} \frac{(-1)^{k-1} k \ln k}{(\alpha-k)!(\alpha+k)!}$$

$$\int_0^{\infty} \frac{\ln x}{x^2-1} \prod_{n=1}^{\alpha} \frac{1}{x^2+n^2}\,dx = -\pi \sum_{k=1}^{\alpha} \frac{(-1)^{k-1} k \ln k}{(k^2+1)(\alpha-k)!(\alpha+k)!} + \frac{\pi^2}{4} \prod_{n=1}^{\alpha} \frac{1}{n^2+1}$$

$$\int_0^{\infty} \frac{\arctan x}{x(x^2+1)} \ln\left(\frac{1+x^2}{x^2}\right) dx = \frac{\pi^3}{24} + \frac{\pi}{2} \ln^2 2$$

$$\int_0^{\infty} \frac{x \sin x + 2\cos x + 1}{2x \sin x + x^2 + 1}\,dx = \frac{\pi}{\Omega+1}$$

$$\int_0^{\infty} \frac{x \sin x + 1}{2x \sin x + x^2 + 1}\,dx = \pi - \frac{\pi}{\Omega+1}$$

$$\int_0^{\infty} \frac{\cos x}{2x \sin x + x^2 + 1}\,dx = \frac{\pi}{\Omega+1} - \frac{\pi}{2}$$

$$\int_0^{\infty} \frac{x \sin x - x^2 \cos x + 1}{(x^2+1)(2x \sin x + x^2 + 1)}\,dx = \frac{\pi\Omega}{1-\Omega^2} - \frac{\pi}{e-1}$$

$$\prod_{n=2}^{\infty} \left(1 + \frac{1}{1-n^2}\right)^{(-1)^n} = -\frac{\pi}{4\sqrt{2}} \tan\left(\frac{\pi}{\sqrt{2}}\right)$$

$$\prod_{n=1}^{\infty} \left(1 + \frac{1}{1+n^2}\right)^{(-1)^n} = \frac{\tanh\left(\frac{\pi}{\sqrt{2}}\right)}{\sqrt{2}\tanh\left(\frac{\pi}{2}\right)}$$

$$\int_{-\infty}^{\infty} \frac{e^{\cos x} \cos(\sin x)}{x^2+1}\,dx = \pi e^{\frac{1}{e}}$$

2 February

2.1 x^x and variants

Different integrals for x^x (First four are in order of Desmos computational speed and accuracy for low number of terms). First two were found from working out, third was using a known derivation of the integral using the method of exhaustion, and fourth was using the definition of the integral using Riemann summation.

$$\int x^x \, dx = \sum_{n=1}^{\infty} \sum_{k=0}^{n-1} \frac{-x^n \ln^k x}{(-n)^{n-k} k!} + C$$

$$\int x^x \, dx = \sum_{n=1}^{\infty} \left(\sum_{k=0}^{\infty} \frac{n^{k+1} (\ln x)^{n+k}}{k! n! (n+k)} - (-n)^{-n} \right) + C$$

$$\int x^x \, dx = \sum_{n=1}^{\infty} \sum_{k=1}^{2^n-1} \frac{x(2^{-n}kx)^{2^{-n}kx}}{(-1)^{k+1} 2^n} + C$$

$$\int x^x \, dx = \lim_{n \to \infty} \frac{x}{n} \sum_{k=1}^{n} \left(\frac{xk}{n} \right)^{\frac{xk}{n}} + C$$

$$\int_1^b x^x \, dx = \sum_{n=1}^{\infty} \sum_{k=0}^{\infty} \frac{n^{k+1} (\ln b)^{n+k}}{k! n! (n+k)}$$

$$\therefore \int_1^e x^x \, dx = \sum_{n=1}^{\infty} \sum_{k=0}^{\infty} \frac{n^{k+1}}{k! n! (n+k)}$$

$$\int_0^1 x^x \, dx = -\sum_{n=1}^{\infty} (-n)^{-n}$$

$$\int_0^1 x^{-x} \, dx = \sum_{n=1}^{\infty} n^{-n}$$

My derivation for the best series expression of

$$\int x^x \, dx$$

$$= \int_0^x t^t \, dt$$

$$= \int_0^x e^{t \ln t} \, dt$$

Using the Maclaurin series for e^x,

$$= \int_0^x \sum_{n=0}^{\infty} \frac{(t \ln t)^n}{n!} \, dt$$

$$= \sum_{n=0}^{\infty} \frac{1}{n!} \int_0^x (t \ln t)^n \, dt$$

$$t = e^{\frac{-u}{n+1}}, dt = \frac{-e^{\frac{-u}{n+1}}}{n+1} \, du$$

$$\therefore \sum_{n=0}^{\infty} \frac{1}{n!} \int_0^x (t \ln t)^n \, dt = \sum_{n=0}^{\infty} \frac{1}{n!} \int_{\infty}^{-(n+1)\ln x} \left(e^{\frac{-u}{n+1}}\right)^{n+1} \left(\frac{-u}{n+1}\right)^n \left(\frac{-1}{n+1}\right) du$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(n+1)^{n+1}} \int_{-(n+1)\ln x}^{\infty} e^{-u} u^n \, du$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(n-1)! n^n} \int_{-n \ln x}^{\infty} e^{-u} u^{n-1} \, du$$

For readability purposes, let $-n \ln x = a$. For now, just consider the integral above.

$$\int_a^{\infty} e^{-u} u^{n-1} \, du$$

IBP with $u = u^{n-1}, dv = e^{-u}$

$$= e^{-a} a^{n-1} + (n-1) \int_a^{\infty} e^{-u} u^{n-2} \, du$$

Using IBP again for second time:

$$= e^{-a} a^{n-1} + e^{-a} a^{n-2}(n-1) + (n-1)(n-2) \int_a^{\infty} e^{-u} u^{n-3} \, du$$

Using IBP again for third time:

$$= e^{-a} a^{n-1} + e^{-a} a^{n-2}(n-1) + e^{-a} a^{n-3}(n-1)(n-2) + (n-1)(n-2)(n-3) \int_a^{\infty} e^{-u} u^{n-4} \, du$$

Thus, after performing IBP a total of $n-1$ times, we get

$$= e^{-a} a^{n-1} + e^{-a} a^{n-2}(n-1) + \cdots + e^{-a} a^2 \frac{(n-1)!}{2} + e^{-a} a(n-1)! + (n-1)! \int_a^{\infty} e^{-u} \, du$$

$$= e^{-a} a^{n-1} + e^{-a} a^{n-2}(n-1) + \cdots + e^{-a} a^2 \frac{(n-1)!}{2} + e^{-a} a(n-1)! + e^{-a}(n-1)!$$

Re-arranging back to front, and rewriting:

$$= e^{-a}(n-1)! + e^{-a} a(n-1)! + e^{-a} a^2 \frac{(n-1)!}{2} + \cdots + e^{-a} a^{n-2} \frac{(n-1)!}{(n-2)!} + e^{-a} a^{n-1}$$

$$= \frac{e^{-a}(n-1)!}{0!} + \frac{e^{-a}a(n-1)!}{1!} + \frac{e^{-a}a^2(n-1)!}{2!} + \cdots + \frac{e^{-a}a^{n-2}(n-1)!}{(n-2)!} + \frac{e^{-a}a^{n-1}(n-1)!}{(n-1)!}$$

$$= e^{-a}(n-1)!\left(\frac{a^0}{0!} + \frac{a^1}{1!} + \frac{a^2}{2!} + \cdots + \frac{a^{n-2}}{(n-2)!} + \frac{a^{n-1}}{(n-1)!}\right)$$

$$= e^{-a}(n-1)!\sum_{k=0}^{n-1}\frac{a^k}{k!}$$

This sum is also known as the incomplete gamma function. As $a_1 = -n\ln x$,

$$e^{-a}(n-1)!\sum_{k=0}^{n-1}\frac{a^k}{k!} = e^{n\ln x}(n-1)!\sum_{k=0}^{n-1}\frac{(-n\ln x)^k}{k!}$$

$$= x^n(n-1)!\sum_{k=0}^{n-1}\frac{(-n\ln x)^k}{k!}$$

Using this result,

$$\therefore \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{(n-1)!n^n}\int_{-n\ln x}^{\infty}e^{-u}u^{n-1}\,du = \sum_{n=1}^{\infty}\frac{(-1)^{n-1}x^n(n-1)!}{(n-1)!n^n}\sum_{k=0}^{n-1}\frac{(-n\ln x)^k}{k!}$$

$$= \sum_{n=1}^{\infty}\sum_{k=0}^{n-1}\frac{(-1)^{n-1}x^n(-n\ln x)^k}{n^n k!}$$

$$= \sum_{n=1}^{\infty}\sum_{k=0}^{n-1}\frac{(-1)^{n+k-1}x^n n^k\ln^k x}{n^n k!}$$

For all values of n and k, $(-1)^{n+k-1} = (-1)^{k-n-1}$

$$= \sum_{n=1}^{\infty}\sum_{k=0}^{n-1}\frac{(-1)^{k-n-1}x^n\ln^k x}{n^{n-k}k!}$$

$$= \sum_{n=1}^{\infty}\sum_{k=0}^{n-1}\frac{-x^n\ln^k x}{(-n)^{n-k}k!}$$

Thus arriving at our solution.

$$\therefore \int x^x\,dx = \sum_{n=1}^{\infty}\sum_{k=0}^{n-1}\frac{-x^n\ln^k x}{(-n)^{n-k}k!} + C$$

Similarly, we can also get

$$\int x^{-x}\,dx = \sum_{n=1}^{\infty}\sum_{k=0}^{n-1}\frac{(-x)^n\ln^k x}{(-n)^{n-k}k!} + C$$

Together, these results allow us to obtain the Sophomore integrals

$$\int_0^1 x^x\,dx = -\sum_{n=1}^{\infty}(-n)^{-n}$$

$$\int_0^1 x^{-x}\,dx = \sum_{n=1}^{\infty}n^{-n}$$

2.2 Five integrals solved via DUIS

Below is a well-known problem with its worked out solution. Solved myself with the prior knowledge (from online sources) of the general method.

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x}{\tan x}\,dx$$

First, consider the following.

$$I(k) = \int_0^{\frac{\pi}{2}} \frac{\arctan\left(k\tan x\right)}{\tan x}\,dx$$

Since the graph of $y = x/\tan x$ is symmetric around the y-axis between $x = -\pi/2$ and $x = \pi/2$, our problem therefore is to solve $2 \times I(1)$. To do so, we first take the derivative of $I(k)$ with respect to k, and solve.

$$\therefore I'(k) = \int_0^{\frac{\pi}{2}} \frac{1}{1+k^2\tan^2 x}\,dx = \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{(1+\tan^2 x)(1+k^2\tan^2 x)}\,dx$$

$$u = \tan x, du = \sec^2 x\,dx$$

$$= \int_0^{\infty} \frac{1}{(1+u^2)(1+k^2u^2)}\,du = \int_0^{\infty} \frac{A(1+k^2u^2)+B(1+u^2)}{(1+u^2)(1+k^2u^2)}\,du$$

$$\therefore A + Ak^2u^2 + B + Bu^2 = 1$$

$$A = \frac{-1}{k^2-1}, B = \frac{k^2}{k^2-1}$$

$$\therefore \int_0^{\infty} \frac{1}{(1+u^2)(1+k^2u^2)}\,du = \int_0^{\infty} \frac{k^2}{(k^2-1)(1+k^2u^2)} - \frac{1}{(k^2-1)(1+u^2)}\,du$$

$$= \frac{1}{k^2-1}\left(k^2\int_0^{\infty}\frac{1}{1+k^2u^2}\,du - \int_0^{\infty}\frac{1}{1+u^2}\,du\right)$$

$$= \frac{\pi}{2}\left(\frac{k-1}{k^2-1}\right) = \frac{\pi}{2}\left(\frac{1}{k+1}\right)$$

Thus, to find $I(k)$, we can simply now integrate this result with respect to k.

$$\therefore I(k) = \int I'(k)\,dk = \frac{\pi}{2}\int \frac{1}{k+1}\,dk$$

$$= \frac{\pi}{2}\ln(|k+1|) + C$$

Substituting $k = 0$

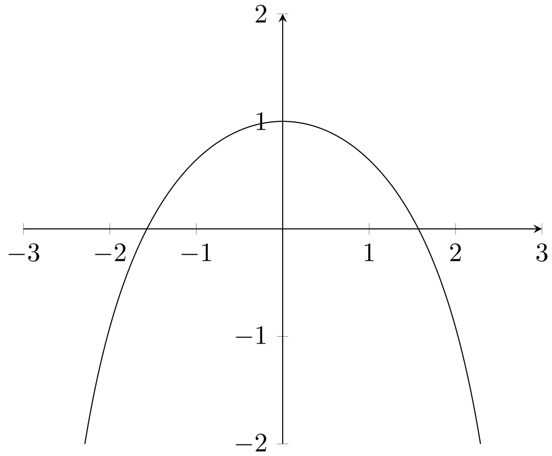
$$I(0) = \int_0^{\frac{\pi}{2}} 0\,dx = 0 = C$$

$$\therefore I(k) = \int_0^{\frac{\pi}{2}} \frac{\arctan(k\tan x)}{\tan x}\,dx = \frac{\pi}{2}\ln(|k+1|)$$

We can now finally substitute $k = 1$ to obtain our result.

$$\therefore I(1) = \int_0^{\frac{\pi}{2}} \frac{x}{\tan x}\,dx = \frac{\pi}{2}\ln 2$$

$$\therefore \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x}{\tan x}\,dx = \pi\ln 2$$


2
1
−3
−2
−1
1
2
3
−1
−2

This one is very well known and appears across the internet. Solved myself with the prior knowledge from online sources of the general method.

$$\int_{-\infty}^{\infty} \frac{\sin x}{x}\, dx$$

$$= 2\int_{0}^{\infty} \frac{\sin x}{x}\, dx$$

First, consider the following.

$$I(k) = \int_{0}^{\infty} e^{-kx}\frac{\sin x}{x}\, dx$$

Our problem is thus to solve for $I(0)$. We first take the derivative of $I(k)$ with respect to k, and solve using IBP twice.

$$I'(k) = \int_{0}^{\infty} -e^{-kx}\sin x\, dx$$

$$\text{IBP with } u = -e^{-kx}, dv = \sin x$$

$$= -1 + k\int_{0}^{\infty} e^{-kx}\cos x\, dx$$

$$\text{(IBP again)}$$

$$= -1 - k^2\int_{0}^{\infty} -e^{-kx}\sin x\, dx$$

$$= -1 - k^2 I'(k)$$

$$\therefore I'(k)(1+k^2) = -1$$

$$\therefore I'(k) = -\frac{1}{1+k^2}$$

In order to find $I(k)$, we can now simply integrate with respect to k.

$$\therefore I(k) = -\int \frac{1}{1+k^2}\, dk = -\arctan(k) + C$$

$$\text{As } k \to \infty, \arctan(k) \to \frac{\pi}{2}$$

$$\lim_{k\to\infty} I(k) = \int_{0}^{\infty} 0\, dx = 0 = -\frac{\pi}{2} + C$$

$$\therefore C = \frac{\pi}{2}$$

We can now set $k = 0$, and obtain our result

$$\therefore I(0) = \int_{0}^{\infty} \frac{\sin x}{x}\, dx = \arctan(0) + \frac{\pi}{2} = \frac{\pi}{2}$$

$$\therefore \int_{-\infty}^{\infty} \frac{\sin x}{x}\, dx = \pi$$

This is a crazy problem. We could do this one quite simply using complex methods, but I want to show off this real-only method I found online [1].

$$\int_{-\infty}^{\infty} \frac{\cos x}{1+x^2}\,dx$$

First consider

$$I(k) = \int_{-\infty}^{\infty} \frac{\cos(kx)}{1+x^2}\,dx$$

Our problem is thus to solve for $I(1)$. Note that as we are purely concerned with $k=1$, this solution omits the use of any absolute value symbols until the final line.

$$\text{IBP with } u = \frac{1}{1+x^2}, dv = \cos(kx)$$

$$\therefore I(k) = \frac{2}{k}\int_{-\infty}^{\infty} \frac{x\sin(kx)}{(1+x^2)^2}\,dx$$

$$\therefore kI(k) = 2\int_{-\infty}^{\infty} \frac{x\sin(kx)}{(1+x^2)^2}\,dx$$

$$\therefore kI'(k) + I(k) = 2\int_{-\infty}^{\infty} \frac{x^2\cos(kx)}{(1+x^2)^2}\,dx$$

$$\int_{-\infty}^{\infty} \frac{x^2\cos(kx)}{(1+x^2)^2}\,dx = \int_{-\infty}^{\infty} \frac{(Ax+B)(1+x^2)+(Cx+D)}{(1+x^2)^2}\,dx$$

$$Ax + Ax^3 + B + Bx^2 + Cx + D = x^2\cos(kx)$$

$$\therefore A = 0, B = \cos(kx), C = 0, D = -\cos(kx)$$

$$\therefore 2\int_{-\infty}^{\infty} \frac{x^2\cos(kx)}{(1+x^2)^2}\,dx = 2\bigg(\int_{-\infty}^{\infty} \frac{\cos(kx)}{1+x^2}\,dx - \int_{-\infty}^{\infty} \frac{\cos(kx)}{(1+x^2)^2}\,dx\bigg)$$

$$= 2I(k) - 2\int_{-\infty}^{\infty} \frac{\cos(kx)}{(1+x^2)^2}\,dx$$

$$\therefore kI'(k) - I(k) = -2\int_{-\infty}^{\infty} \frac{\cos(kx)}{(1+x^2)^2}\,dx$$

$$\therefore kI''(k) = 2\int_{-\infty}^{\infty} \frac{x\sin(kx)}{(1+x^2)^2}\,dx = kI(k)$$

Thus, we can obtain the following differential equation.

$$I''(k) - I(k) = 0$$

To solve, let $I(k) = ce^{\lambda k}$

$$\therefore \lambda^2 ce^{\lambda k} - ce^{\lambda k} = 0$$
$$\therefore ce^{\lambda k}(\lambda^2 - 1) = 0$$
$$\therefore \lambda = \pm 1$$

Thus, we can let

$$I(k) = c_1e^{k} + c_2e^{-k}$$
$$I'(k) = c_1e^{k} - c_2e^{-k}$$

Now, to find c_1 and c_2, we will set $k = 0$.

$$I(0) = \int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \pi = c_1 + c_2$$

We must now find another equation for c_1 and c_2 in order to find their values. To do so, consider $I'(k)$.

$$\begin{aligned}
I'(k) &= -\int_{-\infty}^{\infty} \frac{x\sin(kx)}{1+x^2}\,dx \\
&= -\int_{-\infty}^{\infty} \frac{(x^2+1-1)\sin(kx)}{x(1+x^2)}\,dx \\
&= -\int_{-\infty}^{\infty} \frac{(x^2+1)\sin(kx)}{x(1+x^2)}\,dx + \int_{-\infty}^{\infty} \frac{\sin(kx)}{x(1+x^2)}\,dx \\
&= \int_{-\infty}^{\infty} \frac{\sin(kx)}{x(1+x^2)}\,dx - \int_{-\infty}^{\infty} \frac{\sin(kx)}{x}\,dx
\end{aligned}$$

Now consider

$$\int_{-\infty}^{\infty} \frac{\sin(kx)}{x}\,dx$$

$$u = kx, du = k\,dx$$

As $k = 1$ in our original problem, we do not need to change our boundaries.

$$= \int_{-\infty}^{\infty} \frac{k\sin u}{ku}\,du = \int_{-\infty}^{\infty} \frac{\sin u}{u}\,du = \pi$$

As shown on page 21. Thus,

$$I'(k) = \int_{-\infty}^{\infty} \frac{\sin(kx)}{x(1+x^2)}\,dx - \pi$$

We can now set $k = 0$ for $I'(k)$, in order to gain another equation for c_1 and c_2,

$$I'(0) = \int_{-\infty}^{\infty} 0\,dx - \pi = -\pi = c_1 - c_2$$

Therefore, using the information that

$$\begin{aligned}
c_1 + c_2 &= \pi, \\
c_1 - c_2 &= -\pi,
\end{aligned}$$

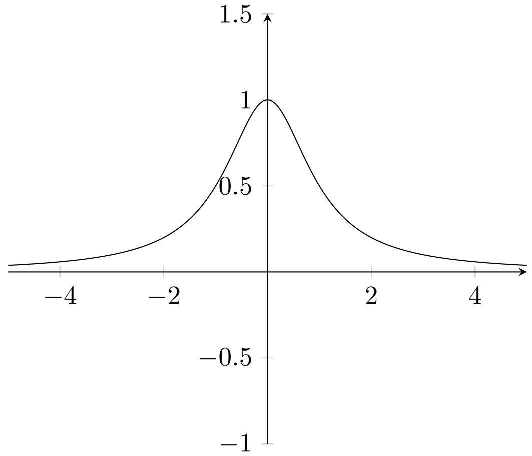
we can determine that

$$c_1 = 0, c_2 = \pi$$

Finally, we can substitute this result into our result for the differential equation, and solve for $I(1)$.

$$I(k) = \int_{-\infty}^{\infty} \frac{\cos(kx)}{1+x^2}\,dx = \frac{\pi}{e^{|k|}}$$

$$\therefore \int_{-\infty}^{\infty} \frac{\cos x}{1+x^2}\,dx = \frac{\pi}{e}$$


1.5
1
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−2
2
4
−0.5
−1

$$\int_0^1 \frac{\ln(x+1)}{x^2+1}\,dx$$

This integral can be quite easily solved using u-sub, but let's do it via Feynman's. Consider

$$I(k) = \int_0^1 \frac{\ln(kx+1)}{x^2+1}\,dx$$

Our problem is thus to solve for $I(1)$. The differentiation here is quite trivial.

$$I'(k) = \int_0^1 \frac{x}{(kx+1)(x^2+1)}\,dx$$

$$\int_0^1 \frac{x}{(kx+1)(x^2+1)}\,dx = \int_0^1 \frac{A(x^2+1)+(Bx+C)(kx+1)}{(kx+1)(x^2+1)}\,dx$$

$$Ax^2 + A + Bkx^2 + Bx + Ckx + C = x$$

$$\therefore A + C = 0, A + Bk = 0, B + Ck = 1$$

$$\therefore A = \frac{-k}{k^2+1}, B = \frac{1}{k^2+1}, C = \frac{k}{k^2+1}$$

$$\therefore \int_0^1 \frac{x}{(kx+1)(x^2+1)}\,dx = \int_0^1 \frac{x+k}{(k^2+1)(x^2+1)}\,dx - \int_0^1 \frac{k}{(k^2+1)(kx+1)}\,dx$$

$$= \int_0^1 \frac{x}{(k^2+1)(x^2+1)}\,dx + \frac{k\pi}{4(k^2+1)} - \frac{\ln(k+1)}{k^2+1}$$

$$= \frac{1}{k^2+1}\left(\int_0^1 \frac{x}{x^2+1}\,dx + \frac{k\pi}{4} - \ln(k+1)\right)$$

Consider

$$\int_0^1 \frac{x}{x^2+1}\,dx$$

$$u = x^2+1, du = 2x\,dx$$

$$= \frac{1}{2}\int_0^1 \frac{1}{u}\,dx = \frac{\ln 2}{2}$$

$$\therefore I'(k) = \frac{1}{k^2+1}\left(\frac{\ln 2}{2} + \frac{k\pi}{4} - \ln(k+1)\right)$$

$$= \frac{\ln 2}{2(k^2+1)} + \frac{k\pi}{4(k^2+1)} - \frac{\ln(k+1)}{k^2+1}$$

$$\therefore I(1) = \frac{\ln 2}{2}\int_0^1 \frac{1}{k^2+1}\,dk + \frac{\pi}{4}\int_0^1 \frac{k}{k^2+1}\,dk - \int_0^1 \frac{\ln(k+1)}{k^2+1}\,dk$$

$$= \frac{\pi \ln 2}{8} + \frac{\pi \ln 2}{8} - I(1)$$

$$\therefore 2I(1) = \frac{\pi}{4}\ln 2$$

$$\therefore I(1) = \int_0^1 \frac{\ln(x+1)}{x^2+1}\,dx = \frac{\pi}{8}\ln 2$$

Below is the u-sub solution btw

$$I = \int_0^1 \frac{\ln(x+1)}{x^2+1}\,dx$$

$$x = \frac{1-u}{1+u}, dx = \frac{-2}{(1+u)^2}\,du$$

$$= \int_1^0 \frac{-2\ln(\frac{1-u}{1+u}+1)}{\left(\frac{(1-u)^2}{((1+u)^2}+1\right)(1+u)^2}\,du$$

$$= 2\int_0^1 \frac{\ln(2)-\ln(1+u)}{(1-u)^2+(1+u^2)}\,du$$

$$\int_0^1 \frac{\ln(2)-\ln(1+u)}{u^2+1}\,du$$

$$= \ln(2)\int_0^1 \frac{1}{u^2+1}\,du - I$$

$$\therefore 2I = \frac{\pi}{4}\ln 2$$

$$\therefore I = \int_0^1 \frac{\ln(x+1)}{x^2+1}\,dx = \frac{\pi}{8}\ln 2$$

$$\int_0^\infty \frac{\ln(x^2+1)}{x^2+1}\,dx$$

Like the previous example, consider

$$I(k) = \int_0^\infty \frac{\ln(k^2x^2+1)}{x^2+1}\,dx$$

Our problem is thus to solve for $I(1)$. Again, the differentiation is quite simple.

$$I'(k) = \int_0^\infty \frac{2kx^2}{(k^2x^2+1)(x^2+1)}\,dx$$

$$\int_0^\infty \frac{2kx^2}{(k^2x^2+1)(x^2+1)}\,dx = \int_0^\infty \frac{(Ax+B)(x^2+1)+(Cx+D)(k^2x^2+1)}{(k^2x^2+1)(x^2+1)}\,dx$$

$$(Ax+B)(x^2+1)+(Cx+D)(k^2x^2+1) = 2kx^2$$

$$\therefore A+Ck^2 = 0, B+Dk^2 = 2k, A+C = 0, B+D = 0$$

$$\therefore A = 0, B = \frac{-2k}{k^2-1}, C = 0, D = \frac{2k}{k^2-1}$$

$$\therefore \int_0^\infty \frac{x^2}{(kx^2+1)(x^2+1)}\,dx = \int_0^\infty \frac{-2k}{(k^2-1)(k^2x^2+1)}\,dx + \int_0^\infty \frac{2k}{(k^2-1)(x^2+1)}\,dx$$

$$= \frac{2k}{k^2-1}\left(\int_0^\infty \frac{1}{x^2+1}\,dx - \int_0^\infty \frac{1}{k^2x^2+1}\,dx\right)$$

$$= \frac{2k}{k^2-1}\left(\frac{\pi}{2} - \frac{\pi}{2k}\right)$$

$$= \frac{\pi(k-1)}{k^2-1} = \frac{\pi}{k+1}$$

$$\therefore I(k) = \pi\int \frac{1}{k+1}\,dk$$

$$= \pi\ln(|k+1|) + C$$

Letting $k = 0$

$$I(0) = \int_0^\infty 0\,dx = 0 = C$$

$$\therefore I(k) = \int_0^\infty \frac{\ln(k^2x^2+1)}{x^2+1}\,dx = \pi\ln(|k+1|)$$

Finally, we may let $k = 1$ in order to obtain our solution.

$$\therefore \int_0^\infty \frac{\ln(x^2+1)}{x^2+1}\,dx = \pi\ln 2$$

3 March

3.1 Two piecewise integrals

$$\int_2^\infty \frac{(-1)^{\lfloor x \rfloor}}{\lfloor x-2 \rfloor! \lceil x-2 \rceil x^2}\, dx$$

To solve, we consider this piecewise function in integer intervals $[n, n+1]$, and evaluate the floors and ceilings for each interval, summing them together.

$$\therefore \int_2^\infty \frac{(-1)^{\lfloor x \rfloor}}{\lfloor x-2 \rfloor! \lceil x-2 \rceil x^2}\, dx$$

$$= \int_2^3 \frac{1}{x^2}\, dx + \int_3^4 \frac{-1}{2x^2}\, dx + \int_4^5 \frac{1}{6x^2}\, dx + \cdots + \int_n^{n+1} \frac{(-1)^n}{(n-1)!x^2}\, dx$$

$$= \sum_{n=2}^\infty \int_n^{n+1} \frac{(-1)^n}{(n-1)!x^2}\, dx$$

$$= \sum_{n=2}^\infty \frac{(-1)^n}{(n-1)!}\left(\frac{1}{n} - \frac{1}{n+1}\right) = \sum_{n=2}^\infty \frac{(-1)^n}{(n-1)!}\left(\frac{(n+1)-n}{n(n+1)}\right)$$

$$= \sum_{n=2}^\infty \frac{(-1)^n}{n(n+1)(n-1)!} = \sum_{n=2}^\infty \frac{(-1)^n}{(n+1)!} = -\sum_{n=3}^\infty \frac{(-1)^n}{n!}$$

$$-\sum_{n=0}^\infty \frac{(-1)^n}{n!} - \frac{(-1)^1}{1!} - \frac{(-1)^2}{2!}$$

$$= -\frac{1}{e} + 1 - \frac{1}{2}$$

$$= \frac{1}{2} - \frac{1}{e}$$

$$\therefore \int_2^\infty \frac{(-1)^{\lfloor x \rfloor}}{\lfloor x-2 \rfloor! \lceil x-2 \rceil x^2}\, dx = \frac{1}{2} - \frac{1}{e}$$

$$\int_0^1 \left\lfloor \frac{1}{x} \right\rfloor^{-1} dx$$

As $\lfloor x \rfloor = n$ for $n \le x < n+1$, $\left\lfloor \frac{1}{x} \right\rfloor = n$ for $\frac{1}{n+1} \le x < \frac{1}{n}$.

$$\therefore \int_0^1 \left\lfloor \frac{1}{x} \right\rfloor^{-1} dx = \int_{\frac{1}{2}}^{1} 1\, dx + \int_{\frac{1}{3}}^{\frac{1}{2}} \frac{1}{2}\, dx + \int_{\frac{1}{4}}^{\frac{1}{3}} \frac{1}{3}\, dx + \cdots + \int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{1}{n}\, dx$$

$$= \sum_{n=1}^{\infty} \int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{1}{n}\, dx$$

$$= \sum_{n=1}^{\infty} \left(\frac{1}{n^2} - \frac{1}{n(n+1)} \right) = \sum_{n=1}^{\infty} \frac{1}{n^2} - \sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

$$= \sum_{n=1}^{\infty} \frac{1}{n^2} - \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right) = \sum_{n=1}^{\infty} \frac{1}{n^2} - 1$$

The sum of the reciprocals of the square integers is known as the Basel problem, also known as $\zeta(2)$, yielding this solution.

$$= \frac{\pi^2}{6} - 1$$

$$\therefore \int_0^1 \left\lfloor \frac{1}{x} \right\rfloor^{-1} dx = \frac{\pi^2}{6} - 1$$

The Basel problem can be solved via equating all terms x^3 and less of the Maclaurin series and the product for $\sin x$, then re-arranging.

3.2 My own collection of integrals (1)

25/3/25 Spent a number of hours on these

$$\int_0^\infty \frac{\cosh x}{\lfloor e^x \rfloor !}\, dx = \frac{1}{2}\left(\sum_{n=1}^{\infty} \frac{1}{n!n} + 1\right) = \frac{1}{2}(\mathrm{Ei}(1) - \gamma + 1)$$

$$\int_0^\infty \frac{\sinh x}{\lfloor e^x \rfloor !}\, dx = \frac{1}{2}\left(2e - 3 - \sum_{n=1}^{\infty} \frac{1}{n!n}\right) = \frac{1}{2}(2e + \gamma - 3 - \mathrm{Ei}(1))$$

And finally (for completion):

$$\int_0^\infty \frac{\tanh x}{\lfloor e^x \rfloor !}\, dx = \sum_{n=1}^{\infty} \frac{1}{n!} \ln\left(\frac{n(n(n+2)+2)}{n(n(n+1)+1)+1}\right)$$

Interestingly, these integrals are also true.

$$\int_0^\infty \frac{\cosh x}{\lfloor e^x + 1 \rfloor !}\, dx = \int_0^\infty \frac{\cosh x}{\lceil e^x \rceil !}\, dx = \frac{1}{2}$$

$$\int_0^\infty \frac{\cosh x}{\lfloor e^x - 1 \rfloor !}\, dx = e - 1$$

$$\int_0^\infty \frac{\cosh x}{\lfloor e^x - 2 \rfloor !}\, dx = \frac{3}{2}$$

$$\int_0^\infty \frac{\sinh x}{\lfloor e^x + 1 \rfloor !}\, dx = e - \frac{5}{2}$$

$$\int_0^\infty \frac{\sinh x}{\lfloor e^x - 1 \rfloor !}\, dx = 1$$

$$\int_0^\infty \frac{\sinh x}{\lfloor e^x - 2 \rfloor !}\, dx = e - \frac{3}{2}$$

There seems to be a pattern between the value being added/subtracted from e^x in the floor function, but it is not well expressed.

Working out for first integral (multiplied by two for nicer solution):

$$2\int_0^\infty \frac{\cosh x}{\lfloor e^x \rfloor!}\,dx$$

As $\lfloor x \rfloor = n$ for $n \leq x < n+1$, $\lfloor e^x \rfloor = n$ for $\ln(n) \leq x < \ln(n+1)$. Further, for $\ln(n)$ to be defined, n must start from 1.

$$\therefore \int_0^\infty \frac{2\cosh x}{\lfloor e^x \rfloor!}\,dx = 2\sum_{n=1}^{\infty}\int_{\ln(n)}^{\ln(n+1)} \frac{\cosh x}{n!}\,dx$$

$$= 2\sum_{n=1}^{\infty}\frac{1}{n!}\int_{\ln(n)}^{\ln(n+1)} \cosh x\,dx = 2\sum_{n=1}^{\infty}\frac{\sinh(\ln(n+1)) - \sinh(\ln(n))}{n!}$$

$$= \sum_{n=1}^{\infty}\frac{e^{\ln(n+1)} - e^{-\ln(n+1)} - e^{\ln(n)} + e^{-\ln(n)}}{n!}$$

$$= \sum_{n=1}^{\infty}\frac{n+1-(n+1)^{-1} - n + n^{-1}}{n!}$$

$$= \sum_{n=1}^{\infty}\frac{1}{n!} - \sum_{n=1}^{\infty}\frac{1}{n!(n+1)} + \sum_{n=1}^{\infty}\frac{1}{n!n}$$

$$= e - 1 - e + 2 + \sum_{n=1}^{\infty}\frac{1}{n!n}$$

$$= 1 + \sum_{n=1}^{\infty}\frac{1}{n!n}$$

$$= \mathrm{Ei}(1) - \gamma + 1$$

3.3 Nested integral of differing roots

$$\lim_{n\to\infty}\int_0^1 x\sqrt{x\sqrt[3]{x\sqrt[4]{x\cdots\sqrt[n]{x}}}}\,dx$$

$$=\lim_{n\to\infty}\int_0^1 x(x(x(x(\cdots(x)^{\frac{1}{n}})^{\frac{1}{4}})^{\frac{1}{3}})^{\frac{1}{2}}\,dx$$

$$=\lim_{n\to\infty}\int_0^1 x(x)^{\frac{1}{2}}(x)^{\frac{1}{6}}(x)^{\frac{1}{24}}\cdots(x)^{\frac{1}{n!}}\,dx$$

$$=\lim_{n\to\infty}\int_0^1 x^{1+\frac{1}{2}+\frac{1}{6}+\frac{1}{24}+\cdots+\frac{1}{n!}}\,dx$$

$$=\int_0^1 x^{\sum_{k=1}^{\infty}\frac{1}{k!}}\,dx$$

$$\text{As }\sum_{k=0}^{\infty}\frac{1}{k!}=e$$

$$\sum_{k=1}^{\infty}\frac{1}{k!}=e-1$$

$$\therefore\int_0^1 x^{\sum_{k=1}^{\infty}\frac{1}{k!}}\,dx=\int_0^1 x^{e-1}\,dx$$

$$=\frac{1}{e}$$

$$\therefore\lim_{n\to\infty}\int_0^1 x\sqrt{x\sqrt[3]{x\sqrt[4]{x\cdots\sqrt[n]{x}}}}\,dx=\frac{1}{e}$$

3.4 Thanks Mr C!

An old one given to me by Mr C (Steps written back in 2024)

$$\int \frac{1}{4+\sin x}\,dx$$

$$u = \tan\left(\frac{x}{2}\right), du = \frac{1}{2}\sec^2\left(\frac{x}{2}\right)dx$$

$$\therefore \int \frac{1}{4+\sin x}\,dx = 2\int \frac{1}{(4+\sin(2\arctan(u))\sec^2(\arctan(u))}\,du$$

$$= 2\int \frac{1}{(4+2\sin(\arctan(u))\cos(\arctan(u)))\sec^2(\arctan(u))}\,du$$

$$= 2\int \frac{\cos^2(\arctan(u))}{4+\frac{2u}{1+u^2}}\,du = \int \frac{\frac{1}{1+u^2}}{\frac{2u^2+u+2}{1+u^2}}\,du$$

$$= \int \frac{1}{2u^2+u+2}\,du = \int \frac{1}{2\left(u+\frac{1}{4}\right)^2+\frac{15}{8}}\,du$$

$$= \frac{1}{2}\int \frac{1}{\left(u+\frac{1}{4}\right)^2+\frac{15}{16}}\,du$$

$$= \frac{2}{\sqrt{15}}\arctan\left(\frac{4u+1}{\sqrt{15}}\right)+C$$

$$= \frac{2}{\sqrt{15}}\arctan\left(\frac{4\tan(\frac{x}{2})+1}{\sqrt{15}}\right)+C$$

A few steps can be skipped by going straight to the direct results of tangent half-angle substitution.

3.5 Power of king's rule

$$I = \int_0^{\frac{\pi}{2}} \frac{\cos x}{\left(1+\sqrt{\sin 2x}\right)^2}\, dx$$

Using king's rule,

$$2I = \int_0^{\frac{\pi}{2}} \frac{\cos x + \sin x}{\left(1+\sqrt{\sin(2x)}\right)^2}\, dx$$

$$u = \sin x - \cos x, du = \sin x + \cos x\, dx$$

$$\therefore u^2 = 1 - \sin 2x$$

$$\therefore I = \frac{1}{2}\int_{-1}^{1} \frac{1}{\left(1+\sqrt{1-u^2}\right)^2}\, du$$

$$t = \sin u, dt = \cos u$$

$$= \frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos t}{\left(1+\sqrt{1-\sin^2 t}\right)^2}\, dt$$

For all values between the bounds, $\sqrt{\cos^2 t} = \cos t$

$$\therefore I = \frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos t}{\left(1+\cos t\right)^2}\, dt$$

$$x = \tan\left(\frac{t}{2}\right), dt = \frac{2}{1+x^2}\, dx$$

$$= \int_{-1}^{1} \frac{\frac{1-x^2}{1+x^2}}{\left(1+\frac{1-x^2}{1+x^2}\right)^2(1+x^2)}\, dx$$

$$= \int_{-1}^{1} \frac{\frac{1-x^2}{1+x^2}}{\frac{4}{1+x^2}}\, dx = \frac{1}{4}\int_{-1}^{1} 1 - x^2\, dx$$

$$= \frac{1}{4}\left(2 - \frac{2}{3}\right)$$

$$= \frac{1}{3}$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{\cos x}{\left(1+\sqrt{\sin 2x}\right)^2}\, dx = \frac{1}{3}$$

3.6 Well known sum expressed as beta function

$$\sum_{n=0}^{\infty} \frac{(k!)^2}{(2k+1)!}$$

This summand can be recognised as $B(k+1, k+1)$, where B is the beta function.

$$\begin{aligned}
\sum_{n=0}^{\infty} \frac{(k!)^2}{(2k+1)!} &= \sum_{n=0}^{\infty} \int_0^1 x^k (1-x)^k \, dx \\
&= \int_0^1 \sum_{n=0}^{\infty} \left(x - x^2\right)^k \, dx
\end{aligned}$$

As $|x - x^2| < 1$ for all $0 \leq x \leq 1$, we can turn the inside of the integral into the Maclaurin series for the geometric series of $x - x^2$.

$$= \int_0^1 \sum_{n=0}^{\infty} \left(x - x^2\right)^k \, dx = \int_0^1 \frac{1}{1 - (x - x^2)} \, dx$$

$$\begin{aligned}
\int_0^1 \frac{1}{x^2 - x + 1} \, dx &= \int_0^1 \frac{1}{\left(x - \frac{1}{2}\right)^2 + \frac{3}{4}} \, dx \\
&= \frac{2}{\sqrt{3}} \arctan\left(\frac{1}{\sqrt{3}}\right) - \frac{2}{\sqrt{3}} \arctan\left(\frac{-1}{\sqrt{3}}\right) \\
&= \frac{2\pi}{3\sqrt{3}}
\end{aligned}$$

$$\therefore \sum_{n=0}^{\infty} \frac{(k!)^2}{(2k+1)!} = \frac{2\pi}{3\sqrt{3}}$$

4 April

4.1 My own collection of integrals (2)

Series of related integrals (first 6 are of even functions)

$$\int_{-\infty}^{\infty} \frac{\cos(kx)}{\cosh(nx)}\, dx = \frac{\pi}{n} \operatorname{sech}\left(\frac{k\pi}{2n}\right)$$

$$\therefore \int_{-\infty}^{\infty} \frac{\cos x}{\cosh x}\, dx = \pi \operatorname{sech} \frac{\pi}{2}$$

$$\int_{0}^{\infty} \frac{\cos x}{\cosh\left(\frac{\pi x}{2}\right)}\, dx = \operatorname{sech} 1$$

$$\int_{-\infty}^{\infty} \frac{\cos^2(kx)}{\cosh(nx)}\, dx = \frac{\pi}{2n}\left(\operatorname{sech}\left(\frac{k\pi}{n}\right) + 1\right)$$

$$\therefore \int_{-\infty}^{\infty} \frac{\cos^2 x}{\cosh x}\, dx = \frac{\pi}{2}(\operatorname{sech}(\pi) + 1)$$

$$\int_{0}^{\pi} \cos\left(k \ln\left(\tan \frac{x}{2}\right)\right) dx = \pi \operatorname{sech}\left(\frac{k\pi}{2}\right)$$

$$\therefore \int_{0}^{\frac{\pi}{2}} \cos(\ln(\tan x))\, dx = \frac{\pi}{2} \operatorname{sech} \frac{\pi}{2}$$

$$\int_{0}^{\pi} x \cos\left(k \ln\left(\tan \frac{x}{2}\right)\right) dx = \frac{\pi^2}{2} \operatorname{sech}\left(\frac{k\pi}{2}\right)$$

$$\therefore \int_{0}^{\frac{\pi}{2}} x \cos(\ln(\tan x))\, dx = \frac{\pi^2}{8} \operatorname{sech} \frac{\pi}{2}$$

As of the 27th of June, I have figured out how to solve the first two batches of integrals properly (without integral expression for Euler numbers). See page 84 for a part two of this collection.

Solution for two of the problems on the above page using Euler numbers.

$$\int_{-\infty}^{\infty} \frac{\cos^2 x}{\cosh x}\, dx$$

$$= \frac{1}{2}\int_{-\infty}^{\infty} \frac{\cos(2x)+1}{\cosh x}\, dx$$

$$= \frac{1}{2}\int_{-\infty}^{\infty} \frac{\cos(2x)}{\cosh x}\, dx + \frac{1}{2}\int_{-\infty}^{\infty} \frac{1}{\cosh x}\, dx$$

On the right integral, $u = \sinh x, du = \cosh x\, dx$

$$= \frac{1}{2}\int_{-\infty}^{\infty} \frac{\cos(2x)}{\cosh x}\, dx + \frac{1}{2}\int_{-\infty}^{\infty} \frac{1}{\cosh^2(\operatorname{arcsinh}(u))}\, du$$

$$= \int_{0}^{\infty} \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n}}{(2n)!\cosh x}\, dx + \frac{1}{2}\int_{-\infty}^{\infty} \frac{1}{u^2+1}\, du$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n}}{(2n)!} \int_{0}^{\infty} \frac{x^{2n}}{\cosh x}\, dx + \frac{\pi}{2}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n}}{(2n)!} \left(\frac{\pi}{2}\right)^{2n+1} \left(\frac{2}{\pi}\right)^{2n+1} \int_{0}^{\infty} \frac{x^{2n}}{\cosh x}\, dx + \frac{\pi}{2}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n}}{(2n)!} \left(\frac{\pi}{2}\right)^{2n+1} (-1)^n E_{2n} + \frac{\pi}{2}$$

$$= \frac{\pi}{2} \sum_{n=0}^{\infty} \frac{E_{2n}\pi^{2n}}{(2n)!} + \frac{\pi}{2}$$

$$= \frac{\pi}{2}(\operatorname{sech}(\pi) + 1)$$

$$\int_{0}^{\frac{\pi}{2}} \cos(\ln(\tan x))\, dx$$

$$= \int_{0}^{\frac{\pi}{2}} \cos\Big(\ln\Big(\tan\frac{x}{2}\Big)\Big)\, dx$$

$$= \int_{0}^{\frac{\pi}{2}} \sum_{n=0}^{\infty} \frac{(-1)^n \ln^{2n}(\tan\frac{x}{2})}{(2n)!}\, dx$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \int_{0}^{\frac{\pi}{2}} \ln^{2n}\Big(\tan\frac{x}{2}\Big)\, dx$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(\frac{\pi}{2}\right)^{2n+1} \left(\frac{2}{\pi}\right)^{2n+1} \int_{0}^{\frac{\pi}{2}} \ln^{2n}\Big(\tan\frac{x}{2}\Big)\, dx$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \left(\frac{\pi}{2}\right)^{2n+1} (-1)^n E_{2n}$$

$$= \frac{\pi}{2} \sum_{n=0}^{\infty} \frac{E_{2n}\left(\frac{\pi}{2}\right)^{2n}}{(2n)!}$$

$$= \frac{\pi}{2}\operatorname{sech}\frac{\pi}{2}$$

4.2 Another DUIS integral

$$\int_0^\infty \frac{\ln x}{x^n+1}\,dx$$

$$\text{Consider } I(k) = \int_0^\infty \frac{x^k}{x^n+1}\,dx$$

$$\therefore I'(k) = \int_0^\infty \frac{x^k \ln x}{x^n+1}\,dx$$

We are thus trying to solve for $I'(0)$.

$$u = x^n, du = nx^{n-1}\,dx$$

$$\begin{aligned}\therefore I(k) &= \frac{1}{n}\int_0^\infty \frac{u^{\frac{k}{n}}}{(u+1)u^{1-\frac{1}{n}}}\,du \\ &= \frac{1}{n}\int_0^\infty \frac{u^{\frac{k+1}{n}-1}}{(u+1)}\,du \\ &= \frac{1}{n}B\left(\frac{k+1}{n}, 1-\frac{k+1}{n}\right)\end{aligned}$$

$$\text{As } B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \frac{(x-1)!(y-1)!}{(x+y-1)!}$$

$$\frac{1}{n}B\left(\frac{k+1}{n}, 1-\frac{k+1}{n}\right) = \frac{1}{n}\left(\frac{k+1}{n}-1\right)!\left(-\frac{k+1}{n}\right)!$$

$$\text{And as } \Gamma(x)\Gamma(1-x) = (x-1)!(-x)! = \pi\csc(\pi x), \text{ for } x \notin \mathbb{Z}$$

$$\frac{1}{n}\left(\frac{k+1}{n}-1\right)!\left(-\frac{k+1}{n}\right)! = \frac{\pi}{n}\csc\left(\frac{\pi(k+1)}{n}\right) \tag{1}$$

Although not the final solution, this gives an interesting result in of itself, as it means that

$$\int_0^\infty \frac{x^k}{x^n+1}\,dx = \frac{\pi}{n}\csc\left(\frac{\pi(k+1)}{n}\right)$$

$$\therefore \int_0^\infty \frac{1}{x^n+1}\,dx = \frac{\pi}{n}\csc\left(\frac{\pi}{n}\right)$$

Which gives some neat results such as

$$\int_0^\infty \frac{1}{x^\pi+1}\,dx = \csc 1$$

$$\int_0^\infty \frac{1}{x^3+1}\,dx = \frac{2\pi}{3\sqrt{3}}$$

$$\int_0^\infty \frac{1}{x^4+1}\,dx = \frac{\pi}{2\sqrt{2}}$$

$$\int_0^\infty \frac{1}{x^6+1}\,dx = \frac{\pi}{3}$$

$$\int_0^\infty \frac{1}{x^{10}+1}\,dx = \frac{\phi\pi}{5}$$

Moving on, we may obtain our final solution by differentiating the result in equation 1.

$$\therefore I'(k) = -\frac{\pi^2}{n^2}\cot\left(\frac{\pi(k+1)}{n}\right)\csc\left(\frac{\pi(k+1)}{n}\right), \text{ for } \frac{k+1}{n} < 1$$

$$\therefore I'(0) = \int_0^\infty \frac{\ln x}{x^n+1}\,dx = -\frac{\pi^2}{n^2}\cot\left(\frac{\pi}{n}\right)\csc\left(\frac{\pi}{n}\right), \text{ for } n > 1$$

Some results:

$$\int_0^\infty \frac{\ln x}{x^\pi+1}\,dx = -\cot(1)\csc(1)$$

$$\int_0^\infty \frac{\ln x}{x^2+1}\,dx = 0$$

$$\int_0^\infty \frac{\ln x}{x^3+1}\,dx = -\frac{2\pi^2}{27}$$

$$\int_0^\infty \frac{\ln x}{x^4+1}\,dx = -\frac{\pi^2}{8\sqrt{2}}$$

$$\int_0^\infty \frac{\ln x}{x^6+1}\,dx = -\frac{\pi^2}{6\sqrt{3}}$$

4.3 Fresnel integrals

The two Fresnel integrals:

$$\int_0^\infty \sin(x^2)\,dx$$

$$\int_0^\infty \cos(x^2)\,dx$$

First consider

$$e^{ix} = \cos(x) + i\sin(x)$$

$$\therefore e^{-ix^2} = \cos(x^2) - i\sin(x^2)$$

Therefore:

$$\Re \int_0^\infty e^{-ix^2}\,dx = \int_0^\infty \cos(x^2)\,dx$$

$$-\Im \int_0^\infty e^{-ix^2}\,dx = \int_0^\infty \sin(x^2)\,dx$$

Consider

$$I = \int_0^\infty e^{-ix^2}\,dx$$

$$\therefore I^2 = \int_0^\infty \int_0^\infty e^{-ix^2} e^{-iy^2}\,dx\,dy$$

$$= \int_0^\infty \int_0^\infty e^{-i(x^2+y^2)}\,dx\,dy$$

$$y = xt, dy = x\,dt$$

$$= \int_0^\infty \int_0^\infty x e^{-ix^2(1+t^2)}\,dx\,dt$$

$$u = -ix^2(1+t^2), du = -2ix\big(1+t^2\big)\,dx$$

$$= \frac{1}{2i}\int_0^\infty \int_{-\infty}^0 \frac{e^u}{1+t^2}\,du\,dt$$

$$= \frac{1}{2i}\int_0^\infty \frac{1}{1+t^2}\,dt$$

$$= \frac{\pi}{4i}$$

$$\therefore I = \frac{1}{2}\sqrt{\frac{\pi}{i}} = \frac{\sqrt{-i\pi}}{2} = \frac{\sqrt{\pi}}{2}\left(\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}\right) = \sqrt{\frac{\pi}{8}} - i\sqrt{\frac{\pi}{8}}$$

Thus, we arrive at our solutions.

$$\Re \int_0^\infty e^{-ix^2}\,dx = \int_0^\infty \cos(x^2)\,dx = \sqrt{\frac{\pi}{8}}$$

$$-\Im \int_0^\infty e^{-ix^2}\,dx = \int_0^\infty \sin(x^2)\,dx = \sqrt{\frac{\pi}{8}}$$

Then, as $\sin(x^2)$ and $\cos(x^2)$ are even functions, therefore

$$\int_{-\infty}^{\infty} \cos(x^2)\,dx = \sqrt{\frac{\pi}{2}}$$
$$\int_{-\infty}^{\infty} \sin(x^2)\,dx = \sqrt{\frac{\pi}{2}}$$

Below is a secondary solution for the integrals, from negative to positive infinity. I wanted to include this as it uses a completely different technique to all of the other problems on this document.

$$I = \int_{-\infty}^{\infty} e^{-ix^2} dx$$
$$\therefore I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-ix^2} e^{-iy^2}\,dx\,dy$$
$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-i(x^2+y^2)}\,dx\,dy$$

$$x = r\cos\theta$$
$$y = r\sin\theta$$

$$\mathbb{J} = \det\begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \det\begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix} = r$$
$$\therefore dx\,dy = r\,dr\,d\theta$$

$$\therefore \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-i(x^2+y^2)}\,dx\,dy = \int_{0}^{2\pi}\int_{0}^{\infty} e^{-ir^2} r\,dr\,d\theta$$

$$u = ir^2, du = 2ir\,dr$$

$$= \frac{1}{2i}\int_{0}^{2\pi}\int_{0}^{\infty} e^{-u}\,du\,d\theta$$
$$= \frac{1}{2i}\int_{0}^{2\pi} 1\,d\theta$$
$$= \frac{\pi}{i}$$

$$\therefore I = \sqrt{\frac{\pi}{i}} = \sqrt{-i\pi} = \sqrt{\pi}\left(\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}\right)$$
$$= \sqrt{\frac{\pi}{2}} - i\sqrt{\frac{\pi}{2}}$$

We thus arrive at our solutions.

$$\Re\int_{-\infty}^{\infty} e^{-ix^2} dx = \int_{-\infty}^{\infty} \cos(x^2)\,dx = \sqrt{\frac{\pi}{2}}$$
$$-\Im\int_{-\infty}^{\infty} e^{-ix^2} dx = \int_{-\infty}^{\infty} \sin(x^2)\,dx = \sqrt{\frac{\pi}{2}}$$

(And since the functions are even, we may go back to our solutions for the original bounds by dividing our answer by two).

4.4 Coefficient generalisation, and example results

$$\int_{-\infty}^{\infty} e^{-\alpha x^2}\cos(kx^2)\,dx = \sqrt{\frac{\pi\left(\sqrt{\alpha^2+k^2}+\alpha\right)}{2\left(\alpha^2+k^2\right)}}$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2}\sin(kx^2)\,dx = \sqrt{\frac{\pi\left(\sqrt{\alpha^2+k^2}-\alpha\right)}{2\left(\alpha^2+k^2\right)}}$$

where $k \in \mathbb{R}$ and $\alpha \in \mathbb{R}^+$. This leads to results such as:

$$\int_{-\infty}^{\infty} e^{-\alpha x^2}\cos(\alpha x^2)\,dx = \sqrt{\frac{\pi\sigma}{4\alpha}}$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2}\sin(\alpha x^2)\,dx = \sqrt{\frac{\pi}{4\sigma\alpha}}$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2}\cos(2\alpha x^2)\,dx = \sqrt{\frac{\phi\pi}{5\alpha}}$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2}\sin(2\alpha x^2)\,dx = \sqrt{\frac{\pi}{5\phi\alpha}}$$

where $\phi = \frac{1+\sqrt{5}}{2}$ (golden ratio), and $\sigma = 1+\sqrt{2}$ (silver ratio). Further results:

$$\int_{-\infty}^{\infty} e^{-x^2}\cos\left(\sqrt{\phi}\,x^2\right)dx = \sqrt{\frac{\pi}{2}}$$

$$\int_{-\infty}^{\infty} e^{-x^2}\sin\left(\sqrt{\phi}\,x^2\right)dx = \sqrt{\frac{\pi}{2\phi^3}}$$

$$\int_{-\infty}^{\infty} e^{-\sqrt{\phi}\,x^2}\cos(\phi x^2)\,dx = \sqrt{\frac{\pi}{2\sqrt{\phi}}}$$

$$\int_{-\infty}^{\infty} e^{-\sqrt{\phi}\,x^2}\sin(\phi x^2)\,dx = \sqrt{\frac{\pi}{2\phi^3\sqrt{\phi}}}$$

$$\int_{-\infty}^{\infty} e^{-x^2}\cos\left(\sqrt{3}\,x^2\right)dx = \sqrt{\frac{3\pi}{8}}$$

$$\int_{-\infty}^{\infty} e^{-x^2}\sin\left(\sqrt{3}\,x^2\right)dx = \sqrt{\frac{\pi}{8}}$$

$$\int_{-\infty}^{\infty} e^{-x^2}\cos\left(\sqrt{8}\,x^2\right)dx = \frac{\sqrt{2\pi}}{3}$$

$$\int_{-\infty}^{\infty} e^{-x^2}\sin\left(\sqrt{8}\,x^2\right)dx = \frac{\sqrt{\pi}}{3}$$

$$\int_{-\infty}^{\infty} e^{-2x^2}\cos\left(\sqrt{5}\,x^2\right)dx = \sqrt{\frac{5\pi}{18}}$$

$$\int_{-\infty}^{\infty} e^{-2x^2}\sin\left(\sqrt{5}\,x^2\right)dx = \sqrt{\frac{\pi}{18}}$$

$$\int_{-\infty}^{\infty} e^{-2x^2}\cos\left(\sqrt{32}\,x^2\right)dx = \frac{\sqrt{\pi}}{3}$$

$$\int_{-\infty}^{\infty} e^{-2x^2}\sin\left(\sqrt{32}\,x^2\right)dx = \sqrt{\frac{\pi}{18}}$$

4.5 Generalisation for power of trig term

Using the property

$$\cos^2 x = \frac{1+\cos(2x)}{2}, \sin^2 x = \frac{1-\cos(2x)}{2},$$

we may also obtain equations like the following:

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} \cos^2(\alpha x^2)\, dx = \sqrt{\frac{\pi}{4\alpha}} \left(1 + \sqrt{\frac{\phi}{5}}\right)$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} \sin^2(\alpha x^2)\, dx = \sqrt{\frac{\pi}{4\alpha}} \left(1 - \sqrt{\frac{\phi}{5}}\right)$$

And going further:

$$\cos^n(x) = \left(\frac{e^{ix}+e^{-ix}}{2}\right)^n = \frac{1}{2^n}\sum_{h=0}^{n}\binom{n}{h} e^{ihx}e^{-i(n-h)x}$$

$$= \frac{1}{2^n}\sum_{h=0}^{n}\binom{n}{h} e^{i(2h-n)x}$$

$$\therefore \cos^n(kx^2) = \frac{1}{2^n}\sum_{h=0}^{n}\binom{n}{h} e^{-ik(2h-n)x^2}$$

Using this information, we can derive the following generalised equations.

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} \cos^n(kx^2)\, dx = \frac{1}{2^n}\sqrt{\frac{\pi}{2}}\sum_{h=0}^{n}\binom{n}{h}\sqrt{\frac{\alpha+\sqrt{\alpha^2+(k(2h-n))^2}}{\alpha^2+(k(2h-n))^2}}$$

$$k \in \mathbb{R}, \alpha \in \mathbb{R}^+, n \in \mathbb{N}.$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} \sin^n(kx^2)\, dx = \frac{1}{2^n}\sqrt{\frac{\pi}{2}}\sum_{h=0}^{n} i^{2h-n}\binom{n}{h}\sqrt{\frac{\alpha+\sqrt{\alpha^2+(k(2h-n))^2}}{\alpha^2+(k(2h-n))^2}}$$

$$k \in \mathbb{R}, \alpha \in \mathbb{R}^+, n \in 2\mathbb{N}.$$

Some basic examples:

$$\int_{-\infty}^{\infty} e^{-3x^2} \cos^3(3x^2)\, dx = \frac{\sqrt{2\pi}}{8}\left(\sqrt{\frac{1+\sqrt{10}}{30}} + 3\sqrt{\frac{\sigma}{6}}\right)$$

$$\int_{-\infty}^{\infty} e^{-4x^2} \cos^4(4x^2)\, dx = \frac{\sqrt{2\pi}}{8}\left(\frac{1}{4}\sqrt{\frac{1+\sqrt{17}}{17}} + \frac{3}{2\sqrt{2}} + \sqrt{\frac{2\phi}{5}}\right)$$

$$\int_{-\infty}^{\infty} e^{-4x^2} \sin^4(4x^2)\, dx = \frac{\sqrt{2\pi}}{8}\left(\frac{1}{4}\sqrt{\frac{1+\sqrt{17}}{17}} + \frac{3}{2\sqrt{2}} - \sqrt{\frac{2\phi}{5}}\right)$$

It is possible to generalise the exponent of x, but the solutions only have a semi-closed form, either involving cosine and arctan, or a fractional factorial.

4.6 Generalisation for power of x

This is based from a YT video, but expanded on by using the previous findings above. These equations are:

$$\int_0^\infty e^{-\alpha x^p}\cos^n(kx^p)\,dx = \frac{1}{2^n}\left(\frac{1}{p}\right)!\sum_{h=0}^{n}\binom{n}{h}\frac{\cos\left(\frac{1}{p}\arctan\left(\frac{k(2h-n)}{\alpha}\right)\right)}{(\alpha^2+(k(2h-n))^2)^{\frac{1}{2p}}}$$

$$\int_0^\infty e^{-\alpha x^p}\sin^n(kx^p)\,dx = \frac{1}{2^n}\left(\frac{1}{p}\right)!\sum_{h=0}^{n}\binom{n}{h}\frac{i^{2h-n}\cos\left(\frac{1}{p}\arctan\left(\frac{k(2h-n)}{\alpha}\right)\right)}{(\alpha^2+(k(2h-n))^2)^{\frac{1}{2p}}}$$

Where $p \in \mathbb{R}$. Secondly, note that the bounds this time are from 0 to ∞, as the domains do not extend over the whole $x-$axis for a general $p \in \mathbb{R}$.

If we were fine with a fractional factorial in our solution, we would need to have p equal to some power of 2. Then, the solutions involve nested radicals that go layer by layer deeper for each power. For $p = 4$, we have the following solutions (using the lemniscate constant)

$$\int_{-\infty}^\infty e^{-\alpha x^4}\cos^n(kx^4)\,dx = \frac{\sqrt{\varpi\sqrt{2\pi}}}{2^{n+1}}\sum_{h=0}^{n}\binom{n}{h}\sqrt{\frac{1+\frac{1}{\sqrt{2}}\sqrt{1+\frac{\alpha}{\sqrt{\alpha^2+(k(2h-n))^2}}}}{\sqrt[4]{(\alpha^2+(k(2h-n))^2)}}}$$

$$\int_{-\infty}^\infty e^{-\alpha x^4}\sin^n(kx^4)\,dx = \frac{\sqrt{\varpi\sqrt{2\pi}}}{2^{n+1}}\sum_{h=0}^{n}\binom{n}{h}i^{2h-n}\sqrt{\frac{1+\frac{1}{\sqrt{2}}\sqrt{1+\frac{\alpha}{\sqrt{\alpha^2+(k(2h-n))^2}}}}{\sqrt[4]{(\alpha^2+(k(2h-n))^2)}}}$$

A couple basic examples (not using lemniscate constant):

$$\int_{-\infty}^\infty e^{-x^4}\cos(x^4)\,dx = \left(\frac{1}{4}\right)!\sqrt{\frac{2+\sqrt{2+\sqrt{2}}}{\sqrt[4]{2}}}$$

$$\int_{-\infty}^\infty e^{-x^4}\sin(x^4)\,dx = \left(\frac{1}{4}\right)!\sqrt{\frac{2-\sqrt{2+\sqrt{2}}}{\sqrt[4]{2}}}$$

$$\int_{-\infty}^\infty e^{-8x^4}\cos^2(2x^4)\,dx = \frac{1}{2\sqrt{2}}\left(\frac{1}{4}\right)!\left(2^{\frac{3}{4}}+\sqrt{\frac{10+\sqrt{10(5+2\sqrt{5})}}{5\sqrt[4]{5}}}\right)$$

$$\int_{-\infty}^\infty e^{-8x^4}\sin^2(2x^4)\,dx = \frac{1}{2\sqrt{2}}\left(\frac{1}{4}\right)!\left(2^{\frac{3}{4}}-\sqrt{\frac{10+\sqrt{10(5+2\sqrt{5})}}{5\sqrt[4]{5}}}\right)$$

If we wanted to do so without either, we would have to 1) Have 'p' be a power of 2 like mentioned above, AND 2) Square the integral and, introducing a new variable y, have the exponent of y be equal to $-\frac{p}{p+1}$. This is to satisfy the identity

$$\Gamma(1-z)\Gamma(z) = \frac{\pi}{\sin(\pi z)}$$

Therefore, if we were to use the next valid value for p, which is 4, the generalised, closed-form formula for the double integral

$$\int_0^\infty \int_0^\infty e^{-\alpha x^4 - \beta y^{-\frac{4}{5}}} \cos^n(kx^4) \sin^r(qy^{-4/5})\, dx\, dy$$

$$= \frac{n!r!\pi}{2^{n+r+\frac{1}{2}}} \sum_{h=0}^{n} \sum_{j=0}^{r} \frac{\sqrt{\left(1+\sqrt{\frac{1+\sqrt{1+\left(\frac{k}{\alpha}(2h-n)\right)^2}}{2\sqrt{1+\left(\frac{k}{\alpha}(2h-n)\right)^2}}}\right)\left(1+\sqrt{\frac{1+\sqrt{1+\left(\frac{q}{\beta}(2j-r)\right)^2}}{2\sqrt{1+\left(\frac{q}{\beta}(2j-r)\right)^2}}}\right)}\left(\frac{2}{\sqrt{1+\left(\frac{q}{\beta}(2j-r)\right)^2}} - 2\sqrt{\frac{1+\sqrt{1+\left(\frac{q}{\beta}(2j-r)\right)^2}}{2\sqrt{1+\left(\frac{q}{\beta}(2j-r)\right)^2}}} + 1\right)}{i^{r+2j} h! j! (n-h)! (r-j)! (\alpha^2 + (k(2h-n))^2)^{\frac{1}{8}} (\beta^2 + (q(2j-r))^2)^{-\frac{5}{8}}}$$

$$\alpha, \beta \in \mathbb{R}^+, k, q \in \mathbb{R}, n \in \mathbb{N}, r \in 2\mathbb{Z}^+.$$

Here are some examples (basic ones):

$$\int_0^\infty \int_0^\infty e^{-(x^4+y^{-4/5})} \cos(x^4) \sin^2(y^{-4/5})\, dx\, dy$$

$$= \frac{\pi}{2\sqrt{2}} \sqrt{\frac{1+\sqrt{\frac{1+\sqrt{2}}{2\sqrt{2}}}}{\sqrt[4]{2}}} \left(\sqrt{2} - \left(\frac{2}{\sqrt{5}} - 2\sqrt{\frac{\phi}{\sqrt{5}}} + 1\right)\sqrt{5\sqrt[4]{5}\left(1+\sqrt{\frac{\phi}{\sqrt{5}}}\right)}\right)$$

$$\int_0^\infty \int_0^\infty e^{-4x^4 - 3y^{-\frac{4}{5}}} \cos^3(4x^4) \sin^2(3y^{-4/5})\, dx\, dy$$

$$= \frac{9\pi\sqrt[4]{3}}{16\sqrt[8]{2}} \left(\sqrt{\sqrt{\phi}+\sqrt[4]{5}}\left(2\sqrt{\phi\sqrt{5}} - \sqrt{5} - 2\right) + \sqrt{2}\right)\left(\frac{1}{3}\sqrt{\frac{1+\sqrt{\frac{1+\sqrt{10}}{2\sqrt{10}}}}{\sqrt[4]{5}}} + \sqrt{1+\sqrt{\frac{\sigma}{2\sqrt{2}}}}\right)$$

$$\int_0^\infty \int_0^\infty e^{-2(x^4+y^{-4/5})} \cos^3(3x^4) \sin^2(3y^{-4/5})\, dx\, dy$$

$$= \frac{3\pi\sqrt[4]{2}}{4} \left(\sqrt{\frac{1+\sqrt{\frac{\sqrt{\frac{13}{4}}+1}{\sqrt{13}}}}{\sqrt[4]{13}}} + \frac{1}{3}\sqrt{\frac{1+\sqrt{\frac{\sqrt{\frac{85}{4}}+1}{\sqrt{85}}}}{\sqrt[4]{85}}}\right)\left(1 - \left(\frac{2}{\sqrt{10}} - 2\sqrt{\frac{1+\sqrt{10}}{2\sqrt{10}}} + 1\right)\sqrt{\frac{10\sqrt[4]{10}}{2}\left(1+\sqrt{\frac{1+\sqrt{10}}{2\sqrt{10}}}\right)}\right)$$

Full solution for an integral with lots of two's (Solution could be written in much less lines but this is to show a full method from start to finish):

$$\int_{-\infty}^{\infty} x^2 e^{-2x^2} \cos^2(2x^2)\, dx$$

$$= \int_{-\infty}^{\infty} x^2 e^{-2x^2} \left(\frac{e^{2ix^2} + e^{-2ix^2}}{2} \right)^2 dx$$

$$= \frac{1}{2} \sum_{h=0}^{2} \frac{1}{h!(2-h)!} \int_{-\infty}^{\infty} x^2 e^{-2x^2} e^{2ihx^2} e^{-2i(2-h)x^2}\, dx$$

$$= \frac{1}{2} \sum_{h=0}^{2} \frac{1}{h!(2-h)!} \int_{-\infty}^{\infty} x^2 e^{-2x^2(1+2i(h-1))}\, dx$$

IBP with $u = x, dv = xe^{-2x^2(1+2i(h-1))}$

$$= \frac{1}{2} \sum_{h=0}^{2} \frac{1}{h!(2-h)!} \left(\lim_{N\to\infty} \left[x \int xe^{-2x^2(1+2i(h-1))}\, dx \right]_{-N}^{N} - \int_{-\infty}^{\infty} \int xe^{-2x^2(1+2i(h-1))}\, dx\, dx \right)$$

Within the integrals, let $u = -2x^2(1+2i(h-1)), du = -4x(1+2i(h-1))\, dx$

$$= \frac{1}{2} \sum_{h=0}^{2} \frac{1}{h!(2-h)!} \left(\lim_{N\to\infty} \left[\frac{-x}{4(1+2i(h-1))} \int e^u\, du \right]_{-N}^{N} + \frac{1}{4(1+2i(h-1))} \int_{-\infty}^{\infty} \int e^u\, du\, dx \right)$$

$$= \frac{1}{2} \sum_{h=0}^{2} \frac{1}{h!(2-h)!} \left(\lim_{N\to\infty} \left[\frac{-xe^{-2x^2(1+2i(h-1))}}{4(1+2i(h-1))} \right]_{-N}^{N} + \frac{1}{4(1+2i(h-1))} \int_{-\infty}^{\infty} e^{-2x^2(1+2i(h-1))} dx \right)$$

$$= \frac{1}{8} \sum_{h=0}^{2} \frac{1}{h!(2-h)!(1+2i(h-1))} \int_{-\infty}^{\infty} e^{-2x^2(1+2i(h-1))} dx \tag{2}$$

As our integral is even, consider

$$I = \frac{1}{2} \int_{-\infty}^{\infty} e^{-2x^2(1+2i(h-1))} dx = \int_{0}^{\infty} e^{-2x^2(1+2i(h-1))} dx$$

$$\therefore I^2 = \int_{0}^{\infty} \int_{0}^{\infty} e^{-2x^2(1+2i(h-1))} e^{-2x^2(1+2i(h-1))}\, dx\, dy$$

$$= \int_{0}^{\infty} \int_{0}^{\infty} e^{-2(1+2i(h-1))(x^2+y^2)}\, dx\, dy$$

$$y = xt, dy = x\, dt$$

$$= \int_{0}^{\infty} \int_{0}^{\infty} xe^{-2(1+2i(h-1))x^2(1+t^2)}\, dx\, dt$$

$$u = -2(1+2i(h-1))x^2\big(1+t^2\big), du = -4(1+2i(h-1))x\big(1+t^2\big)\, dx$$

$$= \frac{1}{4(1+2i(h-1))} \int_{0}^{\infty} \int_{-\infty}^{0} \frac{e^u}{1+t^2}\, du\, dt$$

$$= \frac{1}{4(1+2i(h-1))} \int_{0}^{\infty} \frac{1}{1+t^2}\, dt$$

$$= \frac{\pi}{8(1+2i(h-1))}$$

$$\therefore I = \sqrt{\frac{\pi}{8(1+2i(h-1))}}$$

$$\therefore 2I = \int_{-\infty}^{\infty} e^{-2x^2(1+2i(h-1))} dx = \sqrt{\frac{\pi}{2(1+2i(h-1))}}$$

Subbing into equation 2,

$$\frac{1}{8}\sum_{h=0}^{2} \frac{1}{h!(2-h)!(1+2i(h-1))} \int_{-\infty}^{\infty} e^{-2x^2(1+2i(h-1))} dx$$

$$= \frac{1}{8}\sum_{h=0}^{2} \frac{1}{h!(2-h)!(1+2i(h-1))} \sqrt{\frac{\pi}{2(1+2i(h-1))}}$$

$$= \frac{1}{8}\sqrt{\frac{\pi}{2}}\sum_{h=0}^{2} \frac{1-2i(h-1)}{h!(2-h)!(1+2i(h-1))(1-2i(h-1))} \sqrt{\frac{1-2i(h-1)}{(1+2i(h-1))(1-2i(h-1))}}$$

$$= \frac{1}{8}\sqrt{\frac{\pi}{2}}\sum_{h=0}^{2} \frac{(1-2i(h-1))^{\frac{3}{2}}}{h!(2-h)!(1+4(h-1)^2)^{\frac{3}{2}}}$$

$$= \frac{1}{8}\sqrt{\frac{\pi}{2}}\left(\frac{(1-2i(-1))^{\frac{3}{2}}}{2(1+4(-1)^2)^{\frac{3}{2}}} + \frac{(1-2i(1-1))^{\frac{3}{2}}}{(2-1)!(1+4(1-1)^2)^{\frac{3}{2}}} + \frac{(1-2i)^{\frac{3}{2}}}{2!(1+4)^{\frac{3}{2}}}\right)$$

$$= \frac{1}{8}\sqrt{\frac{\pi}{2}}\left(\frac{(1+2i)^{\frac{3}{2}}}{2(5)^{\frac{3}{2}}} + 1 + \frac{(1-2i)^{\frac{3}{2}}}{2(5)^{\frac{3}{2}}}\right)$$

$$= \Re\,\frac{1}{8}\sqrt{\frac{\pi}{2}}\left(\frac{(1+2i)^{\frac{3}{2}}}{5^{\frac{3}{2}}} + 1\right) \tag{3}$$

Now consider the following

$$\Re\,(1+2i)^{\frac{3}{2}}$$

$$= 5^{\frac{3}{4}}\cos\left(\frac{3}{2}\arctan(2)\right)$$

$$\text{Let } \arctan(2) = \theta$$

$$= 5^{\frac{3}{4}}\cos\left(\frac{3}{2}\theta\right)$$

$$= 5^{\frac{3}{4}}\cos\left(\theta + \frac{1}{2}\theta\right)$$

$$= 5^{\frac{3}{4}}\left(\cos(\theta)\cos\left(\frac{\theta}{2}\right) - \sin(\theta)\sin\left(\frac{\theta}{2}\right)\right)$$

$$\text{As } \frac{O}{A} = 2, H = \sqrt{5}$$

$$\therefore \cos(\theta) = \frac{1}{\sqrt{5}}, \sin(\theta) = \frac{2}{\sqrt{5}}$$

And using the cos/sin half-angle identities:

$$\cos\left(\frac{\theta}{2}\right) = \sqrt{\frac{1+\frac{1}{\sqrt{5}}}{2}}, \sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1-\frac{1}{\sqrt{5}}}{2}}$$

$$\therefore 5^{\frac{3}{4}} \cos\left(\frac{3}{2}\arctan(2)\right) =$$

$$5^{\frac{3}{4}}\left(\sqrt{\frac{1+\frac{1}{\sqrt{5}}}{10}} - 2\sqrt{\frac{1-\frac{1}{\sqrt{5}}}{10}}\right)$$

Subbing into equation 3,

$$\begin{aligned}
&= \frac{1}{8}\sqrt{\frac{\pi}{2}}\left(\frac{5^{\frac{3}{4}}}{5^{\frac{3}{2}}}\left(\sqrt{\frac{1+\frac{1}{\sqrt{5}}}{10}} - 2\sqrt{\frac{1-\frac{1}{\sqrt{5}}}{10}}\right) + 1\right) \\
&= \frac{1}{8}\sqrt{\frac{\pi}{2}}\left(\frac{1}{5^{\frac{3}{4}}\sqrt{10}}\left(\sqrt{1+\frac{1}{\sqrt{5}}} - 2\sqrt{1-\frac{1}{\sqrt{5}}}\right) + 1\right) \qquad (4)
\end{aligned}$$

All we now need to do is simplify this massive expression. Consider

$$\begin{aligned}
a &= \sqrt{1+\frac{1}{\sqrt{5}}} - 2\sqrt{1-\frac{1}{\sqrt{5}}} \\
\therefore a^2 &= 1 + \frac{1}{\sqrt{5}} - \frac{8}{\sqrt{5}} + 4 - \frac{4}{\sqrt{5}} \\
&= 5 - \frac{11}{\sqrt{5}}
\end{aligned}$$

As a is negative,

$$\therefore a = -\sqrt{5 - \frac{11}{\sqrt{5}}}$$

Subbing into equation 4,

$$\begin{aligned}
&= \frac{1}{8}\sqrt{\frac{\pi}{2}}\left(1 - \frac{1}{5^{\frac{3}{4}}\sqrt{10}}\sqrt{5 - \frac{11}{\sqrt{5}}}\right) \\
&= \frac{1}{8}\sqrt{\frac{\pi}{2}}\left(1 - \frac{1}{\sqrt{10}}\sqrt{\frac{5\sqrt{5} - 11}{25}}\right) \\
&= \frac{1}{8}\sqrt{\frac{\pi}{2}}\left(1 - \frac{1}{5\sqrt{2}}\sqrt{\sqrt{5} - \frac{11}{5}}\right) \\
&= \frac{\sqrt{\pi}}{16}\left(\sqrt{2} - \frac{1}{5}\sqrt{\sqrt{5} - \frac{11}{5}}\right)
\end{aligned}$$

We have finally arrived at our solution.

$$\int_{-\infty}^{\infty} x^2 e^{-2x^2} \cos^2(2x^2)\, dx = \frac{\sqrt{\pi}}{16}\left(\sqrt{2} - \frac{1}{5}\sqrt{\sqrt{5} - \frac{11}{5}}\right)$$

Just as an aside piece of info, the general solution for

$$\int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} \cos(kx^2)\, dx = \frac{\alpha\sqrt{\alpha^2+k^2}+\alpha^2-k^2}{2(\alpha^2+k^2)^{\frac{3}{2}}\sqrt{\sqrt{\alpha^2+k^2}+\alpha}}\sqrt{\frac{\pi}{2}}$$

$$\int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} \sin(kx^2)\, dx = \frac{k(\sqrt{\alpha^2+k^2}+2\alpha)}{2(\alpha^2+k^2)^{\frac{3}{2}}\sqrt{\sqrt{\alpha^2+k^2}+\alpha}}\sqrt{\frac{\pi}{2}}$$

And when the power of the trig term is 2,

$$\int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} \cos^2(kx^2)\, dx = \frac{1}{4\alpha}\sqrt{\frac{\pi}{\alpha}} + \frac{\alpha\sqrt{\alpha^2+4k^2}+\alpha^2-4k^2}{4(\alpha^2+4k^2)^{\frac{3}{2}}\sqrt{\sqrt{\alpha^2+4k^2}+\alpha}}\sqrt{\frac{\pi}{2}}$$

$$\int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} \sin^2(kx^2)\, dx = \frac{1}{4\alpha}\sqrt{\frac{\pi}{\alpha}} - \frac{\alpha\sqrt{\alpha^2+4k^2}+\alpha^2-4k^2}{4(\alpha^2+4k^2)^{\frac{3}{2}}\sqrt{\sqrt{\alpha^2+4k^2}+\alpha}}\sqrt{\frac{\pi}{2}}$$

5 May

5.1 An integral including the Lambert-W function

Main component of solution from this citation: [2].

$$\int_0^\infty \frac{W(x)}{x\sqrt{x}}\,dx$$

$$u = W(x)$$

$$\text{Since } x = W(x)e^{W(x)}, dx = (u+1)e^u\,du$$

$$\therefore \int_0^\infty \frac{W(x)}{x\sqrt{x}}\,dx = \int_0^\infty \frac{ue^u(u+1)}{ue^u\sqrt{ue^u}}\,du$$

$$= \int_0^\infty \frac{u+1}{\sqrt{ue^u}}\,du$$

$$= \int_0^\infty \frac{\sqrt{u}}{\sqrt{e^u}}\,du + \int_0^\infty \frac{1}{\sqrt{ue^u}}\,du$$

$$= \int_0^\infty u^{\frac{1}{2}}e^{-\frac{u}{2}}\,du + \int_0^\infty u^{-\frac{1}{2}}e^{-\frac{u}{2}}\,du$$

$$x = \frac{u}{2}, dx = \frac{1}{2}\,du$$

$$= 2\int_0^\infty (2x)^{\frac{1}{2}}e^{-x}\,dx + 2\int_0^\infty (2x)^{-\frac{1}{2}}e^{-x}\,dx$$

$$= 2\sqrt{2}\int_0^\infty x^{\frac{1}{2}}e^{-x}\,dx + \sqrt{2}\int_0^\infty x^{-\frac{1}{2}}e^{-x}\,dx$$

Consider left integral.

$$\int_0^\infty x^{\frac{1}{2}}e^{-x}\,dx$$

$$w = x^{\frac{1}{2}}, dw = \frac{1}{2}x^{-\frac{1}{2}}\,dx$$

$$= 2\int_0^\infty w^2e^{-w^2}\,dw$$

$$\text{IBP with } u = w, dv = we^{-w^2}$$

$$= 2\left(\lim_{N\to\infty}\left[w\int we^{-w^2}\,dw\right]_0^N - \int_0^\infty \int we^{-w^2}\,dw\,dw\right)$$

$$u = -w^2, du = -2w\,dw$$

$$= 2\left(\lim_{N\to\infty}\left[-\frac{w}{2}\int e^u\,du\right]_0^N + \frac{1}{2}\int_0^\infty \int e^u\,du\,dw\right)$$

$$= 2\left(\lim_{N\to\infty}\left[-\frac{w}{2}e^{-w^2}\right]_0^N + \frac{1}{2}\int_0^\infty e^{-w^2}\,dw\right)$$

$$= \int_0^\infty e^{-w^2}\,dw = \frac{\sqrt{\pi}}{2} \text{ (Using method from page 40).}$$

$$\therefore \int_0^\infty x^{\frac{1}{2}} e^{-x}\,dx = \frac{\sqrt{\pi}}{2}$$

IBP with $u = x^{\frac{1}{2}}, dv = e^{-x}$

$$\int_0^\infty x^{\frac{1}{2}} e^{-x}\,dx = \lim_{N\to\infty}\Big[-x^{\frac{1}{2}}e^{-x}\Big]_0^N + \frac{1}{2}\int_0^\infty x^{-\frac{1}{2}} e^{-x}\,dx$$

$$= \frac{1}{2}\int_0^\infty x^{-\frac{1}{2}} e^{-x}\,dx$$

$$\therefore \int_0^\infty x^{-\frac{1}{2}} e^{-x}\,dx = \sqrt{\pi}$$

Therefore, putting the two results together,

$$2\sqrt{2}\int_0^\infty x^{\frac{1}{2}} e^{-x}\,dx + \sqrt{2}\int_0^\infty x^{-\frac{1}{2}} e^{-x}\,dx$$

$$= \int_0^\infty \frac{W(x)}{x\sqrt{x}}\,dx = 2\sqrt{2\pi}$$

5.2 An integral using DUIS twice

Very cool use of complex numbers. Solved with the prior knowledge (found online) of the first step or so.

$$\int_0^1 \frac{\sin^2(\ln\sqrt{x})}{\ln^2 x}\,dx$$

$$\text{Let } I(k) = \int_0^1 \frac{\sin^2(k\ln x)}{\ln^2 x}\,dx$$

We are thus looking to solve for $I(\frac{1}{2})$.

$$= -\frac{1}{4}\int_0^1 \frac{(e^{ik\ln x} - e^{-ik\ln x})^2}{\ln^2 x}\,dx$$

$$= -\frac{1}{4}\int_0^1 \frac{(x^{ik} - x^{-ik})^2}{\ln^2 x}\,dx$$

$$\therefore I'(k) = -\frac{1}{2}\int_0^1 \frac{x^{ik} - x^{-ik}}{\ln^2 x}\frac{\partial}{\partial k}(x^{ik} - x^{-ik})\,dx$$

$$= -\frac{i}{2}\int_0^1 \frac{(x^{ik} - x^{-ik})(x^{ik} + x^{-ik})}{\ln x}\,dx$$

Using reverse-DOPS:

$$= -\frac{i}{2}\int_0^1 \frac{x^{2ik} - x^{-2ik}}{\ln x}\,dx$$

$$\therefore I''(k) = \int_0^1 x^{2ik} + x^{-2ik}\,dx$$

$$= \frac{1}{2ik+1} + \frac{1}{1-2ik}$$

$$\therefore I'(k) = \int \frac{1}{2ik+1} + \frac{1}{1-2ik}\,dk$$

$$u = 2ik, du = 2i\,dk$$

$$= \frac{1}{2i}\int \frac{1}{u+1} + \frac{1}{1-u}\,du$$

$$= -\frac{i}{2}\big(\ln(u+1) - \ln(u-1)\big) + C$$

$$= -\frac{i}{2}\big(\ln(2ik+1) - \ln(2ik-1)\big) + C$$

To find C, sub in $k = 0$ into both expressions for $I'(k)$

$$I'(0) = -\frac{i}{2}\int_0^1 0\,dx = 0 = -\frac{i}{2}\big(\ln(1) - \ln(-1)\big) + C$$

$$\therefore C = -\frac{i}{2}\ln(-1) = \frac{\pi}{2}$$

$$\therefore I'(k) = -\frac{i}{2}\big(\ln(2ik+1) - \ln(2ik-1)\big) + \frac{\pi}{2}$$

$$\therefore I(k) = -\frac{i}{2}\int \ln(2ik+1) - \ln(2ik-1)\,dk + \int \frac{\pi}{2}\,dk$$

$$u = 2ik, du = 2i\,dk$$

$$= -\frac{1}{4}\int \ln(u+1) - \ln(u-1)\,du + \frac{k\pi}{2}$$

Using IBP with $u = \ln(u+1) - \ln(u-1), dv = 1$

$$= \frac{1}{4}\Big(u\big(\ln(u-1) - \ln(u+1)\big) + \int \frac{u}{u+1} - \frac{u}{u-1}\,du\Big) + \frac{k\pi}{2}$$

$$= \frac{ik}{2}\big(\ln(2ik-1) - \ln(2ik+1)\big) + \frac{1}{4}\int \frac{u}{u+1} - \frac{u}{u-1}\,du + \frac{k\pi}{2}$$

$$= \frac{ik}{2}\big(\ln(2ik-1) - \ln(2ik+1)\big) + \frac{1}{4}\int \frac{-2u}{u^2-1}\,du + \frac{k\pi}{2}$$

$$t = u^2 - 1, dt = 2u\,du$$

$$= \frac{ik}{2}\big(\ln(2ik-1) - \ln(2ik+1)\big) - \frac{1}{4}\int \frac{1}{t}\,dt + \frac{k\pi}{2}$$

$$= \frac{ik}{2}\big(\ln(2ik-1) - \ln(2ik+1)\big) - \frac{1}{4}\ln(t) + \frac{k\pi}{2} + C$$

$$= \frac{ik}{2}\big(\ln(2ik-1) - \ln(2ik+1)\big) - \frac{1}{4}\ln(-4k^2-1) + \frac{k\pi}{2} + C$$

To find C, let $k = 0$. Further, as our integral is purely real, we are only looking for the real part of C.

$$I(0) = -\frac{1}{4}\int_0^1 0\,dx = 0 = -\frac{1}{4}\ln(-1) + C$$

$$\therefore C = \frac{1}{4}\ln(-1) = -\frac{i\pi}{4}$$

$$\therefore \Re C = 0$$

We can finally simplify this expression and find the real part. This means that we can discard all of the imaginary parts below.

$$\therefore I(k) = \frac{ik}{2}\big(\ln(2ik-1) - \ln(2ik+1)\big) - \Re\frac{1}{4}\ln(-4k^2-1) + \frac{k\pi}{2}$$

$$= \frac{ik}{2}\big(\ln(2ik-1) - \ln(2ik+1)\big) - \frac{1}{4}\ln(4k^2+1) + \frac{k\pi}{2}$$

$$= \frac{ik}{2}\ln(-1) + \frac{ik}{2}\ln(1-2ik) - \frac{ik}{2}\ln(2ik+1) - \frac{1}{4}\ln(4k^2+1) + \frac{k\pi}{2}$$

$$= -\frac{k\pi}{2} + \frac{ik}{2}\ln\big(\sqrt{4k^2+1}\,e^{i\arctan(-2k)}\big) - \frac{ik}{2}\ln\big(\sqrt{4k^2+1}\,e^{i\arctan(2k)}\big) - \frac{1}{4}\ln(4k^2+1) + \frac{k\pi}{2}$$

$$= \frac{ik\ln(4k^2+1)}{4} - \frac{k\arctan(-2k)}{2} - \frac{ik\ln(4k^2+1)}{4} + \frac{k\arctan(2k)}{2} - \frac{1}{4}\ln(4k^2+1)$$

$$= \frac{k\arctan(2k)}{2} - \frac{k\arctan(-2k)}{2} - \frac{1}{4}\ln(4k^2+1)$$

$$= \frac{k}{2}\big(\arctan(2k) - \arctan(-2k)\big) - \frac{1}{4}\ln(4k^2+1)$$

$$= k\arctan(2k) - \frac{1}{4}\ln(4k^2+1)$$

$$\therefore I(k) = \int_0^1 \frac{\sin^2(k\ln x)}{\ln^2 x}\,dx = k\arctan(2k) - \frac{1}{4}\ln(4k^2+1)$$

Subbing in for $k = \frac{1}{2}$ thus yields our solution.

$$\therefore \int_0^1 \frac{\sin^2(\ln\sqrt{x})}{\ln^2 x}\,dx = \frac{1}{4}\left(\frac{\pi}{2} - \ln 2\right)$$

Similarly, by applying DUIS once instead of twice to the slightly different integral

$$I(k) = \int_0^1 \frac{\sin^2(k\ln x)}{\ln x}\,dx$$

We obtain

$$I'(k) = -\frac{i}{2}\int_0^1 x^{2ik} - x^{-2ik}\,dx$$

$$= -\frac{i}{2}\left(\frac{1}{2ik+1} - \frac{1}{1-2ik}\right)$$

$$\therefore I(k) = -\frac{i}{2}\int \frac{1}{2ik+1} - \frac{1}{1-2ik}\,dk$$

$$\therefore I(k) = -\frac{1}{4}(\ln(2ik+1) + \ln(2ik-1)) + C$$

Letting $k = 0$,

$$\therefore I(0) = \int_0^1 0\,dx = 0 = -\frac{1}{4}\ln(-1) + C$$

As our integral is purely real, $\Re C = 0$. This also means that we can discard all of the imaginary parts below.

$$\therefore I(k) = -\frac{1}{4}(\ln(2ik+1) + \Re\,\ln(2ik-1))$$

$$= -\frac{1}{4}(\ln(2ik+1) + \ln(1-2ik))$$

$$= -\frac{1}{4}\Big(\ln\big(\sqrt{4k^2+1}\,e^{i\arctan(2k)}\big) + \ln\big(\sqrt{4k^2+1}\,e^{i\arctan(-2k)}\big)\Big)$$

$$= -\frac{1}{4}\ln(4k^2+1) - \frac{i\arctan(2k)}{2} - \frac{i\arctan(-2k)}{2}$$

$$= -\frac{1}{4}\ln(4k^2+1)$$

$$\therefore I(k) = \int_0^1 \frac{\sin^2(k\ln x)}{\ln x}\,dx = -\frac{1}{4}\ln(4k^2+1)$$

We may now sub in $k = 1$ to obtain the solution.

$$\therefore \int_0^1 \frac{\sin^2(\ln x)}{\ln x}\,dx = -\frac{1}{4}\ln 5$$

5.3 Four integrals involving series

$$\int_0^\infty \frac{\sin x}{e^x - 1}\,dx$$

$$= \frac{1}{2i}\int_0^\infty \frac{e^{ix} - e^{-ix}}{e^x - 1}\,dx$$

$$= \frac{1}{2i}\int_0^\infty \frac{(e^{ix} - e^{-ix})e^{-x}}{(e^x - 1)e^{-x}}\,dx$$

$$= \frac{1}{2i}\int_0^\infty \frac{e^{-x(1-i)} - e^{-x(1+i)}}{1 - e^{-x}}\,dx$$

Converting to Geometric sum, with $a_1 = e^{-x(1-i)} - e^{-x(1+i)}, r = e^{-x}$.

$$= \frac{1}{2i}\int_0^\infty \sum_{n=1}^\infty (e^{-x(1-i)} - e^{-x(1+i)})(e^{-x})^{n-1}\,dx$$

$$= \frac{1}{2i}\sum_{n=1}^\infty \int_0^\infty e^{-x(n-i)} - e^{-x(n+i)}\,dx$$

$$= \frac{1}{2i}\sum_{n=1}^\infty \left(\frac{1}{n-i} - \frac{1}{n+i}\right)$$

$$= \frac{1}{2i}\sum_{n=1}^\infty \left(\frac{n+i}{(n-i)(n+i)} - \frac{n-i}{(n+i)(n-i)}\right)$$

$$= \frac{1}{2i}\sum_{n=1}^\infty \frac{2i}{n^2+1}$$

$$= \sum_{n=1}^\infty \frac{1}{n^2+1}$$

This sum actually has a solution. Using the Weierstrass product for $\sinh x$:

$$\sinh x = x\prod_{n=1}^\infty \left(1 + \frac{x^2}{n^2\pi^2}\right)$$

$$\therefore \ln(\sinh x) = \ln x + \ln\left(\prod_{n=1}^\infty \left(1 + \frac{x^2}{n^2\pi^2}\right)\right)$$

$$= \ln x + \sum_{n=1}^\infty \ln\left(1 + \frac{x^2}{n^2\pi^2}\right)$$

Differentiating both sides:

$$\therefore \coth x = \frac{1}{x} + \sum_{n=1}^\infty \frac{\frac{2x}{n^2\pi^2}}{1 + \frac{x^2}{n^2\pi^2}}$$

$$= \frac{1}{x} + \sum_{n=1}^\infty \frac{\frac{2x}{n^2\pi^2}}{\frac{n^2\pi^2 + x^2}{n^2\pi^2}}$$

$$= \frac{1}{x} + \sum_{n=1}^\infty \frac{2x}{n^2\pi^2 + x^2}$$

Let $x = \pi$

$$\coth \pi = \frac{1}{\pi} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2+1}$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n^2+1} = \frac{\pi}{2} \coth(\pi) - \frac{1}{2}$$

$$\therefore \int_0^{\infty} \frac{\sin x}{e^x - 1}\, dx = \frac{\pi}{2} \coth(\pi) - \frac{1}{2}$$

In fact, repeating the same method with this integral below yields

$$\int_0^{\infty} \frac{\sin(kx)}{e^{2\pi x} - 1}\, dx = \frac{1}{4} \coth\left(\frac{k}{2}\right) - \frac{1}{2k}$$

$$\int_0^1 \frac{\ln x}{x^2+1}\,dx$$

Consider the Maclaurin series for $\arctan x$.

$$\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1}$$

$$\therefore \frac{1}{x^2+1} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$$

$$\therefore \int_0^1 \frac{\ln x}{x^2+1}\,dx = \int_0^1 \sum_{n=0}^{\infty} (-1)^n x^{2n} \ln x\,dx$$

$$= \sum_{n=0}^{\infty} (-1)^n \int_0^1 x^{2n} \ln x\,dx$$

IBP with $u = \ln x, dv = x^{2n}$

$$= -\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \int_0^1 x^{2n}\,dx$$

$$= -\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2}$$

$$= -G$$

Where G is Catalan's constant, equal to roughly 0.916 (3sf).

$$\therefore \int_0^1 \frac{\ln x}{x^2+1}\,dx = -G$$

$$\int_1^\infty \operatorname{arccsc}^2 x \, dx$$

$$= \int_1^\infty \arcsin^2\left(\frac{1}{x}\right) dx$$

$$u = \frac{1}{x}, du = -\frac{1}{x^2}\, dx$$

$$= \int_0^1 \frac{\arcsin^2 u}{u^2}\, du$$

$$t = \arcsin u, dt = \frac{1}{\sqrt{1-u^2}}\, du$$

$$= \int_0^{\frac{\pi}{2}} \frac{t^2\sqrt{1-\sin^2 t}}{\sin^2 t}\, dt$$

$$= \int_0^{\frac{\pi}{2}} \frac{t^2 \cos t}{\sin^2 t}\, dt$$

$$= \int_0^{\frac{\pi}{2}} t^2 \cot(t)\csc(t)\, dt$$

$$\text{IBP with } u = t^2, dv = \cot(t)\csc(t)$$

$$= \Big[-t^2 \csc t)\Big]_0^{\frac{\pi}{2}} + 2\int_0^{\frac{\pi}{2}} t\csc t\, dt$$

$$= -\frac{\pi^2}{4} + 2\int_0^{\frac{\pi}{2}} t\csc t\, dt$$

$$= 2\int_0^{\frac{\pi}{2}} \frac{t}{\sin t}\, dt - \frac{\pi^2}{4}$$

$$x = \tan\left(\frac{t}{2}\right), dt = \frac{2}{1+x^2}\, dx$$

$$= 2\int_0^1 \frac{2\arctan x}{\frac{2x}{1+x^2}} \cdot \frac{2}{1+x^2}\, dx - \frac{\pi^2}{4}$$

$$= 4\int_0^1 \frac{\arctan x}{x}\, dx - \frac{\pi^2}{4}$$

$$= 4\int_0^1 \sum_{n=0}^\infty \frac{(-1)^n x^{2n}}{2n+1}\, dx - \frac{\pi^2}{4}$$

$$= 4\sum_{n=0}^\infty \frac{(-1)^n}{2n+1}\int_0^1 x^{2n}\, dx - \frac{\pi^2}{4}$$

$$= 4\sum_{n=0}^\infty \frac{(-1)^n}{(2n+1)^2} - \frac{\pi^2}{4}$$

$$= 4G - \frac{\pi^2}{4}$$

$$\therefore \int_1^\infty \operatorname{arccsc}^2 x\, dx = 4G - \frac{\pi^2}{4}$$

This one can be solved in a very unique way! There is a much simpler way of solving this one (via king's rule and symmetry), but here is a very cool method i saw at (CURRENTLY UNKNOWN).

$$\int_0^\pi \ln(\sin x)\,dx$$

First, consider the (lower) Riemann sum for this integral.

$$= \lim_{n\to\infty} \frac{\pi}{n} \sum_{k=1}^{n-1} \ln\left(\sin\left(\frac{k\pi}{n}\right)\right)$$

$$= \lim_{n\to\infty} \frac{\pi}{n} \ln\left(\prod_{k=1}^{n-1} \sin\left(\frac{k\pi}{n}\right)\right)$$

Next, consider

$$\prod_{k=1}^{n-1} \sin\left(\frac{k\pi}{n}\right)$$

$$= \prod_{k=1}^{n-1} \frac{e^{\frac{ik\pi}{n}} - e^{-\frac{ik\pi}{n}}}{2i}$$

$$= \frac{1}{(2i)^{n-1}} \prod_{k=1}^{n-1} (e^{\frac{ik\pi}{n}} - e^{-\frac{ik\pi}{n}})$$

$$= (2i)^{1-n} \prod_{k=1}^{n-1} e^{-\frac{ik\pi}{n}} \left(e^{\frac{2ik\pi}{n}} - 1\right)$$

$$\text{As } \sum_{k=1}^{n-1} k = \frac{n(n-1)}{2}$$

We can take out $e^{-\frac{ik\pi}{n}}$ from the product.

$$= (2i)^{1-n} \left(e^{-\frac{i\pi}{n}}\right)^{\frac{n(n-1)}{2}} \prod_{k=1}^{n-1} (e^{\frac{2ik\pi}{n}} - 1)$$

$$= 2^{1-n} i^{1-n} \left(e^{\frac{i\pi}{2}}\right)^{1-n} \prod_{k=1}^{n-1} (e^{\frac{2ik\pi}{n}} - 1)$$

$$= 2^{1-n} i^{2(1-n)} \prod_{k=1}^{n-1} (e^{\frac{2ik\pi}{n}} - 1)$$

$$= 2^{1-n} (-1)^{1-n} (-1)^{n-1} \prod_{k=1}^{n-1} (1 - e^{\frac{2ik\pi}{n}})$$

$$= 2^{1-n} \prod_{k=1}^{n-1} (1 - e^{\frac{2ik\pi}{n}})$$

Now consider

$$x^n - 1$$

$$= (x-1)(x^{n-1} + x^{n-2} + \cdots + 1)$$

$$= (x-1) \sum_{k=0}^{n-1} x^k$$

And by using the roots of unity:

$$x^n - 1 = \prod_{k=0}^{n-1} \left(x - e^{\frac{2ik\pi}{n}}\right)$$

$$\therefore \prod_{k=0}^{n-1} \left(x - e^{\frac{2ik\pi}{n}}\right) = (x-1) \sum_{k=0}^{n-1} x^k$$

$$\therefore \prod_{k=1}^{n-1} \left(x - e^{\frac{2ik\pi}{n}}\right) = \sum_{k=0}^{n-1} x^k$$

Thus, by letting $x = 1$, we get

$$\prod_{k=1}^{n-1} \left(1 - e^{\frac{2ik\pi}{n}}\right) = \sum_{k=0}^{n-1} 1 = n$$

$$\therefore \prod_{k=1}^{n-1} \sin\left(\frac{k\pi}{n}\right) = 2^{1-n} n$$

$$\therefore \lim_{n\to\infty} \frac{\pi}{n} \ln\left(\prod_{k=1}^{n-1} \sin\left(\frac{k\pi}{n}\right)\right) = \lim_{n\to\infty} \frac{\pi}{n} \ln(2^{1-n} n)$$

$$= \lim_{n\to\infty} \left(\frac{\pi(1-n)}{n} \ln(2) + \frac{\pi}{n} \ln(n)\right)$$

$$= \lim_{n\to\infty} \frac{\pi}{n} \ln(2) - \pi \ln(2)$$

$$= -\pi \ln 2$$

$$\therefore \int_0^{\pi} \ln(\sin x)\, dx = -\pi \ln 2$$

Thus, the integral is solved. However, also notice that when we let $u = \frac{\pi}{2} - x, du = -dx$, we get

$$\int_0^{\pi} \ln(\sin x)\, dx = \int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} - \ln(\sin(\pi/2 - u))\, du$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \ln(\cos u)\, du$$

$$\therefore \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \ln(\cos x)\, dx = -\pi \ln 2$$

5.4 Raabe's integral, and related

This page largely comes with help from [3], with the next page by me. This is a great problem.

$$\int_a^{a+1} \ln\Gamma(x)\,dx$$

Let's call this integral $I(a)$. Using the fundamental theorem of calculus,

$$I'(a) = \ln\Gamma(a+1) - \ln\Gamma(a)$$

$$= \ln\left(\frac{\Gamma(a+1)}{\Gamma(a)}\right) = \ln a$$

$$\therefore I(a) = \int \ln a\,da$$

$$\text{IBP with } u = \ln a, dv = 1$$

$$= a\ln(a) - \int 1\,da$$

$$= a\ln(a) - a + C$$

In order to find C, substitute in $a = 0$.

$$I(0) = \int_0^1 \ln\Gamma(x)\,dx = \lim_{a\to 0}\,(a\ln(a) - a) + C = C$$

Consider this integral. Using king's rule and properties of logarithms,

$$\int_0^1 \ln\Gamma(x)\,dx = \int_0^1 \ln\Gamma(1-x)\,dx$$

$$\therefore 2C = \int_0^1 \ln(\Gamma(x)\Gamma(1-x))\,dx$$

$$= \int_0^1 \ln\left(\frac{\pi}{\sin(\pi x)}\right)dx$$

$$= \int_0^1 \ln\pi\,dx - \int_0^1 \ln(\sin(\pi x))\,dx$$

$$u = \pi x, du = \pi\,dx$$

$$= \ln\pi - \frac{1}{\pi}\int_0^\pi \ln(\sin u)\,du$$

Using our result from the previous integral on page 60,

$$= \ln\pi + \ln 2 = \ln(2\pi)$$

$$\therefore C = \int_0^1 \ln\Gamma(x)\,dx = \frac{1}{2}\ln(2\pi)$$

Our integral is thus solved, for $a \geq 0$.

$$\therefore I(a) = \int_a^{a+1} \ln\Gamma(x)\,dx = \frac{1}{2}\ln(2\pi) + a\ln(a) - a$$

Further, by adding abs val signs around the x in the log term of the solution, this following equation is now true for all $a \in \mathbb{R}$. The integral itself is purely real for $a \geq 0$, and interestingly, for $a \in 2\mathbb{Z}^-$.

$$\Re \int_a^{a+1} \ln\Gamma(x)\,dx = \frac{1}{2}\ln(2\pi) + a\ln(|a|) - a$$

Here is the imaginary part for the integral, for all $a \in \mathbb{R}$.

$$\Im \int_a^{a+1} \ln\Gamma(x)\,dx = \begin{cases} 0 & \text{if } a \geq 0 \\ \pi\Big|a - 2\left\lfloor \frac{a+1}{2} \right\rfloor\Big| & \text{if } a \leq 0 \end{cases}$$

After some playing around on Desmos, I found a way of finding

$$\Re \int_{-a}^{a} \ln\Gamma(x)\,dx$$

for all non-zero multiples of $1/2$, where the integral itself is only purely real for $a = 0$. To find this, one can sum up incrementing values of a from the equation at the top of the page, and see that for positive integer a,

$$\Re \int_0^a \ln\Gamma(x)\,dx = \frac{a}{2}\ln(2\pi) + \sum_{n=1}^{a-1}(n\ln(|n|) - n)$$

$$\Re \int_{-a}^0 \ln\Gamma(x)\,dx = \frac{a}{2}\ln(2\pi) - \sum_{n=1}^{a}(n\ln(|n|) - n)$$

Thus, we can do the following:

$$\therefore \Re \int_{-a}^{a} \ln\Gamma(x)\,dx = a\ln(2\pi) + \sum_{n=1}^{a-1}(n\ln(|n|) - n) - \sum_{n=1}^{a}(n\ln(|n|) - n)$$

$$= a\ln(2\pi) + \sum_{n=1}^{a-1}(n\ln(|n|) - n) - \sum_{n=1}^{a-1}(n\ln(|n|) - n) - a\ln(|a|) + a$$

$$= a\ln\left(\frac{2\pi}{|a|}\right) + a$$

So, for $a \in \mathbb{Z}/2, a \neq 0$ (all non-zero multiples of $1/2$),

$$\Re \int_{-a}^{a} \ln\Gamma(x)\,dx = a\ln\left(\frac{2\pi}{|a|}\right) + a$$

I have not found an equation that holds for all real values for a, but I have found one for the imaginary component of the integral. For all $a \in \mathbb{R}$,

$$\Im \int_{-a}^{a} \ln\Gamma(x)\,dx = \begin{cases} \pi\left(a - \left\lfloor \frac{a}{2} \right\rfloor\right) & \text{if } \lfloor a \rfloor \in 2\mathbb{N} \\ \pi\left\lfloor \frac{a+1}{2} \right\rfloor & \text{if } \lfloor a \rfloor \in 2\mathbb{N}+1 \\ \pi\left(a - \left\lfloor \frac{a+1}{2} \right\rfloor\right) & \text{if } \lfloor a \rfloor \in 2\mathbb{Z}^- + 1 \\ \pi\left\lfloor \frac{a}{2} \right\rfloor & \text{if } \lfloor a \rfloor \in 2\mathbb{Z}^- \end{cases}$$

As expected, this is a repeating, odd piecewise relation.

6 June

6.1 Generalised arccot integral

Here is a quick one that will reveal itself to be related to a previously written out integral on this document.

$$\int_0^\infty \operatorname{arccot}(x^k)\,dx$$

$$\text{IBP with } u = \operatorname{arccot}(x^k), dv = 1$$

$$= \lim_{N\to\infty}\left[x\operatorname{arccot}(x^k)\right]_0^N + k\int_0^\infty \frac{x^k}{x^{2k}+1}\,dx$$

$$= k\int_0^\infty \frac{x^k}{x^{2k}+1}\,dx$$

Using page 38, where we found that (on the route to a later solution)

$$\int_0^\infty \frac{x^k}{x^n+1}\,dx = \frac{\pi}{n}\csc\left(\frac{\pi(k+1)}{n}\right)$$

Therefore, the integral we have in our problem is the case where $n = 2k$

$$\therefore k\int_0^\infty \frac{x^k}{x^{2k}+1}\,dx = \frac{k\pi}{2k}\csc\left(\frac{\pi(k+1)}{2k}\right)$$

$$= \frac{\pi}{2}\csc\left(\frac{\pi}{2}+\frac{\pi}{2k}\right)$$

$$= \frac{\pi}{2}\sec\left(\frac{\pi}{2k}\right)$$

$$\therefore \int_0^\infty \operatorname{arccot}(x^k)\,dx = \frac{\pi}{2}\sec\left(\frac{\pi}{2k}\right)$$

Some results:

$$\int_0^\infty \operatorname{arccot}(x^\pi)\,dx = \frac{\pi}{2}\sec\frac{1}{2}$$

$$\int_0^\infty \operatorname{arccot}(x^{5/2})\,dx = \frac{\pi}{\phi}$$

$$\int_0^\infty \operatorname{arccot}(x^2)\,dx = \frac{\pi}{\sqrt{2}}$$

$$\int_0^\infty \operatorname{arccot}(x^3)\,dx = \frac{\pi}{\sqrt{3}}$$

$$\int_0^\infty \operatorname{arccot}(x^4)\,dx = \frac{\pi}{\sqrt{2+\sqrt{2}}}$$

6.2 Application of several standard integral techniques

(3rd June) This one is a journey, and will also reveal some other interesting answers too. Solution from [4].

$$\int_{-\infty}^{\infty} \arctan(e^x)\arctan(e^{-x})\,dx$$

$$u = e^x, du = e^x\,dx$$

$$= \int_0^{\infty} \frac{\arctan(u)\operatorname{arccot}(u)}{u}\,du$$

$$u = \tan t, du = \sec^2 t\,dt$$

$$= \int_0^{\frac{\pi}{2}} \frac{t\operatorname{arccot}(\tan t)\sec^2 t}{\tan t}\,dt$$

$$= \int_0^{\frac{\pi}{2}} \frac{2t\operatorname{arccot}(\tan t)}{\sin(2t)}\,dt$$

Using king's rule,

$$= \int_0^{\frac{\pi}{2}} \frac{2t\left(\frac{\pi}{2}-t\right)}{\sin(2t)}\,dt$$

$$u = 2t, du = 2\,dt$$

$$= \frac{1}{4}\int_0^{\pi} \frac{u\left(\pi-u\right)}{\sin u}\,du$$

$$\text{IBP with } u = u(\pi-u), dv = \csc u$$

First consider finding the anti-derivative for $\csc x$.

$$\int \csc x\,dx$$

$$= \int \frac{1}{\sin x}\,dx$$

$$u = \tan\left(\frac{x}{2}\right), dx = \frac{2}{1+u^2}\,du$$

$$= \int \frac{1}{\frac{2u}{1+u^2}} \cdot \frac{2}{1+u^2}\,du$$

$$= \int \frac{1}{u}\,du = \ln\left(\tan\frac{x}{2}\right) + C$$

$$\therefore \frac{1}{4}\int_0^{\pi} \frac{u\left(\pi-u\right)}{\sin u}\,du = \frac{1}{4}\left[u(\pi-u)\ln\left(\tan\frac{u}{2}\right)\right]_0^{\pi} - \frac{1}{4}\int_0^{\pi}(\pi-2u)\ln\left(\tan\frac{u}{2}\right)du$$

$$= -\frac{1}{4}\int_0^{\pi}(\pi-2u)\ln\left(\tan\frac{u}{2}\right)du$$

$$= \frac{1}{2}\int_0^{\pi} u\ln\left(\tan\frac{u}{2}\right)du - \frac{\pi}{4}\int_0^{\pi}\ln\left(\tan\frac{u}{2}\right)du$$

$$x = \frac{u}{2}, dx = \frac{1}{2}\,du$$

$$= 2\int_0^{\frac{\pi}{2}} x\ln(\tan x)\,dx - \frac{\pi}{2}\int_0^{\frac{\pi}{2}} \ln(\tan x)\,dx$$

$$= 2\int_0^{\frac{\pi}{2}} x\ln(\tan x)\,dx - \frac{\pi}{2}\int_0^{\frac{\pi}{2}} \ln(\sin x)\,dx + \frac{\pi}{2}\int_0^{\frac{\pi}{2}} \ln(\cos x)\,dx$$

Using a previous result from page 60, and noticing some specific properties of $\ln(\sin x)$ and $\ln(\cos x)$,

$$= 2\int_0^{\frac{\pi}{2}} x\ln(\tan x)\,dx + \frac{\pi^2}{4}\ln 2 - \frac{\pi^2}{4}\ln 2$$

$$= 2\int_0^{\frac{\pi}{2}} x\ln(\tan x)\,dx$$

$$= 2\int_0^{\frac{\pi}{2}} x\ln(\sin x)\,dx - 2\int_0^{\frac{\pi}{2}} x\ln(\cos x)\,dx$$

Using king's rule on the right integral,

$$= 2\int_0^{\frac{\pi}{2}} x\ln(\sin x)\,dx - 2\int_0^{\frac{\pi}{2}} \left(\frac{\pi}{2} - x\right)\ln(\sin x)\,dx$$

$$= 4\int_0^{\frac{\pi}{2}} x\ln(\sin x)\,dx - \pi\int_0^{\frac{\pi}{2}} \ln(\sin x)\,dx$$

Using a previous result again from page 60,

$$= 4\int_0^{\frac{\pi}{2}} x\ln(\sin x)\,dx + \frac{\pi^2}{2}\ln 2$$

Now consider the following known equation

$$\sum_{n=1}^{\infty} \frac{\cos(2nx)}{n} = -\ln(2\sin x)$$

$$\therefore \ln(\sin x) = -\sum_{n=1}^{\infty} \frac{\cos(2nx)}{n} - \ln 2$$

This holds across the region we are integrating over. Therefore,

$$4\int_0^{\frac{\pi}{2}} x\ln(\sin x)\,dx + \frac{\pi^2}{2}\ln 2 = -4\int_0^{\frac{\pi}{2}} x\left(\sum_{n=1}^{\infty} \frac{\cos(2nx)}{n} + \ln(2)\right)dx + \frac{\pi^2}{2}\ln 2$$

$$= -4\sum_{n=1}^{\infty} \frac{1}{n}\int_0^{\frac{\pi}{2}} x\cos(2nx)\,dx - 4\ln(2)\int_0^{\frac{\pi}{2}} x\,dx + \frac{\pi^2}{2}\ln 2$$

$$= -4\sum_{n=1}^{\infty} \frac{1}{n}\int_0^{\frac{\pi}{2}} x\cos(2nx)\,dx - \frac{\pi^2}{2}\ln 2 + \frac{\pi^2}{2}\ln 2$$

$$= -4\sum_{n=1}^{\infty} \frac{1}{n}\int_0^{\frac{\pi}{2}} x\cos(2nx)\,dx$$

IBP with $u = x, dv = \cos(2nx)$

$$= -4\sum_{n=1}^{\infty} \frac{1}{n}\left(\left[\frac{x}{2n}\sin(2nx)\right]_0^{\frac{\pi}{2}} - \frac{1}{2n}\int_0^{\frac{\pi}{2}} \sin(2nx)\,dx\right)$$

$$= 2\sum_{n=1}^{\infty} \frac{1}{n^2}\int_0^{\frac{\pi}{2}} \sin(2nx)\,dx$$

$$= 2\sum_{n=1}^{\infty}\frac{1}{n^2}\left[\frac{-1}{2n}\cos(2nx)\right]_0^{\frac{\pi}{2}}$$

$$= \sum_{n=1}^{\infty}\frac{1-\cos(\pi n)}{n^3}$$

$$= \sum_{n=1}^{\infty}\frac{1-(-1)^n}{n^3}$$

$$= \frac{2}{1^3}+\frac{0}{2^3}+\frac{2}{3^3}+\frac{0}{4^3}+\frac{2}{5^3}+\frac{0}{6^3}+\cdots$$

$$= 2\left(\sum_{n=1}^{\infty}\frac{1}{n^3}-\sum_{n=1}^{\infty}\frac{1}{(2n)^3}\right)$$

$$= 2\left(\sum_{n=1}^{\infty}\frac{1}{n^3}-\frac{1}{8}\sum_{n=1}^{\infty}\frac{1}{n^3}\right)$$

$$= \frac{7}{4}\sum_{n=1}^{\infty}\frac{1}{n^3}$$

$$= \frac{7}{4}\zeta(3)$$

$$\therefore \int_{-\infty}^{\infty}\arctan(e^x)\arctan(e^{-x})\,dx = \frac{7}{4}\zeta(3)$$

Going back to previous steps along the way, this result also means that

$$\int_0^{\frac{\pi}{2}} x\ln(\sin x)\,dx = \frac{7}{16}\zeta(3)-\frac{\pi^2}{8}\ln 2$$

$$\int_0^{\frac{\pi}{2}} x\ln(\cos x)\,dx = -\frac{7}{16}\zeta(3)-\frac{\pi^2}{8}\ln 2$$

$$\int_0^{\frac{\pi}{2}} x\ln(\tan x)\,dx = \frac{7}{8}\zeta(3)$$

6.3 Nested root integral with golden ratio in answer!

$$\int_0^1 \frac{1}{\sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}}\,dx$$

Consider the denominator, and let it equal to some unknown p

$$\sqrt{x+\sqrt{x+\sqrt{x+\cdots}}} = p$$

$$\therefore \sqrt{x+p} = p$$

$$\therefore p^2 - p - x = 0$$

$$\therefore p = \frac{1 \pm \sqrt{1+4x}}{2}$$

We are looking for the positive case

$$\therefore p = \frac{1 + \sqrt{1+4x}}{2}$$

$$\therefore \int_0^1 \frac{1}{\sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}}\,dx = \int_0^1 \frac{2}{1+\sqrt{1+4x}}\,dx$$

$$u = 1+\sqrt{1+4x}, du = \frac{2}{\sqrt{1+4x}}\,dx$$

$$= \int_2^{2\phi} \frac{u-1}{u}\,du$$

$$= 2\phi - 2 - \ln(2\phi) + \ln 2$$

$$= \frac{2}{\phi} - \ln\phi$$

$$\therefore \int_0^1 \frac{1}{\sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}}\,dx = \frac{2}{\phi} - \ln\phi$$

6.4 Neat application of DUIS on a trig integral

Another one using DUIS but its cool so (Great DUIS practice) (6/6/25):

$$I = \int_0^{\frac{\pi}{2}} \frac{1}{\sin^2 x} \ln\left(\frac{1+\sin^2 x}{1-\sin^2 x}\right) dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\ln(1+\sin^2 x)}{\sin^2 x}\, dx - \int_0^{\frac{\pi}{2}} \frac{\ln(1-\sin^2 x)}{\sin^2 x}\, dx$$

First consider the integral on the left.

$$I_1(k) = \int_0^{\frac{\pi}{2}} \frac{\ln(1+k\sin^2 x)}{\sin^2 x}\, dx$$

$$\therefore I_1'(k) = \int_0^{\frac{\pi}{2}} \frac{1}{\sin^2 x} \cdot \frac{\sin^2 x}{1+k\sin^2 x}\, dx = \int_0^{\frac{\pi}{2}} \frac{1}{1+k\sin^2 x}\, dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{1+k\tan^2(x)\cos^2(x)}\, dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{\sec^2 x + k\tan^2 x}\, dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{(k+1)\tan^2 x + 1}\, dx$$

$$u = \tan x, du = \sec^2 x\, dx$$

$$= \int_0^{\infty} \frac{1}{(k+1)u^2+1}\, du$$

$$= \frac{\pi}{2\sqrt{k+1}}$$

$$\therefore I_1(k) = \frac{\pi}{2} \int \frac{1}{\sqrt{k+1}}\, dk = \pi\sqrt{k+1} + C$$

Letting $k = 0$

$$I_1(0) = \int_0^{\frac{\pi}{2}} 0\, dx = 0 = \pi + C$$

$$\therefore C = -\pi$$

$$\therefore I_1(k) = \int_0^{\frac{\pi}{2}} \frac{\ln(1+k\sin^2 x)}{\sin^2 x}\, dx = \pi\sqrt{k+1} - \pi$$

Now consider the integral on the right on the second line from the top.

$$I_2(n) = \int_0^{\frac{\pi}{2}} \frac{\ln(1-n\sin^2 x)}{\sin^2 x}\, dx$$

$$\therefore I_2'(n) = \int_0^{\frac{\pi}{2}} \frac{1}{\sin^2 x} \cdot \frac{-\sin^2 x}{1-n\sin^2 x}\, dx = -\int_0^{\frac{\pi}{2}} \frac{1}{1-n\sin^2 x}\, dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{n\sin^2 x - 1}\, dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{n\tan^2(x)\cos^2(x) - 1}\, dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{n\tan^2 x - \sec^2 x}\,dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{(n-1)\tan^2 x - 1}\,dx$$

$$u = \tan x, du = \sec^2 x\,dx$$

$$= \int_0^{\infty} \frac{1}{(n-1)u^2 - 1}\,du$$

$$= -\int_0^{\infty} \frac{1}{(1-n)u^2 + 1}\,du$$

$$= -\frac{\pi}{2\sqrt{1-n}}$$

$$\therefore I_2(k) = -\frac{\pi}{2}\int \frac{1}{\sqrt{1-n}}\,dk = \pi\sqrt{1-n} + C$$

Letting $n = 0$

$$I_2(0) = \int_0^{\frac{\pi}{2}} 0\,dx = 0 = \pi + C$$

$$\therefore C = -\pi$$

$$\therefore I_2(n) = -\frac{\pi}{2}\int \frac{1}{\sqrt{1-n}}\,dk = \pi\sqrt{1-n} - \pi$$

As our original integral I is equal to $I_1(k) - I_2(n)$, we obtain

$$I = \pi\sqrt{k+1} - \pi - \pi\sqrt{1-n} + \pi$$

$$= \pi\left(\sqrt{k+1} - \sqrt{1-n}\right)$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{1}{\sin^2 x}\ln\left(\frac{1+k\sin^2 x}{1-n\sin^2 x}\right)dx = \pi\left(\sqrt{k+1} - \sqrt{1-n}\right)$$

From the above line, it is clear that $k \geq -1, n \leq 1$. We may now let $k = n = 1$, and arrive at our solution.

$$\therefore \int_0^{\frac{\pi}{2}} \frac{1}{\sin^2 x}\ln\left(\frac{1+\sin^2 x}{1-\sin^2 x}\right)dx = \pi\sqrt{2}$$

6.5 The dreaded $\sqrt{\tan x}$ anti-derivative

Here is my own solution for the integral of $\sqrt{\tan x}$. It is long and requires a lot of algebraic working out, but each technique used is very simple.

$$\int \sqrt{\tan x}\, dx$$

$$u = \sqrt{\tan x}, dx = \frac{2u}{u^4+1}\, du$$

$$= \int \frac{2u^2}{u^4+1}\, du$$

$$u^4 + 1 = (u - e^{\frac{i\pi}{4}})(u - e^{\frac{3i\pi}{4}})(u - e^{-\frac{3i\pi}{4}})(u - e^{-\frac{i\pi}{4}})$$

$$= 2\int \frac{u^2}{(u - e^{\frac{i\pi}{4}})(u - e^{\frac{3i\pi}{4}})(u - e^{-\frac{3i\pi}{4}})(u - e^{-\frac{i\pi}{4}})}\, du$$
$$= 2\int \frac{AA^* + BB^* + CC^* + DD^*}{(u - e^{\frac{i\pi}{4}})(u - e^{\frac{3i\pi}{4}})(u - e^{-\frac{3i\pi}{4}})(u - e^{-\frac{i\pi}{4}})}\, du$$

where

$$A^* = (u - e^{\frac{3i\pi}{4}})(u - e^{-\frac{3i\pi}{4}})(u - e^{-\frac{i\pi}{4}}),\ B^* = (u - e^{\frac{i\pi}{4}})(u - e^{-\frac{3i\pi}{4}})(u - e^{-\frac{i\pi}{4}}),$$
$$C^* = (u - e^{\frac{i\pi}{4}})(u - e^{\frac{3i\pi}{4}})(u - e^{-\frac{i\pi}{4}}),\ D^* = (u - e^{\frac{i\pi}{4}})(u - e^{\frac{3i\pi}{4}})(u - e^{-\frac{3i\pi}{4}}).$$

To solve, let u equal each of the euler form complex numbers, then simplify the products. We get the following.

$$A = \frac{i}{(e^{\frac{i\pi}{4}} - e^{\frac{3i\pi}{4}})(e^{\frac{i\pi}{4}} - e^{-\frac{3i\pi}{4}})(e^{\frac{i\pi}{4}} - e^{-\frac{i\pi}{4}})} = \frac{i}{(2+2i)i\sqrt{2}} = \frac{1}{2\sqrt{2} + 2i\sqrt{2}}$$
$$B = \frac{-i}{(e^{\frac{3i\pi}{4}} - e^{\frac{i\pi}{4}})(e^{\frac{3i\pi}{4}} - e^{-\frac{3i\pi}{4}})(e^{\frac{3i\pi}{4}} - e^{-\frac{i\pi}{4}})} = \frac{-i}{(-2i)(-\sqrt{2} + \sqrt{2}i)} = \frac{1}{-2\sqrt{2} + 2i\sqrt{2}}$$
$$C = \frac{i}{(e^{-\frac{3i\pi}{4}} - e^{\frac{i\pi}{4}})(e^{-\frac{3i\pi}{4}} - e^{\frac{3i\pi}{4}})(e^{-\frac{3i\pi}{4}} - e^{-\frac{i\pi}{4}})} = \frac{i}{(-2+2i)(-\sqrt{2})} = \frac{1}{-2\sqrt{2} - 2i\sqrt{2}}$$
$$D = \frac{-i}{(e^{-\frac{i\pi}{4}} - e^{\frac{i\pi}{4}})(e^{-\frac{i\pi}{4}} - e^{\frac{3i\pi}{4}})(e^{-\frac{i\pi}{4}} - e^{-\frac{3i\pi}{4}})} = \frac{-i}{(-2-2i)\sqrt{2}} = \frac{1}{2\sqrt{2} - 2i\sqrt{2}}$$

Thus,

$$2\int \frac{u^2}{(u - e^{\frac{i\pi}{4}})(u - e^{\frac{3i\pi}{4}})(u - e^{-\frac{3i\pi}{4}})(u - e^{-\frac{i\pi}{4}})}\, du$$

$$= \int \frac{1}{(\sqrt{2} + i\sqrt{2})(u - e^{\frac{i\pi}{4}})}\, du + \int \frac{1}{(-\sqrt{2} + i\sqrt{2})(u - e^{\frac{3i\pi}{4}})}\, du + \int \frac{1}{(-\sqrt{2} - i\sqrt{2})(u - e^{-\frac{3i\pi}{4}})}\, du + \int \frac{1}{(\sqrt{2} - i\sqrt{2})(u - e^{-\frac{i\pi}{4}})}\, du$$

$$\text{Using the formula } \frac{1}{z} = \frac{a - bi}{a^2 + b^2}$$

$$= \int \frac{(\sqrt{2} - i\sqrt{2})}{4(u - e^{\frac{i\pi}{4}})}\, du + \int \frac{(-\sqrt{2} - i\sqrt{2})}{4(u - e^{\frac{3i\pi}{4}})}\, du + \int \frac{(-\sqrt{2} + i\sqrt{2})}{4(u - e^{-\frac{3i\pi}{4}})}\, du + \int \frac{(\sqrt{2} + i\sqrt{2})}{4(u - e^{-\frac{i\pi}{4}})}\, du$$

$$= \frac{1}{2}\int \frac{(\sqrt{2} - i\sqrt{2})}{(2u - \sqrt{2} - i\sqrt{2})}\, du + \frac{1}{2}\int \frac{(-\sqrt{2} - i\sqrt{2})}{(2u + \sqrt{2} - i\sqrt{2})}\, du + \frac{1}{2}\int \frac{(-\sqrt{2} + i\sqrt{2})}{(2u + \sqrt{2} + i\sqrt{2})}\, du + \frac{1}{2}\int \frac{(\sqrt{2} + i\sqrt{2})}{(2u - \sqrt{2} + i\sqrt{2})}\, du$$

$$= \frac{1}{2}\int \frac{(\sqrt{2}-i\sqrt{2})(2u-\sqrt{2}+i\sqrt{2})}{(2u-\sqrt{2})^2+2}\,du + \frac{1}{2}\int \frac{(-\sqrt{2}-i\sqrt{2})(2u+\sqrt{2}+i\sqrt{2})}{(2u+\sqrt{2})^2+2}\,du$$

$$+\frac{1}{2}\int \frac{(-\sqrt{2}+i\sqrt{2})(2u+\sqrt{2}-i\sqrt{2})}{(2u+\sqrt{2})^2+2}\,du + \frac{1}{2}\int \frac{(\sqrt{2}+i\sqrt{2})(2u-\sqrt{2}-i\sqrt{2})}{(2u-\sqrt{2})^2+2}\,du$$

$$= \frac{1}{8}\int \frac{(2u\sqrt{2}-2ui\sqrt{2}+4i)+(2u\sqrt{2}+2ui\sqrt{2}-4i)}{u^2-u\sqrt{2}+1}\,du$$

$$+\frac{1}{8}\int \frac{(-2u\sqrt{2}-2ui\sqrt{2}-4i)+(-2u\sqrt{2}+2ui\sqrt{2}+4i)}{u^2+u\sqrt{2}+1}\,du$$

$$= \frac{1}{2}\int \frac{u\sqrt{2}}{u^2-u\sqrt{2}+1}\,du - \frac{1}{2}\int \frac{u\sqrt{2}}{u^2+u\sqrt{2}+1}\,du$$

$$t = u\sqrt{2}, dt = \sqrt{2}\,du$$

$$= \frac{1}{\sqrt{2}}\left(\int \frac{t}{t^2-2t+2}\,dt - \int \frac{t}{t^2+2t+2}\,dt\right)$$

$$= \frac{1}{\sqrt{2}}\left(\int \frac{2t-2}{2(t^2-2t+2)}\,dt + \int \frac{1}{t^2-2t+2}\,dt - \int \frac{2t+2}{2(t^2+2t+2)}\,dt + \int \frac{1}{t^2+2t+2}\,dt\right)$$

$$= \frac{1}{\sqrt{2}}\left(\int \frac{2t-2}{2(t^2-2t+2)}\,dt + \int \frac{1}{(t-1)^2+1}\,dt - \int \frac{2t+2}{2(t^2+2t+2)}\,dt + \int \frac{1}{(t+1)^2+1}\,dt\right)$$

From left to right,

$$w = t^2-2t+2 = dw, 2t-2\,dt$$
$$w = t-1, dw = dt$$
$$w = t^2+2t+2, dw = 2t+2\,dt$$
$$w = t+1, dw = dt$$

$$= \frac{1}{\sqrt{2}}\left(\int \frac{1}{2w}\,dw + \int \frac{1}{w^2+1}\,dw - \int \frac{1}{2w}\,dw + \int \frac{1}{w^2+1}\,dw\right)$$

$$= \frac{1}{\sqrt{2}}\left(\frac{1}{2}\ln(t^2-2t+2) + \arctan(t-1) - \frac{1}{2}\ln(t^2+2t+2) + \arctan(t+1)\right)$$

$$= \frac{1}{\sqrt{2}}\left(\frac{1}{2}\ln(2u^2-2u\sqrt{2}+2) + \arctan(u\sqrt{2}-1) - \frac{1}{2}\ln(2u^2+2u\sqrt{2}+2) + \arctan(u\sqrt{2}+1)\right)$$

$$= \frac{1}{2\sqrt{2}}\ln\left(\frac{\tan(x)-\sqrt{2\tan x}+1}{\tan(x)+\sqrt{2\tan x}+1}\right) + \frac{1}{\sqrt{2}}(\arctan(\sqrt{2\tan x}-1) + \arctan(\sqrt{2\tan x}+1)) + C$$

The integral is finally solved. Of notice is if we let $x = \frac{\pi}{2}$.

$$\int_0^{\frac{\pi}{2}} \sqrt{\tan x}\,dx = \frac{\pi}{\sqrt{2}}$$

Again, as mentioned above, there are better ways of solving this indefinite integral, but this is a good one for practising your algebraic simplifying skills. The best solution takes x^4+1 and separates it into the two quadratics seen above. Partial fractions is then used to get it into the state seen after applying the substitution $t = u\sqrt{2}$. It is about 7-8 lines shorter of a solution.

6.6 Power towers...

What is this... (11 June). Took a while to figure this one out, but the connection to the Lambert-W function is not to be forgotten.

$$\int_0^1 (x^x)^{(x^x)^{(x^x)^{\cdots}}}\,dx$$

$$\text{Let } y = (x^x)^{(x^x)^{(x^x)^{\cdots}}}$$

$$\therefore y = (x^x)^y$$

$$\therefore -1 = -y(x^x)^{-y}$$

$$\therefore -\ln(x^x) = -y\ln(x^x)e^{-y\ln(x^x)}$$

$$\therefore -xy\ln(x) = W(-x\ln(x))$$

Where $W(x)$ is the Lambert-W function, defined as the inverse of xe^x.

$$\therefore y = -\frac{W(-x\ln x)}{x\ln x}$$

$$\therefore \int_0^1 (x^x)^{(x^x)^{(x^x)^{\cdots}}}\,dx = -\int_0^1 \frac{W(-x\ln x)}{x\ln x}\,dx$$

The Lambert-W function has the series

$$W(x) = \sum_{n=1}^{\infty} \frac{(-n)^{n-1}x^n}{n!}$$

$$\therefore -\int_0^1 \frac{W(-x\ln x)}{x\ln x}\,dx = -\int_0^1 \sum_{n=1}^{\infty} \frac{(-n)^{n-1}(-x\ln x)^n}{n!x\ln x}\,dx$$

$$= -\sum_{n=1}^{\infty} \frac{(-n)^{n-1}(-1)^n}{n!} \int_0^1 (x\ln x)^{n-1}\,dx$$

$$u = -\ln x, du = -\frac{1}{x}\,dx$$

$$= \sum_{n=1}^{\infty} \frac{(-n)^{n-1}(-1)^{n+1}}{n!} \int_0^{\infty} e^{-u}(-e^{-u}u)^{n-1}\,du$$

$$= \sum_{n=1}^{\infty} \frac{(-n)^{n-1}(-1)^{n+1}(-1)^{n-1}}{n!} \int_0^{\infty} e^{-nu}u^{n-1}\,du$$

$$= \sum_{n=1}^{\infty} \frac{n^{n-1}(-1)^{n+1}}{n!} \int_0^{\infty} e^{-nu}u^{n-1}\,du$$

$$t = nu, dt = n\,du$$

$$= \sum_{n=1}^{\infty} \frac{n^{n-2}(-1)^{n+1}}{n!n^{n-1}} \int_0^{\infty} e^{-t}t^{n-1}\,du$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n!n^{n-1}n^{2-n}}(n-1)!$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$$

$$= \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \frac{1}{5^2} - \frac{1}{6^2} + \cdots$$

We observe that this sum is equal to the sum of the reciprocals of all of the square integers, minus two lots of the sum of the reciprocals of all of the square evens.

$$= \sum_{n=1}^{\infty} \frac{1}{n^2} - 2\sum_{n=1}^{\infty} \frac{1}{(2n)^2}$$

As mentioned on page 28, these sums have the following solution.

$$= \frac{\pi^2}{6} - \frac{\pi^2}{12}$$

$$= \frac{\pi^2}{12}$$

$$\therefore \int_0^1 (x^x)^{(x^x)^{(x^x)^{\cdots}}} \, dx = \frac{\pi^2}{12}$$

Similarly, the same method can be used to obtain

$$\int_0^1 (x^{-x})^{(x^{-x})^{(x^{-x})^{\cdots}}} \, dx = \frac{\pi^2}{6}$$

This answer comes about due to an extra $(-1)^{n-1}$ term in the series, cancelling out with the numerator of the sum at the top of this page. Further, with this integral here, the lambert function we get is $W(x\ln x)$, which can be simplified to equal $\ln x$ for $x \geq \frac{1}{e}$. Thus,

$$\int_{\frac{1}{e}}^{k} (x^{-x})^{(x^{-x})^{(x^{-x})^{\cdots}}} \, dx = \int_{\frac{1}{e}}^{k} \frac{W(x\ln x)}{x\ln x} \, dx$$

$$= \int_{\frac{1}{e}}^{k} \frac{\ln x}{x\ln x} \, dx = \int_{\frac{1}{e}}^{k} \frac{1}{x} \, dx = \ln(k) + 1$$

$$= \ln(k) + 1$$

This means that

$$\int_0^k (x^{-x})^{(x^{-x})^{(x^{-x})^{\cdots}}} \, dx = \frac{\pi^2}{6} + \ln k \ \ (1/e \leq k \leq e)$$

We can apply the same methods to other power towers, given that the series is defined over the interval of integration, or otherwise that the lambert-W function term is at one of its special values that can be integrated or simplified. For example, we can use the method on the first half of the previous page to find that (for $k \geq -\frac{1}{e}$)

$$\int_0^k (e^{-x})^{(e^{-x})^{(e^{-x})^{\cdots}}} \, dx = \int_0^k \frac{W(x)}{x} \, dx$$

This can be integrated with a u-sub

$$u = W(x), dx = (u+1)e^u \, du$$

$$= \int_0^{W(k)} \frac{u}{ue^u} \cdot (u+1)e^u \, du$$

$$= \int_0^{W(k)} u + 1 \, du = \frac{W^2(k)}{2} + W(k)$$

$$= \frac{W(k)}{2}(W(k)+2)$$

$$\therefore \int_0^k (e^{-x})^{(e^{-x})^{(e^{-x})^{\cdots}}} dx = \frac{W(k)}{2}(W(k)+2)$$

Letting $k = e$, and as $W(e) = 1$, then

$$\int_0^e (e^{-x})^{(e^{-x})^{(e^{-x})^{\cdots}}} dx = \frac{3}{2}$$

Similarly, for $k \leq \frac{1}{e}$ (Note that $W(-\frac{1}{e}) = -1$),

$$\int_0^k (e^{x})^{(e^{x})^{(e^{x})^{\cdots}}} dx = -\frac{W(-k)}{2}(W(-k)+2)$$

$$\therefore \int_0^{\frac{1}{e}} (e^{x})^{(e^{x})^{(e^{x})^{\cdots}}} dx = \frac{1}{2}$$

Another example is this monster, which can be simplified by taking use of the fact that $W(x^{x+1}\ln x) = x\ln x$

$$\int_0^1 \left(x^{-x^{x+1}}\right)^{\left(x^{-x^{x+1}}\right)^{\cdots}} dx = -\int_0^1 \frac{W(-\ln\left(x^{-x^{x+1}}\right)}{\ln\left(x^{-x^{x+1}}\right)}$$

$$= \int_0^1 \frac{W(x^{x+1}\ln x)}{x^{x+1}\ln x}\, dx = \int_0^1 \frac{x\ln x}{x^{x+1}\ln x}\, dx$$

$$= \int_0^1 x^{-x}\, dx$$

$$= \sum_{n=1}^{\infty} n^{-n}$$

More generally,

$$\int_a^b f(x)^{f(x)^{f(x)^{\cdots}}} dx = \int_a^b \frac{W(-\ln(f(x))}{-\ln(f(x))}\, dx$$

A couple more power tower integrals (third one is for $a > 1$):

$$\int_{-1}^{\infty} \left(e^{-xe^x}\right)^{\left(e^{-xe^x}\right)^{\cdots}} dx = e$$

$$\int_0^1 (x^{-x/2})^{(x^{-x/2})^{(x^{-x/2})^{\cdots}}} dx = \frac{\pi^2}{6} - \ln^2 2$$

$$\int_0^{\infty} (a^{-x})^{(a^{x})^{(a^{-x})^{(a^{x})^{\cdots}}}} dx = \frac{\pi^2}{12\ln a}$$

The first one uses a property of the function, the second one uses the dilogarithm, and the last one's solution can be seen on page 94.

6.7 Three integrals using a Geometric sum

Introducing... Laplace. (17th June). Solution entirely from [5].

$$\int_0^1 \frac{\ln(x)\ln(-\ln x)}{1-x}\,dx$$

$$u = -\ln x, du = -\frac{1}{x}\,dx$$

$$= \int_0^\infty \frac{-ue^{-u}\ln u}{1-e^{-u}}\,du$$

Using geometric sum with $a_1 = -ue^{-u}\ln u, r = e^{-u}$,

$$= \int_0^\infty \sum_{n=1}^\infty -ue^{-u}\ln(u)e^{-u(n-1)}\,du$$

$$= -\sum_{n=1}^\infty \int_0^\infty e^{-nu}u\ln u\,du$$

$$= -\sum_{n=1}^\infty \mathcal{L}\{u\ln u\}(n)$$

$$= -\sum_{n=1}^\infty \frac{1}{n}\mathcal{L}\{1+\ln u\}(n)$$

$$= -\sum_{n=1}^\infty \frac{1}{n}\int_0^\infty e^{-nu}(1+\ln u)\,du$$

$$= -\sum_{n=1}^\infty \frac{1}{n}\left(\frac{1}{n} + \int_0^\infty e^{-nu}\ln u\,du\right)$$

$$= -\frac{\pi^2}{6} - \sum_{n=1}^\infty \frac{1}{n}\int_0^\infty e^{-nu}\ln u\,du$$

$$t = un, dt = n\,du$$

$$= -\frac{\pi^2}{6} - \sum_{n=1}^\infty \frac{1}{n^2}\left(\int_0^\infty e^{-t}\ln t\,dt - \ln n\int_0^\infty e^{-t}\,dt\right)$$

It's known that

$$\int_0^\infty e^{-x}\ln x\,dx = -\gamma$$

Therefore, our expression above is equal to

$$= \frac{\pi^2(\gamma-1)}{6} + \sum_{n=1}^\infty \frac{\ln n}{n^2}$$

The sum is equal to $-\zeta'(2)$, whose solution yields

$$= \frac{\pi^2(\gamma-1)}{6} + \frac{\pi^2}{6}(12\ln(A) - \ln(2\pi) - \gamma)$$

Where A is the Glaisher-Kinkelin constant, equal to roughly 1.282 (4sf).

$$= \frac{\pi^2}{6}(12\ln(A) - \ln(2\pi) - 1)$$

$$\therefore \int_0^1 \frac{\ln(x)\ln(-\ln x)}{1-x}\,dx = \frac{\pi^2}{6}(12\ln(A) - \ln(2\pi) - 1)$$

Nice. (19th June). Solved with the prior knowledge that the method involves a geometric sum followed by IBP.

$$\int_{-\infty}^{\infty} \frac{\sin x \cos x}{\sinh x \cosh x}\, dx$$

$$= 2\int_{-\infty}^{\infty} \frac{\sin(2x)}{(e^x - e^{-x})(e^x + e^{-x})}\, dx$$

$$= 2\int_{-\infty}^{\infty} \frac{\sin(2x)}{e^{2x} - e^{-2x}}\, dx$$

$$u = 2x, du = 2\, dx$$

$$= \int_{-\infty}^{\infty} \frac{\sin u}{e^u - e^{-u}}\, du$$

$$= \int_{-\infty}^{\infty} \frac{e^{-u}\sin u}{1 - e^{-2u}}\, du$$

Converting to geometric sum with $a_1 = e^{-u}\sin u, r = e^{-2u}$, and applying property of even functions

$$= 2\int_0^{\infty} \sum_{n=1}^{\infty} e^{-u}\sin(u) e^{-2u(n-1)}\, du$$

$$= 2\sum_{n=1}^{\infty} \int_0^{\infty} e^{-u(2n-1)} \sin u\, du$$

Consider

$$I = \int_0^{\infty} e^{-u(2n-1)} \sin u\, du$$

$$\text{IBP with } u = e^{-u(2n-1)}, dv = \sin u$$

$$= \lim_{N\to\infty} \big[-e^{-u(2n-1)}\cos u\big]_0^N - (2n-1)\int_0^{\infty} e^{-u(2n-1)}\cos u\, du$$

$$\text{IBP again with } u = e^{-u(2n-1)}, dv = \cos u$$

$$= 1 - (2n-1)\bigg(\lim_{N\to\infty} \big[e^{-u(2n-1)}\sin u\big]_0^N + (2n-1)\int_0^{\infty} e^{-u(2n-1)}\sin u\, du\bigg)$$

$$= 1 - (2n-1)^2 I$$

$$\therefore I = \frac{1}{(2n-1)^2 + 1}$$

$$\therefore 2\sum_{n=1}^{\infty}\int_0^{\infty} e^{-u(2n-1)}\sin u\, du = 2\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2+1}$$

To solve this series, consider the Weierstrass product for $\cosh x$:

$$\cosh x = \prod_{n=1}^{\infty}\left(1 + \frac{4x^2}{(2n-1)^2\pi^2}\right)$$

$$\therefore \ln(\cosh x) = \ln\left(\prod_{n=1}^{\infty}\left(1 + \frac{4x^2}{(2n-1)^2\pi^2}\right)\right)$$

$$= \sum_{n=1}^{\infty} \ln\left(1 + \frac{4x^2}{(2n-1)^2\pi^2}\right)$$

Differentiating both sides:

$$\therefore \tanh x = \sum_{n=1}^{\infty} \frac{\frac{8x}{(2n-1)^2\pi^2}}{1 + \frac{4x^2}{(2n-1)^2\pi^2}}$$

$$= \sum_{n=1}^{\infty} \frac{\frac{8x}{(2n-1)^2\pi^2}}{\frac{(2n-1)^2\pi^2+4x^2}{(2n-1)^2\pi^2}}$$

$$= \sum_{n=1}^{\infty} \frac{8x}{(2n-1)^2\pi^2 + x^2}$$

$$\text{Let } x = \frac{\pi}{2}$$

$$\tanh \frac{\pi}{2} = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2 + 1}$$

$$\therefore 2\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2 + 1} = \frac{\pi}{2} \tanh \frac{\pi}{2}$$

$$\therefore \int_{-\infty}^{\infty} \frac{\sin x \cos x}{\sinh x \cosh x}\, dx = \frac{\pi}{2} \tanh \frac{\pi}{2}$$

Third integral in a row using Geometric sum (19/6).

$$\int_0^\infty \frac{x^2 \ln x}{\cosh x + 1}\, dx$$

Consider

$$I(k) = \int_0^\infty \frac{x^k}{\cosh x + 1}\, dx$$

$$\therefore I'(k) = \int_0^\infty \frac{x^k \ln x}{\cosh x + 1}\, dx$$

We are thus trying to solve for $I'(2)$. Let's thus solve for $I(k)$, then differentiate.

$$\begin{aligned}\int_0^\infty \frac{x^k}{\cosh x + 1}\, dx &= 2\int_0^\infty \frac{x^k}{e^x + e^{-x} + 2}\, dx \\ &= 2\int_0^\infty \frac{e^{-x}x^k}{1 + e^{-2x} + 2e^{-x}}\, dx \\ &= 2\int_0^\infty \frac{e^{-x}x^k}{(e^{-x}+1)^2}\, dx\end{aligned}$$

Consider the geometric sum with ratio $-r$, for $|r| < 1$, differentiate both sides with respect to r, and move the -1 term to the right hand side. We can then re-index the sum to $n = 1$ without any changes to the inside, since the $n = 0$ term will be zero.

$$\begin{aligned}\frac{a_1}{1-(-r)} = \frac{a_1}{1+r} &= \sum_{n=0}^{\infty}(-1)^n a_1 r^n \\ \therefore \frac{a_1}{(1+r)^2} &= -\sum_{n=0}^{\infty} n(-1)^n a_1 r^{n-1} \\ &= -\sum_{n=1}^{\infty} n(-1)^n a_1 r^{n-1}\end{aligned}$$

In our case, $a_1 = e^{-x}x^k, r = e^{-x}$.

$$\begin{aligned}\therefore 2\int_0^\infty \frac{e^{-x}x^k}{(e^{-x}+1)^2}\, dx &= 2\int_0^\infty \sum_{n=1}^{\infty} n(-1)^{n+1} e^{-x}x^k e^{-x(n-1)}\, dx \\ &= 2\sum_{n=1}^{\infty} n(-1)^{n+1}\int_0^\infty e^{-x}x^k e^{-x(n-1)}\, dx \\ &= 2\sum_{n=1}^{\infty} n(-1)^{n+1}\int_0^\infty x^k e^{-nx}\, dx\end{aligned}$$

$$u = xn, du = n\, dx$$

$$= 2\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^k}\int_0^\infty u^k e^{-u}\, du$$

This integral is now the gamma function at $k+1$.

$$= 2\Gamma(k+1)\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^k} \tag{5}$$

Using the method on top of page 73, and then recognising that the sum on the left below is equal to $\zeta(k)$,

$$= 2\Gamma(k+1)\left(\sum_{n=1}^{\infty}\frac{1}{n^k} - 2\sum_{n=1}^{\infty}\frac{1}{(2n)^k}\right)$$

$$= 2\Gamma(k+1)\zeta(k)\big(1-2^{1-k}\big)$$

$$\therefore I(k) = \int_0^{\infty}\frac{x^k}{\cosh x+1}\,dx = 2\Gamma(k+1)\zeta(k)\big(1-2^{1-k}\big)$$

We may now differentiate both sides and let $k = 2$ to obtain our result.

$$\therefore I'(k) = \int_0^{\infty}\frac{x^k\ln x}{\cosh x+1}\,dx = 2\Gamma'(k+1)\zeta(k)\big(1-2^{1-k}\big)+2\Gamma(k+1)\zeta'(k)\big(1-2^{1-k}\big)+2\Gamma(k+1)\zeta(k)2^{1-k}\ln 2$$

$$\therefore \int_0^{\infty}\frac{x^2\ln x}{\cosh x+1}\,dx = \Gamma'(3)\zeta(2)+2\zeta'(2)+2\zeta(2)\ln 2$$

The constants above have solutions previously seen on pages 28 and 74. Furthermore, $\Gamma'(3) = 3-2\gamma$.

$$= \frac{\pi^2}{6}(3-2\gamma)+\frac{\pi^2}{3}\big(\gamma+\ln(2\pi)-12\ln A\big)+\frac{\pi^2}{3}\ln 2$$

$$= \frac{\pi^2}{6}\big(3+2\ln(2\pi)-24\ln(A)+2\ln 2\big)$$

$$= \frac{\pi^2}{6}\big(3+2\ln(4\pi)-24\ln A\big)$$

$$\therefore \int_0^{\infty}\frac{x^2\ln x}{\cosh x+1}\,dx = \frac{\pi^2}{6}\big(3+2\ln(4\pi)-24\ln A\big)$$

When considering a general value for k, we must have $k > 1$ as otherwise we run into non-convergent values. However, the integral is still convergent for all $k > -1$. To solve for these values of k, we may use the analytic continuation of the Riemann zeta function. Thus, now consider when $k = 0$.

$$I'(0) = \int_0^{\infty}\frac{\ln x}{\cosh x+1}\,dx = -2\Gamma'(1)\zeta(0)-2\zeta'(0)+4\zeta(0)\ln 2$$

$$\text{where } \Gamma'(1) = -\gamma, \zeta(0) = -\frac{1}{2}, \zeta'(0) = -\frac{1}{2}\ln(2\pi)$$

$$\therefore \int_0^{\infty}\frac{\ln x}{\cosh x+1}\,dx = \ln\left(\frac{\pi}{2}\right)-\gamma$$

This does not work however for $k = 1$. For this value, we actually need go back to equation 5. The sum seen there is called the eta function, $\eta(k)$. By then differentiating equation 5, a simpler expression for the general solution, $k > 0$, is as follows.

$$\int_0^{\infty}\frac{x^k\ln x}{\cosh x+1}\,dx = 2\Gamma'(k+1)\eta(k)+2\Gamma(k+1)\eta'(k)$$

Then, using the values $\eta(1) = \ln 2$ and $\eta'(1) = \gamma\ln(2)-\frac{1}{2}\ln^2(2)$, we get the following equation. Also, note that $\Gamma'(2) = 1-\gamma$.

$$I'(1) = \int_0^{\infty}\frac{x\ln x}{\cosh x+1}\,dx = 2\Gamma'(2)\eta(1)+2\eta'(1) = 2\ln 2-\ln^2 2$$

$$\therefore \int_0^{\infty}\frac{x\ln x}{\cosh x+1}\,dx = 2\ln 2-\ln^2 2$$

6.8 Integral using Legendre duplication formula

Utilising the Legendre Duplication formula (I solved this one knowing that the method uses this formula), we can obtain a cool answer as the solution to this integral. (20th June)

$$\int_0^1 \sin(\sqrt{-\ln x})\,dx$$

$$u = -\ln x, du = -\frac{1}{x}\,dx$$

$$= \int_0^\infty e^{-u}\sin(\sqrt{u})\,du$$

Using the Maclaurin series for $\sin\sqrt{u}$, we get

$$= \int_0^\infty e^{-u}\sum_{n=0}^{\infty}\frac{(-1)^n u^{n+\frac{1}{2}}}{(2n+1)!}\,du$$

$$= \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}\int_0^\infty e^{-u}u^{n+\frac{1}{2}}\,du$$

$$= \sum_{n=0}^{\infty}\frac{(-1)^n\left(n+\frac{1}{2}\right)!}{(2n+1)!}$$

The Legendre Duplication formula states (where $\Gamma(z) = (z-1)!$) that

$$\therefore \Gamma(2z) = \frac{\Gamma(z)\Gamma\left(z+\frac{1}{2}\right)}{2^{1-2z}\sqrt{\pi}}$$

$$\therefore (2n+1)! = \frac{n!\left(n+\frac{1}{2}\right)!}{2^{-2n-1}\sqrt{\pi}}$$

$$\therefore \sum_{n=0}^{\infty}\frac{(-1)^n\left(n+\frac{1}{2}\right)!}{(2n+1)!} = \sum_{n=0}^{\infty}\frac{(-1)^n 2^{-2n-1}\sqrt{\pi}}{n!}$$

$$= \frac{\sqrt{\pi}}{2}\sum_{n=0}^{\infty}\frac{(-1)^n}{2^{2n}n!}$$

$$= \frac{\sqrt{\pi}}{2}\sum_{n=0}^{\infty}\frac{1}{n!}\left(\frac{-1}{4}\right)^n$$

The sum above is now in the form of the Maclaurin series for $e^{-\frac{1}{4}}$. Thus, the solution becomes

$$= \frac{1}{2}\sqrt{\frac{\pi}{\sqrt{e}}}$$

$$\therefore \int_0^1 \sin(\sqrt{-\ln x})\,dx = \frac{1}{2}\sqrt{\frac{\pi}{\sqrt{e}}}$$

Similarly, we may use the Maclaurin series for $\sinh\sqrt{u}$ to obtain this integral.

$$\int_0^1 \sinh(\sqrt{-\ln x})\,dx = \frac{1}{2}\sqrt{\pi\sqrt{e}}$$

6.9 Constant spam solution

Constant spam (20th June). Done with the prior knowledge that it involves DUIS of a variable k in the exponent of e^{-kx}.

$$\int_0^\infty \frac{e^{-\alpha x}\sin(nx)\ln x}{x}\,dx$$

$$= \Im \int_0^\infty \frac{e^{(ni-\alpha)x}\ln x}{x}\,dx$$

Consider

$$I(k) = \int_0^\infty \frac{e^{-kx}\ln x}{x}\,dx$$

where $k = \alpha - ni$ in our original integral. Differentiating with respect to k,

$$\therefore I'(k) = -\int_0^\infty e^{-kx}\ln x\,dx$$

$$u = kx, du = k\,dx$$

$$= -\frac{1}{k}\int_0^\infty e^{-u}\ln u\,du + \frac{\ln k}{k}\int_0^\infty e^{-u}\,du$$

$$= -\frac{1}{k}\int_0^\infty e^{-u}\ln u\,du + \frac{\ln k}{k}$$

$$\text{As } \int_0^\infty e^{-x}\ln x\,dx = -\gamma$$

$$-\frac{1}{k}\int_0^\infty e^{-u}\ln u\,du + \frac{\ln k}{k} = \frac{\gamma}{k} + \frac{\ln k}{k}$$

$$\therefore I(k) = \gamma\int \frac{1}{k}\,dk + \int \frac{\ln k}{k}\,dk$$

$$= \gamma\ln(k) + \int \frac{\ln k}{k}\,dk$$

$$t = \ln k, dt = \frac{1}{k}dk$$

$$= \gamma\ln(k) + \int t\,dt$$

$$= \gamma\ln(k) + \frac{1}{2}\ln^2(k) + C$$

We will leave the integration constant C there for now. As $k = \alpha - ni$, and we are looking for the imaginary component, we may discard the real parts of the expression below.

$$\therefore I = \Im\left(\gamma\ln(\alpha - ni) + \frac{1}{2}\ln^2(\alpha - ni) + C\right)$$

$$= \Im\left(\gamma\ln\left(\sqrt{\alpha^2+n^2}\,e^{i\arctan(-\frac{n}{\alpha})}\right) + \frac{1}{2}\ln^2\left(\sqrt{\alpha^2+n^2}\,e^{i\arctan(-\frac{n}{\alpha})}\right) + C\right)$$

$$= \gamma\arctan\left(\frac{-n}{\alpha}\right) + \Im\frac{1}{2}\left(\frac{1}{2}\ln(\alpha^2+n^2) + i\arctan\left(\frac{-n}{\alpha}\right)\right)^2 + C$$

$$= -\gamma\arctan\left(\frac{n}{\alpha}\right) - \frac{1}{2}\ln(\alpha^2+n^2)\arctan\left(\frac{n}{\alpha}\right) + C$$

As $\alpha \to \infty$, $-\arctan\left(\frac{n}{\alpha}\right) \to 0$ and $I \to 0$.

$$\therefore C = 0$$

$$\therefore \int_0^\infty \frac{e^{-\alpha x} \sin(nx) \ln x}{x}\, dx = -\gamma \arctan\left(\frac{n}{\alpha}\right) - \frac{1}{2} \ln(\alpha^2 + n^2) \arctan\left(\frac{n}{\alpha}\right)$$

Some results:

$$\int_0^\infty \frac{\sin x \ln x}{x}\, dx = -\frac{\gamma\pi}{2}$$

$$\int_0^\infty \frac{e^{-x} \sin x \ln x}{x}\, dx = -\frac{\pi}{4}\left(\gamma + \frac{1}{2} \ln 2\right)$$

Further, we can make $n = i$ and get solutions like

$$\int_0^\infty \frac{e^{-2x} \sinh x \ln x}{x}\, dx = -\frac{1}{4} \ln^2 3 - \frac{\gamma}{2} \ln 3$$

$$\int_0^\infty \frac{e^{-\sqrt{2}x} \sinh x \ln x}{x}\, dx = \frac{\gamma}{2} \ln\left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)$$

6.10 The lemniscate biscuit

The lemniscate constant ϖ is equal to $\Gamma(1/4)^2/2\sqrt{2\pi}$.

$$\int_0^\infty \frac{1}{\sqrt{\cosh x}}\,dx$$

$$= \frac{\sqrt{2}}{2}\int_{-\infty}^\infty \frac{1}{\sqrt{e^x+e^{-x}}}\,dx$$

$$= \frac{1}{\sqrt{2}}\int_{-\infty}^\infty \frac{e^{-\frac{x}{2}}}{\sqrt{1+e^{-2x}}}\,dx$$

$$u = e^{-2x}, du = -2e^{-2x}\,dx$$

$$= \frac{1}{2\sqrt{2}}\int_0^\infty \frac{u^{-\frac{3}{4}}}{\sqrt{1+u}}\,du$$

This integral is expressible with the beta function

$$= \frac{1}{2\sqrt{2}}B\left(\frac{1}{4},\frac{1}{4}\right)$$

Writing the beta function in terms of the gamma function.

$$= \frac{1}{2\sqrt{2}}\cdot\frac{\Gamma^2\left(\frac{1}{4}\right)}{\Gamma\left(\frac{1}{2}\right)}$$

$$= \frac{\Gamma^2\left(\frac{1}{4}\right)}{2\sqrt{2\pi}}$$

This expression is equal to the Lemniscate constant

$$= \varpi$$

$$\therefore \int_0^\infty \frac{1}{\sqrt{\cosh x}}\,dx = \varpi$$

Similarly, when we perform the same steps on

$$\int_0^\infty \frac{1}{\sqrt{\sinh x}}\,dx$$

We get

$$= \frac{1}{\sqrt{2}}B\left(\frac{1}{4},\frac{1}{2}\right)$$

$$= \frac{1}{\sqrt{2}}\cdot\frac{\Gamma\left(\frac{1}{4}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{3}{4}\right)}$$

$$= \frac{\Gamma\left(\frac{1}{4}\right)}{\Gamma\left(\frac{3}{4}\right)}\sqrt{\frac{\pi}{2}}$$

Using the reflection formula,

$$= \frac{\Gamma^2\left(\frac{1}{4}\right)}{2\sqrt{\pi}}$$

$$= \varpi\sqrt{2}$$

$$\therefore \int_0^\infty \frac{1}{\sqrt{\sinh x}}\,dx = \varpi\sqrt{2}$$

6.11 My own collection of integrals (2), part 2!

Expanded on from page 38. In the list of integrals at the top, these will be alongside the others from page 38.

$$\int_{-\infty}^{\infty} \frac{\cos(kx)}{\cosh^2(nx)}\,dx = \frac{k\pi}{n^2}\operatorname{csch}\left(\frac{k\pi}{2n}\right)$$

$$\therefore \int_{-\infty}^{\infty} \frac{\cos x}{\cosh^2 x}\,dx = \pi\operatorname{csch}\frac{\pi}{2}$$

$$\int_{-\infty}^{\infty} \frac{\cos^2(kx)}{\cosh^2(nx)}\,dx = \frac{k\pi}{n^2}\operatorname{csch}\left(\frac{k\pi}{n}\right) + \frac{1}{n}$$

$$\therefore \int_{-\infty}^{\infty} \frac{\cos^2 x}{\cosh^2 x}\,dx = \pi\operatorname{csch}(\pi) + 1$$

Solution of

$$\int_{-\infty}^{\infty} \frac{\cos^2 x}{\cosh^2 x}\,dx$$

$$= \int_{-\infty}^{\infty} \frac{1+\cos(2x)}{1+\cosh(2x)}\,dx$$

$$u = 2x, du = 2\,dx$$

$$= \frac{1}{2}\int_{-\infty}^{\infty} \frac{1+\cos u}{1+\cosh u}\,du$$

$$= \frac{1}{2}\int_{-\infty}^{\infty} \frac{\cos u}{1+\cosh u}\,du + \frac{1}{2}\int_{-\infty}^{\infty} \frac{1}{1+\cosh u}\,du$$

$$= \int_{-\infty}^{\infty} \frac{\cos u}{2+e^u+e^{-u}}\,du + \frac{1}{4}\int_{-\infty}^{\infty} \operatorname{sech}^2\left(\frac{u}{2}\right)du$$

On the right integral,

$$t = \frac{u}{2}, dt = \frac{1}{2}\,du$$

$$= \int_{-\infty}^{\infty} \frac{\cos u}{2+e^u+e^{-u}}\,du + \frac{1}{2}\int_{-\infty}^{\infty} \operatorname{sech}^2 t\,dt$$

The integral of $\operatorname{sech}^2 x$ is $\tanh x$

$$= \int_{-\infty}^{\infty} \frac{\cos u}{2+e^u+e^{-u}}\,du + 1$$

$$= \int_{-\infty}^{\infty} \frac{e^u\cos u}{2e^u+e^{2u}+1}\,du + 1$$

$$= \int_{-\infty}^{\infty} \frac{e^u\cos u}{(e^u+1)^2}\,du + 1$$

$$t = e^u, dt = e^u\,du$$

$$= \int_{0}^{\infty} \frac{\cos(\ln t)}{(t+1)^2}\,dt + 1$$

$$= \int_{0}^{\infty} \frac{t^i}{(t+1)^2}\,dt + 1$$

This integral can be expressed in terms of the beta function, and thus in terms of the gamma function

$$= B(i + 1, 1 - i) + 1$$

$$= \Gamma(i + 1)\Gamma(1 - i) + 1$$

$$= i\Gamma(i)\Gamma(1 - i) + 1$$

Using the Euler reflection formula,

$$\Gamma(1 - i) = \frac{\pi}{\sin(i\pi)\Gamma(i)}$$

$$\therefore i\Gamma(i)\Gamma(1 - i) + 1 = \frac{i\pi}{\sin(i\pi)} + 1$$

$$= \pi \operatorname{csch}(\pi) + 1$$

$$\therefore \int_{-\infty}^{\infty} \frac{\cos^2 x}{\cosh^2 x}\, dx = \pi \operatorname{csch}(\pi) + 1$$

6.12 Another double DUIS integral (HOLY constant spam)

Very cool solutions come outta this with multiple constants included.

$$\int_0^\infty \ln^2(x)\sin(x^2)\,dx$$

First, let

$$I(k) = \int_0^\infty x^k \sin(x^2)\,dx$$

$$\therefore I'(k) = \int_0^\infty x^k \ln(x)\sin(x^2)\,dx$$

$$\therefore I''(k) = \int_0^\infty x^k \ln^2(x)\sin(x^2)\,dx$$

We are thus solving for $I''(0)$.

$$\int_0^\infty x^k \sin(x^2)\,dx = -\Im\int_0^\infty x^k e^{-ix^2}\,dx$$

$$u = ix^2, du = \ 2ix\,dx$$

$$= -\Im\frac{1}{2i^{\frac{k+1}{2}}}\int_0^\infty u^{\frac{k-1}{2}}e^{-u}\,du$$

$$= -\Im\frac{1}{2}\Gamma\left(\frac{k+1}{2}\right)e^{\frac{-i\pi(k+1)}{4}}$$

$$= \frac{1}{2}\Gamma\left(\frac{k+1}{2}\right)\sin\left(\frac{\pi(k+1)}{4}\right)$$

$$\therefore I'(k) = \frac{1}{4}\Gamma'\left(\frac{k+1}{2}\right)\sin\left(\frac{\pi(k+1)}{4}\right) + \frac{\pi}{8}\Gamma\left(\frac{k+1}{2}\right)\cos\left(\frac{\pi(k+1)}{4}\right)$$

Providing in of itself some cool solutions for $I'(k)$, at values of k where the integral is convergent. The solution above can be written in terms of the digamma function $\psi_0(z)$.

$$= \frac{1}{8}\Gamma\left(\frac{k+1}{2}\right)\left(2\psi_0\left(\frac{k+1}{2}\right)\sin\left(\frac{\pi(k+1)}{4}\right) + \pi\cos\left(\frac{\pi(k+1)}{4}\right)\right)$$

Letting $k=0$, (where one can find that $\psi_0\left(\frac{1}{2}\right) = -\gamma - 2\ln 2$ by using Gauss's digamma theorem), we get

$$I'(0) = \int_0^\infty \ln(x)\sin(x^2)\,dx = \frac{1}{8}\sqrt{\frac{\pi}{2}}(\pi - 2\gamma - 4\ln 2)$$

To final our solution for the integral at the top of the page, $I''(k)$, we then differentiate the solution for $I'(k)$

$$\therefore I''(k) = \frac{1}{8}\Gamma''\left(\frac{k+1}{2}\right)\sin\left(\frac{\pi(k+1)}{4}\right) + \frac{\pi}{8}\Gamma'\left(\frac{k+1}{2}\right)\cos\left(\frac{\pi(k+1)}{4}\right) - \frac{\pi^2}{32}\Gamma\left(\frac{k+1}{2}\right)\sin\left(\frac{\pi(k+1)}{4}\right)$$

This can then be written in terms of the digamma function and the trigamma function $\psi_1(z)$.

$$\therefore \int_0^\infty x^k \ln^2(x)\sin(x^2)\,dx =$$

$$\frac{1}{8}\Gamma\left(\frac{k+1}{2}\right)\left(\left(\psi_0^2\left(\frac{k+1}{2}\right) + \psi_1\left(\frac{k+1}{2}\right)\right)\sin\left(\frac{\pi(k+1)}{4}\right) + \pi\psi_0\left(\frac{k+1}{2}\right)\cos\left(\frac{\pi(k+1)}{4}\right) - \frac{\pi^2}{4}\sin\left(\frac{\pi(k+1)}{4}\right)\right)$$

Setting $k = 0$, we get the following amazing solution, (where one can find that $\psi_1\left(\frac{1}{2}\right) = \frac{\pi^2}{2}$ by using the trigamma reflection formula).

$$\begin{aligned}
I''(0) &= \frac{1}{8}\sqrt{\frac{\pi}{2}}\left(\psi_0^2\left(\frac{1}{2}\right) + \psi_1\left(\frac{1}{2}\right) + \pi\psi_0\left(\frac{1}{2}\right) - \frac{\pi^2}{4}\right) \\
&= \frac{1}{8}\sqrt{\frac{\pi}{2}}\left((\gamma + 2\ln 2)^2 - \pi(\gamma + 2\ln 2) + \frac{\pi^2}{4}\right) \\
&= \frac{1}{32}\sqrt{\frac{\pi}{2}}(2\gamma - \pi + 4\ln 2)^2
\end{aligned}$$

$$\therefore \int_0^\infty \ln^2(x)\sin(x^2)\,dx = \frac{1}{32}\sqrt{\frac{\pi}{2}}(2\gamma - \pi + 4\ln 2)^2$$

Another nice solution comes about when letting $k = -2$. The constants at this k value can be found using each functions respective recurrence relations. Thus, we obtain

$$\int_0^\infty \frac{\ln^2(x)\sin(x^2)}{x^2}\,dx = \frac{1}{16}\sqrt{\frac{\pi}{2}}\left((4 - 2\gamma - \pi - 4\ln 2)^2 + 16\right)$$

Note in general that the integral seems to be divergent for $-3 < k < 1$. Two other great solutions involving Catalan's constant comes about when letting $k = \frac{1}{2}$ and $k = -\frac{1}{2}$. This is due to $\psi_1\left(\frac{3}{4}\right) = \pi^2 - 8G$ and $\psi_1\left(\frac{1}{4}\right) = 8G + \pi^2$. These solutions also involve the gamma function at 3/4 and 1/4, but I'll be using the lemniscate constant instead.

$$\int_0^\infty \sqrt{x}\ln^2(x)\sin(x^2)\,dx = \frac{\pi}{64}\sqrt{\frac{2+\sqrt{2}}{\varpi\sqrt{2\pi}}}\left((\pi - 2\gamma - 6\ln 2)^2 + 2\pi(\sqrt{2}-1)(\pi - 2\gamma - 6\ln 2) + 3\pi^2 - 32G\right)$$

$$\int_0^\infty \frac{\ln^2(x)\sin(x^2)}{\sqrt{x}}\,dx = \frac{1}{16}\sqrt[4]{\frac{\pi}{2}}\sqrt{\varpi(2-\sqrt{2})}\left(2(\gamma + 3\ln 2)^2 - 2\pi\sqrt{2}(\gamma + 3\ln 2) + (1-\sqrt{2})\pi^2 + 16G\right)$$

These great solutions include Pi, the natural logarithm, the Euler-Mascheroni constant, Catalan's constant, and the lemniscate constant.

6.13 Abusing Cauchy's integral formula

If $f(x)$ is an analytic function on the unit disk, then the following equation holds.

$$\Re\int_0^{2\pi} f(e^{ix})\,dx = \Re\big(2\pi f(0)\big)$$

This can be used to create some truly horrifying integrals. Firstly, when $f(x) = e^{\frac{1+x}{2+x}}$, we obtain

$$\int_0^{\pi} e^{\frac{3(1+\cos x)}{5+4\cos x}} \cos\left(\frac{\sin x}{5+4\cos x}\right) dx = \pi\sqrt{e}$$

If $f(x) = \arctan(x+1)$, we can find that

$$\int_0^{2\pi} \arctan\left(\frac{1}{1+2\cos x}+1\right) dx = \frac{\pi^2}{3}$$

If $f(x) = \ln(\ln(x+1)-2)$, we get

$$\int_0^{\frac{\pi}{2}} \ln\big((\ln(2\cos x)-2)^2 + x^2\big)\,dx = \pi\ln 2$$

If $f(x) = \cosh(e^{ix})$, we obtain

$$\int_0^{2\pi} \cosh\big(e^{-\sin x}\cos(\cos x)\big)\cos\big(e^{-\sin x}\sin(\cos x)\big)\,dx = \pi e + \frac{\pi}{e}$$

When $f(x) = \cosh\big(\frac{\cos x}{1+x}\big)$, we get the following:

$$\int_{-2\pi}^{2\pi} \cosh\left(\frac{A(1+\cos x) - B\sin x}{2(1+\cos x)}\right)\cos\left(\frac{B(1+\cos x)+A\sin x}{2(1+\cos x)}\right) dx = \pi e + \frac{\pi}{e}$$

where $A = \cos(\cos x)\cosh(\sin x)$ and $B = \sin(\cos x)\sinh(\sin x)$.

When $f(x) = \cosh(x\ln(x+1))$, we have

$$\int_0^{2\pi} \cosh\left(\frac{A}{2}\cos(x) - \sin(x)\arctan\left(\frac{\sin x}{1+\cos x}\right)\right)\cos\left(\frac{A}{2}\sin(x) + \cos(x)\arctan\left(\frac{\sin x}{1+\cos x}\right)\right) dx = \frac{3\pi}{2}$$

where $A = \ln(2(1+\cos x))$.

If $f(x) = (x+1)^x$, we get

$$\int_{-\pi}^{\pi} e^{\frac{1}{2}\cos(x)\ln(2(1+\cos x)) - \frac{x}{2}\sin x}\cos\left(\frac{x}{2}\cos(x) + \frac{1}{2}\sin(x)\ln\big(2(1+\cos x)\big)\right) dx = \pi$$

When $f(x) = \ln\big(\csc\big(x+\frac{\pi}{3}\big)\big)$,

$$\int_0^{\pi} \ln\left(\sin^2\left(\cos(x)+\frac{\pi}{3}\right)\cosh^2(\sin x) + \cos^2\left(\cos(x)+\frac{\pi}{3}\right)\sinh^2(\sin x)\right) dx = \pi\ln\frac{3}{4}$$

If $f(x) = \tanh(\arctan(x+1))$,

$$\int_0^{\pi} \frac{\sinh\big(\arctan\big(\frac{1+\cos x}{1+\sin x}\big) + \arctan\big(\frac{1+\cos x}{1-\sin x}\big)\big)}{\cosh\big(\arctan\big(\frac{1+\cos x}{1+\sin x}\big) + \arctan\big(\frac{1+\cos x}{1-\sin x}\big)\big) + \cos\Big(\frac{1}{2}\ln\Big(\frac{3+2(\cos x+\sin x)}{3+2(\cos x-\sin x)}\Big)\Big)}\,dx = \pi\tanh\frac{\pi}{4}$$

And lastly, I wanted to show off this beast. When $f(x) = \sinh(\cos^{\frac{1}{x}}(x))$, this integral can be obtained.

$$\int_0^{2\pi} \sinh\big(e^{A\cos x - B\sin x}\cos(A\sin x + B\cos x)\big)\cos\big(e^{A\cos x - B\sin x}\sin(A\sin x + B\cos x)\big)\,dx = \pi e - \frac{\pi}{e}$$

where:

$$A = \frac{1}{2}\ln\left(\frac{1}{2}(\cos(2\cos x) + \cosh(2\sin x))\right),$$
$$B = \arctan(\tanh(\sin x)\tan(\cos x)).$$

7 July

7.1 More work related to Raabe's integral

Update originally on 28th May 2025 after making the first section on Raabe's integral, then later expanded on the 8th of July.

$$\Re \int_{-a}^{a} \ln\Gamma(x)\,dx$$

for all $a \in \mathbb{R}$. The working out is below. First, we need to find the answer to

$$\int_{0}^{a} \ln\Gamma(x)\,dx$$

To do so, we substitute in the Weierstrass product for the gamma function, then take the natural log of both sides.

$$\Gamma(z) = \frac{1}{ze^{\gamma z}} \prod_{n=1}^{\infty} \frac{ne^{\frac{z}{n}}}{z+n}$$

$$\therefore \ln\Gamma(z) = \sum_{n=1}^{\infty}\left(\ln\left(\frac{n}{z+n}\right) + \frac{z}{n}\right) - \gamma z - \ln z$$

$$\therefore \int_{0}^{a} \ln\Gamma(x)\,dx = \int_{0}^{a}\sum_{n=1}^{\infty}\left(\ln\left(\frac{n}{x+n}\right) + \frac{x}{n}\right) - \gamma x - \ln x\,dx$$

$$= \sum_{n=1}^{\infty}\int_{0}^{a} \ln(n) - \ln(x+n) + \frac{x}{n}\,dx - \frac{\gamma a^2}{2} + a - a\ln a$$

$$= \sum_{n=1}^{\infty}\left(a\ln(n) - \int_{0}^{a} \ln(x+n)\,dx + \frac{a^2}{2n}\right) - \frac{\gamma a^2}{2} + a - a\ln a$$

$$= \sum_{n=1}^{\infty}\left(a\ln(n) - ((a+n)\ln(a+n) - (a+n) - n\ln(n) + n) + \frac{a^2}{2n}\right) - \frac{\gamma a^2}{2} + a - a\ln a$$

$$= \sum_{n=1}^{\infty}\left(a\ln\left(\frac{n}{a+n}\right) + n\ln\left(\frac{n}{a+n}\right) + a + \frac{a^2}{2n}\right) - \frac{\gamma a^2}{2} + a - a\ln a \tag{6}$$

Using the expression above for $\ln\Gamma(z)$, we can rearrange, set $z = a$, and find that

$$\sum_{n=1}^{\infty}\ln\left(\frac{n}{a+n}\right) = \ln\Gamma(a) + \ln(a) - \gamma a - \sum_{k=1}^{\infty}\frac{a}{k}$$

Thus, equation 6 is equal to

$$= a\ln\Gamma(a) + a\ln(a) + \gamma a^2 - \sum_{k=1}^{\infty}\frac{a^2}{k} + \sum_{n=1}^{\infty}\left(n\ln\left(\frac{n}{a+n}\right) + a + \frac{a^2}{2n}\right) - \frac{\gamma a^2}{2} + a - a\ln a$$

$$= \sum_{n=1}^{\infty}\left(n\ln\left(\frac{n}{a+n}\right) + a - \frac{a^2}{2n}\right) + a\ln\Gamma(a) + \frac{\gamma a^2}{2} + a \tag{7}$$

To find an 'answer' for the other logarithm under the sum, we turn to the Barnes-G function, denoted as $G(z)$. The function satisfies the equation $G(1+z) = \Gamma(z)\Gamma(z)$. The Weierstrass product for $G(1+z)$ is the following.

$$G(1+z) = (2\pi)^{\frac{z}{2}} e^{-\frac{z+z^2(1+\gamma)}{2}} \prod_{n=1}^{\infty}\left(\frac{z+n}{n}\right)^{n} e^{\frac{z^2}{2n}-z}$$

$$\therefore \ln G(1+z) = \sum_{n=1}^{\infty}\left(\frac{z^2}{2n} - z - n\ln\left(\frac{n}{z+n}\right)\right) - \frac{(\gamma+1)z^2}{2} - \frac{z}{2} + \frac{z}{2}\ln(2\pi)$$

Using this expression, we can rearrange, set $z = a$, and find that

$$\sum_{n=1}^{\infty} n \ln\left(\frac{n}{a+n}\right) = \frac{a}{2}\ln(2\pi) - \ln G(1+a) - \frac{(\gamma+1)a^2}{2} - \frac{a}{2} + \sum_{k=1}^{\infty}\left(\frac{a^2}{2k} - a\right)$$

Thus, equation 7 is equal to

$$= \frac{a}{2}\ln(2\pi) - \ln G(1+a) - \frac{(\gamma+1)a^2}{2} - \frac{a}{2} + \sum_{k=1}^{\infty}\left(\frac{a^2}{2k} - a\right) + \sum_{n=1}^{\infty}\left(a - \frac{a^2}{2n}\right) + a\ln\Gamma(a) + \frac{\gamma a^2}{2} + a$$

$$= \frac{a}{2}\ln(2\pi) - \ln G(1+a) - \frac{(\gamma+1)a^2}{2} - \frac{a}{2} + a\ln\Gamma(a) + \frac{\gamma a^2}{2} + a$$

$$= \frac{a}{2}\ln(2\pi) + a\ln\Gamma(a) - \ln G(1+a) + \frac{a(1-a)}{2}$$

$$\therefore \int_0^a \ln\Gamma(x)\,dx = \frac{a}{2}\ln(2\pi) + a\ln\Gamma(a) - \ln G(1+a) + \frac{a(1-a)}{2}$$

This thus solves the first part of the problem. Some example solutions can be seen below, using the respective values of the Barnes-G function. Interestingly and quite nicely, none of the equations contain a value of the gamma function, as the Barnes-G function term above always contains a respective term of the gamma function of at the same value that cancels out. The V constant in the third equation is known at the Gieseking manifold constant.

$$\int_0^{\frac{1}{2}} \ln\Gamma(x)\,dx = \frac{3}{2}\ln(A) + \frac{1}{4}\ln(\pi) + \frac{5}{24}\ln(2)$$

$$\int_0^{\frac{1}{3}} \ln\Gamma(x)\,dx = \frac{4}{3}\ln(A) + \frac{1}{6}\ln(2\pi) - \frac{1}{72}\ln(3) + \frac{V}{6\pi}$$

$$\int_0^{\frac{1}{4}} \ln\Gamma(x)\,dx = \frac{9}{8}\ln(A) + \frac{1}{8}\ln(2\pi) + \frac{G}{4\pi}$$

$$\int_0^{\frac{1}{6}} \ln\Gamma(x)\,dx = \frac{5}{6}\ln(A) + \frac{1}{12}\ln(2\pi) + \frac{1}{144}\ln(3) + \frac{1}{72}\ln(2) + \frac{V}{4\pi}$$

Moving on, to find the real part of the integral from $-a$ to a, we make the following simplifications (assume we are finding the real part of each side).

$$\therefore \int_{-a}^0 \ln\Gamma(x)\,dx = \frac{a(1+a)}{2} + \frac{a}{2}\ln(2\pi) + a\ln\Gamma(-a) - \ln G(1-a)$$

$$\therefore \int_{-a}^a \ln\Gamma(x)\,dx = \frac{a(1-a)}{2} + \frac{a}{2}\ln(2\pi) + a\ln\Gamma(a) - \ln G(1+a) + \frac{a(1+a)}{2} + \frac{a}{2}\ln(2\pi) + a\ln\Gamma(-a) + \ln G(1-a)$$

$$= \frac{a(1-a)+a(1+a)}{2} + a\ln(2\pi) + a\ln(\Gamma(a)\Gamma(-a)) + \ln\left(\frac{G(1-a)}{G(1+a)}\right)$$

$$= a + a\ln(2\pi) + a\ln\left(\frac{\Gamma(a)\Gamma(1-a)}{-a}\right) + \ln\left(\frac{G(1-a)}{G(1+a)}\right)$$

$$= a\ln(2\pi) + a\ln\left(\frac{-\pi}{a\sin(\pi a)}\right) + \ln\left(\frac{G(1-a)}{G(1+a)}\right) + a$$

As we are finding the real part, the negative symbol in the log in the middle can be removed, and an abs symbol is put on the denominator.

$$= a\ln\left(\frac{2\pi^2}{|a\sin(\pi a)|}\right) + \ln\left(\frac{G(1-a)}{G(1+a)}\right) + a$$

We could be done here, but there are actually a few simplifications that could be made. First, to simplify the log that is in terms of the Barnes-G function, I did some research and found this equation

$$\ln\left(\frac{G(1-z)}{G(1+z)}\right) = z\ln\left(\frac{\sin(\pi z)}{\pi}\right) + \frac{1}{2\pi}\sum_{n=1}^{\infty}\frac{\sin(2\pi nz)}{n^2}$$

$$\therefore a\ln\left(\frac{2\pi^2}{|a\sin(\pi a)|}\right) + \ln\left(\frac{G(1-a)}{G(1+a)}\right) + a = a\ln\left(\frac{2\pi^2}{|a\sin(\pi a)|}\right) + a\ln\left(\frac{\sin(\pi a)}{\pi}\right) + \frac{1}{2\pi}\sum_{n=1}^{\infty}\frac{\sin(2\pi na)}{n^2} + a$$

$$= a\ln\left(\frac{2\pi}{|a|}\right) + \frac{1}{2\pi}\sum_{n=1}^{\infty}\frac{\sin(2\pi na)}{n^2} + a$$

This sum on the right is known as the Clausen function of order two at the value $2\pi a$. Thus, the above expression is equal to

$$= a\ln\left(\frac{2\pi}{|a|}\right) + \frac{1}{2\pi}\mathrm{Cl}_2(2\pi a) + a$$

Therefore, for all $a \in \mathbb{R}$,

$$\Re\int_{-a}^{a}\ln\Gamma(x)\,dx = a\ln\left(\frac{2\pi}{|a|}\right) + \frac{1}{2\pi}\mathrm{Cl}_2(2\pi a) + a$$

First note that when a is a multiple of $1/2$, the Clausen function above is equal to zero (look at the terms in the sum). In these cases, the equation reduces back down to the one seen on page 62. This gives us nice equations like

$$\Re\int_{-\frac{1}{2}}^{\frac{1}{2}}\ln\Gamma(x)\,dx = \frac{1}{2}\ln(4\pi) + \frac{1}{2}$$

Another, more interesting special case of this integral, outside of a equalling a multiple of $1/2$, is the following.

$$\Re\int_{-\frac{1}{4}}^{\frac{1}{4}}\ln\Gamma(x)\,dx = \frac{1}{4}\ln(8\pi) + \frac{1}{2\pi}\sum_{n=1}^{\infty}\frac{\sin\left(\frac{\pi n}{2}\right)}{n^2} + \frac{1}{4}$$

$$= \frac{1}{4}\ln(8\pi) + \frac{1}{2\pi}\left(\frac{1}{1^2} + \frac{0}{2^2} - \frac{1}{3^2} + \frac{0}{4^2} + \frac{1}{5^2} + \frac{0}{6^2} - \frac{1}{7^2} + \cdots\right) + \frac{1}{4}$$

$$= \frac{1}{4}\ln(8\pi) + \frac{1}{2\pi}\left(\frac{1}{1^2} - \frac{1}{3^2} + \frac{1}{5^2} - \frac{1}{7^2} + \cdots\right) + \frac{1}{4}$$

$$= \frac{1}{4}\ln(8\pi) + \frac{1}{2\pi}\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2} + \frac{1}{4}$$

$$= \frac{1}{4}\ln(8\pi) + \frac{G}{2\pi} + \frac{1}{4}$$

$$\therefore \Re\int_{-\frac{1}{4}}^{\frac{1}{4}}\ln\Gamma(x)\,dx = \frac{1}{4}\ln(8\pi) + \frac{G}{2\pi} + \frac{1}{4}$$

If we let $a = \frac{1}{6}$, the Clausen function yields the Gieseking manifold constant. Thus,

$$\Re\int_{-\frac{1}{6}}^{\frac{1}{6}}\ln\Gamma(x)\,dx = \frac{1}{6}\ln(12\pi) + \frac{V}{2\pi} + \frac{1}{6}$$

Then finally, using the relationship $\mathrm{Cl}_2(2\theta) = 2\mathrm{Cl}_2(\theta) - 2\mathrm{Cl}_2(\pi - \theta)$, we have

$$\Re\int_{-\frac{1}{3}}^{\frac{1}{3}}\ln\Gamma(x)\,dx = \frac{1}{3}\ln(6\pi) + \frac{V}{3\pi} + \frac{1}{3}$$

7.2 Filler IBP problem

Really nice solution for not super complicated working out (11/7).

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\tan^2 x \cosh x}{\cos x}\,dx$$

$$= \frac{1}{2}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{(e^x + e^{-x})\tan^2 x}{\cos x}\,dx$$

$$= \frac{1}{2}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{e^x \tan^2 x}{\cos x}\,dx + \frac{1}{2}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{e^{-x}\tan^2 x}{\cos x}\,dx$$

On the right integral, let change of variables $x \to -x$

$$= \frac{1}{2}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{e^x \tan^2 x}{\cos x}\,dx + \frac{1}{2}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{e^x \tan^2 x}{\cos x}\,dx$$

$$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} e^x \tan^2 x \sec x\,dx$$

Lets call this integral above I.

$$\text{IBP with } u = e^x \tan x, dv = \tan x \sec x$$

$$= \left[e^x \tan x \sec x\right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} e^x \sec(x)(\tan x + \sec^2 x)\,dx$$

$$= 2\sqrt{2}\cosh\left(\frac{\pi}{4}\right) - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} e^x \tan x \sec x\,dx - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} e^x \sec^3 x\,dx$$

On the left integral,

$$\text{IBP with } u = e^x, dv = \tan x \sec x$$

$$= 2\sqrt{2}\cosh\left(\frac{\pi}{4}\right) - \left(\left[e^x \sec x\right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} e^x \sec x\,dx\right) - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} e^x \sec^3 x\,dx$$

$$= 2\sqrt{2}\cosh\left(\frac{\pi}{4}\right) - 2\sqrt{2}\sinh\left(\frac{\pi}{4}\right) + \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} e^x \sec x\,dx - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} e^x \sec(x)(1 + \tan^2 x)\,dx$$

As $\cosh(x) - \sinh(x) = e^{-x}$,

$$= \frac{2\sqrt{2}}{e^{\frac{\pi}{4}}} + \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} e^x \sec x\,dx - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} e^x \sec x\,dx - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} e^x \tan^2 x \sec x\,dx$$

$$= \frac{2\sqrt{2}}{e^{\frac{\pi}{4}}} - I$$

$$\therefore 2I = \frac{2\sqrt{2}}{e^{\frac{\pi}{4}}}$$

As I is equal to the original integral, we therefore find that

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\tan^2 x \cosh x}{\cos x}\,dx = \frac{\sqrt{2}}{e^{\frac{\pi}{4}}}$$

Similarly, we can find an answer for

$$\int_0^{\frac{\pi}{4}} \frac{\tan^2 x \sinh x}{\cos x}\, dx$$

The method is largely the same, but with some changes.

$$= \frac{1}{2}\int_0^{\frac{\pi}{4}} \frac{(e^x - e^{-x})\tan^2 x}{\cos x}\, dx$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{4}} \frac{e^x \tan^2 x}{\cos x}\, dx - \frac{1}{2}\int_0^{\frac{\pi}{4}} \frac{e^{-x} \tan^2 x}{\cos x}\, dx$$

On the right integral, and let change of variables $x \to -x$

$$= \frac{1}{2}\int_0^{\frac{\pi}{4}} \frac{e^x \tan^2 x}{\cos x}\, dx - \frac{1}{2}\int_{-\frac{\pi}{4}}^{0} \frac{e^x \tan^2 x}{\cos x}\, dx$$

Lets call the left integral above I_1 and the right integral I_2.

$$\text{IBP with } u = e^x \tan x, dv = \tan x \sec x$$

Similar process as the integral on the previous page, so some lines will be skipped.

$$= \frac{1}{2}\left(e^{\frac{\pi}{4}}\sqrt{2} - \int_0^{\frac{\pi}{4}} e^x \sec(x)(\tan x + \sec^2 x)\, dx\right) - \frac{1}{2}\left(e^{-\frac{\pi}{4}}\sqrt{2} - \int_{-\frac{\pi}{4}}^{0} e^x \sec(x)(\tan x + \sec^2 x)\, dx\right)$$

$$= \frac{1}{2}\left(e^{\frac{\pi}{4}}\sqrt{2} - \int_0^{\frac{\pi}{4}} e^x \tan x \sec x\, dx - \int_0^{\frac{\pi}{4}} e^x \sec^3 x\, dx\right) - \frac{1}{2}\left(e^{-\frac{\pi}{4}}\sqrt{2} - \int_{-\frac{\pi}{4}}^{0} e^x \tan x \sec x\, dx - \int_{-\frac{\pi}{4}}^{0} e^x \sec^3 x\, dx\right)$$

$$\text{As } \sec^3 x = \sec(x)(1 + \tan^2 x)$$

$$= \frac{1}{2}\left(e^{\frac{\pi}{4}}\sqrt{2} - \int_0^{\frac{\pi}{4}} e^x \tan x \sec x\, dx - \int_0^{\frac{\pi}{4}} e^x \sec x\, dx - I_1\right) - \frac{1}{2}\left(e^{-\frac{\pi}{4}}\sqrt{2} - \int_{-\frac{\pi}{4}}^{0} e^x \tan x \sec x\, dx - \int_{-\frac{\pi}{4}}^{0} e^x \sec x\, dx - I_2\right)$$

After re-arranging, the above line is equal to

$$= \frac{1}{4}\left(e^{\frac{\pi}{4}}\sqrt{2} - \int_0^{\frac{\pi}{4}} e^x \tan x \sec x\, dx - \int_0^{\frac{\pi}{4}} e^x \sec x\, dx\right) - \frac{1}{4}\left(e^{-\frac{\pi}{4}}\sqrt{2} - \int_{-\frac{\pi}{4}}^{0} e^x \tan x \sec x\, dx - \int_{-\frac{\pi}{4}}^{0} e^x \sec x\, dx\right)$$

On the first and third integrals from the left,

$$\text{IBP with } u = e^x, dv = \tan x \sec x$$

$$= \frac{1}{4}\left(1 + \int_0^{\frac{\pi}{4}} e^x \sec x\, dx - \int_0^{\frac{\pi}{4}} e^x \sec x\, dx\right) - \frac{1}{4}\left(2e^{-\frac{\pi}{4}}\sqrt{2} - 1 + \int_{-\frac{\pi}{4}}^{0} e^x \sec x\, dx - \int_{-\frac{\pi}{4}}^{0} e^x \sec x\, dx\right)$$

$$= \frac{1}{4} - \frac{1}{4}\left(2e^{-\frac{\pi}{4}}\sqrt{2} - 1\right)$$

$$= \frac{1}{2} - \frac{\sqrt{2}}{2e^{\frac{\pi}{4}}}$$

$$\therefore \int_0^{\frac{\pi}{4}} \frac{\tan^2 x \sinh x}{\cos x}\, dx = \frac{1}{2} - \frac{\sqrt{2}}{2e^{\frac{\pi}{4}}}$$

7.3 A continuation of power towers

Here is the working out (from page 74) (method by Tyma Gaidash [6]) for

$$\int_0^\infty (a^{-x})^{(a^x)^{(a^{-x})^{(a^x)^{\cdots}}}} dx$$

Let's call this integral I. It is known that for $x^{y^z} = z$

$$z = \frac{1}{\ln y}\sum_{n=1}^{\infty}\frac{(\ln(y)x)^n}{n!}B_{n-1}(\ln(x)n)$$

where $B_n(x)$ is the Bell Polynomial. As our integrand tells us that $y = (a^{-x})^{(a^x)^y}$, we get

$$I = \int_0^\infty \frac{1}{x\ln a}\sum_{n=1}^{\infty}\frac{(\ln(a)xa^{-x})^n}{n!}B_{n-1}(-x\ln(a)n)\,dx$$

Then as

$$B_n(x) = \sum_{k=0}^{n} s_n^{(k)}x^k$$

where $s_n^{(k)}$ are Stirling numbers of the second kind, the integral can then be re-written as

$$= \int_0^\infty \frac{1}{x\ln a}\sum_{n=1}^{\infty}\sum_{k=0}^{n-1}\frac{(\ln(a)xa^{-x})^n(-x\ln(a)n)^k s_{n-1}^{(k)}}{n!}\,dx$$

$$= \sum_{n=1}^{\infty}\sum_{k=0}^{n-1}\frac{s_{n-1}^{(k)}(\ln a)^{n+k-1}(-n)^k}{n!}\int_0^\infty x^{n+k-1}a^{-nx}\,dx$$

$$t = n(\ln a)x, dt = n\ln(a)\,dx$$

$$= \frac{1}{\ln a}\sum_{n=1}^{\infty}\sum_{k=0}^{n-1}\frac{(-1)^k s_{n-1}^{(k)}}{n!n^n}\int_0^\infty x^{n+k-1}e^{-t}\,dt$$

$$= \frac{1}{\ln a}\sum_{n=1}^{\infty}\sum_{k=0}^{n-1}\frac{(-1)^k s_{n-1}^{(k)}(n+k-1)!}{n!n^n}$$

$$= \frac{1}{\ln a}\sum_{n=0}^{\infty}\sum_{k=0}^{n}\frac{(-1)^k s_n^{(k)}(n+k)!}{(n+1)!(n+1)^{n+1}}$$

Using the Pochhammer symbol for the rising factorial, we get

$$= \frac{1}{\ln a}\sum_{n=0}^{\infty}\sum_{k=0}^{n}\frac{(-1)^k(n+1)_k s_n^{(k)}}{(n+1)^{n+2}}$$

Using a finite series for the symbol, it is known that

$$\sum_{k=0}^{n}(-1)^k(n+1)_k s_n^{(k)} = (-1)^n(n+1)^n$$

Thus, our double sum is equal to

$$= \frac{1}{\ln a}\sum_{n=0}^{\infty}\frac{(-1)^n}{(n+1)^2} = \frac{1}{\ln a}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}$$

$$= \frac{\pi^2}{12\ln a}$$

$$\therefore \int_0^\infty (a^{-x})^{(a^x)^{(a^{-x})^{(a^x)^{\cdots}}}} dx = \frac{\pi^2}{12\ln a} \quad (a > 1)$$

Then, by using the same method, I found this thing. When the integral converges,

$$\int_0^\infty \left(a^{-bx^p}\right)^{\left(a^{bx^p}\right)^{\cdots}} dx = \frac{1}{p(b\ln a)^{\frac{1}{p}}} \sum_{n=0}^{\infty} \frac{\Gamma\left(n+\frac{1}{p}\right)\left(-\left(n+\frac{1}{p}\right)\right)^n}{\Gamma(n+2)(n+1)^{n+\frac{1}{p}}}$$

And if we wanted alternating coefficients for x (where b and c must both be positive for the integral to converge),

$$\int_0^\infty \left(a^{-bx^p}\right)^{\left(a^{cx^p}\right)^{\cdots}} dx = \frac{1}{p\ln^{\frac{1}{p}} a} \sum_{n=0}^{\infty} \frac{\Gamma\left(n+\frac{1}{p}\right)\left(-c\left(n+\frac{1}{p}\right)\right)^n}{\Gamma(n+2)(b(n+1))^{n+\frac{1}{p}}}$$

One could also have alternating powers and/or bases, but the sum(s) just gets all the more needlessly complicated, so I won't bother. It's clear from this sum that closed solutions only occur when $p = 1$. When this is so, we get

$$\int_0^\infty (a^{-bx})^{(a^{cx})^{\cdots}} dx = -\frac{1}{c\ln a}\mathrm{Li}_2\left(-\frac{c}{b}\right)$$

where $\mathrm{Li}_2(x)$ is the dilogarithm. This has a few closed solutions for special values of b and c, given that the integral converges. Here is one such solution:

$$\int_0^\infty (e^{-x})^{(e^{\phi x})^{(e^{-x})^{(e^{\phi x})^{\cdots}}}} dx = \frac{1}{\phi}\left(\frac{\pi^2}{10} + \ln^2\phi\right)$$

8 August

8.1 Neat application of several less-standard techniques

This integral involves DUIS, use of geometric series, IBP twice, nice rearrangement, and use of the Weierstrass product for $\sinh x$ (Solution from [7]).

$$\int_0^\infty \frac{1-\cos x}{x(1-e^x)}\,dx$$

To start, let

$$I(k) = \int_0^\infty \frac{1-\cos(kx)}{x(1-e^x)}\,dx$$

We are thus trying to solve for $I(1)$. Now, we differentiate both sides with respect to k.

$$\begin{aligned}\therefore I'(k) &= \int_0^\infty \frac{\sin(kx)}{1-e^x}\,dx \\ &= \int_0^\infty \frac{-e^{-x}\sin(kx)}{1-e^{-x}}\,dx\end{aligned}$$

Note that $I'(k)$ is the negative of the integral from page 55, but more generalised. Instead of going straight to the solution using that page, let's do this slightly differently. We may now use a geometric series with $a_1 = -e^{-x}\sin(kx)$ and $r = e^{-x}$.

$$\begin{aligned}&= -\int_0^\infty \sum_{n=1}^{\infty} e^{-x}\sin(kx)e^{-x(n-1)}\,dx \\ &= -\sum_{n=1}^{\infty}\int_0^\infty \sin(kx)e^{-xn}\,dx\end{aligned}$$

Let us call this integral I_1.

$$\text{IBP with } u = \sin(kx), dv = e^{-xn}$$

$$= -\sum_{n=1}^{\infty}\frac{k}{n}\int_0^\infty \cos(kx)e^{-xn}\,dx$$

$$\text{IBP again with } u = \cos(kx), dv = e^{-xn}$$

$$\begin{aligned}&= -\sum_{n=1}^{\infty}\left(\frac{k}{n^2} - \frac{k^2}{n^2}\int_0^\infty \sin(kx)e^{-xn}\,dx\right) \\ &= -\sum_{n=1}^{\infty}\left(\frac{k}{n^2} - \frac{k^2}{n^2}I_1\right)\end{aligned}$$

This allows to make rearrangements and say that

$$\therefore I_1 = \frac{k}{n^2+k^2}$$

$$\therefore -\sum_{n=1}^{\infty}\int_0^\infty \sin(kx)e^{-xn}\,dx = -\sum_{n=1}^{\infty}\frac{k}{n^2+k^2}$$

This gives us our solution to $I'(k)$. Note again how it is a more generalised version of page 55. To find our target integral, We now integrate with respect to k to find that

$$\therefore I(k) = -\sum_{n=1}^{\infty}\int \frac{k}{n^2+k^2}\,dk$$

$$u = k^2 + n^2, du = 2k\,dk$$

$$= -\frac{1}{2}\sum_{n=1}^{\infty}\int \frac{1}{u}\,du = -\frac{1}{2}\sum_{n=1}^{\infty}\ln u + C$$

$$= -\frac{1}{2}\sum_{n=1}^{\infty}\ln(n^2 + k^2) + C$$

Letting $k = 0$, we find that

$$I(0) = \int_0^{\infty} 0\,dx = 0 = -\frac{1}{2}\sum_{n=1}^{\infty}\ln(n^2) + C$$

$$\therefore C = \frac{1}{2}\sum_{n=1}^{\infty}\ln(n^2)$$

$$\therefore I(k) = -\frac{1}{2}\sum_{n=1}^{\infty}\ln\left(1 + \frac{k^2}{n^2}\right)$$

To find the value for this sum, consider the Weierstrass product for $\sinh x$, then log both sides.

$$\sinh x = x\prod_{n=1}^{\infty}\left(1 + \frac{x^2}{n^2\pi^2}\right)$$

$$\therefore \ln(\sinh x) = \ln x + \ln\left(\prod_{n=1}^{\infty}\left(1 + \frac{x^2}{n^2\pi^2}\right)\right)$$

$$= \ln x + \sum_{n=1}^{\infty}\ln\left(1 + \frac{x^2}{n^2\pi^2}\right)$$

After rearranging and letting $x = k\pi$, we find that

$$\sum_{n=1}^{\infty}\ln\left(1 + \frac{k^2}{n^2}\right) = \ln\left(\frac{\sinh(k\pi)}{k\pi}\right)$$

$$\therefore I(k) = \int_0^{\infty}\frac{1 - \cos(kx)}{x(1 - e^x)}\,dx = \frac{1}{2}\ln\left(\frac{k\pi}{\sinh(k\pi)}\right)$$

Thus arriving at our solution when letting $k = 1$.

$$\therefore \int_0^{\infty}\frac{1 - \cos x}{x(1 - e^x)}\,dx = \frac{1}{2}\ln\left(\frac{\pi}{\sinh \pi}\right)$$

Interestingly, this solution is also equal to $\ln(|i!|)$, which will be shown on the next page.

8.2 Solution to $|i!|$

Here, we will solve for

$$|i!|$$

$$= |\Gamma(i+1)| = |i\Gamma(i)|$$

$$= |i||\Gamma(i)| = |\Gamma(i)|$$

$$\therefore |\Gamma(1-i)\Gamma(i)| = |i!(-i)!|$$

But, we also know that

$$|\Gamma(1-i)\Gamma(i)| = \left|\frac{\pi}{\sin(i\pi)}\right|$$

$$= \left|\frac{\pi}{i \sinh \pi}\right| = \frac{\pi}{\sinh \pi}$$

Putting these pieces of info together, we get

$$|i!(-i)!| = \frac{\pi}{\sinh \pi}$$

Finally, as $(-i)!$ is the conjugate of $i!$ (which can be found using the property that the conjugate of an integral is equal to the conjugate of the integrand, and relating that property to the integral definition for the gamma function), and $zz^* = |z|^2$,

$$|i!(-i)!| = |i!|^2$$

$$\therefore |i!| = \sqrt{\frac{\pi}{\sinh \pi}}$$

8.3 Reverse solving an integral from page 88

(22/8) Is it possible to take this integral and get it back to the cauchy's integral formula input?

$$\int_0^\pi \frac{\sinh\left(\arctan\left(\frac{1+\cos x}{1+\sin x}\right)+\arctan\left(\frac{1+\cos x}{1-\sin x}\right)\right)}{\cosh\left(\arctan\left(\frac{1+\cos x}{1+\sin x}\right)+\arctan\left(\frac{1+\cos x}{1-\sin x}\right)\right)+\cos\left(\frac{1}{2}\ln\left(\frac{3+2(\cos x+\sin x)}{3+2(\cos x-\sin x)}\right)\right)}\,dx$$

First, i will make a u-sub, for the purposes of getting things in terms of algebraic functions where possible instead of trigonometric. This u-sub will be undone later. Note that this process is fully possible without this substitution, but I personally think it's slightly easier this way.

$$t=\tan\frac{x}{2},\,dx=\frac{2}{1+t^2}\,dt$$

$$=\int_0^\infty \frac{\sinh\left(\arctan\left(\frac{1+\frac{1-t^2}{1+t^2}}{1+\frac{2t}{1+t^2}}\right)+\arctan\left(\frac{1+\frac{1-t^2}{1+t^2}}{1-\frac{2t}{1+t^2}}\right)\right)}{\cosh\left(\arctan\left(\frac{1+\frac{1-t^2}{1+t^2}}{1+\frac{2t}{1+t^2}}\right)+\arctan\left(\frac{1+\frac{1-t^2}{1+t^2}}{1-\frac{2t}{1+t^2}}\right)\right)+\cos\left(\frac{1}{2}\ln\left(\frac{3+2\left(\frac{1-t^2}{1+t^2}+\frac{2t}{1+t^2}\right)}{3+2\left(\frac{1-t^2}{1+t^2}-\frac{2t}{1+t^2}\right)}\right)\right)}\cdot\frac{2}{1+t^2}\,dt$$

$$=\int_0^\infty \frac{\sinh\left(\arctan\left(\frac{2}{t^2+2t+1}\right)+\arctan\left(\frac{2}{t^2-2t+1}\right)\right)}{\cosh\left(\arctan\left(\frac{2}{t^2+2t+1}\right)+\arctan\left(\frac{2}{t^2-2t+1}\right)\right)+\cos\left(\frac{1}{2}\ln\left(\frac{3+\frac{-2t^2+4t+2}{1+t^2}}{3+\frac{-2t^2-4t+2}{1+t^2}}\right)\right)}\cdot\frac{2}{1+t^2}\,dt$$

$$=\int_{-\infty}^\infty \frac{1}{1+t^2}\cdot\frac{\sinh\left(\arctan\left(\frac{2}{(t+1)^2}\right)+\arctan\left(\frac{2}{(t-1)^2}\right)\right)}{\cosh\left(\arctan\left(\frac{2}{(t+1)^2}\right)+\arctan\left(\frac{2}{(t-1)^2}\right)\right)+\cos\left(\frac{1}{2}\ln\left(\frac{t^2+4t+5}{t^2-4t+5}\right)\right)}\,dt$$

Since the two expressions in each arctan component have a break at $\sqrt{3}$, we will use atan2 instead, then combine the two arctans with the formula $\arctan(x)+\arctan(y)=\arctan\left(\frac{x+y}{1-xy}\right)$.

$$=\int_{-\infty}^\infty \frac{1}{1+t^2}\cdot\frac{\sinh\left(\operatorname{atan2}\left(\frac{4}{t^2-3}\right)\right)}{\cosh\left(\operatorname{atan2}\left(\frac{4}{t^2-3}\right)\right)+\cos\left(\frac{1}{2}\ln\left(\frac{t^2+4t+5}{t^2-4t+5}\right)\right)}\,dt$$

Now, we must find a complex number z in terms of t that can be morphed into the expressions found in the integrand, hopefully bringing it back to the original function used to create the integral using the Cauchy integral formula. As the integral can be quickly checked to be >0, we know that $f(0)>0$. We can also assume that the function involves arctan due to the high presence of it in the integrand, but $\arctan(0)=0$. Thus, the function may involve $\arctan(x+1)$. From this, we know that $f(e^{ix})$ is what creates the integral. Since $e^{ix}=\cos x+i\sin x$, and we're adding an extra $+1$ on the end because of the $f(0)$ (assumed) issue, let $z_0=1+\cos x+i\sin x$. Then, after performing the u-sub from above on this expression to form z_0, we get

$$z_0=\frac{2}{1+t^2}+i\frac{2t}{1+t^2}=x+iy$$

And let (knowing a likely presence of arctan in the original key for the integral).

$$\arctan(z_0)=u+iv$$

We now wish to find u and v in terms of t. Since

$$\arctan(z)=\frac{1}{2i}\ln\left(\frac{1+iz}{1-iz}\right)$$

$$\therefore u+iv=\frac{1}{2i}\ln\left(\frac{1+i(x+iy)}{1-i(x+iy)}\right)=\frac{1}{2i}\ln\left(\frac{1-y+ix}{1+y-ix}\right)$$

$$=\frac{1}{2i}\ln\left(\frac{1-y^2-x^2+2ix}{(1+y)^2+x^2}\right)$$

$$=\frac{1}{2i}\ln\left(\frac{1-y^2-x^2}{(1+y)^2+x^2}+i\frac{2x}{(1+y)^2+x^2}\right)$$

Let us then write the expression inside the log as $a+bi=|r|e^{i\theta}$, where

$$a=\frac{1-y^2-x^2}{(1+y)^2+x^2}, b=\frac{2x}{(1+y)^2+x^2}$$

Thus,

$$\begin{aligned}u+iv&=\frac{1}{2i}\ln\left(\frac{1-y^2-x^2}{(1+y)^2+x^2}+i\frac{2x}{(1+y)^2+x^2}\right)\\&=\frac{1}{2i}(\ln(|r|)+i\theta)\\&=\frac{\theta}{2}-\frac{i}{2}\ln(|r|)\end{aligned}$$

Calculating r and θ using the definition of the polar form for a complex number ($|r|=\sqrt{a^2+b^2}, \theta=$ atan2(y/x), we get

$$r=\sqrt{\left(\frac{1-y^2-x^2}{(1+y)^2+x^2}\right)^2+\left(\frac{2x}{(1+y)^2+x^2}\right)^2}=\sqrt{\frac{(1-y^2-x^2)+4x^2}{((1+y)^2+x^2)^2}}$$

$$\theta=\text{atan2}\left(\frac{2x}{1-y^2-x^2}\right)$$

Therefore, we can now write u and v in terms of t.

$$\begin{aligned}u=\frac{1}{2}\text{atan2}\left(\frac{2x}{1-y^2-x^2}\right)&=\frac{1}{2}\text{atan2}\left(\frac{\frac{4}{1+t^2}}{1-\left(\frac{2t}{1+t^2}\right)^2-\left(\frac{2}{1+t^2}\right)^2}\right)\\&=\frac{1}{2}\text{atan2}\left(\frac{\frac{4}{1+t^2}}{1-\frac{4}{(1+t^2)}}\right)\\&=\frac{1}{2}\text{atan2}\left(\frac{4}{t^2-3}\right)\end{aligned}$$

$$\begin{aligned}v=-\frac{1}{4}\ln\left(\frac{(1-y^2-x^2)+4x^2}{((1+y)^2+x^2)^2}\right)&=\frac{1}{4}\ln\left(\frac{\left(\left(1+\frac{2t}{1+t^2}\right)^2+\left(\frac{2}{1+t^2}\right)^2\right)^2}{\left(1-\left(\frac{2t}{1+t^2}\right)^2-\left(\frac{2}{1+t^2}\right)^2\right)^2+4\left(\frac{2}{1+t^2}\right)^2}\right)\\&=\frac{1}{4}\ln\left(\frac{((t+1)^4+4)^2}{(1+t^2)^2((t^2-3)^2+16)}\right)\\&=\frac{1}{4}\ln\left(\frac{(t^2+4t+5)^2}{t^4-6t^2+25}\right)\\&=\frac{1}{4}\ln\left(\frac{t^2+4t+5}{t^2-4t+5}\right)\end{aligned}$$

Note the sheer amount of expansions then factorisations it takes to simplify this (a few of the steps for working out v have been omitted). We now see a familiar expression for u and v, as they appear in our integrand on the first page! This indicates that our guess for z_0 was correct. Thus, our integral can be written as

$$\int_{-\infty}^{\infty}\frac{1}{1+t^2}\cdot\frac{\sinh(2u)}{\cosh(2u)+\cos(2v)}\,dt$$

Now, since

$$u=\Re\arctan z_0=\Re\arctan\left(\frac{2(1+it)}{1+t^2}\right)$$

$$v=\Im\arctan z_0=\Im\arctan\left(\frac{2(1+it)}{1+t^2}\right)$$

We can write our integral as

$$= \int_{-\infty}^{\infty} \frac{1}{1+t^2} \cdot \frac{\sinh\left(2\Re \arctan\left(\frac{2(1+it)}{1+t^2}\right)\right)}{\cosh\left(2\Re \arctan\left(\frac{2(1+it)}{1+t^2}\right)\right) + \cos\left(2\Im \arctan\left(\frac{2(1+it)}{1+t^2}\right)\right)}\, dt$$

To clean this up, we can sneakily use the formula for $\tanh w$, where $w = c + di$, which is

$$\tanh w = \frac{\sinh(2c) + i\sin(2d)}{\cosh(2c) + \cos(2d)}$$

$$\therefore \Re \tanh w = \frac{\sinh(2c)}{\cosh(2c) + \cos(2d)}$$

This now perfectly matches the integrand, where

$$w = \arctan\left(\frac{2(1+it)}{1+t^2}\right)$$

Therefore, the integrand now becomes

$$= \Re \int_{-\infty}^{\infty} \frac{1}{1+t^2} \times \tanh\left(\arctan\left(\frac{2(1+it)}{1+t^2}\right)\right) dt$$

We can now undo the u-sub made at the start, and get back to the Cauchy integral formula from where this integral was created from in the first place.

$$t = \tan u,\ dt = \sec^2 u\, du$$

$$= \Re \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \tanh\left(\arctan\left(\frac{2(1+i\tan u)}{\sec^2 u}\right)\right) du$$

$$= \Re \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \tanh\big(\arctan\big(1 + \cos(2u) + i\sin(2u)\big)\big)\, du$$

$$x = 2u, dx = 2\, du$$

$$= \Re \int_{0}^{\pi} \tanh\big(\arctan\big(1 + \cos(x) + i\sin(x)\big)\big)\, dx$$

$$= \Re \int_{0}^{\pi} \tanh\big(\arctan\big(e^{ix} + 1\big)\big)\, dx$$

We have now gotten the integrand back into the form of Cauchy's integral formula

$$\Re \int_{0}^{2\pi} f(e^{ix})\, dx = \Re\big(2\pi f(0)\big)$$

for $f(x) = \tanh(\arctan(x+1))$.

$$\therefore \Re \int_{0}^{\pi} \tanh\big(\arctan\big(e^{ix} + 1\big)\big)\, dx = \pi f(0) = \pi \tanh \frac{\pi}{4}$$

And so, we arrive at our solution:

$$\int_{0}^{\pi} \frac{\sinh\left(\arctan\left(\frac{1+\cos x}{1+\sin x}\right) + \arctan\left(\frac{1+\cos x}{1-\sin x}\right)\right)}{\cosh\left(\arctan\left(\frac{1+\cos x}{1+\sin x}\right) + \arctan\left(\frac{1+\cos x}{1-\sin x}\right)\right) + \cos\left(\frac{1}{2}\ln\left(\frac{3+2(\cos x + \sin x)}{3+2(\cos x - \sin x)}\right)\right)}\, dx = \pi \tanh \frac{\pi}{4}$$

What. a. problem.

8.4 Double integral?!

First double integral on this document, making an appearance.

$$\int_0^\pi \int_0^\pi \frac{\arctan(\sin x \sin y)}{\sin x}\,dx\,dy$$

To start, we use the Taylor series for arctan.

$$= \int_0^\pi \int_0^\pi \sum_{n=0}^\infty \frac{(-1)^n (\sin x)^{2n} (\sin y)^{2n+1}}{2n+1}\,dx\,dy$$

$$= \sum_{n=0}^\infty \frac{(-1)^n}{2n+1} \int_0^\pi (\sin y)^{2n+1}\,dy \int_0^\pi (\sin x)^{2n}\,dx$$

One for the definitions of the beta function is

$$B(z_1, z_2) = \int_0^\pi (\sin x)^{2z_1-1} (\cos x)^{2z_2-1}\,dx$$

Using this, we can write the two separated integrals in terms of the beta function, with differing values for m, but both with $n = 1/2$. Therefore, our integral becomes

$$= \sum_{n=0}^\infty \frac{(-1)^n B\left(n+\frac{1}{2}, \frac{1}{2}\right) B\left(n+1, \frac{1}{2}\right)}{2n+1}$$

Then, we can write the beta functions in terms of factorials.

$$= \sum_{n=0}^\infty \frac{(-1)^n \left(n-\frac{1}{2}\right)! \left(-\frac{1}{2}\right)!^2 n!}{(2n+1)\left(n+\frac{1}{2}\right)! n!}$$

Which can then be simplified. Note that $\left(-\frac{1}{2}\right)! = \sqrt{\pi}$

$$= \pi \sum_{n=0}^\infty \frac{(-1)^n}{(2n+1)\left(n+\frac{1}{2}\right)}$$

$$= 2\pi \sum_{n=0}^\infty \frac{(-1)^n}{4n^2+4n+1}$$

$$= 2\pi \sum_{n=0}^\infty \frac{(-1)^n}{(2n+1)^2}$$

$$= 2G\pi$$

Where G is Catalan's constant.

$$\therefore \int_0^\pi \int_0^\pi \frac{\arctan(\sin x \sin y)}{\sin x}\,dx\,dy = 2G\pi$$

8.5 Similar Integrals involving the zeta and gamma functions

$$\int_0^{2\pi} \zeta(e^{ix} + s)\, dx$$

This integral can be solved using the Cauchy integral formula (page 88), which tells us that it is equal to $2\pi\zeta(s)$. However, i want to solve this without the formula. To do so, I first turn the zeta function into its series form.

$$= \int_0^{2\pi} \sum_{n=1}^{\infty} \frac{1}{n^{e^{ix}+s}}\, dx$$

$$= \sum_{n=1}^{\infty} \frac{1}{n^s} \int_0^{2\pi} n^{-e^{ix}}\, dx$$

$$= \sum_{n=1}^{\infty} \frac{1}{n^s} \int_0^{2\pi} e^{-\ln(n)e^{ix}}\, dx$$

Now we will use the Taylor series for e^x to express the integrand as a secondary summation.

$$= \sum_{n=1}^{\infty} \frac{1}{n^s} \int_0^{2\pi} \sum_{k=0}^{\infty} \frac{(-\ln(n)e^{ix})^k}{k!}\, dx$$

$$= \sum_{n=1}^{\infty} \sum_{k=0}^{\infty} \frac{(-\ln n)^k}{n^s k!} \int_0^{2\pi} e^{kix}\, dx$$

$$= \sum_{n=1}^{\infty} \sum_{k=0}^{\infty} \frac{(-\ln n)^k}{n^s k!} \left(\frac{e^{2ik\pi} - 1}{ik} \right)$$

Notice that as k can only be an integer, and $e^{2ik\pi} = 1$ for $k \neq 0, k \in \mathbb{Z}$, the expression above collapses to zero. However, when $k = 0$, we can use L'hopitals rule to find the limit.

$$\lim_{k \to 0} \left(\frac{e^{2ik\pi} - 1}{ik} \right) = \lim_{k \to 0} (2\pi e^{2ik\pi}) = 2\pi$$

$$\therefore \sum_{n=1}^{\infty} \sum_{k=0}^{\infty} \frac{(-\ln n)^k}{n^s k!} \left(\frac{e^{2ik\pi} - 1}{ik} \right) = 2\pi \sum_{n=1}^{\infty} \frac{(-\ln n)^0}{n^s 0!}$$

$$= 2\pi \sum_{n=1}^{\infty} \frac{1}{n^s}$$

$$= 2\pi\zeta(s)$$

$$\therefore \int_0^{2\pi} \zeta(e^{ix} + s)\, dx = 2\pi\zeta(s)$$

Here are some example results, with the first three using analytic continuation.

$$\int_0^{2\pi} \zeta(e^{ix} - 1)\, dx = -\frac{\pi}{6}$$

$$\int_0^{2\pi} \zeta(e^{ix})\, dx = -\pi$$

$$\int_0^{2\pi} \zeta(e^{ix} + 1)\, dx = 2\pi\gamma$$

$$\int_0^{2\pi} \zeta(e^{ix} + 2)\, dx = \frac{\pi^3}{3}$$

(Technically 1st of Sept but treating it as August still) Now, what if the integral was

$$\int_0^{2\pi} \Gamma(e^{ix} + s)\, dx$$

Here we need to do some special tricks involving contour integration to get an answer, given that we dont directly use Cauchy's integral formula which would give the answer $2\pi\Gamma(s), s \geq 1$. First, take the u-sub

$$t = e^{ix}, dt = it\, dx$$

$$= \frac{1}{i} \oint_{|t|=1} \frac{\Gamma(t+s)}{t}\, dt$$

To evaluate, consider the derivatives of the gamma function using it's Taylor series around s to find that

$$\Gamma(t+s) = \Gamma(s) + \Gamma'(s)t + \frac{1}{2}\Gamma''(s)t^2 + \cdots$$

$$\therefore \frac{\Gamma(t+s)}{t} = \frac{\Gamma(s)}{t} + \Gamma'(s) + \frac{1}{2}\Gamma''(s)t + \cdots$$

Basic pole for $s \geq 1$ gives the residue $\Gamma(s)$. Thus, using the residue theorem,

$$\frac{1}{i} \oint_{|t|=1} \frac{\Gamma(t+s)}{t}\, dt = \frac{1}{i} 2\pi i \Gamma(s)$$

$$= 2\pi\Gamma(s)$$

$$\therefore \int_0^{2\pi} \Gamma(e^{ix} + s)\, dx = 2\pi\Gamma(s)$$

However, for $s < 1$, we must consider some more poles, of $\Gamma(t+s)$, being where $t+s = -n$ for $n \geq 0, n \in \mathbb{Z}$.

$$\therefore t = -(n+s)$$

These lie in the unit disk for $|n+s| < 1$. The residue of the gamma function at these values can be found using the recurrence formula

$$\Gamma(z) = \frac{\Gamma(z+n+1)}{z(z+1)\cdots(z+n)}$$

As $\text{Res}(f, c) = \lim_{z\to c}(z-c)f(z)$, we have for each pole $z = -n, n \geq 0, n \in \mathbb{Z}$ that

$$(z+n)\Gamma(z) = \frac{\Gamma(z+n+1)}{z(z+1)\cdots(z+n-1)}$$

$$\therefore \lim_{z\to -n}(z-(-n))\Gamma(z) = \frac{1}{-n(1-n)\cdots(-1)}$$

$$= \frac{(-1)^n}{n!}$$

$$\therefore \text{Res}(\Gamma, -n) = \frac{(-1)^n}{n!}$$

Therefore, the residues of the integrand can be written as

$$\text{Res}\, \frac{\Gamma(t+s)}{t} = \frac{\text{Res}(\Gamma, -n)}{-(n+s)} = \frac{(-1)^{n+1}}{n!(n+s)}$$

Thus, using the residue theorem, we get our solution for all s-values apart from the poles of the gamma function (zero and every negative integer).

$$\therefore \int_0^{2\pi} \Gamma(e^{ix} + s)\, dx = 2\pi\left(\Gamma(s) + \sum_{\substack{n\geq 0 \\ |n+s|<1}} \frac{(-1)^{n+1}}{n!(n+s)}\right)$$

Some example results:

$$\int_0^{2\pi} \Gamma\left(e^{ix} + \frac{1}{2}\right) dx = 2\pi(\sqrt{\pi} - 2)$$

$$\int_0^{2\pi} \Gamma(e^{ix} + 1)\, dx = 2\pi$$

$$\int_0^{2\pi} \Gamma\left(e^{ix} + \frac{3}{2}\right) dx = \pi^{\frac{3}{2}}$$

For $s = 0$, we first consider the Laurent series for the gamma function, $\Gamma(s) = \frac{1}{s} - \gamma + \cdots$.

$$\therefore \lim_{s\to 0} \int_0^{2\pi} \Gamma(e^{ix} + s)\, dx = 2\pi \lim_{s\to 0} \left(\frac{1}{s} - \gamma + \cdots + \sum_{\substack{n\geq 0 \\ |n+s|<1}} \frac{(-1)^{n+1}}{n!(n+s)} \right)$$

All the latter terms in the Laurent series go to zero as $s \to 0$. Then, as the sum of the residues is for $n \geq 0, |n + s| < 1$, the only term used is $n = 0$, which gives the result of $-\frac{1}{s}$.

$$= 2\pi \lim_{s\to 0} \left(\frac{1}{s} - \gamma - \frac{1}{s} \right)$$

$$= -2\pi\gamma$$

$$\therefore \int_0^{2\pi} \Gamma(e^{ix})\, dx = -2\pi\gamma$$

Interestingly, this gives the cool equation

$$\int_0^{2\pi} \Gamma(e^{ix})\, dx = -\int_0^{2\pi} \zeta(e^{ix} + 1)\, dx = -2\pi\gamma$$

9 September

9.1 I did zeta and gamma, why not others?

Here is a small collection of integrals related to the zeta and gamma ones from pages 103-105. Only a few of these integrals will be on the list at the top of the document.

$$\int_0^{2\pi} \psi_0(e^{ix}+1)\,dx = -2\pi\gamma$$

$$\int_0^{2\pi} \beta(e^{ix}+2)\,dx = 2\pi G$$

$$\int_0^{2\pi} \beta(e^{ix}+3)\,dx = \frac{\pi^4}{16}$$

$$\int_0^{2\pi} \mathrm{Li}_2(e^{ix}-1)\,dx = -\frac{\pi^3}{6}$$

$$\int_0^{2\pi} \mathrm{Li}_2\left(e^{ix}+\frac{1}{2}\right)dx = \frac{\pi^3}{6} - \pi\ln^2 2$$

$$\int_0^{2\pi} \mathrm{Li}_2\left(e^{ix}+\frac{1}{\phi}\right)dx = \frac{\pi^3}{5} - 2\pi\ln^2\phi$$

$$\int_0^{2\pi} \mathrm{Li}_3(e^{ix}-1)\,dx = -\frac{3\pi}{2}\zeta(3)$$

$$\int_0^{2\pi} \mathrm{Li}_3\left(e^{ix}+\frac{1}{2}\right)dx = \frac{\pi}{12}\big(21\zeta(3)+4\ln^3 2-2\pi^2\ln 2\big)$$

$$\int_0^{2\pi} \mathrm{Li}_3\left(e^{ix}+\frac{1}{\phi^2}\right)dx = 4\pi\left(\frac{2}{5}\zeta(3)+\frac{1}{3}\ln^3(\phi)-\frac{\pi^2}{15}\ln\phi\right)$$

$$\int_0^{2\pi} G\left(e^{ix}+\frac{1}{2}\right)dx = \frac{2^{\frac{25}{24}}\pi^{\frac{3}{4}}e^{\frac{1}{8}}}{A^{\frac{3}{2}}} = \sqrt{\frac{\sqrt{\pi^3\sqrt{2^{\frac{25}{3}}e}}}{A^3}}$$

9.2 Getting contour with it

$$\int_0^\pi e^{\cos x}\,dx$$

This is a cool integral that can be solved using contour integration.

$$=\frac{1}{2}\int_0^{2\pi} e^{\frac{e^{ix}+e^{-ix}}{2}}\,dx$$

$$=\frac{1}{2}\int_0^{2\pi} e^{\frac{e^{2ix}+1}{2e^{ix}}}\,dx$$

$$t=e^{ix}, dt=it\,dx$$

$$=\frac{1}{2i}\oint_{|t|=1}\frac{1}{t}e^{\frac{t^2+1}{2t}}\,dt$$

$$=\frac{1}{2i}\oint_{|t|=1}\frac{1}{t}e^{\frac{t}{2}}e^{\frac{1}{2t}}\,dt$$

Expanding these two exponential functions in the integrand in terms of their Maclaurin series.

$$=\frac{1}{2i}\oint_{|t|=1}\frac{1}{t}\left(\sum_{n=0}^{\infty}\frac{t^n}{n!2^n}\right)\left(\sum_{k=0}^{\infty}\frac{1}{n!(2t)^n}\right)dt$$

Since the singularity $t=0$ is in the $1/t$ term out the front, we want to collect all the $1/t$ terms in the expansion of the brackets.

$$=\frac{1}{2i}\oint_{|t|=1}\frac{1}{t}\left(\frac{1}{0!2^0}+\frac{t}{1!2^1}+\frac{t^2}{2!2^2}+\frac{t^3}{3!2^3}+\cdots\right)\left(\frac{1}{0!2^0}+\frac{1}{1!2^1t}+\frac{1}{2!2^2t^2}+\frac{1}{3!2^3t^3}+\cdots\right)dt$$

$$=\frac{1}{2i}\oint_{|t|=1}\left(\frac{1}{0!2^0t}+\frac{1}{1!2^1}+\frac{t}{2!2^2}+\frac{t^2}{3!2^3}+\cdots\right)\left(\frac{1}{0!2^0}+\frac{1}{1!2^1t}+\frac{1}{2!2^2t^2}+\frac{1}{3!2^3t^3}+\cdots\right)dt$$

$$=\frac{1}{i}\oint_{|t|=1}\frac{1}{(0!2^0)^2t}+\frac{1}{(1!2^1)^2t}+\frac{1}{(2!2^2)^2t}+\frac{1}{(3!2^3)^2t}+\cdots\,dt$$

$$=\frac{1}{2i}\oint_{|t|=1}\frac{1}{t}\sum_{n=0}^{\infty}\frac{1}{(n!)^24^n}+\cdots\,dt$$

This sum above is the sum of the residues. Thus, using residue theorem, our integral is equal to

$$=\frac{1}{2i}2\pi i\sum_{n=0}^{\infty}\frac{1}{(n!)^24^n}$$

$$=\pi\sum_{n=0}^{\infty}\frac{1}{(n!)^24^n}$$

$$=\pi I_0(1)$$

Where $I_0(z)$ is the modified Bessel function of the first kind with $n=0$.

$$\therefore\int_0^\pi e^{\cos x}\,dx=\pi I_0(1)$$

Similarly, we can also find that

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^{\sin x}\,dx=\pi I_0(1)$$

10 October

10.1 Variant of early integral

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(x)\cos(n+k\tan x)\,dx$$

To solve, we first use the expansion of $\cos(\theta+\psi)$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(x)(\cos(n)\cos(k\tan x) - \sin(n)\sin(k\tan x))\,dx$$

$$= \cos(n)\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(x)\cos(k\tan x)\,dx - \sin(n)\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(x)\sin(k\tan x)\,dx$$

The second integrand is odd, and therefore the integral is equal to zero since the bounds are of equal magnitude. Thus, the original integral becomes

$$= \cos(n)\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(x)\cos(k\tan x)\,dx$$

$$= \cos(n)\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(k\tan x)\frac{\sec^2 x}{\sec^4 x}\,dx$$

$$= \cos(n)\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(k\tan x)\frac{\sec^2 x}{(1+\tan^2 x)^2}\,dx$$

$$u = \tan x, du = \sec^2 x\,dx$$

$$= \cos(n)\int_{-\infty}^{\infty} \frac{\cos(ku)}{(1+u^2)^2}\,du$$

Here comes the sneaky trick (essentially Feynman's).

$$= -\cos(n)\frac{d}{dt}\left(\int_{-\infty}^{\infty} \frac{\cos(ku)}{t+u^2}\,du\right)\bigg|_{t=1}$$

With some adjustments, this integrand becomes the integral seen on pages 22-24.

$$= -\cos(n)\frac{d}{dt}\left(\frac{1}{t}\int_{-\infty}^{\infty} \frac{\cos(ku)}{1+\frac{u^2}{t}}\,du\right)\bigg|_{t=1}$$

$$u = x\sqrt{t}, du = \sqrt{t}\,dx$$

$$= -\cos(n)\frac{d}{dt}\left(\frac{1}{\sqrt{t}}\int_{-\infty}^{\infty} \frac{\cos(kx\sqrt{t})}{1+x^2}\,dx\right)\bigg|_{t=1}$$

Using page 23,

$$= -\pi\cos(n)\frac{d}{dt}\left(\frac{1}{e^{\sqrt{t}|k|}\sqrt{t}}\right)\bigg|_{t=1}$$

$$= \frac{\pi\cos(n)(|k|t^{\frac{3}{2}}+t)}{2e^{\sqrt{t}|k|}t^{\frac{5}{2}}}\bigg|_{t=1}$$

$$= \frac{\pi\cos(n)(|k|+1)}{2e^{|k|}}$$

$$\therefore \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(x)\cos(n+k\tan x)\,dx = \frac{\pi\cos(n)(|k|+1)}{2e^{|k|}}$$

Basic example solution with $n=0, k=1$:

$$\therefore \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(x)\cos(\tan x)\,dx = \frac{\pi}{e}$$

11 November

11.1 Cleo's integral

A legendary integral, coming from this post: https://math.stackexchange.com/questions/562694. The first few steps comes from the answer by user xpaul [8].

$$\int_{-1}^{1} \frac{1}{x}\sqrt{\frac{1+x}{1-x}} \ln\left(\frac{2x^2+2x+1}{2x^2-2x+1}\right) dx$$

Let this integral equal I. First, we will use king's rule.

$$\begin{aligned}\therefore I &= \int_{-1}^{1} -\frac{1}{x}\sqrt{\frac{1-x}{1+x}} \ln\left(\frac{2x^2-2x+1}{2x^2+2x+1}\right) dx \\ &= \int_{-1}^{1} \frac{1}{x}\sqrt{\frac{1-x}{1+x}} \ln\left(\frac{2x^2+2x+1}{2x^2-2x+1}\right) dx\end{aligned}$$

Then, using these two expressions for I, we can add them together and divide by two.

$$\begin{aligned}\therefore I &= \frac{1}{2}\int_{-1}^{1} \frac{1}{x}\sqrt{\frac{1+x}{1-x}} \ln\left(\frac{2x^2+2x+1}{2x^2-2x+1}\right) + \frac{1}{x}\sqrt{\frac{1-x}{1+x}} \ln\left(\frac{2x^2+2x+1}{2x^2-2x+1}\right) dx \\ &= \frac{1}{2}\int_{-1}^{1} \frac{1}{x} \ln\left(\frac{2x^2+2x+1}{2x^2-2x+1}\right)\left(\sqrt{\frac{1+x}{1-x}} + \sqrt{\frac{1-x}{1+x}}\right) dx \\ &= \frac{1}{2}\int_{-1}^{1} \frac{1}{x} \ln\left(\frac{2x^2+2x+1}{2x^2-2x+1}\right)\left(\frac{1+x+1-x}{\sqrt{1-x}\sqrt{1+x}}\right) dx \\ &= \int_{-1}^{1} \frac{1}{x\sqrt{1-x^2}} \ln\left(\frac{2x^2+2x+1}{2x^2-2x+1}\right) dx\end{aligned}$$

Due to that square root term on the denominator, we will use the following substitution.

$$x = \sin t, dx = \cos t \, dt$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{\sin t} \ln\left(\frac{2\sin^2(t)+2\sin(t)+1}{2\sin^2(t)-2\sin(t)+1}\right) dt$$

We will now factor the numerator and denominator inside of the log, as so:

$$\begin{aligned}2x^2+2x+1 &= ((1-i)x+1)((1+i)x+1) \\ 2x^2-2x+1 &= ((1-i)x-1)((1+i)x-1)\end{aligned}$$

Thus, the integral can be written as

$$\begin{aligned}&= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{\sin t} \ln\left(\frac{((1-i)\sin(t)+1)((1+i)\sin(t)+1)}{((1-i)\sin(t)-1)((1+i)\sin(t)-1)}\right) dt \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{\sin t} \ln\left(\frac{((1-i)\sin(t)+1)((1+i)\sin(t)+1)}{(-(1-i)\sin(t)+1)(-(1+i)\sin(t)+1)}\right) dt\end{aligned}$$

Now, we will let $k = 1-i$. Thus, $1+i = k+2i$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{\sin t} \ln\left(\frac{(k\sin(t)+1)((k+2i)\sin(t)+1)}{(-k\sin(t)+1)(-(k+2i)\sin(t)+1)}\right) dt$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln(k\sin(t)+1)}{\sin t} dt + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln((k+2i)\sin(t)+1)}{\sin t} dt - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln(-k\sin(t)+1)}{\sin t} dt - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln(-(k+2i)\sin(t)+1)}{\sin t} dt$$

Call these integrals $I_1(k), I_2(k), I_3(k)$, and $I_4(k)$ respectively. We will now differentiate each integral (Feynman's trick), with respect to k. Thus,

$$I' = \frac{d}{dk}\left(\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln(k\sin(t)+1)}{\sin t}\,dt + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln((k+2i)\sin(t)+1)}{\sin t}\,dt - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln(-k\sin(t)+1)}{\sin t}\,dt - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\ln(-(k+2i)\sin(t)+1)}{\sin t}\,dt\right)$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1+k\sin t}\,dt + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1+(k+2i)\sin t}\,dt + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1-k\sin t}\,dt + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1-(k+2i)\sin t}\,dt$$

On $I_3'(k)$ and $I_4'(k)$, we make the following substitution

$$u = -t, du = -\,dt$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1+k\sin t}\,dt + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1+(k+2i)\sin t}\,dt + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1+k\sin u}\,du + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1+(k+2i)\sin u}\,du$$

$$= 2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1+k\sin t} + \frac{1}{1+(k+2i)\sin t}\,dt$$

which can be written as

$$= 4\Re\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1+k\sin t}\,dt$$

This is finally in a very solvable state, and just takes a couple u-substitutions.

$$u = \tan\frac{t}{2}, dt = \frac{2}{1+u^2}\,du$$

$$= 4\Re\int_{-1}^{1} \frac{1}{1+\frac{2ku}{1+u^2}} \cdot \frac{2}{1+u^2}\,du$$

$$= 8\Re\int_{-1}^{1} \frac{1}{u^2+2ku+1}\,du$$

$$= 8\Re\int_{-1}^{1} \frac{1}{(u+k)^2-k^2+1}\,du$$

$$w = \frac{u+k}{\sqrt{1-k^2}}, dw = \frac{1}{\sqrt{1-k^2}}\,du$$

$$= 8\Re\int_{\frac{k-1}{\sqrt{1-k^2}}}^{\frac{1+k}{\sqrt{1-k^2}}} \frac{\sqrt{1-k^2}}{(1-k^2)w^2+(1-k^2)}\,dw$$

$$= 8\Re\left(\frac{1}{\sqrt{1-k^2}}\int_{\frac{k-1}{\sqrt{1-k^2}}}^{\frac{1+k}{\sqrt{1-k^2}}} \frac{1}{w^2+1}\,dw\right)$$

$$= 8\Re\left(\frac{\arctan\left(\frac{k+1}{\sqrt{1-k^2}}\right) - \arctan\left(\frac{k-1}{\sqrt{1-k^2}}\right)}{\sqrt{1-k^2}}\right)$$

$$= 8\Re\left(\frac{\arctan\left(\frac{k+1}{\sqrt{1-k^2}}\right) + \arctan\left(\frac{1-k}{\sqrt{1-k^2}}\right)}{\sqrt{1-k^2}}\right)$$

Using the formula $\arctan(x)+\arctan(y) = \arctan\left(\frac{x+y}{1-xy}\right)$, we see that the arctan terms on the top are equal to $\pi/2$. This is because the denominator of $1-xy$ is $1-\frac{(k+1)(1-k)}{\sqrt{1-k^2}\sqrt{1-k^2}} = 1-\frac{1-k^2}{1-k^2} = 1-1 = 0$. So, the expression becomes

$$= 4\pi\Re\left(\frac{1}{\sqrt{1-k^2}}\right)$$

As this is equal to I', specifically in terms of k, we can integrate with respect to k to get our solution.

$$\therefore I = 4\pi\Re\int \frac{1}{\sqrt{1-k^2}}\,dk$$

$$= 4\pi\Re(\arcsin(k) + C)$$

To find $\Re C$, we sub in $k = 0$ in the following expression from earlier, before we made the differentiation.

$$\Re C = \Re\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{\sin t}\ln\left(\frac{(k\sin(t)+1)((k+2i)\sin(t)+1)}{(-k\sin(t)+1)(-(k+2i)\sin(t)+1)}\right)dt\Big|_{k=0}$$

$$= \Re\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{\sin t}\ln\left(\frac{2i\sin(t)+1}{-2i\sin(t)+1}\right)dt = 0$$

$$\therefore \Re C = 0$$

$$\therefore I = 4\pi\Re(\arcsin(k))$$

Then, since $\arcsin z = -i\ln(iz+\sqrt{1-z^2})$ and $\ln z = \ln(|z|) + i\arg z$, the line above becomes

$$= 4\pi\Re\big(-i\ln\big(ik+\sqrt{1-k^2}\big)\big)$$

$$= 4\pi\Re\big(\arg\big(ik+\sqrt{1-k^2}\big) - i\ln\big(\big|ik+\sqrt{1-k^2}\big|\big)\big)$$

$$= 4\pi\arg\big(ik+\sqrt{1-k^2}\big)$$

We can now finally write k back in terms of its complex number form, $k = 1-i$. Thus, $1-k^2 = 1+2i$.

$$= 4\pi\arg\big(1+i+\sqrt{1+2i}\big)$$

$$= 4\pi\arg\left(1+i+\sqrt{e^{i\arctan 2}\sqrt{5}}\right)$$

$$= 4\pi\arg\left(1+i+\sqrt{\sqrt{5}}\left(\cos\left(\frac{1}{2}\arctan 2\right)+i\sin\left(\frac{1}{2}\arctan 2\right)\right)\right)$$

$$= 4\pi\arg\left(1+\sqrt{\sqrt{5}}\cos\left(\frac{1}{2}\arctan 2\right)+i\left(1+\sqrt{\sqrt{5}}\sin\left(\frac{1}{2}\arctan 2\right)\right)\right)$$

Now we have separated the number into its real and imaginary components, we can use the definition $\arg z = \arctan(y/x)$, where $z = x+iy$. This allows us to remove all mentions of i and finally solve the integral.

$$= 4\pi\arctan\left(\frac{1+\sqrt{\sqrt{5}}\sin\left(\frac{1}{2}\arctan 2\right)}{1+\sqrt{\sqrt{5}}\cos\left(\frac{1}{2}\arctan 2\right)}\right)$$

Recall that $\sin(x/2) = \sqrt{\frac{1-\cos x}{2}}$ and $\cos(x/2) = \sqrt{\frac{1+\cos x}{2}}$

$$= 4\pi\arctan\left(\frac{1+\sqrt{\frac{\sqrt{5}(1-\cos(\arctan 2))}{2}}}{1+\sqrt{\frac{\sqrt{5}(1+\cos(\arctan 2))}{2}}}\right)$$

Then, as $\cos(\arctan x) = \frac{1}{\sqrt{1+x^2}}$, we can write the above line purely in algebraic forms.

$$= 4\pi\arctan\left(\frac{1+\sqrt{\frac{\sqrt{5}(1-\frac{1}{\sqrt{5}})}{2}}}{1+\sqrt{\frac{\sqrt{5}(1+\frac{1}{\sqrt{5}})}{2}}}\right)$$

$$= 4\pi \arctan\left(\frac{1+\sqrt{\frac{\sqrt{5}-1}{2}}}{1+\sqrt{\frac{\sqrt{5}+1}{2}}}\right)$$

Phi is appearing?!?!?!?

$$= 4\pi \arctan\left(\frac{1+\sqrt{\frac{1}{\phi}}}{1+\sqrt{\phi}}\right)$$

$$= 4\pi \arctan\left(\frac{\sqrt{\phi}+1}{\sqrt{\phi}(1+\sqrt{\phi})}\right)$$

$$= 4\pi \arctan\left(\frac{1}{\sqrt{\phi}}\right)$$

$$= 4\pi \operatorname{arccot}\left(\sqrt{\phi}\right)$$

And we have finally arrived at our solution to this crazy and legendary problem

$$\therefore \int_{-1}^{1} \frac{1}{x}\sqrt{\frac{1+x}{1-x}} \ln\left(\frac{2x^2+2x+1}{2x^2-2x+1}\right) dx = 4\pi \operatorname{arccot}\left(\sqrt{\phi}\right)$$

11.2 Algebraic working out be like:

Probably not the quickest method of solving the integral but it works. Solution method composed together from several bits found on Math stack exchange. This gave me the knowledge that I had to use the log definition of arctan(x), then later on to use DUIS.

$$\int_0^\pi \arctan(1+\cos x)\,dx$$

First, using the identity $\arctan z = \frac{i}{2}\ln\left(\frac{i+z}{i-z}\right)$, the integral can be written as

$$= \frac{i}{2}\int_0^\pi \ln\left(\frac{i+1+\cos x}{i-1-\cos x}\right)dx$$

$$= \frac{i}{2}\int_0^\pi \ln\left(\frac{\frac{1-i}{2}(i+1+\cos x)}{\frac{1-i}{2}(i-1-\cos x)}\right)dx$$

$$= \frac{i}{2}\int_0^\pi \ln\left(\frac{1+\frac{1-i}{2}\cos x}{i-\frac{1-i}{2}\cos x}\right)dx$$

$$= \frac{i}{2}\int_0^\pi \ln\left(\frac{1+\frac{1-i}{2}\cos x}{i\left(1+\frac{1+i}{2}\cos x\right)}\right)dx$$

$$= \frac{i}{2}\int_0^\pi \ln\left(1+\frac{1-i}{2}\cos x\right) - \ln\left(1+\frac{1+i}{2}\cos x\right) - \frac{i\pi}{2}\,dx$$

$$= \frac{i}{2}\int_0^\pi \ln\left(1+\frac{1-i}{2}\cos x\right) - \ln\left(1+\frac{1+i}{2}\cos x\right)dx + \frac{\pi^2}{4}$$

Now, let $k = \frac{1-i}{2}$. Thus, the integral can be written as

$$= \frac{i}{2}\int_0^\pi \ln(1+k\cos x) - \ln(1+(k+i)\cos x)\,dx + \frac{\pi^2}{4}$$

We will call this expression $I(k)$. Thus,

$$I'(k) = \frac{d}{dk}\left(\frac{i}{2}\int_0^\pi \ln(1+k\cos x) - \ln(1+(k+i)\cos x)\,dx + \frac{\pi^2}{4}\right)$$

$$= \frac{i}{2}\int_0^\pi \frac{\cos x}{1+k\cos x}\,dx - \frac{i}{2}\int_0^\pi \frac{\cos x}{1+(k+i)\cos x}\,dx$$

$$= \frac{i}{2k}\int_0^\pi \frac{k\cos x}{1+k\cos x}\,dx - \frac{i}{2(k+i)}\int_0^\pi \frac{(k+i)\cos x}{1+(k+i)\cos x}\,dx$$

$$= \frac{i}{2k}\int_0^\pi 1-\frac{1}{1+k\cos x}\,dx - \frac{i}{2(k+i)}\int_0^\pi 1-\frac{1}{1+(k+i)\cos x}\,dx$$

$$= \frac{i\pi}{2k} - \frac{i}{2k}\int_0^\pi \frac{1}{1+k\cos x}\,dx - \frac{i\pi}{2(k+i)} + \frac{i}{2(k+i)}\int_0^\pi \frac{1}{1+(k+i)\cos x}\,dx$$

$$= -\frac{i}{2k}\int_0^\pi \frac{1}{1+k\cos x}\,dx + \frac{i}{2(k+i)}\int_0^\pi \frac{1}{1+(k+i)\cos x}\,dx - \frac{\pi}{2k(k+i)}$$

We will now consider the integral

$$\int_0^\pi \frac{1}{1+k\cos x}\,dx$$

$$t = \tan\frac{x}{2}, dx = \frac{2}{1+u^2}\,dt$$

$$= \int_0^\infty \frac{1}{1+k\frac{(1-t^2)}{1+t^2}}\cdot\frac{2}{1+t^2}\,dt$$

$$= 2\int_0^\infty \frac{1}{(1-k)t^2+(k+1)}\,dt$$

$$u^2 = (1-k)t^2, 2u\,du = 2(1-k)t\,dt$$

$$= \frac{2}{\sqrt{1-k}}\int_0^\infty \frac{1}{u^2+(k+1)}\,du$$

$$= \frac{2}{\sqrt{1-k}}\cdot\frac{\pi}{2\sqrt{1+k}}$$

$$= \frac{\pi}{\sqrt{1+k}\sqrt{1-k}}$$

Using this information, we can conclude that

$$-\frac{i}{2k}\int_0^\pi \frac{1}{1+k\cos x}\,dx + \frac{i}{2(k+i)}\int_0^\pi \frac{1}{1+(k+i)\cos x}\,dx - \frac{\pi}{2k(k+i)}$$

$$= -\frac{i\pi}{2k\sqrt{1+k}\sqrt{1-k}} + \frac{i\pi}{2(k+i)\sqrt{1+k+i}\sqrt{1-k-i}} - \frac{\pi}{2k(k+i)}$$

$$= -\frac{\pi}{2k\sqrt{k^2-1}} - \frac{\pi}{2(k+i)\sqrt{k^2+2ik-2}} - \frac{\pi}{2k(k+i)}$$

We can now integrate each term to get $I(k)$ back.

$$\therefore I(k) = -\int \frac{\pi}{2k\sqrt{k^2-1}}\,dk - \int \frac{\pi}{2(k+i)\sqrt{k^2+2ik-2}}\,dk - \int \frac{\pi}{2k(k+i)}\,dk$$

From left to right:

$$u = \sqrt{k^2-1}, du = \frac{k}{\sqrt{k^2-1}}\,dk$$

$$t = \sqrt{k^2+2ik-2}, dt = \frac{k+i}{\sqrt{k^2+2ik-2}}\,dk$$

$$= -\frac{\pi}{2}\int \frac{1}{u^2+1}\,du - \frac{\pi}{2}\int \frac{1}{t^2+1}\,dt - \frac{i\pi}{2}\ln\left(\left|\frac{i}{k}+1\right|\right) + C$$

$$= -\frac{\pi}{2}\arctan\big(\sqrt{k^2-1}\big) - \frac{\pi}{2}\arctan\big(\sqrt{k^2+2ik-2}\big) - \frac{i\pi}{2}\ln\left(\left|\frac{i}{k}+1\right|\right) + C$$

To find C, we let $k \to 0$ on $I(k)$

$$\therefore I(0) = \lim_{k\to 0}\left(-\frac{\pi}{2}\arctan\big(\sqrt{k^2-1}\big) - \frac{\pi}{2}\arctan\big(\sqrt{k^2+2ik-2}\big) - \frac{i\pi}{2}\ln\left(\left|\frac{i}{k}+1\right|\right)\right) + C$$

Using a sub of $k = 0$ into the original $I(k)$, then taking the real part, we get

$$C = \frac{\pi^2}{4} - \Re\lim_{k\to 0}\left(-\frac{\pi}{2}\arctan\big(\sqrt{k^2-1}\big) - \frac{\pi}{2}\arctan\big(\sqrt{k^2+2ik-2}\big) - \frac{i\pi}{2}\ln\left(\left|\frac{i}{k}+1\right|\right)\right)$$

Above, the last term goes to zero, as $\left|\frac{i}{k}+1\right| = \left|\frac{i+k}{k}\right| = \left|\frac{2i+1-i}{1-i}\right| = \left|\frac{(1+i)^2}{(1-i)(1+i)}\right| = |i| = 1$, and $\ln 1 = 0$. On the other terms, we use the formula $\arctan(x) + \arctan(y) = \arctan\big(\frac{x+y}{1-xy}\big)$

$$\therefore C = \frac{\pi^2}{4} + \frac{\pi}{2}\times\Re\lim_{k\to 0}\left(\arctan\left(\frac{\sqrt{k^2-1}+\sqrt{k^2+2ik-2}}{1-\sqrt{(k^2-1)(k^2+2ik-2)}}\right)\right)$$

Subbing in $k = 0$ we get

$$C = \frac{\pi^2}{4} - \frac{\pi}{2}\times\Re\arctan\left(\frac{(1+\sqrt{2})i}{1-\sqrt{2}}\right)$$

As the real part of the arctan of a complex number with no real part and negative complex part is equal to $-\pi/2$, we can say that

$$\therefore C = \frac{\pi^2}{2}$$

$$\therefore I(k) = -\frac{\pi}{2}\arctan\big(\sqrt{k^2-1}\big) - \frac{\pi}{2}\arctan\big(\sqrt{k^2+2ik-2}\big) + \frac{\pi^2}{2}$$

Then, since $k = \frac{1-i}{2}$, we get

$$= -\frac{\pi}{2}\arctan\left(\sqrt{-1-\frac{i}{2}}\right) - \frac{\pi}{2}\arctan\left(\sqrt{-1+\frac{i}{2}}\right) + \frac{\pi^2}{2}$$

We can then again use the $\arctan(x)+\arctan(y)$ formula. However, note the shift up by π, which becomes $-\pi^2/2$ when expanded out. Thus, the line above becomes

$$= -\frac{\pi}{2}\arctan\left(\frac{\sqrt{-1-\frac{i}{2}} + \sqrt{-1+\frac{i}{2}}}{1-\sqrt{\left(-1-\frac{i}{2}\right)\left(-1+\frac{i}{2}\right)}}\right)$$

$$= -\frac{\pi}{2}\arctan\left(\frac{\sqrt{-1-\frac{i}{2}} + \sqrt{-1+\frac{i}{2}}}{\frac{2-\sqrt{5}}{2}}\right)$$

Notice on the numerator inside of the arctan a complex number and its conjugate. As $z+\bar{z} = 2\Re(z)$, the line above becomes

$$= -\frac{\pi}{2}\arctan\left(\frac{4\Re\sqrt{-1-\frac{i}{2}}}{2-\sqrt{5}}\right)$$

Now, let

$$\sqrt{-1-\frac{i}{2}} = x+iy$$

$$\therefore -1-\frac{i}{2} = x^2-y^2+2iyx$$

$$\therefore -1 = x^2-y^2, -\frac{1}{4} = yx$$

Solving these simulataneous equations, we get

$$x+iy = \sqrt{-1-\frac{i}{2}} = \frac{\sqrt{\sqrt{5}-2}}{2} - \frac{i}{2\sqrt{\sqrt{5}-2}}$$

$$\therefore -\frac{\pi}{2}\arctan\left(\frac{4\Re\sqrt{-1-\frac{i}{2}}}{2-\sqrt{5}}\right) = -\frac{\pi}{2}\arctan\left(\frac{2\sqrt{\sqrt{5}-2}}{2-\sqrt{5}}\right)$$

$$= \frac{\pi}{2}\arctan\left(\frac{2}{\sqrt{\sqrt{5}-2}}\right)$$

Switching to using the golden ratio $\phi = \frac{1+\sqrt{5}}{2}$,

$$= \frac{\pi}{2}\arctan\left(\frac{2}{\sqrt{2(\phi-1)-2}}\right)$$

Then, as $\phi - 1 = \frac{1}{\phi}$, we can multiply the top and bottom to get

$$= \frac{\pi}{2}\arctan\left(\frac{2\sqrt{\phi}}{\sqrt{2-\phi}}\right)$$

Where (since $\phi^2 = \phi + 1$),

$$\sqrt{\phi - 2} = \sqrt{\phi + 1 - 2\phi + 1} = \sqrt{\phi^2 - 2\phi + 1}$$

$$= \sqrt{(\phi - 1)^2} = \phi - 1$$

$$\therefore \frac{\pi}{2}\arctan\left(\frac{2\sqrt{\phi}}{\sqrt{2-\phi}}\right) = \frac{\pi}{2}\arctan\left(\frac{2\sqrt{\phi}}{\phi - 1}\right)$$

Dividing top and bottom by ϕ,

$$= \frac{\pi}{2}\arctan\left(\frac{\frac{2}{\sqrt{\phi}}}{1 - \frac{1}{\phi}}\right)$$

Now we can use the identity $2\arctan x = \arctan\left(\frac{2x}{1-x^2}\right), -1 < x < 1$, going from RHS to LHS where $x = 1/\sqrt{\phi}$, which works since $\left|1/\sqrt{\phi}\right| < 1$.

$$= \pi\arctan\left(\frac{1}{\sqrt{\phi}}\right)$$

$$= \pi\operatorname{arccot}\left(\sqrt{\phi}\right)$$

We are now finally done!

$$\therefore \int_0^\pi \arctan(1 + \cos x)\,dx = \pi\operatorname{arccot}\left(\sqrt{\phi}\right)$$

With just a little bit more work, we can also solve for

$$\int_0^\pi \operatorname{arccot}(\cos x - 1)\,dx$$

First, we use the fact that $\arctan x + \arctan\frac{1}{x} = \frac{\pi}{2}$. Therefore,

$$\int_0^\pi \arctan(1 + \cos x)\,dx + \int_0^\pi \arctan\left(\frac{1}{1 + \cos x}\right)dx = \frac{\pi^2}{2}$$

Using kings rule on the second integral, and taking out a factor of -1, we get

$$\therefore \int_0^\pi \arctan(1 + \cos x)\,dx - \int_0^\pi \arctan\left(\frac{1}{\cos x - 1}\right)dx = \frac{\pi^2}{2}$$

$$\therefore \int_0^\pi \arctan\left(\frac{1}{\cos x - 1}\right)dx = \pi\operatorname{arccot}\left(\sqrt{\phi}\right) - \frac{\pi^2}{2}$$

When reciprocating the arctan in the integral to become arccot, we have a shift of $-\pi$.

$$\therefore \int_0^\pi \operatorname{arccot}(\cos x - 1) - \pi\,dx = \pi\operatorname{arccot}\left(\sqrt{\phi}\right) - \frac{\pi^2}{2}$$

And finally, we can split the integral into two, solve the right integral, and move it over to the other side.

$$\therefore \int_0^\pi \operatorname{arccot}(\cos x - 1)\,dx = \pi\operatorname{arccot}\left(\sqrt{\phi}\right) + \frac{\pi^2}{2}$$

11.3 Another lambert boi!

Epico

$$\int_0^\infty W\left(\frac{1}{x^k}\right) dx$$

This integral converges for $k > 1$. Since $\frac{1}{x^k} = W\left(\frac{1}{x^k}\right)e^{W\left(\frac{1}{x^k}\right)}$, we make this u-sub.

$$u = W\left(\frac{1}{x^k}\right), -kx^{-k-1}\, dx = (u+1)e^u\, du$$

$$= \frac{1}{k}\int_0^\infty \frac{u(u+1)e^u}{x^{-k-1}}\, du$$

$$= \frac{1}{k}\int_0^\infty \frac{(u+1)}{u^{\frac{1}{k}}e^{\frac{u}{k}}}\, du$$

$$= \frac{1}{k}\int_0^\infty u^{1-\frac{1}{k}}e^{-\frac{u}{k}}\, du + \frac{1}{k}\int_0^\infty u^{-\frac{1}{k}}e^{-\frac{u}{k}}\, du$$

$$t = \frac{1}{k}u, dt = \frac{1}{k}\, du$$

$$= k^{1-\frac{1}{k}}\int_0^\infty t^{1-\frac{1}{k}}e^{-t}\, dt + k^{-\frac{1}{k}}\int_0^\infty t^{-\frac{1}{k}}e^{-t}\, dt$$

Using the gamma function, this becomes

$$= k^{1-\frac{1}{k}}\Gamma\left(2-\frac{1}{k}\right) + k^{-\frac{1}{k}}\Gamma\left(1-\frac{1}{k}\right)$$

$$= k^{-\frac{1}{k}}\Gamma\left(1-\frac{1}{k}\right)\left(k\left(1-\frac{1}{k}\right)+1\right)$$

$$= k^{1-\frac{1}{k}}\Gamma\left(1-\frac{1}{k}\right)$$

Our integral is thus solved, for $k > 1$.

$$\therefore \int_0^\infty W\left(\frac{1}{x^k}\right) dx = k^{1-\frac{1}{k}}\Gamma\left(1-\frac{1}{k}\right)$$

Example solution case: $k = 2$. Here, $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$

$$\int_0^\infty W\left(\frac{1}{x^2}\right) dx = \sqrt{2\pi}$$

11.4 Multiple gamma derivative

$$\int_{-\infty}^{\infty} e^{-e^x} x^2 e^{2x}\, dx$$

This will involve not only the gamma function, but the digamma and trigamma too. Solution from [9].

$$t = e^x, dt = t\, dx$$

$$= \int_{0}^{\infty} e^{-t} \ln^2(t) t\, dt$$

Now, we take the gamma function, and get its second derivative

$$\Gamma(k) = \int_{0}^{\infty} e^{-t} t^{k-1}\, dt$$

$$\therefore \Gamma''(k) = \int_{0}^{\infty} e^{-t} \ln^2(t) t^{k-1}\, dt$$

Thus,

$$\therefore \int_{-\infty}^{\infty} e^{-e^x} x^2 e^{2x}\, dx = \Gamma''(2)$$

I could just say what this value is in terms of other constants, but i want to find it myself. To do so, we use the trigamma function, which is

$$\psi_1(k) = \frac{d^2}{dk^2} \ln \Gamma(k) = \frac{d}{dk} \frac{\Gamma'(k)}{\Gamma(k)}$$

$$= \frac{\Gamma''(k)\Gamma(k) - \Gamma'(k)^2}{\Gamma(k)^2}$$

$$\therefore \Gamma''(k) = \psi_1(k)\Gamma(k) + \frac{\Gamma'(k)^2}{\Gamma(k)}$$

Another definition of the trigamma function is the equation

$$\psi_1(k) = \sum_{n=0}^{\infty} \frac{1}{(n+k)^2}$$

And thus, our equation from before becomes

$$\therefore \Gamma''(k) = \Gamma(k) \sum_{n=0}^{\infty} \frac{1}{(n+k)^2} + \frac{\Gamma'(k)^2}{\Gamma(k)}$$

Subbing in $k = 2$, we get

$$\therefore \Gamma''(2) = \sum_{n=0}^{\infty} \frac{1}{(n+2)^2} + \Gamma'(2)^2$$

And, since $\Gamma'(2) = \Gamma(2)\psi_0(2) = \psi_0(2) = \psi_0(1) + 1 = 1 - \gamma$, we can re-index the series to find that

$$\therefore \Gamma''(2) = \sum_{n=2}^{\infty} \frac{1}{n^2} + (1-\gamma)^2$$

$$= \frac{\pi^2}{6} + \gamma^2 - 2\gamma$$

$$\therefore \int_{-\infty}^{\infty} e^{-e^x} x^2 e^{2x}\, dx = \frac{\pi^2}{6} + \gamma^2 - 2\gamma$$

Very cool solution.

11.5 Crazy cancellations ahead

This problem also gives a crazyyy answer. For some of the steps I had to use [10], but otherwise done by me.

$$\int_0^\infty \ln\left(\frac{\tanh x}{x}\right)\frac{1}{x^2(x^2+1)}\,dx$$

Let us call this integral I. First we will break up this integral into two like so, and call them J and L.

$$= \int_0^\infty \ln\left(\frac{\sinh x}{x}\right)\frac{1}{x^2(x^2+1)}\,dx - \int_0^\infty \ln(\cosh x)\frac{1}{x^2(x^2+1)}\,dx$$

$$= J - L$$

We will first focus on J. To start, recall from page 55 the Weierstrass product for $\sinh x$, and divide it by x.

$$\therefore \frac{\sinh x}{x} = \prod_{n=1}^{\infty}\left(1+\frac{x^2}{n^2\pi^2}\right)$$

$$\therefore J = \int_0^\infty \ln\left(\frac{\sinh x}{x}\right)\frac{1}{x^2(x^2+1)}\,dx = \int_0^\infty \ln\left(\prod_{n=1}^{\infty}\left(1+\frac{x^2}{n^2\pi^2}\right)\right)\frac{1}{x^2(x^2+1)}\,dx$$

$$= \sum_{n=1}^{\infty}\int_0^\infty \ln\left(1+\frac{x^2}{n^2\pi^2}\right)\frac{1}{x^2(x^2+1)}\,dx$$

Now, let $k = \frac{1}{n^2\pi^2}$, and call this whole expression above $J(k)$

$$\therefore J(k) = \sum_{n=1}^{\infty}\int_0^\infty \frac{\ln(kx^2+1)}{x^2(x^2+1)}\,dx$$

We will know differentiate with respect to k.

$$\therefore J'(k) = \sum_{n=1}^{\infty}\int_0^\infty \frac{1}{(kx^2+1)(x^2+1)}\,dx$$

Partial fractions

$$= \sum_{n=1}^{\infty}\int_0^\infty \frac{A+B}{(kx^2+1)(x^2+1)}\,dx$$

$$Ax^2 + A + Bkx^2 + B = 1$$

$$\therefore A = \frac{k}{k-1}, B = -\frac{1}{k-1}$$

$$\therefore \sum_{n=1}^{\infty}\int_0^\infty \frac{A+B}{(kx^2+1)(x^2+1)}\,dx = \sum_{n=1}^{\infty}\left(\frac{k}{k-1}\int_0^\infty \frac{1}{kx^2+1}\,dx - \frac{1}{k-1}\int_0^\infty \frac{1}{x^2+1}\,dx\right)$$

$$= \frac{\pi}{2}\sum_{n=1}^{\infty}\left(\frac{\sqrt{k}}{k-1} - \frac{1}{k-1}\right)$$

We can now integrate with respect to k to get our solution.

$$\therefore J(k) = \frac{\pi}{2}\sum_{n=1}^{\infty}\left(\int \frac{\sqrt{k}}{k-1}\,dk - \int \frac{1}{k-1}\,dk\right)$$

$$= \frac{\pi}{2}\sum_{n=1}^{\infty}\left(\int \frac{\sqrt{k}}{k-1}\,dk - \ln(|k-1|)\right)$$

$$u = \sqrt{k}, du = \frac{1}{2\sqrt{k}}\, dk$$

$$= \frac{\pi}{2}\sum_{n=1}^{\infty}\left(2\int \frac{u^2}{u^2-1}\, du - \ln(|k-1|)\right)$$

$$= \frac{\pi}{2}\sum_{n=1}^{\infty}\left(2\int \frac{u^2-1}{u^2-1} + \frac{1}{u^2-1}\, du - \ln(|k-1|)\right)$$

$$= \frac{\pi}{2}\sum_{n=1}^{\infty}\left(2\sqrt{k} + 2\int \frac{1}{(u-1)(u+1)}\, du - \ln(|k-1|)\right)$$

We do simple partial fraction decomposition above, where $A = \frac{1}{2}, B = -\frac{1}{2}$.

$$= \frac{\pi}{2}\sum_{n=1}^{\infty}\left(2\sqrt{k} + \int \frac{1}{u-1}\, du - \int \frac{1}{u+1}\, du - \ln(|k-1|)\right)$$

$$= \frac{\pi}{2}\sum_{n=1}^{\infty}\left(2\sqrt{k} + \ln\left(|\sqrt{k}-1|\right) - \ln\left(|\sqrt{k}+1|\right) - \ln(|k-1|) + C\right)$$

Since $I(0) = 0$ by inspection of the integrand,

$$\therefore C = 0$$

$$\therefore J(k) = \frac{\pi}{2}\sum_{n=1}^{\infty}\left(2\sqrt{k} + \ln\left(|\sqrt{k}-1|\right) - \ln\left(|\sqrt{k}+1|\right) - \ln(|k-1|)\right)$$

$$= \frac{\pi}{2}\sum_{n=1}^{\infty}\left(2\sqrt{k} + \ln\left(\frac{(\sqrt{k}-1)}{(k-1)\left(\sqrt{k}+1\right)}\right)\right)$$

Then, as $k-1 = (\sqrt{k}-1)(\sqrt{k}+1)$,

$$= \frac{\pi}{2}\sum_{n=1}^{\infty}\left(2\sqrt{k} + \ln\left(\frac{1}{\left(\sqrt{k}+1\right)^2}\right)\right)$$

$$= \pi\sum_{n=1}^{\infty}\left(\sqrt{k} - \ln\left(\sqrt{k}+1\right)\right)$$

We can now finally write k back in terms of n again.

$$= \pi\sum_{n=1}^{\infty}\left(\frac{1}{n\pi} - \ln\left(\frac{1}{n\pi}+1\right)\right)$$

$$= \sum_{n=1}^{\infty}\left(\frac{1}{n} - \pi\ln\left(\frac{1}{n\pi}+1\right)\right)$$

Here we will try to get the equation in a way that we can use the equation $\gamma = \lim_{N\to\infty}\left(\sum_{n=1}^{N}\frac{1}{n} - \ln N\right)$

$$= \lim_{N\to\infty}\left(\sum_{n=1}^{N}\frac{1}{n} - \pi\sum_{n=1}^{N}\ln\left(\frac{1}{n\pi}+1\right)\right)$$

$$= \lim_{N\to\infty}\left(\sum_{n=1}^{N}\frac{1}{n} - \pi\ln\left(\prod_{n=1}^{N}\frac{1+n\pi}{n\pi}\right)\right)$$

$$= \lim_{N\to\infty}\left(\sum_{n=1}^{N}\frac{1}{n} - \pi\ln\left(\prod_{n=1}^{N}\frac{\frac{1}{\pi}+n}{n}\right)\right)$$

Note this expression above for later when we are solving for L, we will call it expression (1). Continuing,

$$= \lim_{N\to\infty}\left(\sum_{n=1}^{N}\frac{1}{n} - \pi\ln\left(\frac{1}{\Gamma(N+1)}\prod_{n=1}^{N}(1+n\pi)\right)\right)$$

$$= \lim_{N\to\infty}\left(\sum_{n=1}^{N}\frac{1}{n} - \pi\ln\left(\frac{\Gamma\left(N+\frac{1}{\pi}+1\right)}{\Gamma(N+1)\Gamma\left(\frac{1}{\pi}+1\right)}\right)\right)$$

$$= \lim_{N\to\infty}\left(\sum_{n=1}^{N}\frac{1}{n} - \pi\ln\left(\frac{\Gamma\left(N+\frac{1}{\pi}+1\right)}{\Gamma(N+1)}\right)\right) + \pi\ln\Gamma\left(\frac{1}{\pi}+1\right)$$

As $N\to\infty$, we know that $\Gamma(x+a)\sim\sqrt{2\pi x}\left(\frac{x}{e}\right)^{x}x^{a-1}$. Therefore, the equation becomes

$$= \lim_{N\to\infty}\left(\sum_{n=1}^{N}\frac{1}{n} - \pi\ln\left(\frac{\sqrt{2\pi N}\left(\frac{N}{e}\right)^{N}N^{\frac{1}{\pi}}}{\sqrt{2\pi N}\left(\frac{N}{e}\right)^{N}}\right)\right) + \pi\ln\Gamma\left(\frac{1}{\pi}+1\right)$$

$$= \lim_{N\to\infty}\left(\sum_{n=1}^{N}\frac{1}{n} - \ln N\right) + \pi\ln\Gamma\left(\frac{1}{\pi}+1\right)$$

We can finally use that equation mentioned earlier for γ

$$= \pi\ln\Gamma\left(\frac{1}{\pi}+1\right) + \gamma$$

What a solution.

$$\therefore J = \int_0^\infty \ln\left(\frac{\sinh x}{x}\right)\frac{1}{x^2(x^2+1)}\,dx = \pi\ln\Gamma\left(\frac{1}{\pi}+1\right)+\gamma$$

Now we will solve L. To start, recall the Weierstrass product for $\cosh x$.

$$\cosh x = \prod_{n=0}^{\infty}\left(1+\frac{4x^2}{(2m+1)^2\pi^2}\right)$$

$$\therefore L = \int_0^\infty \ln(\cosh x)\frac{1}{x^2(x^2+1)}\,dx = \int_0^\infty \ln\left(\prod_{n=0}^{\infty}\left(1+\frac{4x^2}{(2n+1)^2\pi^2}\right)\right)\frac{1}{x^2(x^2+1)}\,dx$$

$$= \sum_{n=0}^{\infty}\int_0^\infty \ln\left(1+\frac{4x^2}{(2n+1)^2\pi^2}\right)\frac{1}{x^2(x^2+1)}\,dx$$

Now, let $p = \frac{4}{(2n+1)^2\pi^2}$, and call the whole expression $L(p)$.

$$\therefore L(p) = \sum_{n=0}^{\infty}\int_0^\infty \frac{\ln(1+px^2)}{x^2(x^2+1)}\,dx$$

Notice the similarity to $J(k)$ and its solution. Here, is just in terms of p instead of k. So, we can take the solution of $J(k)$ and write it replacing all k with p (and with the sum starting from $n=0$).

$$\therefore L(p) = \pi\sum_{n=0}^{\infty}\left(\sqrt{p}-\ln\left(\sqrt{p}+1\right)\right)$$

And thus we can write p back in terms of n. Note that

$$\sqrt{p} = \frac{2}{(2n+1)\pi} = \frac{1}{(n+1/2)\pi}$$

Which is the same as $\sqrt{k}$ but with $n+1/2$ instead of n. So, we can write L in the same form as expression (1) from earlier, but with n replaced by $n+1/2$ and with the product and sum starting from $n=0$.

$$= \lim_{N\to\infty}\left(\sum_{n=0}^{N}\frac{1}{n+\frac{1}{2}} - \pi\ln\left(\prod_{n=0}^{N}\frac{\frac{1}{\pi}+\left(n+\frac{1}{2}\right)}{\left(n+\frac{1}{2}\right)}\right)\right)$$

This different starting index of the product also creates some extra gamma function terms compared to when we solved a similar product when solving J. Thus, we get

$$= \lim_{N\to\infty}\left(\sum_{n=0}^{N}\frac{2}{2n+1} - \pi\ln\left(\frac{\Gamma\left(N+\frac{1}{\pi}+\frac{3}{2}\right)\Gamma\left(\frac{1}{2}\right)}{\Gamma\left(N+\frac{3}{2}\right)\Gamma\left(\frac{1}{\pi}+\frac{1}{2}\right)}\right)\right)$$

Note that $\Gamma(1/2) = \sqrt{\pi}$

$$= \lim_{N\to\infty}\left(\sum_{n=0}^{N}\frac{2}{2n+1} - \pi\ln\left(\frac{\Gamma\left(N+\frac{1}{\pi}+\frac{3}{2}\right)}{\Gamma\left(N+\frac{3}{2}\right)}\right)\right)+\pi\ln\Gamma\left(\frac{1}{\pi}+\frac{1}{2}\right) - \frac{\pi}{2}\ln\pi$$

Using the same asymptotic expression for $\Gamma(x+a)$ as $N\to\infty$, this is equal to

$$= \lim_{N\to\infty}\left(2\sum_{n=0}^{N}\frac{1}{2n+1} - \pi\ln\left(\frac{\sqrt{2\pi N}\left(\frac{N}{e}\right)^{N}N^{\frac{1}{\pi}+\frac{3}{2}-1}}{\sqrt{2\pi N}\left(\frac{N}{e}\right)^{N}N^{\frac{3}{2}-1}}\right)\right)+\pi\ln\Gamma\left(\frac{1}{\pi}+\frac{1}{2}\right) - \frac{\pi}{2}\ln\pi$$

$$= \lim_{N\to\infty}\left(2\sum_{n=0}^{N}\frac{1}{2n+1} - \ln N\right)+\pi\ln\Gamma\left(\frac{1}{\pi}+\frac{1}{2}\right) - \frac{\pi}{2}\ln\pi$$

Now, we can break this sum up by using the fact that it is equivalent to the sum of the reciprocals of the positive integrals up to $2N+1$, minus all the even terms. We can then use the definition of γ but rearranged in order to solve the limit.

$$= \lim_{N\to\infty}\left(2\sum_{n=1}^{2N+1}\frac{1}{n} - \sum_{n=1}^{N}\frac{1}{N} - \ln N\right)+\pi\ln\Gamma\left(\frac{1}{\pi}+\frac{1}{2}\right) - \frac{\pi}{2}\ln\pi$$

$$= \lim_{N\to\infty}\Big(2(\gamma+\ln(2N+1)) - (\gamma+\ln(N)) - \ln(N)\Big)+\pi\ln\Gamma\left(\frac{1}{\pi}+\frac{1}{2}\right) - \frac{\pi}{2}\ln\pi$$

$$= \lim_{N\to\infty}\left(\gamma+2\ln\left(\frac{2N+1}{N}\right)\right)+\pi\ln\Gamma\left(\frac{1}{\pi}+\frac{1}{2}\right) - \frac{\pi}{2}\ln\pi$$

Using simple L'hopitals rule, the limit of $(2N+1)/N$ as $N\to\infty$ is equal to 2.

$$= \pi\ln\Gamma\left(\frac{1}{\pi}+\frac{1}{2}\right) - \frac{\pi}{2}\ln(\pi)+\gamma+2\ln 2$$

We are now done with the solution for L.

$$\therefore L = \int_0^\infty \ln(\cosh x)\frac{1}{x^2(x^2+1)}\,dx = \pi\ln\Gamma\left(\frac{1}{\pi}+\frac{1}{2}\right) - \frac{\pi}{2}\ln(\pi)+\gamma+2\ln 2$$

And thus, since $I = J - L$, we can finally solve the original integral.

$$\therefore I = \gamma+\pi\ln\Gamma\left(\frac{1}{\pi}+1\right) - \pi\ln\Gamma\left(\frac{1}{\pi}+\frac{1}{2}\right)+\frac{\pi}{2}\ln(\pi) - \gamma - 2\ln 2$$

$$= \pi\ln\left(\frac{\Gamma\left(\frac{1}{\pi}+1\right)}{\Gamma\left(\frac{1}{\pi}+\frac{1}{2}\right)}\right)+\frac{\pi}{2}\ln(\pi) - 2\ln 2$$

$$\therefore \int_0^\infty \ln\left(\frac{\tanh x}{x}\right)\frac{1}{x^2(x^2+1)}\,dx = \pi\ln\left(\frac{\Gamma\left(\frac{1}{\pi}+1\right)}{\Gamma\left(\frac{1}{\pi}+\frac{1}{2}\right)}\right)+\frac{\pi}{2}\ln(\pi) - 2\ln 2$$

11.6 RMT: Bracket integration method (3 examples)

Example 1:

(27/11) This method applies Ramanujan's master theorem (which i learnt via wikipedia) to a variety of integrals. Here is the first (simplest) example.

$$\int_0^\infty x^p e^{-kx^q}\,dx$$

To solve this generalised integral, we first use the Maclaurin series for e^x.

$$= \int_0^\infty x^p \sum_{n=0}^{\infty} \frac{(-kx^q)^n}{n!}\,dx$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} k^n \int_0^\infty x^{qn+p}\,dx$$

Now, this integral diverges, but the bracket integration technique introduces some notation to deal with this, called 'bracket series notation'. Thus, we write the expression above now as

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} k^n \langle qn+p+1\rangle$$

This is the form that we want our expression in before proceeding. Notation wise, this form can be written as

$$= \sum_{n=0}^{\infty} \phi_n \varphi(n) \langle qn+p+1\rangle$$

Where $\phi_n = (-1)^n/n!$ and $\varphi(n) = k^n$. Now, we check something called the 'complexity index', which is the number of sums minus the number of bracket pairs. Here, that index is equal to zero, and thus this is the method we proceed with: We wish to find n^*, which is the solution of following linear equation.

$$qn^* + p + 1 = 0$$

$$\therefore n^* = -\frac{1+p}{q}$$

We can now finally use the integral formula for zero complexity index in single integral cases. This formula is $a^{-1}\varphi(n^*)\Gamma(-n^*)$, where a is the coefficient of n^* in the linear equation. Recall that $\varphi(n^*) = k^{n^*}$.

$$a^{-1}\Gamma(-n^*)\varphi(n^*) = \frac{1}{q}k^{-\frac{1+p}{q}}\Gamma\left(\frac{1+p}{q}\right)$$

$$\therefore \int_0^\infty x^p e^{-kx^q}\,dx = \frac{1}{q}k^{-\frac{1+p}{q}}\Gamma\left(\frac{1+p}{q}\right)$$

This integral is valid for $k, q > 0, p \geq 0$. u-sub would be a quicker way of doing this specific integral, but this example using this method is meant to introduce the next two problems.

Example 2:

Now we take an integral with a complexity index of zero but with multiple sums and (the same amount of) multiple brackets.

$$\int_0^\infty \frac{1}{(cx^p + kx^q)^s}\,dx$$

Using the sum raised to a power formula according to the RMT, this integrand becomes

$$\frac{1}{\Gamma(s)}\int_0^\infty \sum_{n=0}^\infty \sum_{m=0}^\infty \frac{(-1)^{n+m}}{n!m!} c^n k^m x^{pn+qm} \langle n+m+s\rangle\,dx$$

Which, using the bracket notation, becomes

$$= \frac{1}{\Gamma(s)} \sum_{n=0}^\infty \sum_{m=0}^\infty \frac{(-1)^{n+m}}{n!m!} c^n k^m \langle pn+qm+1\rangle\langle n+m+s\rangle$$

$$= \sum_{n=0}^\infty \sum_{m=0}^\infty \phi_{n,m}\varphi(n,m)\langle pn+qm+1\rangle\langle n+m+s\rangle$$

Now we have it in the correct form, we have two simultaneous equations to solve (similar to the last example as the complexity index is $2-2=0$).

$$pn^* + qm^* + 1 = 0,\; n^* + m^* + s = 0$$

Solving, we get

$$n^* = \frac{1-sq}{q-p},\; m^* = \frac{sp-1}{q-p}$$

For these cases of 'multiple indicators', our formula is $\det|A|^{-1}\varphi(n^*,m^*)\Gamma(-n^*)\Gamma(-m^*)$, where

$$A = \begin{bmatrix} 1 & 1 \\ p & q \end{bmatrix}$$

$$\therefore \det|A|^{-1} = \frac{1}{q-p}$$

And thus our integral is solved:

$$\det|A|^{-1}\Gamma(-n^*)\Gamma(-m^*)\varphi(n^*,m^*) = \frac{1}{(q-p)\Gamma(s)} c^{\frac{sq-1}{p-q}} k^{\frac{1-sp}{p-q}} \Gamma\left(\frac{sq-1}{q-p}\right)\Gamma\left(\frac{1-sp}{q-p}\right)$$

$$\therefore \int_0^\infty \frac{1}{(cx^p+kx^q)^s}\,dx = \frac{1}{(q-p)\Gamma(s)} c^{\frac{sq-1}{p-q}} k^{\frac{1-sp}{p-q}} \Gamma\left(\frac{sq-1}{q-p}\right)\Gamma\left(\frac{1-sp}{q-p}\right)$$

Example 3

But what if the complexity index is greater than zero? This example will have a complexity index of one. The example comes from wikipedia, with a couple edits.

$$\int_0^\infty \frac{1}{(cx^p + kx^q + tx^g)^s}\,dx$$

Using the sum raised to a power formula according to the RMT, this integrand becomes

$$= \frac{1}{\Gamma(s)}\int_0^\infty \sum_{n=0}^\infty\sum_{m=0}^\infty\sum_{h=0}^\infty \frac{(-1)^{n+m+h}}{n!m!h!}c^n k^m t^h x^{pn+qm+gh}\langle n+m+h+s\rangle\,dx$$

$$= \frac{1}{\Gamma(s)}\sum_{n=0}^\infty\sum_{m=0}^\infty\sum_{h=0}^\infty \frac{(-1)^{n+m+h}}{n!m!h!}c^n k^m t^h\langle pn+qm+gh+1\rangle\langle n+m+h+s\rangle$$

$$= \sum_{n=0}^\infty\sum_{m=0}^\infty\sum_{h=0}^\infty \phi_{n,m,h}\varphi(n,m,h)\langle pn+qm+gh+1\rangle\langle n+m+h+s\rangle$$

Now, since the complexity index is 1, we need to 'solve' these simultaneous equations in a different way. To do so, we select a 'free index'. We will choose m, as choosing the other options will lead to a divergent (incorrect) answer . Then, we solve for n^* and h^* in terms of $\bar{m}$ (free index for m).

$$pn^* + q\bar{m} + gh^* + 1 = 0,\ n^* + \bar{m} + h^* + s = 0$$

$$\therefore n^* = \frac{\bar{m}(q-g) - sg + 1}{g-p},\ h^* = \frac{\bar{m}(p-q) + sp - 1}{g-p}$$

Finally, for these cases of a complexity index of one, our formula is

$$\det|A|^{-1}\sum_{\bar{m}=0}^\infty \frac{(-1)^{\bar{m}}}{\bar{m}!}\Gamma(-n^*)\Gamma(-h^*)\varphi(n^*,h^*,\bar{m})$$

where

$$A = \begin{bmatrix} 1 & 1 \\ g & p \end{bmatrix}$$

$$\therefore \det|A|^{-1} = \frac{1}{p-g}$$

We have thus arrived at our solution, and write it all in terms of $\bar{m}$ using the equations above. I will then change the index variable back to m for simplicity.

$$\therefore \det|A|^{-1}\sum_{\bar{m}=0}^\infty \frac{(-1)^{\bar{m}}}{\bar{m}!}\varphi(n^*,h^*,\bar{m})\Gamma(-n^*)\Gamma(-h^*)$$

$$= \frac{1}{(p-g)\Gamma(s)}\sum_{m=0}^\infty \frac{(-k)^m}{m!}c^{\frac{m(q-g)-sg+1}{g-p}}t^{\frac{m(p-q)+sp-1}{g-p}}\Gamma\left(\frac{m(q-g)-sg+1}{p-g}\right)\Gamma\left(\frac{m(p-q)+sp-1}{p-g}\right)$$

$$\therefore \int_0^\infty \frac{1}{(cx^p + kx^q + tx^g)^s}\,dx = \frac{1}{(p-g)\Gamma(s)}\sum_{m=0}^\infty \frac{(-k)^m}{m!}c^{\frac{m(q-g)-sg+1}{g-p}}t^{\frac{m(p-q)+sp-1}{g-p}}\Gamma\left(\frac{m(q-g)-sg+1}{p-g}\right)\Gamma\left(\frac{m(p-q)+sp-1}{p-g}\right)$$

Here are a couple examples for integer powers, constants equal to one, and $s = 1/2$

$$\int_0^\infty \frac{1}{\sqrt{x^3+x^2+x}}\,dx = \frac{1}{2\sqrt{\pi}}\sum_{m=0}^\infty \frac{(-1)^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right)$$

$$\int_0^\infty \frac{1}{\sqrt{x^5+x^4+1}}\,dx = \frac{1}{5\sqrt{\pi}}\sum_{m=0}^\infty \frac{(-1)^m}{m!}\Gamma\left(\frac{4m+1}{5}\right)\Gamma\left(\frac{2m+3}{10}\right)$$

11.7 Complete elliptic integral of the first kind

Interestingly, we can take the complete elliptic integral of the first kind, and write it in a way that the formula from page 125 can be used.

$$K(k) = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1-k^2\sin^2 x}}\,dx$$

$$u = \sin x, du = \cos x\,dx$$

$$= \int_0^1 \frac{1}{\sqrt{1-u^2}\sqrt{1-k^2u^2}}\,du$$

$$t = \frac{u}{\sqrt{1-u^2}}, du = \frac{1}{(1+t^2)^{\frac{3}{2}}}\,dt$$

$$= \int_0^\infty \frac{1}{\sqrt{(1+t^2)(1+(1-k^2)t^2)}}\,dt$$

$$= \int_0^\infty \frac{1}{\sqrt{(1-k^2)t^4+(2-k^2)t^2+1}}\,dt$$

Which, using the formula from page 125, then making a few simplifications, we get the result

$$\therefore K(k) = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1-k^2\sin^2 x}}\,dx = \frac{1}{4(1-k^2)^{\frac{1}{4}}\sqrt{\pi}} \sum_{m=0}^{\infty} \frac{1}{m!}\left(\frac{k^2-2}{\sqrt{1-k^2}}\right)^m \Gamma^2\left(\frac{2m+1}{4}\right)$$

The usual formula given for $K(k)$ online is

$$K(k) = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1-k^2\sin^2 x}}\,dx = \frac{\pi}{2}\sum_{m=0}^{\infty}\left(\frac{\Gamma(2m+1)}{4^m\Gamma^2(m+1)}\right)^2 k^{2m} = \frac{1}{2}\sum_{m=0}^{\infty}\frac{\Gamma^2\left(m+\frac{1}{2}\right)}{\Gamma^2(m+1)}k^{2m}$$

Therefore, we get

$$\sum_{m=0}^{\infty} \frac{1}{m!}\left(\frac{k^2-2}{\sqrt{1-k^2}}\right)^m \Gamma^2\left(\frac{2m+1}{4}\right) = 2(1-k^2)^{\frac{1}{4}}\sqrt{\pi}\sum_{m=0}^{\infty}\frac{\Gamma^2\left(m+\frac{1}{2}\right)}{\Gamma^2(m+1)}k^{2m}$$

Setting $k = 0$ on both sides (or simply plugging $k = 0$ into the integral for $K(k)$ itself, we get the interesting equality

$$\sum_{m=0}^{\infty} \frac{(-2)^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right) = 2\pi^{\frac{3}{2}}$$

As of current, I'm not sure about the domain of k that this holds, because i do not know the domain upon which my formula for $K(k)$ holds either.

Some values of $K(k)$ do actually have a solution in terms of the gamma function. Three such solutions are $K\left(\sqrt{2}-1\right)$, $K\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)$, $K\left(3-2\sqrt{2}\right)$, and $K\left(\frac{3-\sqrt{7}}{4\sqrt{2}}\right)$. Plugging these into our formula, and rearranging, we obtain (respectively)

$$\sum_{m=0}^{\infty}\frac{1}{m!}\left(\frac{1-2\sqrt{2}}{\sqrt{2}\sqrt{\sqrt{2}-1}}\right)^m\Gamma^2\left(\frac{2m+1}{4}\right)=\frac{1}{2}\Gamma\left(\frac{1}{8}\right)\Gamma\left(\frac{3}{8}\right)\sqrt{(\sqrt{2}+1)\sqrt{\sqrt{2}-1}}$$

$$\sum_{m=0}^{\infty}\frac{1}{m!}\left(\frac{3-5\sqrt{3}}{2\sqrt{2}}\right)^m\Gamma^2\left(\frac{2m+1}{4}\right)=\frac{\left(3\left(2+\sqrt{3}\right)\right)^{\frac{1}{4}}}{2^{\frac{1}{3}}\sqrt{2\pi}}\Gamma^3\left(\frac{1}{3}\right)$$

$$\sum_{m=0}^{\infty}\frac{1}{m!}\left(\frac{3\left(5-4\sqrt{2}\right)}{2\sqrt{3\sqrt{2}-4}}\right)^m\Gamma^2\left(\frac{2m+1}{4}\right)=\frac{\sqrt{2}+1}{2}\left(3\sqrt{2}-4\right)^{\frac{1}{4}}\Gamma^2\left(\frac{1}{4}\right)$$

$$\sum_{m=0}^{\infty}\frac{1}{m!}\left(-\frac{3\left(8+\sqrt{7}\right)}{4\sqrt{8+3\sqrt{7}}}\right)^m\Gamma^2\left(\frac{2m+1}{4}\right)=\frac{\left(8+3\sqrt{7}\right)^{\frac{1}{4}}}{2\sqrt{\pi\sqrt{7}}}\Gamma\left(\frac{1}{7}\right)\Gamma\left(\frac{2}{7}\right)\Gamma\left(\frac{4}{7}\right)$$

However, these sums no not actually converge!!! If they did, under analytic continuation those would be their results. In reality, the sum

$$\sum_{m=0}^{\infty}\frac{\alpha^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right)$$

converges for $|\alpha|\leq 2$, while the sums at the bottom of the previous page have an $|\alpha|$ of just over 2. In general,

$$\sum_{m=0}^{\infty}\frac{\alpha^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right)=4\sqrt{\pi}K\left(\frac{\sqrt{2+\alpha}}{2}\right)$$

Using this formula, and again using $K\left(\sqrt{2}-1\right)$, $K\left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)$, $K(3-2\sqrt{2})$, and $K\left(\frac{3-\sqrt{7}}{4\sqrt{2}}\right)$, we get respective values of α such that $|\alpha|\leq 2$!!! This means that these following sums DO converge, and equal the following.

$$\sum_{m=0}^{\infty}\frac{\left(2\left(5-4\sqrt{2}\right)\right)^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right)=\frac{\sqrt{\sqrt{2}+1}}{2^{\frac{5}{4}}}\Gamma\left(\frac{1}{8}\right)\Gamma\left(\frac{3}{8}\right)$$

$$\sum_{m=0}^{\infty}\frac{\left(-\sqrt{3}\right)^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right)=\frac{3^{\frac{1}{4}}}{2^{\frac{1}{3}}\sqrt{\pi}}\Gamma^3\left(\frac{1}{3}\right)$$

$$\sum_{m=0}^{\infty}\frac{\left(6\left(11-8\sqrt{2}\right)\right)^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right)=\frac{\sqrt{2}+1}{2\sqrt{2}}\Gamma^2\left(\frac{1}{4}\right)$$

$$\sum_{m=0}^{\infty}\frac{1}{m!}\left(-\frac{3\sqrt{7}}{4}\right)^m\Gamma^2\left(\frac{2m+1}{4}\right)=\frac{1}{\sqrt{\pi\sqrt{7}}}\Gamma\left(\frac{1}{7}\right)\Gamma\left(\frac{2}{7}\right)\Gamma\left(\frac{4}{7}\right)$$

12 December

12.1 Derivation of formula for gamma sum

Here is the derivation for this equation

$$\sum_{m=0}^{\infty}\frac{\alpha^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right)=4\sqrt{\pi}K\left(\frac{\sqrt{2+\alpha}}{2}\right)$$

We start from the left hand side. Using the formula for the gamma function,

$$\sum_{m=0}^{\infty}\frac{\alpha^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right)=\sum_{m=0}^{\infty}\frac{\alpha^m}{m!}\int_0^{\infty}\int_0^{\infty}x^{\frac{2m+1}{4}-1}y^{\frac{2m+1}{4}-1}e^{-(x+y)}\,dx\,dy$$

$$=\sum_{m=0}^{\infty}\frac{\alpha^m}{m!}\int_0^{\infty}\int_0^{\infty}x^{\frac{m}{2}}x^{-\frac{3}{4}}y^{\frac{m}{2}}y^{-\frac{3}{4}}e^{-(x+y)}\,dx\,dy$$

$$=\int_0^{\infty}\int_0^{\infty}x^{-\frac{3}{4}}y^{-\frac{3}{4}}e^{-(x+y)}\sum_{m=0}^{\infty}\frac{\left(\alpha\sqrt{xy}\right)^m}{m!}\,dx\,dy$$

Using the Maclaurin series for e^x, this sum can be solved to get

$$=\int_0^{\infty}\int_0^{\infty}x^{-\frac{3}{4}}y^{-\frac{3}{4}}e^{-(x+y)+\alpha\sqrt{xy}}\,dx\,dy$$

$$x=t^4,dx=4t^3\,dt,y=u^4,dy=4u^3\,du$$

$$=16\int_0^{\infty}\int_0^{\infty}e^{-(t^4+u^4)+\alpha t^2u^2}\,dt\,du$$

Now we will switch to the polar coordinates plane, remember that $dt\,du=r\,dr\,d\theta$. We let $t=r\cos\theta$ and $u=r\sin\theta$

$$=16\int_0^{\frac{\pi}{2}}\int_0^{\infty}e^{-r^4(\cos^4\theta+\sin^4\theta)+\alpha r^4\cos^2(\theta)\sin^2(\theta)}r\,dr\,d\theta$$

Then, as $\cos^4\theta+\sin^4\theta=1-2\cos^2(\theta)\sin^2(\theta)$, and making a few factorisations,

$$=16\int_0^{\frac{\pi}{2}}\int_0^{\infty}e^{-r^4(1-2\cos^2(\theta)\sin^2(\theta))+\alpha r^4\cos^2(\theta)\sin^2(\theta)}r\,dr\,d\theta$$

$$=16\int_0^{\frac{\pi}{2}}\int_0^{\infty}e^{-r^4(1-2\cos^2(\theta)\sin^2(\theta)-\alpha\cos^2(\theta)\sin^2(\theta))}r\,dr\,d\theta$$

$$=16\int_0^{\frac{\pi}{2}}\int_0^{\infty}e^{-r^4(1-(2+\alpha)\cos^2(\theta)\sin^2(\theta))}r\,dr\,d\theta$$

And, as $\cos^2(x)\sin^2(x)=\frac{1}{4}\sin^2(2x)$

$$=16\int_0^{\frac{\pi}{2}}\int_0^{\infty}e^{-r^4(1-(2+\alpha)\frac{1}{4}\sin^2(2\theta))}r\,dr\,d\theta$$

$$v=r^2,dv=2r\,dr$$

$$=8\int_0^{\frac{\pi}{2}}\int_0^{\infty}e^{-v^2(1-(2+\alpha)\frac{1}{4}\sin^2(2\theta))}\,dv\,d\theta$$

Here we can use the formula for the integral from zero to infinity of e^{-kx^2}, which has been seen quite a few times on this document by now.

$$= 4\sqrt{\pi}\int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1 - \frac{2+\alpha}{4}\sin^2(2\theta)}}\, d\theta$$

$$w = 2\theta, dw = 2\, d\theta$$

$$= 2\sqrt{\pi}\int_0^{\pi} \frac{1}{\sqrt{1 - \frac{2+\alpha}{4}\sin^2 w}}\, dw$$

$$= 4\sqrt{\pi}\int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1 - \frac{2+\alpha}{4}\sin^2 w}}\, dw$$

The integrand is finally in the form of the elliptic integral, where $k^2 = \frac{2+\alpha}{4}$. As we can just take the modulus of the square root as our input.

$$\therefore k = \frac{\sqrt{2+\alpha}}{2}$$

$$\therefore 4\sqrt{\pi}\int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1 - \frac{2+\alpha}{4}\sin^2 w}}\, dw = 4\sqrt{\pi}K\left(\frac{\sqrt{2+\alpha}}{2}\right)$$

And thus we are done, having reached the right hand side. As mentioned earlier, this converges for $|\alpha| \leq 2$

$$\therefore \sum_{m=0}^{\infty} \frac{\alpha^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right) = 4\sqrt{\pi}K\left(\frac{\sqrt{2+\alpha}}{2}\right)$$

12.2 Integral of Euler function

(8/12) The Euler function is

$$\phi(q) = \prod_{n=1}^{\infty}(1 - q^n)$$

Using this, we want to solve the following integral.

$$\int_0^1 \phi(x)\,dx$$

$$= \int_0^1 \prod_{n=1}^{\infty}(1 - x^n)\,dx$$

To solve this, we use the pentagonal number theorem, which states that

$$\prod_{n=1}^{\infty}(1 - x^n) = \sum_{n=-\infty}^{\infty}(-1)^n x^{\frac{n(3n-1)}{2}}$$

$$\therefore \int_0^1 \prod_{n=1}^{\infty}(1 - x^n)\,dx = \int_0^1 \sum_{n=-\infty}^{\infty}(-1)^n x^{\frac{n(3n-1)}{2}}\,dx$$

$$= \sum_{n=-\infty}^{\infty}(-1)^n \int_0^1 x^{\frac{n(3n-1)}{2}}\,dx$$

Solving the integral and taking out a factor of 2, we get

$$= 2\sum_{n=-\infty}^{\infty}\frac{(-1)^n}{3n^2 - n + 2}$$

To deal with this $(-1)^n$ term on the numerator, we consider the sum as the sum of all the even n terms minus the sum of all the odd n terms.

$$= 2\sum_{n=-\infty}^{\infty}\frac{1}{3(2n)^2 - (2n) + 2} - 2\sum_{n=-\infty}^{\infty}\frac{1}{3(2n+1)^2 - (2n+1) + 2}$$

$$= \sum_{n=-\infty}^{\infty}\frac{1}{6n^2 - n + 1} - \sum_{n=-\infty}^{\infty}\frac{1}{6n^2 + 5n + 2}$$

Now, we complete the square on each sum, where $ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + \frac{4ac-b^2}{4a}$

$$= \sum_{n=-\infty}^{\infty}\frac{1}{6\left(x - \frac{1}{12}\right)^2 + \frac{23}{24}} - \sum_{n=-\infty}^{\infty}\frac{1}{6\left(x + \frac{5}{12}\right)^2 + \frac{23}{24}}$$

$$= \frac{1}{6}\sum_{n=-\infty}^{\infty}\frac{1}{\left(x - \frac{1}{12}\right)^2 + \frac{23}{144}} - \frac{1}{6}\sum_{n=-\infty}^{\infty}\frac{1}{\left(x + \frac{5}{12}\right)^2 + \frac{23}{144}}$$

We will now use the following theorem to evaluate these sums. It was surprising to find online that all sums in the form 1/quadratic (where there coefficient of x^2 is not equal to zero) converges. I got this equation from section 1.445 of the table of integrals, series, and products (seventh ed) by Gradshteyn and Ryzhik. However, with the help of knowing the answer, I managed to figure out how to get the solution by starting from the LHS. The formula is:

$$\sum_{n=-\infty}^{\infty}\frac{1}{(n+\beta)^2 + \alpha^2} = \frac{\pi\sinh(2\pi\alpha)}{\alpha(\cosh(2\pi\alpha) - \cos(2\pi\beta))}$$

Which interestingly (as an aside) also means that

$$\sum_{n=-\infty}^{\infty} \frac{1}{an^2+bn+c} = \frac{2\pi}{\sqrt{\Delta}} \frac{\sin\left(\frac{\pi}{a}\sqrt{\Delta}\right)}{\cos\left(\frac{\pi}{a}\sqrt{\Delta}\right) - \cos\left(\frac{b\pi}{a}\right)}$$

Where $\Delta = b^2 - 4ac$ is the discriminant of a quadratic with coefficients a, b, c.

However, instead of simply using the theorem, I wanted to solve it for myself. Starting from the LHS of the first formula at the bottom of the previous page, we separate the sum into halves, then do a change of variables $n \to -n$ to the right sum. We then can thus see that they are equivalent, as the n values simply swap signs, but are still summed across all n from $-\infty$ to ∞.

$$\therefore \sum_{n=-\infty}^{\infty} \frac{1}{(n+\beta)^2+\alpha^2} = \frac{1}{2} \sum_{n=-\infty}^{\infty} \left(\frac{1}{(n+\beta)^2+\alpha^2} + \frac{1}{(n-\beta)^2+\alpha^2} \right)$$

$$= \frac{1}{2} \sum_{n=-\infty}^{\infty} \frac{(n+\beta)^2+(n-\beta)^2+2\alpha^2}{((n+\beta)^2+\alpha^2)((n-\beta)^2+\alpha^2)}$$

$$= \sum_{n=-\infty}^{\infty} \frac{n^2+\alpha^2+\beta^2}{(n-\beta)^2(n+\beta)^2+\alpha^2(n+\beta)^2+\alpha^2(n-\beta)^2+\alpha^4}$$

$$= \sum_{n=-\infty}^{\infty} \frac{n^2+\alpha^2+\beta^2}{n^4-2n^2(\alpha^2-\beta^2)+2\alpha^2\beta^2+\alpha^4+\beta^4}$$

Let the summand be $f(n)$. Since $f(n) = f(-n)$, we can write

$$\sum_{n=-\infty}^{\infty} f(n) = \sum_{n=1}^{\infty} f(n) + \sum_{n=1}^{\infty} f(-n) + f(0) = f(0) + 2\sum_{n=1}^{\infty} f(n)$$

Therefore, our sum becomes

$$= \frac{1}{\alpha^2+\beta^2} + 2\sum_{n=1}^{\infty} \frac{n^2+\alpha^2+\beta^2}{n^4-2n^2(\alpha^2-\beta^2)+2\alpha^2\beta^2+\alpha^4+\beta^4}$$

We can now write these in terms of the real part of some complex numbers. This can be verified by doing the following three steps in reverse order.

$$= \Re\frac{1}{\alpha(\alpha+\beta i)} + 2\sum_{n=1}^{\infty} \frac{n^2+\alpha^2+\beta^2}{n^4-2n^2(\alpha^2-\beta^2)+2\alpha^2\beta^2+\alpha^4+\beta^4}$$

$$= \Re\left(\frac{1}{\alpha(\alpha+\beta i)} + \frac{2}{\alpha}\sum_{n=1}^{\infty} \frac{\alpha+\beta i}{(n^2+\alpha^2-\beta^2)+2\alpha\beta i} \right)$$

$$= \Re\left(\frac{1}{\alpha(\alpha+\beta i)} + \frac{2}{\alpha}\sum_{n=1}^{\infty} \frac{\alpha+\beta i}{n^2+(\alpha+\beta i)^2} \right)$$

This expression inside the $\Re$ symbol looks similar to $\coth(\alpha+\beta i)$, which via using page 35, is equal to the following:

$$\coth(\alpha+\beta i) = \frac{1}{\alpha+\beta i} + 2\sum_{n=1}^{\infty} \frac{\alpha+\beta i}{n^2\pi^2+(\alpha+\beta i)^2}$$

We thus make the following adjustments to both sides to get the RHS the same as what we have in our problem:

$$\therefore \frac{\pi}{\alpha}\coth(\alpha+\beta i) = \frac{\pi}{\alpha(\alpha+\beta i)} + 2\pi\sum_{n=1}^{\infty} \frac{\alpha+\beta i}{n^2\pi^2+(\alpha+\beta i)^2}$$

$$\therefore \frac{\pi}{\alpha}\coth(\pi(\alpha+\beta i)) = \frac{1}{\alpha(\alpha+\beta i)} + 2\sum_{n=1}^{\infty} \frac{\alpha+\beta i}{n^2+(\alpha+\beta i)^2}$$

This now matches the expression that we have inside of the $\Re$ symbol.

$$\therefore \Re\left(\frac{1}{\alpha(\alpha+\beta i)} + \frac{2}{\alpha}\sum_{n=1}^{\infty}\frac{\alpha+\beta i}{n^2+(\alpha+\beta i)^2}\right) = \frac{\pi}{\alpha}\Re\coth(\pi(\alpha+\beta i))$$

Similar to the formula for $\Re\tanh z$ on page 101, the formula for $\Re\coth z$, where $z=\alpha+\beta i$, is

$$\Re\coth z = \frac{\sinh(2\alpha)}{\cosh(2\alpha)-\cos(2\beta)}$$

$$\therefore \frac{\pi}{\alpha}\Re\coth(\pi(\alpha+\beta i)) = \frac{\pi\sinh(2\pi\alpha)}{\alpha(\cosh(2\pi\alpha)-\cos(2\pi\beta))}$$

As this is now the RHS, we have thus completed the 'proof' (un-rigorously) of the theorem. So, by using the respective values of α and β in each sum of our problem, and plugging them into this formula, we get

$$\therefore \frac{1}{6}\sum_{n=-\infty}^{\infty}\frac{1}{\left(x-\frac{1}{12}\right)^2+\frac{23}{144}} - \frac{1}{6}\sum_{n=-\infty}^{\infty}\frac{1}{\left(x+\frac{5}{12}\right)^2+\frac{23}{144}} = \frac{2\pi\sinh\left(\frac{\pi\sqrt{23}}{6}\right)}{\sqrt{23}\left(\cosh\left(\frac{\pi\sqrt{23}}{6}\right)-\frac{\sqrt{3}}{2}\right)} - \frac{2\pi\sinh\left(\frac{\pi\sqrt{23}}{6}\right)}{\sqrt{23}\left(\cosh\left(\frac{\pi\sqrt{23}}{6}\right)+\frac{\sqrt{3}}{2}\right)}$$

$$= \frac{2\pi}{\sqrt{23}}\sinh\left(\frac{\pi\sqrt{23}}{6}\right)\left(\frac{1}{\cosh\left(\frac{\pi\sqrt{23}}{6}\right)-\frac{\sqrt{3}}{2}} - \frac{1}{\cosh\left(\frac{\pi\sqrt{23}}{6}\right)+\frac{\sqrt{3}}{2}}\right)$$

Now, we add the two fractions together by bringing them under the same denominator, then expand the brackets to get

$$= \frac{2\pi}{\sqrt{23}}\sinh\left(\frac{\pi\sqrt{23}}{6}\right)\frac{\sqrt{3}}{\left(\cosh\left(\frac{\pi\sqrt{23}}{6}\right)-\frac{\sqrt{3}}{2}\right)\left(\cosh\left(\frac{\pi\sqrt{23}}{6}\right)+\frac{\sqrt{3}}{2}\right)}$$

$$= \frac{2\pi\sqrt{\frac{3}{23}}\sinh\left(\frac{\pi\sqrt{23}}{6}\right)}{\cosh^2\left(\frac{\pi\sqrt{23}}{6}\right)-\frac{3}{4}}$$

Then finally, using $\cosh^2 x = \frac{\cosh(2x)+1}{2}$, $\sinh(2x) = 2\sinh(x)\cosh(x)$, and $\cosh(3x) = 4\cosh^3 x - 3\cosh x$, we can simplify the expression above and obtain the following solution.

$$= \frac{8\pi\sqrt{\frac{3}{23}}\sinh\left(\frac{\pi\sqrt{23}}{6}\right)}{2\cosh\left(\frac{\pi\sqrt{23}}{3}\right)-1}$$

$$= \frac{4\pi\sqrt{\frac{3}{23}}\sinh\left(\frac{\pi\sqrt{23}}{3}\right)}{\cosh\left(\frac{\pi\sqrt{23}}{6}\right)\left(2\cosh\left(\frac{\pi\sqrt{23}}{3}\right)-1\right)}$$

$$= \frac{4\pi\sqrt{\frac{3}{23}}\sinh\left(\frac{\pi\sqrt{23}}{3}\right)}{\cosh\left(\frac{\pi\sqrt{23}}{6}\right)\left(4\cosh^2\left(\frac{\pi\sqrt{23}}{6}\right)-3\right)}$$

$$= \frac{4\pi\sqrt{\frac{3}{23}}\sinh\left(\frac{\pi\sqrt{23}}{3}\right)}{\cosh\left(\frac{\pi\sqrt{23}}{2}\right)}$$

This completes the solution for the integral.

$$\therefore \int_0^1 \phi(x)\,dx = \int_0^1 \prod_{n=1}^{\infty}(1-x^n)\,dx = \frac{4\pi\sqrt{\frac{3}{23}}\sinh\left(\frac{\pi\sqrt{23}}{3}\right)}{\cosh\left(\frac{\pi\sqrt{23}}{2}\right)}$$

It's actually crazy that this has a closed form solution. In fact, it can even be easily generalised (using the same method) further to the integral from 0 to 1 of $\phi(x^k)$ for some $k \geq 0$, with solution seen below.

$$\int_0^1 \phi(x^k)\,dx = \int_0^1 \prod_{n=1}^{\infty}(1-x^{kn})\,dx = \frac{4\pi\sqrt{\frac{3}{k(24-k)}}\sinh\left(\frac{\pi}{3}\sqrt{\frac{24-k}{k}}\right)}{\cosh\left(\frac{\pi}{2}\sqrt{\frac{24-k}{k}}\right)}$$

Of particular interest from this is when $k = 24$, where we get an $0/0$ indeterminate form when plugging in. Instead, we may use the fact that $\sinh(x) \sim x$ as $x \to 0$, to obtain

$$\int_0^1 \phi(x^{24})\,dx = \int_0^1 \prod_{n=1}^{\infty}(1-x^{24n})\,dx = \frac{\pi^2}{6\sqrt{3}}$$

We can also make integrands of this form multiplied with the Rogers-Ramanujan identities $G(x)$ and $H(x)$, where

$$G(x) = \prod_{n=1}^{\infty}\frac{1}{(1-x^{5n-4})(1-x^{5n-1})}$$

$$H(x) = \prod_{n=1}^{\infty}\frac{1}{(1-x^{5n-3})(1-x^{5n-2})}$$

Thus, we can find that

$$\int_0^1 G(x)\phi(x)\,dx = \sqrt{\frac{10+2\sqrt{5}}{39}}\,\frac{8\pi\sinh\left(\frac{\pi\sqrt{39}}{10}\right)}{4\cosh\left(\frac{\pi\sqrt{39}}{5}\right)-1-\sqrt{5}}$$

$$\int_0^1 H(x)\phi(x)\,dx = \sqrt{\frac{10-2\sqrt{5}}{31}}\,\frac{8\pi\sinh\left(\frac{\pi\sqrt{31}}{10}\right)}{4\cosh\left(\frac{\pi\sqrt{31}}{5}\right)-1+\sqrt{5}}$$

12.3 Malmsten

The Malmsten integral is very well known. I thus certainly didn't do this one myself. Consider the following generalised integral.

$$\int_1^\infty \frac{\ln(\ln x)}{x^2+2x\cos a+1}\,dx$$

To solve, we start with the following substitution.

$$x=\frac{1}{u},dx=-\frac{1}{u^2}\,du$$

$$=\int_0^1 \frac{\ln(-\ln u)}{u^2+2u\cos a+1}\,du$$

$$=\int_0^1 \frac{\ln(-\ln u)}{u^2+u(e^{ia}+e^{-ia})+1}\,du$$

$$=\int_0^1 \frac{\ln(-\ln u)}{(1+ue^{ia})(1+ue^{-ia})}\,du$$

$$=\int_0^1 \frac{\ln(-\ln u)}{(1-(-ue^{ia}))(1-(-ue^{-ia}))}\,du$$

Consider the denominators as two geometric series multiplied together.

$$=\int_0^1 \ln(-\ln u)\sum_{n=0}^\infty(-ue^{ia})^n\sum_{m=0}^\infty(-ue^{-ia})^m\,du$$

$$=\int_0^1 \ln(-\ln u)\sum_{n=0}^\infty\sum_{m=0}^\infty(-u)^{n+m}e^{ia(n-m)}\,du$$

Let $k=n+m$. When holding k constant, we see n run from 0 to k. Thus, the expression above can we written as

$$=\int_0^1 \ln(-\ln u)\sum_{k=0}^\infty\sum_{n=0}^k(-u)^k e^{ia(2n-k)}\,du$$

$$=\int_0^1 \ln(-\ln u)\sum_{k=0}^\infty\frac{(-u)^k}{e^{iak}}\sum_{n=0}^k(e^{2ia})^n\,du$$

Writing the second sum as a (non-infinite) geometric series, we get

$$=\int_0^1 \ln(-\ln u)\sum_{k=0}^\infty\frac{(-u)^k(1-e^{2ia(k+1)})}{e^{iak}(1-e^{2ia})}\,du$$

$$=\int_0^1 \ln(-\ln u)\sum_{k=0}^\infty\frac{(-u)^k(e^{-ia(k+1)}-e^{ia(k+1)})}{e^{-ia}-e^{ia}}\,du$$

Simplifying and re-indexing the series to $k=1$, we get

$$=\frac{1}{\sin a}\int_0^1 \ln(-\ln u)\sum_{k=1}^\infty(-u)^{k-1}\sin(ak)\,du$$

$$=\frac{1}{\sin a}\sum_{k=1}^\infty(-1)^{k-1}\sin(ak)\int_0^1 \ln(-\ln u)u^{k-1}\,du$$

$$t=-\ln u, dt=-\frac{1}{u}\,du$$

$$= \frac{1}{\sin a} \sum_{k=1}^{\infty} (-1)^{k-1} \sin(ak) \int_0^{\infty} \ln(t) e^{-tk}\, dt$$

$$w = tk, dw = k\, dt$$

$$= \frac{1}{\sin a} \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \sin(ak)}{k} \left(\int_0^{\infty} \ln(w) e^{-w}\, dw - \ln k \right)$$

The remaining integral is known to be $-\gamma$.

$$= \frac{1}{\sin a} \sum_{k=1}^{\infty} \frac{(-1)^k \sin(ak)(\gamma + \ln k)}{k}$$

$$= \frac{\gamma}{\sin a} \sum_{k=1}^{\infty} \frac{(-1)^k \sin(ak)}{k} + \frac{1}{\sin a} \sum_{k=1}^{\infty} \frac{(-1)^k \sin(ak) \ln k}{k}$$

$$= \Im\left(\frac{\gamma}{\sin a} \sum_{k=1}^{\infty} \frac{(-e^{ia})^k}{k} \right) + \frac{1}{\sin a} \sum_{k=1}^{\infty} \frac{(-1)^k \sin(ak) \ln k}{k}$$

For now we will be focusing on the left sum only. First, recognise how it is the taylor series of $-\ln(1+z)$ where $z = e^{ia}$. Thus, the expression becomes

$$= \Im\left(-\frac{\gamma}{\sin a} \ln(1 + e^{ia}) \right) + \frac{1}{\sin a} \sum_{k=1}^{\infty} \frac{(-1)^k \sin(ak) \ln k}{k}$$

$$= \Im\left(-\frac{\gamma}{\sin a} \ln(1 + \cos a + i \sin a) \right) + \frac{1}{\sin a} \sum_{k=1}^{\infty} \frac{(-1)^k \sin(ak) \ln k}{k}$$

$$= \Im\left(-\frac{\gamma}{\sin a} \ln\left(2\cos^2\left(\frac{a}{2}\right) + 2i \sin\left(\frac{a}{2}\right) \cos\left(\frac{a}{2}\right) \right) \right) + \frac{1}{\sin a} \sum_{k=1}^{\infty} \frac{(-1)^k \sin(ak) \ln k}{k}$$

$$= \Im\left(-\frac{\gamma}{\sin a} \ln\left(2\cos\left(\frac{a}{2}\right) e^{\frac{ia}{2}} \right) \right) + \frac{1}{\sin a} \sum_{k=1}^{\infty} \frac{(-1)^k \sin(ak) \ln k}{k}$$

$$= -\frac{\gamma a}{2 \sin a} + \frac{1}{\sin a} \sum_{k=1}^{\infty} \frac{(-1)^k \sin(ak) \ln k}{k}$$

We will now solve the remaining sum. First, we use a formula by Malmsten that

$$\frac{1}{\pi} \sum_{k=1}^{\infty} \frac{\sin(2\pi kx) \ln k}{k} = \ln \Gamma(x) - \frac{1}{2} \ln(2\pi) + \frac{1}{2} \ln(2 \sin(\pi x)) - \frac{1}{2}(\gamma + \ln(2\pi))(1 - 2x)$$

By re-arranging then letting $x = \frac{1}{2} - \frac{a}{2\pi}$, we get

$$\sum_{k=1}^{\infty} \frac{\sin(k(\pi - a)) \ln k}{k} = \pi \ln \Gamma\left(\frac{1}{2} - \frac{a}{2\pi} \right) - \frac{\pi}{2} \ln(2\pi) + \frac{\pi}{2} \ln\left(2 \sin\left(\frac{\pi - a}{2} \right) \right) - \frac{a}{2}(\gamma + \ln(2\pi))$$

Then, as (for integer k from $k = 1$)

$$\sin(k(\pi - a)) = \sin(k\pi)\cos(ka) - \cos(k\pi)\sin(ka)$$

$$= -(-1)^k \sin(ka)$$

we can sub this into the expressions and obtain

$$\therefore -\frac{\gamma a}{2 \sin a} + \frac{1}{\sin a} \sum_{k=1}^{\infty} \frac{(-1)^k \sin(ak) \ln k}{k}$$

$$= -\frac{\gamma a}{2\sin a} - \frac{1}{\sin a}\left(\pi \ln\Gamma\left(\frac{1}{2} - \frac{a}{2\pi}\right) - \frac{\pi}{2}\ln(2\pi) + \frac{\pi}{2}\ln\left(2\sin\left(\frac{\pi - a}{2}\right)\right) - \frac{a}{2}(\gamma + \ln(2\pi))\right)$$

$$= \frac{1}{\sin a}\left(-\pi\ln\Gamma\left(\frac{1}{2} - \frac{a}{2\pi}\right) - \frac{\pi}{2}\ln\left(2\sin\left(\frac{\pi - a}{2}\right)\right) + \left(\frac{\pi + a}{2}\right)\ln(2\pi)\right)$$

$$= \frac{1}{2\sin a}\left(\pi\ln\left(\frac{\pi}{\Gamma^2\left(\frac{1}{2} - \frac{a}{2\pi}\right)\sin\left(\frac{\pi - a}{2}\right)}\right) + a\ln(2\pi)\right)$$

Finally, since $\Gamma(1-x)\Gamma(x) = \frac{\pi}{\sin(\pi x)}$, we get

$$= \frac{1}{2\sin a}\left(\pi\ln\left(\frac{\Gamma\left(\frac{1}{2} + \frac{a}{2\pi}\right)}{\Gamma\left(\frac{1}{2} - \frac{a}{2\pi}\right)}\right) + a\ln(2\pi)\right)$$

And thus we are done.

$$\therefore \int_1^\infty \frac{\ln(\ln x)}{x^2 + 2x\cos a + 1}\,dx = \frac{1}{2\sin a}\left(\pi\ln\left(\frac{\Gamma\left(\frac{1}{2} + \frac{a}{2\pi}\right)}{\Gamma\left(\frac{1}{2} - \frac{a}{2\pi}\right)}\right) + a\ln(2\pi)\right)$$

Particular solutions (the last one can be derived via l'hopitals and using given values of the digamma function):

$$\int_1^\infty \frac{\ln(\ln x)}{x^2 + 1}\,dx = \pi\ln\left(\frac{\sqrt{2\sqrt{\pi^3}}}{\Gamma\left(\frac{1}{4}\right)}\right)$$

$$\int_1^\infty \frac{\ln(\ln x)}{x^2 + x + 1}\,dx = \frac{2\pi}{\sqrt{3}}\ln\left(\frac{(2\pi)^{\frac{2}{3}}}{3^{\frac{1}{4}}\Gamma\left(\frac{1}{3}\right)}\right)$$

$$\int_1^\infty \frac{\ln(\ln x)}{x^2 - x + 1}\,dx = \frac{2\pi}{\sqrt{3}}\ln\left(\frac{(2\pi)^{\frac{5}{6}}}{\Gamma\left(\frac{1}{6}\right)}\right)$$

$$\int_1^\infty \frac{\ln(\ln x)}{(x+1)^2}\,dx = \frac{1}{2}\left(\ln\left(\frac{\pi}{2}\right) - \gamma\right)$$

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13.1 More on sum of 1/quadratic

From page 130/131, we have that

$$\sum_{n=-\infty}^{\infty} \frac{1}{(n+\beta)^2+\alpha^2} = \frac{\pi \sinh(2\pi\alpha)}{\alpha(\cosh(2\pi\alpha)-\cos(2\pi\beta))}$$

Which means that

$$\sum_{n=-\infty}^{\infty} \frac{1}{an^2+bn+c} = \frac{2\pi}{\sqrt{\Delta}} \frac{\sin\left(\frac{\pi}{a}\sqrt{\Delta}\right)}{\cos\left(\frac{\pi}{a}\sqrt{\Delta}\right)-\cos\left(\frac{b\pi}{a}\right)}$$

where $\Delta = b^2 - 4ac$.

Here are some further related results.

When $a = b$, this becomes

$$\sum_{n=-\infty}^{\infty} \frac{1}{a(n^2+n)+c} = \frac{2\pi \tan\left(\frac{\pi}{2}\sqrt{\frac{a-4c}{a}}\right)}{\sqrt{a(a-4c)}}$$

where we can also find that

$$\sum_{n=0}^{\infty} \frac{1}{a(n^2+n)+c} = \frac{\pi \tan\left(\frac{\pi}{2}\sqrt{\frac{a-4c}{a}}\right)}{\sqrt{a(a-4c)}}$$

Then, when $b = 0$, we can obtain

$$\sum_{n=-\infty}^{\infty} \frac{1}{a^2n^2+c^2} = \frac{\pi \sinh\left(\frac{2\pi c}{a}\right)}{ac\left(\cosh\left(\frac{2\pi c}{a}\right)-1\right)}$$

Since $f(n) = f(-n)$ above, this series can also be turned into a sum from 0 instead of infinity.

$$\sum_{n=0}^{\infty} \frac{1}{a^2n^2+c^2} = \frac{\pi \sinh\left(\frac{2\pi c}{a}\right)}{2ac\left(\cosh\left(\frac{2\pi c}{a}\right)-1\right)} + \frac{1}{2c^2}$$

Lastly, we have the following alternating series:

$$\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{(n+\beta)^2+\alpha^2} = \frac{2\pi \sinh(\pi\alpha)\cos(\pi\beta)}{\alpha(\cosh(2\pi\alpha)-\cos(2\pi\beta))}$$

Which means in turn that

$$\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{an^2+bn+c} = \frac{4\pi}{\sqrt{\Delta}} \frac{\sin\left(\frac{\pi}{2a}\sqrt{\Delta}\right)\cos\left(\frac{\pi b}{2a}\right)}{\cos\left(\frac{\pi}{a}\sqrt{\Delta}\right)-\cos\left(\frac{\pi b}{a}\right)}$$

13.2 List of sums related to elliptic integral

Coming from page 127, we can extend our list to a full one here. This list (in order), is for $r = 2$ to $r = 16$, then as $r \to \infty$, with the exceptions of $r = 11, 13, 14$ because they're too complicated to write down in a single line.

$$\sum_{m=0}^{\infty} \frac{\left(2\left(5-4\sqrt{2}\right)\right)^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right) = \frac{\sqrt{\sqrt{2}+1}}{2^{\frac{5}{4}}}\Gamma\left(\frac{1}{8}\right)\Gamma\left(\frac{3}{8}\right)$$

$$\sum_{m=0}^{\infty} \frac{\left(-\sqrt{3}\right)^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right) = \frac{3^{\frac{1}{4}}}{2^{\frac{1}{3}}\sqrt{\pi}}\Gamma^3\left(\frac{1}{3}\right)$$

$$\sum_{m=0}^{\infty} \frac{\left(6\left(11-8\sqrt{2}\right)\right)^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right) = \frac{\sqrt{2}+1}{2\sqrt{2}}\Gamma^2\left(\frac{1}{4}\right)$$

$$\sum_{m=0}^{\infty} \frac{\left(-4\sqrt{\sqrt{5}-2}\right)^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right) = \sqrt{\frac{\sqrt{\sqrt{5}+2}}{10}\Gamma\left(\frac{1}{20}\right)\Gamma\left(\frac{3}{20}\right)\Gamma\left(\frac{7}{20}\right)\Gamma\left(\frac{9}{20}\right)}$$

$$\sum_{m=0}^{\infty} \frac{\left(4\left(\left(2-\sqrt{3}\right)\left(\sqrt{3}-\sqrt{2}\right)\right)^2-2\right)^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right) = \frac{\sqrt{\left(\sqrt{2}-1\right)\left(\sqrt{3}+\sqrt{2}\right)\left(2+\sqrt{3}\right)\Gamma\left(\frac{1}{24}\right)\Gamma\left(\frac{5}{24}\right)\Gamma\left(\frac{7}{24}\right)\Gamma\left(\frac{11}{24}\right)}}{2\sqrt{6}}$$

$$\sum_{m=0}^{\infty} \frac{1}{m!}\left(-\frac{3\sqrt{7}}{4}\right)^m\Gamma^2\left(\frac{2m+1}{4}\right) = \frac{1}{\sqrt{\pi\sqrt{7}}}\Gamma\left(\frac{1}{7}\right)\Gamma\left(\frac{2}{7}\right)\Gamma\left(\frac{4}{7}\right)$$

$$\sum_{m=0}^{\infty} \frac{\left(4\left(\sqrt{2}+1-\sqrt{2\sqrt{2}+2}\right)^4-2\right)^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right) = \frac{1}{4}\left(\frac{\left(2\sqrt{2}+\left(1+5\sqrt{2}\right)^{\frac{1}{2}}\right)\left(\sqrt{2}+1\right)}{\sqrt{2}}\right)^{\frac{1}{2}}\Gamma\left(\frac{1}{8}\right)\Gamma\left(\frac{3}{8}\right)$$

$$\sum_{m=0}^{\infty} \frac{\left(2^{\frac{5}{2}}3^{\frac{1}{4}}\left(\sqrt{3}-2\right)\right)^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right) = \frac{\sqrt{2+\sqrt{3}}}{3^{\frac{3}{4}}}\Gamma^2\left(\frac{1}{4}\right)$$

$$\sum_{m=0}^{\infty} \frac{\left(4\left(\sqrt{10}-3\right)^2\left(\sqrt{2}-1\right)^4-2\right)^m}{m!}\Gamma^2\left(\frac{2m+1}{4}\right) = \frac{\sqrt{\left(2+3\sqrt{2}+\sqrt{5}\right)\Gamma\left(\frac{1}{40}\right)\Gamma\left(\frac{7}{40}\right)\Gamma\left(\frac{9}{40}\right)\Gamma\left(\frac{11}{40}\right)\Gamma\left(\frac{13}{40}\right)\Gamma\left(\frac{19}{40}\right)\Gamma\left(\frac{23}{40}\right)\Gamma\left(\frac{37}{40}\right)}}{4\pi\sqrt{10}}$$

$$\sum_{m=0}^{\infty} \frac{1}{m!}\left(4\frac{\left(4-\sqrt{6}-\sqrt{2}\right)^2}{\left(4+\sqrt{6}+\sqrt{2}\right)^2}-2\right)^m\Gamma\left(\frac{2m+1}{4}\right)^2 = \frac{3^{\frac{1}{4}}\left(4+\sqrt{6}+\sqrt{2}\right)}{2^{\frac{10}{3}}\sqrt{\pi}}\Gamma^3\left(\frac{1}{3}\right)$$

$$\sum_{m=0}^{\infty} \frac{1}{m!}\left(-\frac{7\sqrt{3}+\sqrt{15}}{8}\right)^m\Gamma\left(\frac{2m+1}{4}\right)^2 = \sqrt{\frac{2\phi}{15}\Gamma\left(\frac{1}{15}\right)\Gamma\left(\frac{2}{15}\right)\Gamma\left(\frac{4}{15}\right)\Gamma\left(\frac{8}{15}\right)}$$

$$\sum_{m=0}^{\infty} \frac{1}{m!}\left(\frac{64(2^{\frac{1}{4}}-1)^4}{(1+\sqrt{2})2^{\frac{29}{4}}+16(2^{\frac{1}{4}}-1)^4}-2\right)^m\Gamma\left(\frac{2m+1}{4}\right)^2 = \frac{(2^{\frac{1}{4}}+1)^2}{2^{\frac{5}{2}}}\Gamma^2\left(\frac{1}{4}\right)$$

$$\sum_{m=0}^{\infty} \frac{(-2)^m}{m!}\Gamma\left(\frac{2m+1}{4}\right)^2 = 2\pi^{\frac{3}{2}}$$

14 February 26

14.1 Polygamma integral

(20/2/26) Very cool polygamma related integral.

$$\int_0^\infty x^{-s}\psi_k(x+1)\,dx$$

One equation for the polygamma function is

$$\psi_k(x) = (-1)^{k+1}k!\sum_{n=0}^\infty \frac{1}{(n+x)^{k+1}}$$

This can be substituted into the integral.

$$\therefore \int_0^\infty x^{-s}\psi_k(x+1)\,dx = (-1)^{k+1}k!\int_0^\infty x^{-s}\sum_{n=0}^\infty \frac{1}{(n+x+1)^{k+1}}\,dx$$

$$= (-1)^{k+1}k!\sum_{n=0}^\infty \int_0^\infty \frac{x^{-s}}{(n+x+1)^{k+1}}\,dx$$

$$= (-1)^{k+1}k!\sum_{n=1}^\infty \int_0^\infty \frac{x^{-s}}{(n+x)^{k+1}}\,dx$$

$$= (-1)^{k+1}k!\sum_{n=1}^\infty \frac{1}{n^{k+1}}\int_0^\infty \frac{x^{-s}}{(1+x/n)^{k+1}}\,dx$$

$$t = \frac{x}{n}, dt = \frac{1}{n}\,dx$$

$$= (-1)^{k+1}k!\sum_{n=1}^\infty \frac{1}{n^{k+s}}\int_0^\infty \frac{t^{-s}}{(1+t)^{k+1}}\,dt$$

The sum can be recognised as the zeta function of $k+s$, and the integral can be recognised as the beta function of $1-s, k+s$.

$$= (-1)^{k+1}k!\zeta(k+s)\beta(1-s,k+s)$$

Using the identity for the beta function in terms of the gamma function, we obtain

$$= (-1)^{k+1}\zeta(k+s)\Gamma(1-s)\Gamma(k+s)$$

Which is our answer.

$$\therefore \int_0^\infty x^{-s}\psi_k(x+1)\,dx = (-1)^{k+1}\zeta(k+s)\Gamma(1-s)\Gamma(k+s)$$

This integral seems to converge for $s > 1, k \in \mathbb{N}$, but I'm not 100% sure yet. Anyways, Here are some closed form results:

$$\int_0^\infty x\psi_3(x+1)\,dx = \frac{\pi^2}{6}$$

$$\int_0^\infty x^2\psi_4(x+1)\,dx = -\frac{\pi^2}{3}$$

$$\int_0^\infty x\psi_5(x+1)\,dx = \frac{\pi^4}{15}$$

$$\int_0^\infty x^2\psi_6(x+1)\,dx = -\frac{2\pi^4}{15}$$

We can notice that the power of the π is always $k+s$.

15 March 26

15.1 Shortcut method for solving sum

$$\sum_{n=0}^{\infty}\left(\frac{1}{3n+1}-\frac{1}{3n+2}\right)$$

To solve this, we can turn the summands into integrals, swap integral and summation, and solve.

$$= \sum_{n=0}^{\infty}\int_0^1 x^{3n}-x^{3n+1}\,dx$$

$$= \int_0^1\sum_{n=0}^{\infty}x^{3n}\,dx - \int_0^1\sum_{n=0}^{\infty}x^{3n+1}\,dx$$

$$= \int_0^1 \frac{1}{1-x^3}-\frac{x}{1-x^3}\,dx$$

$$= \int_0^1 \frac{1-x}{(1-x)(1+x+x^2)}\,dx$$

$$= \int_0^1 \frac{1}{1+x+x^2}\,dx$$

$$= \int_0^1 \frac{1}{\left(x+\frac{1}{2}\right)^2+\frac{3}{4}}\,dx$$

This can simply be solved in terms of arctans.

$$= \frac{2}{\sqrt{3}}\arctan\left(\frac{3}{\sqrt{3}}\right)-\frac{2}{\sqrt{3}}\arctan\left(\frac{1}{\sqrt{3}}\right)$$

$$= \frac{\pi}{3\sqrt{3}}$$

And thus we are done.

$$\therefore \sum_{n=0}^{\infty}\left(\frac{1}{3n+1}-\frac{1}{3n+2}\right)=\frac{\pi}{3\sqrt{3}}$$

Interestingly this sum can also be solved using the formula (from page 137)

$$\sum_{n=0}^{\infty}\frac{1}{a(n^2+n)+c}=\frac{\pi\tan\left(\frac{\pi}{2}\sqrt{\frac{a-4c}{a}}\right)}{\sqrt{a(a-4c)}}$$

Where in the case of our problem, the sum has $a=9$ and $c=2$.

15.2 Difficult looking integral with simple solution

The integral in question here is (for $\alpha > 0$)

$$\int_0^1 x^{\alpha-1} \ln(x) \sin(\ln x)\, dx$$

First, let

$$I(\alpha) = \int_0^1 x^{\alpha-1} \sin(\ln x)\, dx$$

$$\therefore I'(\alpha) = \int_0^1 x^{\alpha-1} \ln(x) \sin(\ln x)\, dx$$

So $I'(\alpha)$ is what we are trying to solve. To do so, we can use the fact that $\sin(x) = \Im(e^{ix})$

$$\therefore I(\alpha) \int_0^1 x^{\alpha-1} \sin(\ln x)\, dx = \int_0^1 x^{\alpha-1} \Im(x^i)\, dx$$

$$= \int_0^1 \Im(x^{\alpha+i-1})\, dx$$

$$= \Im \frac{1}{\alpha+i}$$

$$= \Im \frac{\alpha-i}{\alpha^2+1}$$

$$= \frac{-1}{\alpha^2+1}$$

Thus, to get our answer, we simply need to take the derivative of this with respect to α

$$\therefore I'(\alpha) = \frac{2\alpha}{(\alpha^2+1)^2}$$

And we are done. Thus, for $\alpha > 0$, we have found that

$$\therefore \int_0^1 x^{\alpha-1} \ln(x) \sin(\ln x)\, dx = \frac{2\alpha}{(\alpha^2+1)^2}$$

How simple. The indefinite integral for general α does indeed have a solution (the function has an antiderivative), but its quite a long process involving u-sub and a LOT of IBP (like 5 times or something).

15.3 Sophomore's Dream type integral

Recall from page 19 the two sophomore integrals.

$$\int_0^1 x^x \, dx = -\sum_{n=1}^{\infty} (-n)^{-n}$$

$$\int_0^1 x^{-x} \, dx = \sum_{n=1}^{\infty} n^{-n}$$

In this section, I will solve the similar type integral. The general direction for the solution was found online (to try and use the Lambert W function), but solution otherwise by me.

$$\int_0^{\frac{1}{2}} \frac{(1-x)^{x-1}}{x^x} \, dx$$

Which will also turn out to also be a Sophomore type integral! First, notice by symmetry that we can rewrite the bounds as

$$= \int_{\frac{1}{2}}^1 \frac{(1-x)^{x-1}}{x^x} \, dx$$

$$= \int_{\frac{1}{2}}^1 \frac{1}{1-x} \left(\frac{1-x}{x} \right)^x \, dx$$

$$t = \frac{1-x}{x}, x = \frac{1}{t+1}, dx = \frac{-1}{(t+1)^2} \, dt$$

$$= \int_0^1 \frac{t^{\frac{1}{t+1}}}{t(t+1)} \, dt$$

$$u = t^{\frac{1}{t+1}}, du = t^{\frac{1}{t+1}} \left(\frac{1+t-t\ln t}{t(t+1)^2} \right) dt$$

$$= \int_0^1 \frac{t+1}{1+t-t\ln t} \, du$$

Since $u = t^{\frac{1}{t+1}}$,

$$\therefore \ln u = \frac{\ln t}{t+1}$$

Therefore we can rewrite the denominator as

$$= \int_0^1 \frac{t+1}{t+1-(t+1)t\ln u} \, du$$

$$= \int_0^1 \frac{1}{1-t\ln u} \, du$$

Now we must write t in terms of u. Since $u = t^{\frac{1}{t+1}}$,

$$\therefore u^{t+1} = t$$

$$u = tu^{-t} = te^{-t\ln u}$$

$$-u\ln u = -t\ln(u)e^{-t\ln u}$$

$$W(-u\ln u) = -t\ln(u)$$

$$\therefore t = \frac{-W(-u\ln u)}{\ln u}$$

Thus, subbing this into the integral, we get

$$= \int_0^1 \frac{1}{1+W(-u\ln u)}\,du$$

Now we want to find the Lagrange inversion summation of $1/(1+W(x))$. First, let (where $w = W(x)$)

$$g(w) = \frac{1}{1+w}, f(w) = we^w$$

The Lagrange inversion sum for $g(w)$ is thus

$$\frac{1}{1+w} = g(0) + \sum_{n=1}^{\infty} \frac{(x-f(0))^n}{n!} \lim_{w\to 0} \frac{d^{n-1}}{dw^{n-1}}\left(g'(w)\left(\frac{w}{f(w)-f(0)}\right)^n\right)$$

$$\frac{1}{1+w} = 1 + \sum_{n=1}^{\infty} \frac{x^n}{n!} \lim_{w\to 0} \frac{d^{n-1}}{dw^{n-1}}\left(\frac{-1}{(1+w)^2}e^{-nw}\right)$$

Now we need to find this limit of the $(n-1)$th derivative. In general the kth derivative of a product of two functions $u(w)$ and $v(w)$ is

$$\frac{d^k}{dx^k}u(x)v(x) = \sum_{j=0}^{k} \binom{k}{j} \frac{d^{k-j}}{dx^{k-j}}u(x)\frac{d^j}{dx^j}v(x)$$

Thus, in our case we are working with w instead of x, with

$$u(w) = \frac{-1}{(1+w)^2}, v(w) = e^{-nw}$$

and $k = n-1$. We are then looking for the limit of this as $w \to 0$. Thus, the formula becomes

$$\lim_{w\to 0} \frac{d^{n-1}}{dw^{n-1}}\left(\frac{-1}{(1+w)^2}e^{-nw}\right) = \lim_{w\to 0} \sum_{j=0}^{n-1} \binom{n-1}{j} \frac{d^{n-j-1}}{dx^{n-j-1}}\left(\frac{-1}{(1+w)^2}\right)\frac{d^j}{dx^j}e^{-nw}$$

Since

$$\lim_{w\to 0} \frac{d^j}{dw^j}e^{-nw} = (-n)^j$$

We can now write the expression as

$$= \lim_{w\to 0} \sum_{j=0}^{n-1} \binom{n-1}{j} \frac{d^{n-j-1}}{dx^{n-j-1}}\left(\frac{-1}{(1+w)^2}\right)(-n)^j$$

Then, since (by analysing derivatives)

$$\lim_{w\to 0} \frac{d^{n-j-1}}{dw^{n-j-1}}\left(\frac{-1}{(1+w)^2}\right) = (-1)^{n-j}(n-j)!$$

Letting us now write the expression as

$$= \sum_{j=0}^{n-1} \binom{n-1}{j} (-1)^{n-j}(n-j)!(-n)^j$$

$$= \sum_{j=0}^{n-1} \frac{(n-1)!}{j!(n-j-1)!}(-1)^{n-j}(n-j)!(-n)^j$$

$$= (-1)^n(n-1)! \sum_{j=0}^{n-1} \frac{n^j}{j!}(n-j)$$

Splitting the sum into two, we get

$$= (-1)^n(n-1)!\left(n\sum_{j=0}^{n-1}\frac{n^j}{j!} - \sum_{j=0}^{n-1}\frac{n^j}{(j-1)!}\right)$$

Now, by letting $m = j-1$ on the right sum, we can reindex the sum and terms to match it with the left sum and subtract the right sum from the left sum. We are left with simply the $j = (n-1)$ term.

$$= (-1)^n(n-1)!\left(n\sum_{j=0}^{n-1}\frac{n^j}{j!} - n\sum_{m=0}^{n-2}\frac{n^m}{m!}\right)$$

$$= (-1)^n(n-1)!\left(n\frac{n^{n-1}}{(n-1)!}\right)$$

$$= (-n)^n$$

Great! We can finally sub this back into the Lagrange inversion sum for $g(w) = 1/(1+w)$. The radius of convergence is $|x| < 1/e$.

$$\therefore \lim_{w\to 0}\frac{d^{n-1}}{dw^{n-1}}\left(\frac{-1}{(1+w)^2}e^{-nw}\right) = (-n)^n$$

$$\therefore \frac{1}{1+w} = \frac{1}{1+W(x)} = 1 + \sum_{n=1}^{\infty}\frac{(-xn)^n}{n!}$$

Now that we have a summation expression for $1/(1+W(x))$, we can sub in $x = -u\ln u$ in order to write the integrand in terms of the sum. This sum converges for the entire region over the bounds of the integral (0 to 1), so we're all good here.

$$\therefore \int_0^1 \frac{1}{1+W(-u\ln u)}\,du = \int_0^1 1 + \sum_{n=1}^{\infty}\frac{(-u\ln u)^n(-n)^n}{n!}\,du$$

$$= 1 + \sum_{n=1}^{\infty}\frac{n^n}{n!}\int_0^1 (u\ln u)^n\,du$$

$$u = e^{-p}, du = -e^{-p}\,dp$$

$$= 1 + \sum_{n=1}^{\infty}\frac{(-n)^n}{n!}\int_0^{\infty} e^{-(n+1)p}p^n\,dp$$

$$q = (n+1)p, dq = (n+1)\,dp$$

$$= 1 + \sum_{n=1}^{\infty}\frac{(-n)^n}{n!(n+1)^{n+1}}\int_0^{\infty} e^{-q}q^n\,dq$$

This remaining integral is the gamma function at $\Gamma(n+1)$, which is equal to $n!$. Thus, we obtain

$$= 1 + \sum_{n=1}^{\infty}\frac{(-n)^n}{(n+1)^{n+1}}$$

$$= 1 + \sum_{n=2}^{\infty}\frac{(1-n)^{n-1}}{n^n}$$

$$= \sum_{n=1}^{\infty}\frac{(1-n)^{n-1}}{n^n}$$

And thus we are complete. Have a look at this result and how the integrand matches the summand!

$$\therefore \int_0^{\frac{1}{2}}\frac{(1-x)^{x-1}}{x^x}\,dx = \sum_{n=1}^{\infty}\frac{(1-n)^{n-1}}{n^n}$$

15.4 So complex its real (2 methods)

Similar to the last integral, but has a closed solution this time?!

$$\int_0^1 \sin(\pi x) x^x (1-x)^{1-x}\, dx$$

Method one is largely from [11] (residue calculation by me), but the second method is my method. Method one will be to using the residue at infinity method with a dog bone contour. First, let

$$f(z) = e^{z\ln z + (1-z)\ln(1-z) + i\pi z}$$

Where one can see that the imaginary part of $f(z)$ is equal to the original integrand, and that the integral of the real part is equal to zero. Thus, let's move into contour integration. We note and use that $\ln z$ is defined for $-\pi \le \arg z < \pi$, and let $\ln(1-z)$ be defined for $0 \le \arg(1-z) < 2\pi$. These will be our branch cut when applicable, as these branches make the function continuous everywhere apart from $[0,1]$. Now, using a dog bone contour, we can see that the integrals over the circles at 0 and 1 vanish when the radius tends to zero, since the function is bounded there. We thus need to solve for the integrals over the upper (Above real axis from 0 to 1) and lower (Below real axis from 1 back to 0) segments (we typically go clockwise for dog bone contours).

We now evaluate the values on the cut:

Lower segment ($z = x - i\epsilon$): For $z\ln z$, since the branch cut is on the negative real axis (due to our defined arg of $\ln z$), the angle of z on the positive real axis is zero.

$$\therefore z\ln z \to x\ln x$$

For $(1-z)\ln(1-z)$, recall that from $\ln(1-z)$ the branch is defined as $0 \le \arg(1-z) < 2\pi$. Now let $w = 1-z$. Thus,

$$w = 1 - x + i\epsilon = (1-x) + i\epsilon$$

As the imaginary part is slightly positive, the angle of $w = 1-z$ is zero.

$$\therefore (1-z)\ln(1-z) \to (1-x)\ln(1-x)$$

$$\therefore f_{lower}(x) = e^{x\ln x + (1-x)\ln(1-x) + i\pi x} = e^{i\pi x} x^x (1-x)^{1-x}$$

Upper segment ($z = x + i\epsilon$): For $z\ln z$, the angle of z is still zero.

$$\therefore z\ln z \to x\ln x$$

For $(1-z)\ln(1-z)$, again let $w = 1-z$. Thus,

$$w = 1 - x - i\epsilon = (1-x) - i\epsilon$$

Since the imaginary part is slightly negative this time, and the branch is $[0, 2\pi)$, we cannot use an angle of zero and thus must wrap around the angle to 2π.

$$\therefore (1-z)\ln(1-z) \to (1-x)(\ln(1-x) + 2i\pi)$$

$$\therefore f_{upper}(x) = e^{x\ln x + (1-x)(\ln(1-x) + 2i\pi) + i\pi x}$$

$$= e^{x\ln x + (1-x)\ln(1-x) + (1-x)2i\pi + i\pi x}$$

$$= e^{x\ln x + (1-x)\ln(1-x) + 2i\pi - i\pi x}$$

$$= e^{-i\pi x} x^x (1-x)^{1-x}$$

Thus, the integral around the contour is

$$\oint f(z)\,dz = \oint e^{z\ln z+(1-z)\ln(1-z)+i\pi z}\,dz = \int_0^1 e^{i\pi x}x^x(1-x)^{1-x}\,dx + \int_1^0 e^{-i\pi x}x^x(1-x)^{1-x}\,dx$$

$$= \int_0^1 (e^{i\pi x} - e^{-i\pi x})x^x(1-x)^{1-x}\,dx$$

$$= 2i\int_0^1 \sin(\pi x)x^x(1-x)^{1-x}\,dx$$

Now it is time to solve the contour integral. Due to these branch cuts, the way to go is by finding the residue at infinity. This is the following.

$$\text{Res}(f(z),\infty) = -\text{Res}\left(\frac{1}{z^2}f\left(\frac{1}{z}\right),0\right)$$

Let the exponent of $f(z)$ be $g(z)$. Thus,

$$g(z) = z\ln z + (1-z)\ln(1-z) + i\pi z = z\ln z + (1-z)\ln(-z(1-1/z)) + i\pi z$$

$$= z\ln z + (1-z)\ln(-z) + (1-z)\ln(1-1/z) + i\pi z$$

$$= z\ln z + (1-z)\ln(z) + (1-z)\ln(1-1/z) + i\pi z + (1-z)i\pi$$

$$= \ln(z) + (1-z)\ln(1-1/z) + i\pi$$

$$\therefore g(1/z) = \left(1-\frac{1}{z}\right)\ln(1-z) - \ln(z) + i\pi$$

Since

$$\ln(1-x) = -\sum_{n=1}^{\infty}\frac{x^n}{n}$$

$$\therefore (1-1/z)\ln(1-z) = -(1-1/z)\sum_{n=1}^{\infty}\frac{z^n}{n} = -(1-1/z)\left(z+\frac{z^2}{2}+\frac{z^3}{3}+\cdots\right)$$

$$= \left(1+\frac{z}{2}+\frac{z^2}{3}+\cdots\right) - \left(z+\frac{z^2}{2}+\frac{z^3}{3}+\cdots\right)$$

$$= 1 - \frac{z}{2} - \frac{z^2}{6} + \cdots$$

$$\therefore g(1/z) = 1 + i\pi - \ln(z) - \frac{z}{2} - \frac{z^2}{6} + \cdots$$

$$\therefore f(1/z) = e^{g(1/z)} = e^{1+i\pi-\ln(z)-\frac{z}{2}-\frac{z^2}{6}+\cdots}$$

$$= -\frac{e}{z}\cdot e^{-\frac{z}{2}-\frac{z^2}{6}+\cdots}$$

We had to take out the first few terms from the exponent in order to properly approximate the function by a Maclaurin series for e^x. Since we know that

$$e^x = 1 + x + \frac{x^2}{2} + \cdots$$

$$\therefore \frac{-e}{z}\cdot e^{-\frac{z}{2}-\frac{z^2}{6}+\cdots} = \frac{-e}{z}\left(1+\left(-\frac{z}{2}-\frac{z^2}{6}+\cdots\right)+\frac{1}{2}\left(-\frac{z}{2}-\frac{z^2}{6}+\cdots\right)^2+\cdots\right)$$

$$= \frac{-e}{z}\left(1 - \frac{z}{2} - \frac{z^2}{24} + \cdots\right)$$

$$= -\frac{e}{z} + \frac{e}{2} + \frac{ez}{24} + \cdots$$

Finally, as we are looking for $-\text{Res}\big(\frac{1}{z^2}f\big(\frac{1}{z}\big), 0\big)$, we are looking for the coefficient of the $1/z$ term of $\frac{1}{z^2}f\big(\frac{1}{z}\big)$

$$\therefore \frac{1}{z^2}f\left(\frac{1}{z}\right) = -\frac{e}{z^3} + \frac{e}{2z^2} + \frac{e}{24z} + \cdots$$

$$\therefore -\text{Res}\left(\frac{1}{z^2}f\left(\frac{1}{z}\right), 0\right) = -\frac{e}{24}$$

$$\therefore \text{Res}(f(z), \infty) = -\frac{e}{24}$$

Finally, plugging this into the contour integral formula when using the residue at infinity (adding in an extra -1 factor due to integrating clockwise), the value of

$$\oint f(z)\, dz = -2\pi i \text{Res}(f, \infty) = \frac{\pi e i}{12}$$

So, as

$$\oint f(z)\, dz = 2i \int_0^1 \sin(\pi x) x^x (1-x)^{1-x}\, dx$$

We obtain our solution.

$$\therefore 2i \int_0^1 \sin(\pi x) x^x (1-x)^{1-x}\, dx = \frac{\pi e i}{12}$$

$$\therefore \int_0^1 \sin(\pi x) x^x (1-x)^{1-x}\, dx = \frac{\pi e}{24}$$

What a solution!

However, see the next page for a slightly simpler method of solving this integral

Before moving on I wanted to write out a different complex way of solving this integral, that might be considered simpler. This is my method!

$$\int_0^1 \sin(\pi x)x^x(1-x)^{1-x}\,dx$$

$$= \Im\int_0^1 (1-x)e^{i\pi x + x(\ln x - \ln(1-x))}\,dx$$

$$u = \ln x - \ln(1-x), x = \frac{e^u}{e^u+1}, dx = \frac{e^u}{(e^u+1)^2}\,du$$

$$= \Im\int_{-\infty}^{\infty} \exp\left(\frac{e^u(i\pi+u)}{e^u+1}\right)\frac{e^u}{(e^u+1)^3}\,du$$

$$t = u + i\pi$$

$$= \Im\int_{-\infty+i\pi}^{\infty+i\pi} \exp\left(\frac{-e^t t}{1-e^t}\right)\frac{-e^t}{(1-e^t)^3}\,dt$$

$$= \Im\int_{-\infty+i\pi}^{\infty+i\pi} \exp\left(\frac{e^t t}{e^t-1}+t\right)\frac{1}{(e^t-1)^3}\,dt$$

Let the above integrand be $f(t)$, and the above integral be I. Our final answer will thus be to find $\Im(I)$. We can then see that $f(t)$ has a third order pole at $t=0$ as it is a simple pole (via inspecting the denominator) raised to third power. Using a box contour with vertices $-R-i\pi, -R+i\pi, R-i\pi, R+i\pi$, the other poles lie outside this contour. Thus, for complex t,

$$\text{Res}_0(f) = \lim_{t\to 0}\frac{1}{2}\frac{d^2}{dt^2}t^3 f(t)$$

To find this residue, it will be easiest to use polynomial approximations of the function instead of directly taking the second derivative of $t^3 f(t)$ and using l'hopitals multiple times in order to find the limit. Thus, in the following section, we will only be keeping terms up to the second power, as any higher powers will become zero due to the limit.

$$t^3 f(t) = \exp\left(\frac{e^t t}{e^t-1}+t\right)\left(\frac{t}{e^t-1}\right)^3$$

$$\frac{t}{e^t-1} = \sum_{n=0}^{\infty}\frac{B_n t^n}{n!} = 1 - \frac{t}{2} + \frac{t^2}{12} + \cdots$$

$$\therefore \left(\frac{t}{e^t-1}\right)^3 \approx \left(1-\frac{t}{2}+\frac{t^2}{12}\right)^3 \approx 1 - \frac{3t}{2} + t^2$$

$$\frac{e^t t}{e^t-1} + t \approx \left(1-\frac{t}{2}+\frac{t^2}{12}\right)\left(1+t+\frac{t^2}{2}\right) + t$$

$$\approx 1 + \frac{3t}{2} + \frac{t^2}{12}$$

$$\therefore t^3 f(t) = \exp\left(\frac{e^t t}{e^t-1}+t\right)\left(\frac{t}{e^t-1}\right)^3 \approx e^{1+\frac{3t}{2}+\frac{t^2}{12}}\left(1-\frac{3t}{2}+t^2\right)$$

Here, we take out a factor of e to find the correct coefficient in the Laurent expansion. This is because the exponent at $t=0$ is equal to one, when we want it to equal zero so that we are able to use a maclaurin expansion. After taking out the factor of e, the remaining linear and quadratic terms in the exponent become zero at $t=0$, meaning we can now use the Maclaurin series for e^t.

$$\approx e\left(1+\frac{3t}{2}+\frac{t^2}{12}+\frac{1}{2}\left(\frac{3t}{2}+\frac{t^2}{12}\right)^2\right)\left(1-\frac{3t}{2}+t^2\right)$$

$$\approx e\left(1+\frac{3t}{2}+\frac{29t^2}{24}\right)\left(1-\frac{3t}{2}+t^2\right)$$

$$\approx e-\frac{eu^2}{24}$$

We now take the second derivative of this, let t approach 0, and divide the result by two.

$$\frac{d^2}{dt^2}\left(e-\frac{eu^2}{24}\right)=-\frac{e}{12}$$

$$\therefore \operatorname{Res}_0(f)=\lim_{t\to 0}\frac{1}{2}\left(-\frac{e}{12}\right)=-\frac{e}{24}$$

This is our residue. Thus, as the integral around the contour (anticlockwise) is equal to $2\pi i$ times the sum of the residues (in this case just one pole/residue lies in the contour), we get

$$\therefore \oint f(z)\,dz=-\frac{\pi ie}{12}$$

Going around a box contour, we can solve for the left hand side in terms of I.

$$\oint f(z)\,dz=\int_{-\infty-i\pi}^{\infty-i\pi}f(z)\,dz+\int_{\infty-i\pi}^{\infty+i\pi}f(z)\,dz+\int_{\infty+i\pi}^{-\infty+i\pi}f(z)\,dz+\int_{-\infty+i\pi}^{-\infty-i\pi}f(z)\,dz=\bar{I}+0-I+0=\bar{I}-I$$

Letting $I=a+bi$, this is equivalent to

$$\bar{I}-I=a-bi-a-bi=-2bi=-2i\Im(I)$$

Where $\Im(I)$ is exactly the integral we are looking for!

$$\therefore -2i\Im(I)=-\frac{\pi ie}{12}$$

$$\therefore \Im(I)=\frac{\pi e}{24}$$

As our original integral is exactly this, we arrive at our answer.

$$\therefore \int_0^1 \sin(\pi x)x^x(1-x)^{1-x}\,dx=\frac{\pi e}{24}$$

Certainly a simpler method than the first one we used. Interestingly, this new method can be used to also find that

$$\int_0^1 \frac{\sin(\pi x)}{x^x(1-x)^{1-x}}\,dx=\frac{\pi}{e}$$

Where this new integral above actually ends up being easier to solve since we only need to find the residue of a first order pole, instead of third order.

15.5 Nice feynman and lobachevsky tricks

17/3/26. Let's consider the following integral. Solved by me with prior knowledge (Hints from online) that it uses DUIS and the Lobachevsky integral trick.

$$\int_0^\infty \frac{\sin(x)\ln(1+\cos x)}{x\cos x}\,dx$$

We will now let $I(\alpha)$ be the following integral, which will match the original integral when $\alpha = 1$.

$$I(\alpha) = \int_0^\infty \frac{\sin(x)\ln(1+\alpha\cos x)}{x\cos x}\,dx$$

Thus, differentiating with respect to α we get

$$I'(\alpha) = \frac{d}{d\alpha}\int_0^\infty \frac{\sin(x)\ln(1+\alpha\cos x)}{x\cos x}\,dx = \int_0^\infty \frac{\sin(x)}{x}\cdot\frac{1}{1+\alpha\cos x}\,dx$$

Now, let

$$f(x) = \frac{1}{1+\alpha\cos x}$$

Thus, our integral becomes

$$= \int_0^\infty \frac{\sin(x)}{x} f(x)\,dx$$

We can almost now use Lobachevsky's integral formula/trick, which says that

$$\int_0^\infty \frac{\sin x}{x} f(x)\,dx = \int_0^{\frac{\pi}{2}} f(x)\,dx$$

when $f(x) = f(x\pm\pi)$, and

$$= \int_0^{\frac{\pi}{2}} f(x)\cos(x)\,dx$$

when $f(x) = -f(x\pm\pi)$. However, our current $f(x)$ is in neither of these forms. To get it into the form, we can perform the following adjustments.

$$\int_0^\infty \frac{\sin(x)}{x}\cdot\frac{1}{1+\alpha\cos x}\,dx = \int_0^\infty \frac{\sin(x)}{x}\cdot\frac{1-\alpha\cos x}{(1+\alpha\cos x)(1-\alpha\cos x)}\,dx$$

$$= \int_0^\infty \frac{\sin(x)}{x}\cdot\frac{1}{1-\alpha^2\cos^2 x}\,dx - \int_0^\infty \frac{\sin(x)}{x}\cdot\frac{\alpha\cos x}{1-\alpha^2\cos^2 x}\,dx$$

This gives us two different $f(x)$, the second we will name $g(x)$ to avoid confusion. It can now be seen that $f(x) = f(x\pm\pi)$ and $g(x) = -g(x\pm\pi)$. Thus, using Lobachevsky's integral formula, we get

$$\therefore \int_0^\infty \frac{\sin(x)}{x}\cdot\frac{1}{1-\alpha^2\cos^2 x}\,dx - \int_0^\infty \frac{\sin(x)}{x}\cdot\frac{\alpha\cos x}{1-\alpha^2\cos^2 x}\,dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{1-\alpha^2\cos^2 x}\,dx - \int_0^{\frac{\pi}{2}} \frac{\alpha\cos^2 x}{1-\alpha^2\cos^2 x}\,dx$$

$$t = \tan x,\ dx = \frac{1}{t^2+1}\,dt$$

Noting that $\cos^2(\arctan t) = 1/(t^2+1)$, we thus get

$$= \int_0^\infty \frac{1}{t^2+1-\alpha^2}\,dt - \alpha\int_0^\infty \frac{1}{(t^2+1-\alpha^2)(t^2+1)}\,dt$$

To split the second integrand into two partial fractions, we must solve for

$$(A+B)t^2 + (A+B) - \alpha^2 B = 1$$

$$\therefore A = -B, B = -\frac{1}{\alpha^2}, \therefore A = \frac{1}{\alpha^2}$$

Thus the right integral becomes

$$= \int_0^\infty \frac{1}{t^2+1-\alpha^2}\,dt - \frac{1}{\alpha}\left(\int_0^\infty \frac{1}{t^2+1-\alpha^2}\,dt - \int_0^\infty \frac{1}{t^2+1}\,dt\right)$$

$$= \frac{\pi}{2\sqrt{1-\alpha^2}} - \frac{\pi}{2\alpha\sqrt{1-\alpha^2}} + \frac{\pi}{2\alpha}$$

Since this is $I'(\alpha)$, we can thus integrate each term and solve for the constant of integration C.

$$\therefore I(\alpha) = \int \frac{\pi}{2\sqrt{1-\alpha^2}}\,d\alpha - \int \frac{\pi}{2\alpha\sqrt{1-\alpha^2}}\,d\alpha + \int \frac{\pi}{2\alpha}\,d\alpha$$

$$= \frac{\pi}{2}\arcsin(\alpha) + \frac{\pi}{2}\operatorname{arcsech}(|\alpha|) + \frac{\pi}{2}\ln(|\alpha|) + C$$

Using $\operatorname{arcsech}(x) = \ln\big(\frac{1}{x}\big(1+\sqrt{1-x^2}\big)\big)$, this can be written as

$$= \frac{\pi}{2}\arcsin(\alpha) + \frac{\pi}{2}\ln\left(\frac{1}{|\alpha|}\left(1+\sqrt{1-\alpha^2}\right)\right) + \frac{\pi}{2}\ln(|\alpha|) + C$$

$$= \frac{\pi}{2}\arcsin(\alpha) + \frac{\pi}{2}\ln\big(1+\sqrt{1-\alpha^2}\big) + C$$

Looking back at $I(\alpha)$, it is obvious that $I(0) = 0$. Then, as $\arcsin(0) = 0$, we get

$$C = -\frac{\pi}{2}\ln 2$$

This thus completes our solution for $I(\alpha)$.

$$\therefore I(\alpha) = \int_0^\infty \frac{\sin(x)\ln(1+\alpha\cos x)}{x\cos x}\,dx = \frac{\pi}{2}\big(\arcsin(\alpha) + \ln\big(1+\sqrt{1-\alpha^2}\big) - \ln 2\big)$$

Then, as $\alpha = 1$ in the original integral, we can conclude that

$$\therefore \int_0^\infty \frac{\sin(x)\ln(1+\cos x)}{x\cos x}\,dx = \frac{\pi}{2}\left(\frac{\pi}{2} - \ln 2\right)$$

15.6 Fractional part integral

The symbol used for the fractional part of a number a is $\{a\}$. For example, if $a = 5.55$ then $\{a\} = 0.55$. The integral we will be looking at is

$$\int_0^\infty \frac{\{e^x\}}{e^x}\,dx$$

To solve, we can first use a u-sub and use the property that $\{a\} = a - \lfloor a \rfloor$ for $a \geq 0$.

$$t = e^x, dt = t\,dx$$

$$= \int_1^\infty \frac{\{t\}}{t^2}\,dt$$

$$= \int_1^\infty \frac{t - \lfloor t \rfloor}{t^2}\,dt$$

$$= \int_1^\infty \frac{1}{t} - \frac{\lfloor t \rfloor}{t^2}\,dt$$

We can now use the trick we used for integrals involving floor functions, where we change the bounds to a discrete cn that is summed.

$$= \sum_{n=1}^\infty \int_n^{n+1} \frac{1}{t} - \frac{n}{t^2}\,dt$$

$$= \sum_{n=1}^\infty \left(\ln(n+1) - \ln(n) - n\int_n^{n+1} \frac{1}{t^2}\,dt\right)$$

$$= \sum_{n=1}^\infty \left(\ln(n+1) - \ln(n) + n\left(\frac{1}{n+1} - \frac{1}{n}\right)\right)$$

$$= \sum_{n=1}^\infty \left(\ln(n+1) - \ln(n) - \frac{1}{n+1}\right)$$

We can now write this sum as

$$= \lim_{N\to\infty} \sum_{n=1}^N \left(\ln(n+1) - \ln(n) - \frac{1}{n+1}\right)$$

Observing the log terms, the sum of those would be $(\ln(2) - \ln(1)) + (\ln(3) - \ln(2)) + \cdots$, where we can see that the first term of each pair cancels out with the second term of the next pair. Thus, the solution to this part of the sum up to N is $\ln(N+1)$. Thus we write

$$= \lim_{N\to\infty} \left(\ln(N+1) - \sum_{n=1}^N \frac{1}{n+1}\right)$$

As N gets large, $\ln(N+1)$ is essentially just $\ln(N)$.

$$= \lim_{N\to\infty} \left(\ln(N) - \sum_{n=1}^N \frac{1}{n+1}\right)$$

Now, to the sum, we can do the following.

$$= \lim_{N\to\infty} \left(\ln(N) - \left(\sum_{n=1}^N \frac{1}{n+1} + 1 - 1\right)\right)$$

Take in the $n = 0$ term

$$= \lim_{N\to\infty}\left(\ln(N) - \left(\sum_{n=0}^{N}\frac{1}{n+1} - 1\right)\right)$$

Re-index the sum back to $n = 1$, noticing that the upper bound is essentially the same for large N. We can also expand the brackets.

$$= \lim_{N\to\infty}\left(\ln(N) - \sum_{n=1}^{N}\frac{1}{n}\right) + 1$$

Finally, notice that the limit is the negative of the Euler-Mascheroni constant γ. Thus, we obtain our answer.

$$= 1 - \gamma$$

$$\therefore \int_0^\infty \frac{\{e^x\}}{e^x}\, dx = 1 - \gamma$$

Very cool!

16 April 26

16.1 Estimation of integral without good closed form

(5/4/26) Today we will be looking at the following integral

$$\int_{-1}^{0} \sqrt{x - \ln(x+1)}\, dx$$

In order to work on this integral, lets the following u-sub

$$t^2 = x - \ln(x+1)$$

However, we'd like to find out what x is in terms of u. To do this, we need the lambert-W function.

$$\therefore e^{t^2} = \frac{e^x}{x+1}$$

$$e^{-t^2} = (x+1)e^{-x}$$

$$-e^{-1-t^2} = -(x+1)e^{-(x+1)}$$

$$-(x+1) = W\left(-e^{-1-t^2}\right)$$

$$\therefore x = -1 - W\left(-e^{-1-t^2}\right)$$

Let $f(t) = W\left(-e^{-1-t^2}\right)$. Thus,

$$dx = -df(t)$$

$$\therefore \int_{-1}^{0} \sqrt{x - \ln(x+1)}\, dx = \int_{0}^{\infty} t\, df(t)$$

$$\text{IBP with } u = t, dv = df(t)$$

$$= \lim_{N\to\infty} \left[tf(t)\right]_0^N - \int_0^\infty f(t)\, dt$$

Since $f(t)$ goes to zero much faster than t goes to infinity, the limit is zero.

$$= -\int_0^\infty f(t)\, dt$$

$$= -\int_0^\infty W\left(-e^{-1-t^2}\right) dt$$

Using the summation expression for $W(x)$, we get

$$= -\int_0^\infty \sum_{n=1}^{\infty} \frac{(-n)^{n-1}}{n!} \left(-e^{-1-t^2}\right)^n dt$$

$$= \sum_{n=1}^{\infty} \frac{n^{n-1}}{n!e^n} \int_0^\infty e^{-nt^2}\, dt$$

Finally, we are left with the familiar Gaussian integral, and obtain our solution.

$$= \frac{\sqrt{\pi}}{2} \sum_{n=1}^{\infty} \frac{n^{n-\frac{3}{2}}}{n!e^n}$$

$$\therefore \int_{-1}^{0} \sqrt{x - \ln(x+1)}\, dx = \frac{\sqrt{\pi}}{2} \sum_{n=1}^{\infty} \frac{n^{n-\frac{3}{2}}}{n!e^n}$$

However, as this sum is unsolvable in some sort of closed form, we could go further to provide a decent estimation of the integral. To do so, consider Stirling's approximation for $n!$

$$n! = \sqrt{2\pi n}\left(\frac{n}{e}\right)^n\left(1+\frac{1}{12n}+\frac{1}{288n^2}-\frac{139}{51840n^3}+\cdots\right)$$

For our purposes, we will be using the second order approximation (up to $1/12n$). Plugging this into our solution for the integral, we get

$$\begin{aligned}\frac{\sqrt{\pi}}{2}\sum_{n=1}^{\infty}\frac{n^{n-\frac{3}{2}}}{n!e^n} &\approx \frac{\sqrt{\pi}}{2}\sum_{n=1}^{\infty}\frac{n^{n-\frac{3}{2}}}{e^n\sqrt{2\pi n}(n/e)^n(1+1/12n)} \\ &= \frac{1}{2\sqrt{2}}\sum_{n=1}^{\infty}\frac{1}{n^2+\frac{n}{12}} \\ &= \frac{1}{2\sqrt{2}}\sum_{n=1}^{\infty}\frac{1}{n(n+\frac{1}{12})}\end{aligned}$$

Finally, we can use the following equation

$$\psi(x+1) = \sum_{n=1}^{\infty}\frac{x}{n(n+x)} - \gamma$$

Where $\psi(x)$ is the digamma function (also defined as the derivative of the gamma function divided by the gamma function. In our case, $x = \frac{1}{12}$, where we obtain the following result.

$$= \frac{6}{\sqrt{2}}\left(\psi\left(\frac{13}{12}\right)+\gamma\right)$$

We can thus approximate the integral by the following:

$$\therefore \int_{-1}^{0}\sqrt{x-\ln(x+1)}\,dx \approx \frac{6}{\sqrt{2}}\left(\psi\left(\frac{13}{12}\right)+\gamma\right)$$

Using numerical approximation, the left and right sides of this approximation differ by 0.0003944 (7dp) (The digamma approximation is slightly greater than the integral itself).

16.2 Very cool floor integral

Second floor/ceiling/fractional integral of the year.

$$\int_0^1 \sin\left\lfloor \frac{1}{x} \right\rfloor dx$$

Putting this function into desmos shows a really crazy graph from 0 to 1. To begin solving this integral, we can use the properties of floor functions. Using them, the integral can be written as

$$= \sum_{n=1}^{\infty} \int_{\frac{1}{n+1}}^{\frac{1}{n}} \sin(n)\, dx$$

$$= \sum_{n=1}^{\infty} \frac{\sin n}{n} - \sum_{n=1}^{\infty} \frac{\sin n}{n+1}$$

$$= \sum_{n=1}^{\infty} \frac{\sin n}{n} - \sum_{n=2}^{\infty} \frac{\sin(n-1)}{n}$$

Using the formula for $\sin(x-y)$,

$$= \sum_{n=1}^{\infty} \frac{\sin n}{n} - \sum_{n=2}^{\infty} \frac{\sin(n)\cos(1) - \cos(n)\sin(1)}{n}$$

$$= \sum_{n=1}^{\infty} \frac{\sin n}{n} - \cos(1) \sum_{n=2}^{\infty} \frac{\sin n}{n} + \sin(1) \sum_{n=2}^{\infty} \frac{\cos n}{n}$$

Moving the sums to start from $n = 1$, then taking off the $n = 1$ term in the right two sums we have a cancellation, meaning that

$$= \sum_{n=1}^{\infty} \frac{\sin n}{n} - \cos(1) \sum_{n=1}^{\infty} \frac{\sin n}{n} + \sin(1) \sum_{n=1}^{\infty} \frac{\cos n}{n}$$

To find the value of these sums, consider the following.

$$-\ln(1-z) = \sum_{n=1}^{\infty} \frac{x^n}{n}$$

$$\therefore -\ln(1-e^i) = \sum_{n=1}^{\infty} \frac{e^{in}}{n}$$

$$= \sum_{n=1}^{\infty} \frac{\cos n}{n} + i \sum_{n=1}^{\infty} \frac{\sin n}{n}$$

We thus have to expand $-\ln(1-e^i)$ into its real and imaginary parts

$$-\ln(1-e^i) = -\ln(1 - \cos(1) + i\sin(1))$$

$$= -\ln\left(\sqrt{(1-\cos 1)^2 + \sin^2 1}\right) - i \arctan\left(\frac{-\sin 1}{1-\cos 1}\right)$$

$$= -\frac{1}{2}\ln\left((1-\cos 1)^2 + \sin^2 1\right) + i \arctan\left(\frac{\sin 1}{1-\cos 1}\right)$$

The expression inside of the arctan on the right is known to be $\cot\left(\frac{1}{2}\right)$.

$$= -\frac{1}{2}\ln\left(1 - 2\cos(1) + \cos^2 1 + \sin^2 1\right) + i \arctan\left(\cot\left(\frac{1}{2}\right)\right)$$

$$= -\frac{1}{2}\ln(2 - 2\cos 1) + i \arctan\left(\tan\left(\frac{\pi}{2} - \frac{1}{2}\right)\right)$$

$$= -\frac{1}{2}\ln(2(1-\cos 1)) + i\left(\frac{\pi-1}{2}\right)$$

We have found the imaginary part. To finish with the real part, use the formula for $\sin^2 x$ in terms of $\cos x$

$$= -\frac{1}{2}\ln\left(4\sin^2\left(\frac{1}{2}\right)\right) + i\left(\frac{\pi-1}{2}\right)$$

$$= -\ln\left(2\sin\left(\frac{1}{2}\right)\right) + i\left(\frac{\pi-1}{2}\right)$$

And thus we are done with both the real and imaginary parts. We thus have found all the results for the sums in our working out for the original integral. Plugging them in,

$$\therefore \sum_{n=1}^{\infty}\frac{\sin n}{n} - \cos(1)\sum_{n=1}^{\infty}\frac{\sin n}{n} + \sin(1)\sum_{n=1}^{\infty}\frac{\cos n}{n}$$

$$= \frac{\pi-1}{2}(1-\cos 1) - \sin(1)\ln\left(2\sin\left(\frac{1}{2}\right)\right)$$

$$\therefore \int_0^1 \sin\left\lfloor\frac{1}{x}\right\rfloor dx = \frac{\pi-1}{2}(1-\cos 1) - \sin(1)\ln\left(2\sin\left(\frac{1}{2}\right)\right)$$

Nice!

16.3 Airy fairy integral

The Airy functions are special linearly independent solutions of the following second order linear ODE.

$$\frac{d^2y}{dx^2} - xy = 0$$

They are defined as

$$\mathrm{Ai}(x) = \frac{1}{\pi}\int_0^\infty \cos\left(\frac{t^3}{3} + xt\right)dt$$

$$\mathrm{Bi}(x) = \frac{1}{\pi}\int_0^\infty \exp\left(-\frac{t^3}{3} + xt\right) + \sin\left(\frac{t^3}{3} + xt\right)dt$$

The integral we will be solving in this section is the following. It has a very known solution, so the solution is not my own.

$$\int_0^\infty \frac{1}{\mathrm{Ai}^2(x) + \mathrm{Bi}^2(x)}\,dx$$

To begin, we make the following adjustment to the integrand.

$$= \int_0^\infty \frac{1}{\frac{\mathrm{Ai}^2(x)}{\mathrm{Bi}^2(x)} + 1}\cdot\frac{1}{\mathrm{Bi}^2(x)}\,dx$$

We now wish to do the following u-sub.

$$u = \frac{\mathrm{Ai}(x)}{\mathrm{Bi}(x)}, du = \frac{\mathrm{Ai}'(x)\mathrm{Bi}(x) - \mathrm{Ai}(x)\mathrm{Bi}'(x)}{\mathrm{Bi}^2(x)}$$

To find our lower bound of the integral after the u-sub, we use the values of the functions at $x = 0$. These values can be found by using techniques used on this document from previous sections (such as on page 123).

$$\mathrm{Ai}(0) = \frac{1}{\pi}\int_0^\infty \cos\left(\frac{t^3}{3}\right)dt = \frac{1}{3^{\frac{2}{3}}\Gamma\left(\frac{2}{3}\right)}$$

$$\mathrm{Bi}(0) = \frac{1}{\pi}\int_0^\infty \exp\left(-\frac{t^3}{3}\right) + \sin\left(\frac{t^3}{3}\right)dt = \frac{1}{3^{\frac{1}{6}}\Gamma\left(\frac{2}{3}\right)}$$

$$\therefore \frac{\mathrm{Ai}(0)}{\mathrm{Bi}(0)} = \frac{1}{\sqrt{3}}$$

Then, to find the upper bound, we look at the asymptotic formulae for the Airy functions.

$$\mathrm{Ai}(x) \sim \frac{1}{4e^{\frac{2}{3}x^{\frac{3}{2}}}\sqrt{\pi\sqrt{x}}}$$

$$\mathrm{Bi}(x) \sim \frac{e^{\frac{2}{3}x^{\frac{3}{2}}}}{\sqrt{\pi\sqrt{x}}}$$

As seen above, as each of these functions goes to infinity, it becomes trivial that

$$\lim_{x\to\infty}\frac{\mathrm{Ai}(x)}{\mathrm{Bi}(x)} = 0$$

Thus, our new integral after the u-sub is the following.

$$\therefore \int_0^\infty \frac{1}{\frac{\mathrm{Ai}^2(x)}{\mathrm{Bi}^2(x)} + 1}\cdot\frac{1}{\mathrm{Bi}^2(x)}\,dx = \int_{\frac{1}{\sqrt{3}}}^0 \frac{1}{u^2+1}\cdot\frac{1}{\mathrm{Ai}'(x)\mathrm{Bi}(x) - \mathrm{Ai}(x)\mathrm{Bi}'(x)}\,du$$

But what is

$$\mathrm{Ai}'(x)\mathrm{Bi}(x) - \mathrm{Ai}(x)\mathrm{Bi}'(x)$$

Notice that this is the Wronskian of the two Airy functions, as they're solutions to the second order DE at the top of the previous page. This allows us to use Abel's identity, which states that for a homogeneous linear second-order ODE

$$y'' + p(x)y' + q(x)y = 0$$

the Wronskian of two real or complex solutions y_1, y_2, has the following formula

$$W(y_1, y_2)(x) = W(y_1, y_2)(x_0) \cdot \exp\left(-\int_{x_0}^{x} p(t)\,dt\right)$$

For each point x_0 in the interval of the real line with the functions $p(x)$ and $q(x)$. In our case,$y_1 = \mathrm{Ai}(x)$ and $y_2 = \mathrm{Bi}(x)$, and $p(x) = 0$. As we know the values for the Airy functions and their derivatives at $x = 0$, we will choose $x_0 = 0$. Thus, Abel's identity tells us that

$$\mathrm{Ai}'(x)\mathrm{Bi}(x) - \mathrm{Ai}(x)\mathrm{Bi}'(x) = \mathrm{Ai}'(0)\mathrm{Bi}(0) - \mathrm{Ai}(0)\mathrm{Bi}'(0)$$

After plugging in the known values of each of these constants (again we can use method from page 123 for this), we get

$$\mathrm{Ai}'(0)\mathrm{Bi}(0) - \mathrm{Ai}(0)\mathrm{Bi}'(0) = -\frac{1}{\pi}$$

This means that

$$\begin{aligned}
\therefore \int_{\frac{1}{\sqrt{3}}}^{0} \frac{1}{u^2+1} \cdot \frac{1}{\mathrm{Ai}'(x)\mathrm{Bi}(x) - \mathrm{Ai}(x)\mathrm{Bi}'(x)}\,du &= -\pi \int_{\frac{1}{\sqrt{3}}}^{0} \frac{1}{u^2+1}\,du \\
&= \pi \arctan\left(\frac{1}{\sqrt{3}}\right) \\
&= \frac{\pi^2}{6}
\end{aligned}$$

And we are done.

$$\therefore \int_0^\infty \frac{1}{\mathrm{Ai}^2(x) + \mathrm{Bi}^2(x)}\,dx = \frac{\pi^2}{6}$$

16.4 Khinchin's constant

The following integral has a nice solution involving Khinchin's constant K_0. Khinchin's constant is very cool. To explain what it is, we note that any number x has a continued fraction expansion in the following form.

$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{\cdots}}}$$

Very interestingly, for almost all real numbers x (usually when the continued fraction expansion for x is infinite and non-increasing), the geometric mean of all of the a_i will equal the same number, and that number is Khinchin's constant! In other words, for almost all real numbers,

$$\lim_{n\to\infty} (a_0 a_1 a_2 \cdots)^{\frac{1}{n}} = K_0 \approx 2.68545...$$

Crazy! Anyways, the integral that will be solved in this section is the following.

$$\int_0^1 \frac{\log_2(\Gamma(2+x)\Gamma(2-x))}{x(1+x)}\,dx$$

Using the Gamma reflection formula, we'd have

$$\Gamma(2+x)\Gamma(-1-x) = \frac{\pi}{\sin(\pi(x+2)} = \frac{\pi}{\sin(\pi x)}$$

Thus,

$$\Gamma(2+x)\Gamma(2-x) = \Gamma(2+x)\Gamma(-1-x)(1-x)(-x)(-1-x) = \frac{x\pi(1-x^2)}{\sin(\pi x)}$$

Thus, our integral can become

$$= \int_0^1 \frac{1}{x(1+x)}\log_2\left(\frac{\pi x(1-x^2)}{\sin(\pi x)}\right)dx$$

Now consider the Weierstrass product formula for $\sin x$.

$$\sin x = x\prod_{n=1}^{\infty}\left(1 - \frac{x^2}{n^2\pi^2}\right)$$

$$\therefore \sin(\pi x) = \pi x \prod_{n=1}^{\infty}\left(1 - \frac{x^2}{n^2}\right)$$

$$= \pi x(1-x^2)\prod_{n=2}^{\infty}\left(1 - \frac{x^2}{n^2}\right)$$

$$\therefore \int_0^1 \frac{1}{x(1+x)}\log_2\left(\frac{\pi x(1-x^2)}{\sin(\pi x)}\right)dx = \int_0^1 \frac{1}{x(1+x)}\log_2\left(\prod_{n=2}^{\infty}\left(1-\frac{x^2}{n^2}\right)^{-1}\right)dx$$

$$= \frac{1}{\ln 2}\int_0^1 \left(\frac{1}{x} - \frac{1}{x+1}\right)\ln\left(\prod_{n=2}^{\infty}\left(1-\frac{x^2}{n^2}\right)^{-1}\right)dx$$

$$= \frac{1}{\ln 2}\int_0^1 \left(\frac{1}{x+1} - \frac{1}{x}\right)\sum_{n=2}^{\infty}\ln\left(1-\frac{x^2}{n^2}\right)dx$$

Using the Maclaurin series for $\ln(1-x)$,

$$= \frac{1}{\ln 2}\int_0^1 \left(\frac{1}{x} - \frac{1}{x+1}\right)\sum_{n=2}^{\infty}\sum_{k=1}^{\infty}\frac{1}{k}\left(\frac{x}{n}\right)^{2k}dx$$

$$= \frac{1}{\ln 2}\sum_{n=2}^{\infty}\sum_{k=1}^{\infty}\frac{1}{kn^{2k}}\int_0^1 x^{2k-1} - \frac{x^{2k}}{x+1}\,dx$$

This sum of $1/n^{2k}$ term is the zeta function of $2k$, minus 1 due to the index starting at $n = 2$. Meanwhile, the integrand can be simplified to

$$= \frac{1}{\ln 2} \sum_{k=1}^{\infty} \frac{\zeta(2k) - 1}{k} \int_0^1 \frac{x^{2k-1}}{x+1} \, dx$$

To evaluate this integral, consider the geometric series formula (not to infinity) of $-x$.

$$\sum_{j=0}^{n-1} (-x)^j = \frac{1 - (-x)^n}{1+x}$$

In our case we wish to use $n = 2k - 1$

$$\therefore \sum_{j=0}^{2k-2} (-x)^j = \frac{1 - (-x)^{2k-1}}{1+x} = \frac{1}{1+x} - \frac{(-x)^{2k-1}}{1+x}$$

$$\therefore \frac{x^{2k-1}}{1+x} = \frac{1}{(1+x)(-1)^{2k-1}} - \sum_{j=0}^{2k-2} \frac{(-x)^j}{(-1)^{2k-1}}$$

$$= \sum_{j=0}^{2k-2} (-x)^j - \frac{1}{1+x}$$

This is our integrand.

$$\therefore \int_0^1 \frac{x^{2k-1}}{x+1} \, dx = \int_0^1 \sum_{j=0}^{2k-2} (-x)^j - \frac{1}{1+x} \, dx$$

$$= \sum_{j=0}^{2k-2} (-1)^j \int_0^1 x^j \, dx - \ln 2$$

$$= \sum_{j=1}^{2k-1} \frac{(-1)^{j+1}}{j} - \ln 2$$

Putting this back into our overall expression for the original integral, we get

$$\therefore \frac{1}{\ln 2} \sum_{k=1}^{\infty} \frac{\zeta(2k) - 1}{k} \int_0^1 \frac{x^{2k-1}}{x+1} \, dx = \frac{1}{\ln 2} \sum_{k=1}^{\infty} \frac{\zeta(2k) - 1}{k} \left(\sum_{j=1}^{2k-1} \frac{(-1)^{j+1}}{j} - \ln 2 \right)$$

$$= \frac{1}{\ln 2} \sum_{k=1}^{\infty} \frac{\zeta(2k) - 1}{k} \sum_{j=1}^{2k-1} \frac{(-1)^{j+1}}{j} - \sum_{k=1}^{\infty} \frac{\zeta(2k) - 1}{k}$$

The left term of two sums is known to be equal to the natural log of Khinchin's constant. We thus still just need to evaluate the sum on the right.

$$= \ln(K_0) - \sum_{k=1}^{\infty} \frac{\zeta(2k) - 1}{k}$$

$$= \ln(K_0) - \sum_{n=2}^{\infty} \sum_{k=1}^{\infty} \frac{1}{k} \left(\frac{1}{n^2} \right)^k$$

The sum in terms of k is the Maclaurin series for $-\ln(1 - x)$ at $x = 1/n^2$.

$$= \ln(K_0) + \sum_{n=2}^{\infty} \ln\left(1 - \frac{1}{n^2}\right)$$

$$= \ln(K_0) + \ln\left(\prod_{n=2}^{\infty} \frac{n^2 - 1}{n^2} \right)$$

$$= \ln(K_0) + \ln\left(\prod_{n=2}^{\infty} \frac{(n-1)(n+1)}{n^2}\right)$$

$$= \ln(K_0) + \ln\left(\frac{1 \cdot 3}{2 \cdot 2} \times \frac{2 \cdot 4}{3 \cdot 3} \times \frac{3 \cdot 5}{4 \cdot 4} \times \cdots\right)$$

Looking at the products in side of the natural log, we can see that everything cancels out with each other apart from the 1/2 at the start.

$$= \ln(K_0) + \ln\left(\frac{1}{2}\right)$$

$$= \ln\left(\frac{K_0}{2}\right)$$

And we're done!

$$\therefore \int_0^1 \frac{\log_2(\Gamma(2+x)\Gamma(2-x))}{x(1+x)}\, dx = \ln\left(\frac{K_0}{2}\right)$$

16.5 A beast of an integral: method 1

Surely this rivals Cleo's integral. Here is my solution.

$$\int_0^1 \ln(1-x)\ln(x)\ln(1+x)\,dx$$

Let this integral be I. Using the Maclaurin series' for $\ln(1-x)$ and $\ln(1+x)$,

$$= \int_0^1 \sum_{n=1}^{\infty}\sum_{k=1}^{\infty} \frac{(-1)^n}{nk} x^{n+k} \ln(x)\,dx$$

$$= \sum_{n=1}^{\infty}\sum_{k=1}^{\infty} \frac{(-1)^n}{nk} \int_0^1 x^{n+k} \ln(x)\,dx$$

IBP with $u = \ln(x), dv = x^{n+k}$

$$= \sum_{n=1}^{\infty}\sum_{k=1}^{\infty} \frac{(-1)^n}{nk}\left(-\frac{1}{n+k+1}\int_0^1 x^{n+k}\,dx\right)$$

$$= \sum_{n=1}^{\infty}\sum_{k=1}^{\infty} \frac{-(-1)^n}{nk(n+k+1)^2}$$

Reindex with $m = k+n$, thus $k = m-n$

$$= \sum_{m=1}^{\infty}\sum_{n=1}^{m-1} \frac{-(-1)^n}{n(m-n)(m+1)^2}$$

$$= \sum_{m=1}^{\infty} \frac{-1}{(m+1)^2} \sum_{n=1}^{m-1} \frac{(-1)^n}{n(m-n)}$$

Partial fractions

$$= \sum_{m=1}^{\infty} \frac{-1}{m(m+1)^2} \sum_{n=1}^{m-1} (-1)^n \left(\frac{1}{n} + \frac{1}{m-n}\right)$$

$$= \sum_{m=1}^{\infty} \frac{-1}{m(m+1)^2} \left(\sum_{n=1}^{m-1} \frac{(-1)^n}{n} + \sum_{n=1}^{m-1} \frac{(-1)^n}{m-n}\right)$$

Turn sum on the right back in terms of $k = m-n$

$$= \sum_{m=1}^{\infty} \frac{-1}{m(m+1)^2} \left(\sum_{n=1}^{m-1} \frac{(-1)^n}{n} + (-1)^m \sum_{k=1}^{m-1} \frac{(-1)^k}{k}\right)$$

$$= \sum_{m=1}^{\infty} \frac{-1}{m(m+1)^2} (1+(-1)^m) \sum_{n=1}^{m-1} \frac{(-1)^n}{n}$$

Notice how this vanishes when m is even. Thus, let $m = 2j$

$$= \sum_{j=1}^{\infty} \frac{-1}{j(2j+1)^2} \sum_{n=1}^{2j-1} \frac{(-1)^n}{n}$$

$$= \sum_{j=1}^{\infty} \frac{-\tilde{H}_{2j-1}}{j(2j+1)^2}$$

$$= \sum_{j=1}^{\infty} \frac{H_{2j-1} - H_{j-1}}{j(2j+1)^2}$$

It 'just' remains to solve this sum..

$$= \sum_{j=1}^{\infty} \frac{H_{2j-1}}{j(2j+1)^2} - \sum_{j=1}^{\infty} \frac{H_{j-1}}{j(2j+1)^2}$$

We will call these two sums S_1 and S_2

$$\therefore I = S_1 - S_2$$

Let us first solve S_1. By partial fractions,

$$S_1 = \sum_{j=1}^{\infty} \frac{H_{2j-1}}{j(2j+1)^2} = \sum_{j=1}^{\infty} \frac{H_{2j-1}}{j(2j+1)} - 2\sum_{j=1}^{\infty} \frac{H_{2j-1}}{(2j+1)^2} = S_a - 2S_b$$

Now we will first solve S_a, followed by S_b.

$$S_a = \sum_{j=1}^{\infty} \frac{H_{2j-1}}{j(2j+1)}$$

Using the fact that

$$\frac{1}{j(2j+1)} = 2\int_0^1 x^{2j-1}(1-x)\,dx$$

$$\therefore S_a = 2\sum_{j=1}^{\infty} H_{2j-1} \int_0^1 x^{2j-1}(1-x)\,dx$$

$$= 2\int_0^1 (1-x)\sum_{j=1}^{\infty} H_{2j-1}x^{2j-1}\,dx$$

The generating function for harmonic numbers is

$$\sum_{k=1}^{\infty} H_k x^k = -\frac{\ln(1-x)}{1-x}$$

Using its odd part gives

$$\therefore 2\int_0^1 (1-x)\sum_{j=1}^{\infty} H_{2j-1}x^{2j-1}\,dx = -\int_0^1 (1-x)\left(\frac{\ln(1-x)}{1-x} - \frac{\ln(1+x)}{1+x}\right)dx$$

$$= -\int_0^1 \ln(1-x)\,dx + \int_0^1 \frac{(1-x)\ln(1+x)}{1+x}\,dx$$

$$= 1 + \int_0^1 \frac{1-(x+1)+1}{1+x}\ln(1+x)\,dx$$

$$= 1 + \int_0^1 \left(\frac{2}{x+1} - 1\right)\ln(1+x)\,dx$$

$$= 1 + 2\int_0^1 \frac{\ln(x+1)}{x+1}\,dx - \int_0^1 \ln(x+1)\,dx$$

$$= 1 + 2\int_0^1 \frac{\ln(x+1)}{x+1}\,dx - 2\ln 2 + 1$$

$$u = \ln(x+1), du = \frac{1}{x+1}\,dx$$

$$= 1 + 2\int_0^{\ln 2} u\,du - 2\ln 2 + 1$$

$$= \ln^2 2 - 2\ln 2 + 2$$

$$\boxed{\therefore S_a = \ln^2 2 - 2\ln 2 + 2}$$

Now we solve for S_b

$$S_b = \sum_{j=1}^{\infty} \frac{H_{2j-1}}{(2j+1)^2} = \sum_{j=1}^{\infty} \frac{H_{2j}}{(2j+1)^2} - \sum_{j=1}^{\infty} \frac{1}{2j(2j+1)^2} = S_{b_1} - S_{b_2}$$

For S_{b_1}, Use the fact that

$$-\int_0^1 x^{2n} \ln(x)\, dx = \frac{1}{(2n+1)^2}$$

$$\therefore S_{b_1} = -\sum_{j=1}^{\infty} H_{2j} \int_0^1 x^{2j} \ln(x)\, dx$$

$$= -\int_0^1 \ln(x) \sum_{j=1}^{\infty} H_{2j} x^{2j}\, dx$$

Now using the even part of the generating function for harmonic numbers, this becomes

$$= \frac{1}{2}\int_0^1 \ln(x)\left(\frac{\ln(1-x)}{1-x} + \frac{\ln(1+x)}{1+x}\right) dx$$

$$= \frac{1}{2}\int_0^1 \frac{\ln(1-x)\ln(x)}{1-x}\, dx + \frac{1}{2}\int_0^1 \frac{\ln(1+x)\ln(x)}{1+x}\, dx = \frac{1}{2}I_1 + \frac{1}{2}I_2$$

Starting with I_1 and performing the u-sub

$$u = 1-x, du = -\, dx$$

$$\therefore I_1 = \int_0^1 \frac{\ln(u)\ln(1-u)}{u}\, du$$

$$= -\int_0^1 \sum_{n=1}^{\infty} \frac{u^{n-1}}{n} \ln(u)\, du$$

$$= -\sum_{n=1}^{\infty} \frac{1}{n} \int_0^1 u^{n-1} \ln(u)\, du$$

The remaining integral is like what we had on the first page, but with $n-1$ instead of $n+k$. Thus we get the result

$$= \sum_{n=1}^{\infty} \frac{1}{n^3} = \zeta(3)$$

$$\therefore I_1 = \zeta(3)$$

Now moving onto I_2.

$$I_2 = \int_0^1 \frac{\ln(1+x)\ln(x)}{1+x}\, dx$$

$$t = \ln(1+t),\ du = \frac{1}{1+t}\, dt$$

$$= \int_0^{\ln 2} t\ln(e^t - 1)\, dt$$

$$= \int_0^{\ln 2} t^2 + t\ln(1-e^{-t})\, dt$$

$$= \frac{1}{3}\ln^3 2 + \int_0^{\ln 2} t\ln(1-e^{-t})\,dt$$

$$= \frac{1}{3}\ln^3 2 - \int_0^{\ln 2} t\sum_{n=1}^{\infty}\frac{e^{-tn}}{n}\,dt$$

$$= \frac{1}{3}\ln^3 2 - \sum_{n=1}^{\infty}\frac{1}{n}\int_0^{\ln 2} te^{-tn}\,dt$$

IBP with $u = t, dv = e^{-tn}$

$$= \frac{1}{3}\ln^3 2 - \sum_{n=1}^{\infty}\frac{1}{n}\left(-\frac{\ln 2}{n2^n} + \frac{1}{n}\int_0^{\ln 2} e^{-tn}\,dt\right)$$

$$= \frac{1}{3}\ln^3 2 - \sum_{n=1}^{\infty}\frac{1}{n}\left(-\frac{\ln 2}{n2^n} - \frac{1}{n^2 2^n} + \frac{1}{n^2}\right)$$

$$= \frac{1}{3}\ln^3 2 + \ln(2)\sum_{n=1}^{\infty}\frac{1}{n^2 2^n} + \sum_{n=1}^{\infty}\frac{1}{n^3 2^n} - \zeta(3)$$

$$= \frac{1}{3}\ln^3 2 + \ln(2)\mathrm{Li}_2\left(\frac{1}{2}\right) + \mathrm{Li}_3\left(\frac{1}{2}\right) - \zeta(3)$$

These dilog and trilog values have known solutions.

$$= \frac{1}{3}\ln^3 2 + \ln(2)\left(\frac{\pi^2}{12} - \frac{1}{2}\ln^2 2\right) + \frac{7}{8}\zeta(3) - \frac{\pi^2}{12}\ln 2 + \frac{1}{6}\ln^3 2 - \zeta(3)$$

$$= -\frac{1}{8}\zeta(3)$$

$$\therefore I_2 = -\frac{1}{8}\zeta(3)$$

Now, as

$$S_{b_1} = \frac{1}{2}I_1 + \frac{1}{2}I_2$$

$$\therefore S_{b_1} = \frac{7}{16}\zeta(3)$$

We now can solve S_{b_2} in order to finish up our solution for S_b and thus S_1 as a whole!

$$S_{b_2} = \sum_{j=1}^{\infty}\frac{1}{2j(2j+1)^2}$$

Using partial fractions, this becomes

$$= \sum_{j=1}^{\infty}\left(\frac{1}{2j} - \frac{1}{2j+1}\right) - \sum_{j=1}^{\infty}\frac{1}{(2j+1)^2}$$

$$= \sum_{j=1}^{\infty}\left(\frac{1}{2j} - \frac{1}{2j-1}\right) + 1 - \frac{\pi^2}{8} + 1$$

$$= \sum_{j=1}^{\infty}\frac{(-1)^j}{j} - \frac{\pi^2}{8} + 2$$

$$= -\ln 2 - \frac{\pi^2}{8} + 2$$

As $S_b = S_{b_1} - S_{b_2}$

$$\boxed{\therefore S_b = \frac{7}{16}\zeta(3) + \frac{\pi^2}{8} + \ln 2 - 2}$$

And finally, as $S_1 = S_a - 2S_b$

$$\boxed{\therefore S_1 = \sum_{j=1}^{\infty} \frac{H_{2j-1}}{j(2j+1)^2} = -\frac{7}{8}\zeta(3) - \frac{\pi^2}{4} + \ln^2 2 - 4\ln 2 + 6}$$

It is time to solve S_2, which will allow us to complete the solution for the original integral! Using partial fractions,

$$S_2 = \sum_{j=1}^{\infty} \frac{H_{j-1}}{j(2j+1)^2} = \sum_{j=1}^{\infty} \frac{H_{j-1}}{j(2j+1)} - 2\sum_{j=1}^{\infty} \frac{H_{j-1}}{(2j+1)^2} = S_c - 2S_d$$

More sums to solve! Starting with S_c,

$$S_c = \sum_{j=1}^{\infty} \frac{H_{j-1}}{j(2j+1)} = \sum_{j=1}^{\infty} \frac{H_j}{j(2j+1)} - \sum_{j=1}^{\infty} \frac{1}{j^2(2j+1)} = S_{c_1} - S_{c_2}$$

First we must solve S_{c_1}. First, use again the fact that

$$\frac{1}{j(2j+1)} = 2\int_0^1 x^{2j-1}(1-x)\,dx$$

$$\therefore S_{c_1} = 2\sum_{j=1}^{\infty} H_j \int_0^1 x^{2j-1}(1-x)\,dx$$

$$= 2\int_0^1 \frac{1-x}{x} \sum_{j=1}^{\infty} H_j (x^2)^j\,dx$$

Using the generating function, we have x^2 instead of x. Thus, this is equal to

$$= -2\int_0^1 \frac{(1-x)\ln(1-x^2)}{x(1-x^2)}\,dx$$

$$= -2\int_0^1 \frac{\ln(1-x^2)}{x(1+x)}\,dx$$

Using partial fractions,

$$= -2\int_0^1 \frac{\ln(1-x^2)}{x}\,dx + 2\int_0^1 \frac{\ln(1-x^2)}{1+x}\,dx = -2I_3 + 2I_4$$

Solving I_3 first

$$I_3 = \int_0^1 \frac{\ln(1-x)}{x}\,dx + \int_0^1 \frac{\ln(1+x)}{x}\,dx$$

$$= -\int_0^1 \sum_{n=1}^{\infty} \frac{x^{n-1}}{n}\,dx + \int_0^1 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}x^{n-1}}{n}\,dx$$

$$= -\sum_{n=1}^{\infty} \frac{1}{n^2} + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$$

$$= -\frac{\pi^2}{6} + \frac{\pi^2}{12} = -\frac{\pi^2}{12}$$

$$\therefore I_3 = -\frac{\pi^2}{12}$$

$$I_4 = \int_0^1 \frac{\ln(1-x)}{1+x}\,dx + \int_0^1 \frac{\ln(1+x)}{1+x}\,dx$$

On the right integral

$$u = \ln(1+x), du = \frac{1}{1+x}\,dx$$

$$= \int_0^1 \frac{\ln(1-x)}{1+x}\,dx + \int_0^{\ln 2} u\,du$$

$$= \int_0^1 \frac{\ln(1-x)}{1+x}\,dx + \frac{1}{2}\ln^2 2$$

Now, for the remaining integral, let

$$t = 1+x, dt = dx$$

$$= \int_1^2 \frac{\ln(2-t)}{t}\,dt + \frac{1}{2}\ln^2 2$$

$$= \int_1^2 \frac{\ln 2 + \ln(1-t/2)}{t}\,dt + \frac{1}{2}\ln^2 2$$

$$= \ln^2 2 + \int_1^2 \frac{\ln(1-t/2)}{t}\,dt + \frac{1}{2}\ln^2 2$$

$$u = \frac{t}{2}, du = \frac{1}{2}\,dt$$

$$= \int_{1/2}^1 \frac{\ln(1-u)}{u}\,du + \frac{3}{2}\ln^2 2$$

$$= -\int_{1/2}^1 \sum_{n=1}^{\infty} \frac{u^{n-1}}{n}\,du + \frac{3}{2}\ln^2 2$$

$$= -\sum_{n=1}^{\infty} \frac{1}{n^2} + \sum_{n=1}^{\infty} \frac{1}{n^2 2^n} + \frac{3}{2}\ln^2 2$$

$$= -\frac{\pi^2}{6} + \mathrm{Li}_2\left(\frac{1}{2}\right) + \frac{3}{2}\ln^2 2$$

Using its known solution,

$$= -\frac{\pi^2}{6} + \frac{\pi^2}{12} - \frac{1}{2}\ln^2 2 + \frac{3}{2}\ln^2 2$$

$$= -\frac{\pi^2}{12} + \ln^2 2$$

We have thus solved I_4. Now, as $S_{c_1} = -2I_3 + 2I_4$

$$\therefore S_{c_1} = 2\ln^2 2$$

Time to solve S_{c_2}.

$$S_{c_2} = \sum_{j=1}^{\infty} \frac{1}{j^2(2j+1)}$$

Using partial fractions,

$$= \sum_{j=1}^{\infty} \frac{1}{j^2} - 2\sum_{j=1}^{\infty}\left(\frac{1}{j} - \frac{2}{2j+1}\right)$$

$$= \frac{\pi^2}{6} - 2\sum_{j=1}^{\infty}\left(\frac{1}{j} - \frac{2}{2j-1}\right) - 4$$

$$= \frac{\pi^2}{6} - 4\sum_{j=1}^{\infty}\left(\frac{1}{2j} - \frac{1}{2j-1}\right) - 4$$

$$= \frac{\pi^2}{6} - 4\sum_{j=1}^{\infty}\frac{(-1)^j}{j} - 4$$

$$= \frac{\pi^2}{6} + 4\ln 2 - 4$$

As $S_c = S_{c_1} - S_{c_2}$

$$\boxed{\therefore S_c = -\frac{\pi^2}{6} + 2\ln^2 2 - 4\ln 2 + 4}$$

Finally, let's do S_d.

$$S_d = \sum_{j=1}^{\infty} \frac{H_{j-1}}{(2j+1)^2} = \sum_{j=1}^{\infty} \frac{H_j}{(2j+1)^2} - \sum_{j=1}^{\infty} \frac{1}{j(2j+1)^2} = S_{d_1} - S_{d_2}$$

First notice that $S_{d_2} = 2S_{b_2}$

$$S_{d_2} = 2S_{b_2} = -\frac{\pi^2}{4} - 2\ln 2 + 4$$

Alright, now time for S_{d_1}

$$S_{d_1} = \sum_{j=1}^{\infty} \frac{H_j}{(2j+1)^2}$$

Let us again use the fact that

$$-\int_0^1 x^{2n}\ln(x)\,dx = \frac{1}{(2n+1)^2}$$

$$\therefore S_{d_1} = -\sum_{j=1}^{\infty} H_j \int_0^1 x^{2j}\ln(x)\,dx$$

$$= -\int_0^1 \ln(x)\sum_{j=1}^{\infty} H_j x^{2j}\,dx$$

We may again use the generating function with x^2 instead of x.

$$= \int_0^1 \frac{\ln(x)\ln(1-x^2)}{1-x^2}\,dx$$

To solve this integral, we will use the following u-sub

$$x = \frac{1-u}{1+u}, dx = -\frac{2}{(1+u)^2}\,du$$

$$= \int_0^1 \frac{\ln\left(\frac{1-u}{1+u}\right)\ln\left(1 - \frac{1-2u+u^2}{(1+u)^2}\right)}{1 - \frac{1-2u+u^2}{(1+u)^2}} \cdot \frac{2}{(1+u)^2}\,du$$

$$= \frac{1}{2}\int_0^1 \frac{(\ln(1-u) - \ln(1+u))(2\ln 2 + \ln(u) - 2\ln(1+u))}{u}\,du$$

$$= \ln(2)\int_0^1 \frac{\ln(1-u)}{u}\,du - \ln(2)\int_0^1 \frac{\ln(1+u)}{u}\,du + \frac{1}{2}\int_0^1 \frac{\ln(1-u)\ln(u)}{u}\,du - \frac{1}{2}\int_0^1 \frac{\ln(1+u)\ln(u)}{u}\,du$$

$$-\int_0^1 \frac{\ln(1+u)\ln(1-u)}{u}\,du + \int_0^1 \frac{\ln^2(1+u)}{u}\,du$$

In order let's label these integrals J_1 to J_6. Thus,

$$S_{d_1} = \ln(2)J_1 - \ln(2)J_2 + \frac{1}{2}J_3 - \frac{1}{2}J_4 - J_5 + J_6$$

From earlier parts of this integral, we know that

$$J_1 = -\frac{\pi^2}{6}$$

$$J_2 = \frac{\pi^2}{12}$$

$$J_3 = \zeta(3)$$

J_1 and J_2 were obtained by looking at the working out for I_3. J_3 was obtained by noticing that $J_3 = I_1$, which can be seen after doing $u = 1-x$ on I_1. Now we must solve for J_4, J_5, and J_6. To solve J_4, we can use J_3 by adding them together

$$J_3 + J_4 = \int_0^1 \frac{\ln(u)\ln(1-u^2)}{u}\,du$$

$$u = \sqrt{t}, du = \frac{1}{2\sqrt{t}}\,dt$$

$$= \frac{1}{4}\int_0^1 \frac{\ln(t)\ln(1-t)}{t}\,dt = \frac{1}{4}J_3 = \frac{1}{4}\zeta(3)$$

$$\therefore J_4 = -\frac{3}{4}\zeta(3)$$

Now we will solve J_5

$$J_5 = \int_0^1 \frac{\ln(1+u)\ln(1-u)}{u}\,du$$

$$= \int_0^1 \ln(1-u)\sum_{n=1}^{\infty}\frac{(-1)^{n+1}u^{n-1}}{n}\,du$$

$$= -\int_0^1 \sum_{k=1}^{\infty}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}u^{n+k-1}}{nk}\,du$$

$$= -\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sum_{k=1}^{\infty}\frac{1}{k(n+k)}$$

After partial fractions, this becomes

$$= -\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}\sum_{k=1}^{\infty}\left(\frac{1}{k} - \frac{1}{n+k}\right)$$

$$= -\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}\lim_{N\to\infty}(H_N - (H_{n+N} - H_N))$$

$$= -\sum_{n=1}^{\infty}\frac{(-1)^{n+1}H_n}{n^2}$$

Now let's use the fact that

$$\int_0^1 x^{n-1}\ln(x)\,dx = -\frac{1}{n^2}$$

And thus our expression becomes

$$= \sum_{n=1}^{\infty}(-1)^{n+1}H_n\int_0^1 x^{n-1}\ln(x)\,dx$$

$$= -\int_0^1 \frac{\ln x}{x}\sum_{n=1}^{\infty}H_n(-x)^n\,dx$$

Using the generating function, substituting $-x$ for x,

$$= \int_0^1 \frac{\ln(x)\ln(1+x)}{x(1+x)}\,dx$$

Using partial fractions,

$$= \int_0^1 \frac{\ln(x)\ln(1+x)}{x}\,dx - \int_0^1 \frac{\ln(x)\ln(1+x)}{x+1}\,dx$$

$$= J_4 - I_2 = -\frac{5}{8}\zeta(3)$$

$$\therefore J_5 = -\frac{5}{8}\zeta(3)$$

Lastly, for J_6, we can do the following .

$$J_6 = \int_0^1 \frac{\ln^2(1+u)}{u}\,du$$

$$u = \frac{1-t}{t}, dx = -\frac{1}{t^2}\,dt$$

$$\therefore \ln(1+u) = \ln t$$

$$\therefore J_6 = \int_{1/2}^1 \frac{\ln^2 t}{t(1-t)}\,dt$$

$$= \int_{1/2}^1 \frac{\ln^2 t}{t}\,dt + \int_{1/2}^1 \frac{\ln^2 t}{1-t}\,dt$$

The left integrand has anti derivative $\frac{1}{3}\ln^3 t$. Then on the right we can use a geometric series

$$= \frac{1}{3}\ln^3 2 + \sum_{n=1}^{\infty}\int_{1/2}^1 t^{n-1}\ln^2(t)\,dt$$

IBP twice with $u = \ln^2 t, dv = t^{n-1}$, then $u = \ln t, dv = t^{n-1}$

$$= \frac{1}{3}\ln^3 2 + \sum_{n=1}^{\infty}\left(-\frac{\ln^2 2}{n2^n} - \frac{2}{n}\int_{1/2}^1 t^{n-1}\ln(t)\,dt\right)$$

$$= \frac{1}{3}\ln^3 2 + \sum_{n=1}^{\infty}\left(-\frac{\ln^2 2}{n2^n} - \frac{2}{n}\left(\frac{\ln 2}{n2^n} - \frac{1}{n}\int_{1/2}^1 t^{n-1}\,dt\right)\right)$$

$$= \frac{1}{3}\ln^3 2 + \sum_{n=1}^{\infty}\left(-\frac{\ln^2 2}{n2^n} - \frac{2}{n}\left(\frac{\ln 2}{n2^n} - \frac{1}{n^2} + \frac{1}{n^2 2^n}\right)\right)$$

$$= \frac{1}{3}\ln^3 2 + \sum_{n=1}^{\infty}\left(-\frac{\ln^2 2}{n2^n} - \frac{2\ln 2}{n^2 2^n} + \frac{2}{n^3} - \frac{2}{n^3 2^n}\right)$$

$$= -\frac{2}{3}\ln^3 2 - 2\ln(2)\mathrm{Li}_2\left(\frac{1}{2}\right) + 2\zeta(3) - 2\mathrm{Li}_3\left(\frac{1}{2}\right)$$

Using known values, this simplifies to

$$= -\frac{2}{3}\ln^3 2 - 2\ln(2)\left(\frac{\pi^2}{12} - \frac{1}{2}\ln^2 2\right) + 2\zeta(3) - \frac{7}{4}\zeta(3) + \frac{\pi^2}{6}\ln 2 - \frac{1}{3}\ln^3 2$$

$$= \frac{1}{4}\zeta(3)$$

$$\therefore J_6 = \frac{1}{4}\zeta(3)$$

We thus have found all of the J integrals, and can solve for S_{d_1}.

$$\therefore S_{d_1} = -\frac{\pi^2}{4}\ln 2 + \frac{1}{2}\zeta(3) + \frac{3}{8}\zeta(3) + \frac{5}{8}\zeta(3) + \frac{1}{4}\zeta(3)$$

$$= \frac{7}{4}\zeta(3) - \frac{\pi^2}{4}\ln 2$$

Finally, as $S_d = S_{d_1} - S_{d_2}$

$$\boxed{S_d = \frac{7}{4}\zeta(3) - \frac{\pi^2}{4}\ln 2 + \frac{\pi^2}{4} + 2\ln 2 - 4}$$

And now we can solve for S_2, where $S_2 = S_c - 2S_d$. After simplifying the terms, we get

$$\boxed{\therefore S_2 = \sum_{j=1}^{\infty}\frac{H_{j-1}}{j(2j+1)^2} = -\frac{7}{2}\zeta(3) - \frac{2\pi^2}{3} + 2\ln^2 2 + \left(\frac{\pi^2}{2} - 8\right)\ln 2 + 12}$$

And at last we can solve for the original integral I, which was equal to $S_1 - S_2$.

$$\therefore I = S_1 - S_2 = -\frac{7}{8}\zeta(3) - \frac{\pi^2}{4} + \ln^2 2 - 4\ln 2 + 6 - \left(-\frac{7}{2}\zeta(3) - \frac{2\pi^2}{3} + 2\ln^2 2 + \left(\frac{\pi^2}{2} - 8\right)\ln 2 + 12\right)$$

$$= \frac{21}{8}\zeta(3) + \frac{5\pi^2}{12} - \ln^2 2 - \left(\frac{\pi^2}{2} - 4\right)\ln 2 - 6$$

AND WE'RE DONE!

$$\boxed{\boxed{\therefore \int_0^1 \ln(1-x)\ln(x)\ln(1+x)\,dx = \frac{21}{8}\zeta(3) + \frac{5\pi^2}{12} - \ln^2 2 - \left(\frac{\pi^2}{2} - 4\right)\ln 2 - 6}}$$

What a beast.

16.6 A beast of an integral: method 2

$$\int_0^1 \ln(1-x)\ln(x)\ln(1+x)$$

Comes from [12]. Call the integral I. This solution will involve 'Goncharov Polylogarithms'. Define

$$G(a_1,\ldots,a_n;z) = \int_0^z \frac{1}{t-a_1} G(a_2,\ldots,a_n;t)\,dt$$

Thus,

$$\ln x = G(0;x),\ \ln(1-x) = G(1;x),\ \ln(1+x) = G(-1;x)$$

with

$$G(0;1) = G(1;1) = 0$$

$$\therefore I = \int_0^1 G(0;x)G(1;x)G(-1;x)\,dx$$

Now we use something called the shuffle product to these weight 1 factors, with 6 permutations in total (6 terms on the RHS of the equation below).

$$G(0;x)G(1;x)G(-1;x) = \sum_{\sigma\in S_3} G(\sigma(0,1,-1);x)$$

$$\therefore I = \int_0^1 \sum_{\sigma\in S_3} G(\sigma(0,1,-1);x)\,dx$$

Now, a property of the Goncharov polylog's is that

$$\int_0^1 G(a_1,\ldots,a_n;x)\,dx = \mathcal{J}(a_1,\ldots,a_n)$$

And thus

$$\mathcal{J}(a_1,\ldots,a_n) = \sum_{k=1}^{n}(-1)^{k-1}(1-a_k)G(a_k,\ldots,a_n;1) + (-1)^n$$

Finally, for our purposes we need to consider Goncharov values where $a_n = 0$. When this happens, we use the following

$$G(a_1,\ldots,a_{n-1},0;1) = -G(0,a_1,\ldots,a_{n-1};1) - \sum_{k=1}^{n-1} G(a_1,\ldots,a_k,0,a_{k+1},\ldots,a_{n-1};1)$$

A similar thing is used for when $a_1 = 1$ but those wont appear in our integral solution so we'll ignore that. Now, using our formula for $\mathcal{J}(a_1,\ldots,a_n)$, we get

$$I = \sum_{\sigma\in S_3} \mathcal{J}(\sigma(0,1,-1))$$

Using our formulas, this becomes

$$I = G(0,1,-1;1) + 2G(-1;1) + G(0,-1,1;1) - 2G(-1,1;1) - 2G(-1,0;1) + G(0;1)$$

$$-G(0,-1;1) + 2G(-1;1) + 2G(-1,0,1;1) - G(0,1;1) + 2G(-1,1,0;1) + G(0;1) - 6$$

As we can see there are two terms where the last entry is a zero, using our formula for those cases,

$$G(-1,0;1) = -G(0,-1;1)$$

$$G(-1,1,0;1) = -G(-1,1,0;1) - G(-1,0,1;1)$$

$$\therefore I = G(0,1,-1;1) + 2G(-1;1) + G(0,-1,1;1) - 2G(-1,1;1) + 2G(0,-1;1) + G(0;1)$$
$$-G(0,-1;1) + 2G(-1;1) + 2G(-1,0,1;1) - G(0,1;1) - 2G(0,-1,1;1) - 2G(-1,0,1;1) + G(0;1) - 6$$

$$= G(0,1,-1;1) - G(0,-1,1;1) - 2G(-1,1;1) + G(0,-1;1) - G(0,1;1) + 4\ln 2 - 6$$

Using the definition of the Goncharov Polylog, the right three polylog's become

$$= G(0,1,-1;1) - G(0,-1,1;1) - 2\int_0^1 \frac{\ln(1-t)}{t+1}\,dt + \int_0^1 \frac{\ln(t+1)}{t}\,dt - \int_0^1 \frac{\ln(1-t)}{t}\,dt + 4\ln 2 - 6$$

From the previous method, we have solved these integrals before. From left to right they are: left part of I_4, right part of I_3, left part of I_3.

$$= G(0,1,-1;1) - G(0,-1,1;1) + \frac{\pi^2}{6} - \ln^2 2 + \frac{\pi^2}{12} + \frac{\pi^2}{6} + 4\ln 2 - 6$$

$$= G(0,1,-1;1) - G(0,-1,1;1) + \frac{5\pi^2}{12} - \ln^2 2 + 4\ln 2 - 6$$

These last two are more complex but also stem from some of the integrals solved in the previous method.

$$G(0,-1,1;1) = \frac{\pi^2}{4}\ln 2 - \frac{13}{8}\zeta(3)$$

$$G(0,1,-1;1) = -\frac{\pi^2}{4}\ln 2 + \zeta(3)$$

Thus our answer is

$$I = -\frac{\pi^2}{4}\ln 2 + \zeta(3) - \frac{\pi^2}{4}\ln 2 + \frac{13}{8}\zeta(3) + \frac{5\pi^2}{12} - \ln^2 2 + 4\ln 2 - 6$$

$$= \frac{21}{8}\zeta(3) + \frac{5\pi^2}{12} - \ln^2 2 - \left(\frac{\pi^2}{2} - 4\right)\ln 2 - 6$$

$$\therefore \int_0^1 \ln(1-x)\ln(x)\ln(1+x) = \frac{21}{8}\zeta(3) + \frac{5\pi^2}{12} - \ln^2 2 - \left(\frac{\pi^2}{2} - 4\right)\ln 2 - 6$$

17 May 26

17.1 An example problem from RMT example 3

The integral in question for this one is

$$\int_0^\infty \frac{1}{(x^4 - x^2 + 1)^3}\, dx$$

In order to solve this problem we can first use the result from page 125 of example 3 using Ramanujan's master theorem. In our case, $c = t = 1$, $k = -1$, $p = 4$, $q = 2$, $g = 0$, $s = 3$. From this, we get the following summation expression for the integral

$$\int_0^\infty \frac{1}{(x^4 - x^2 + 1)^3}\, dx = \frac{1}{8}\sum_{n=0}^{\infty} \frac{1}{n!}\Gamma\left(\frac{2n+1}{4}\right)\Gamma\left(\frac{2n+11}{4}\right)$$

It now remains to solve this sum.

$$= \frac{1}{128}\sum_{n=0}^{\infty} \frac{1}{n!}\Gamma\left(\frac{2n+1}{4}\right)\Gamma\left(\frac{2n+3}{4}\right)(2n+7)(2n+3)$$

We can now use legendre's duplication formula, which states that

$$\Gamma(x)\Gamma\left(x + \frac{1}{2}\right) = 2^{1-2x}\Gamma(2x)\sqrt{\pi}$$

In our case, $x = \frac{2n+1}{4}$.

$$= \frac{\sqrt{2\pi}}{128}\sum_{n=0}^{\infty} \frac{1}{n!2^n}\Gamma\left(n + \frac{1}{2}\right)(2n+7)(2n+3)$$

Now, using the fact that

$$\Gamma\left(x + \frac{1}{2}\right) = \frac{(2n)!}{n!4^n}\sqrt{\pi}$$

our sum can be written as

$$= \frac{\pi\sqrt{2}}{128}\sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2 8^n}(2n+7)(2n+3)$$

$$= \frac{\pi\sqrt{2}}{128}\sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2 8^n}(4n + 20n + 21)$$

$$= \frac{\pi\sqrt{2}}{128}\sum_{n=0}^{\infty} \binom{2n}{n}\left(\frac{1}{8}\right)^n (4n^2 + 20n + 21)$$

We now envoke the generating function for the central binomial coefficients, which states that

$$S(x) = \sum_{n=0}^{\infty}\binom{2n}{n}x^n = \frac{1}{\sqrt{1-4x}}$$

Taking derivatives with respect to x, then multiplying each derivative by x so that the power of x stays at n, we find that

$$xS'(x) = \sum_{n=0}^{\infty}\binom{2n}{n}nx^n = \frac{2x}{(1-4x)^{\frac{3}{2}}}$$

$$xS'(x) + x^2S''(x) = \sum_{n=0}^{\infty}\binom{2n}{n}n^2x^n = \frac{2x}{(1-4x)^{\frac{3}{2}}} + \frac{12x^2}{(1-4x)^{\frac{5}{2}}}$$

In our case, $x = \frac{1}{8}$, and our summation expression for the original integral is a linear combination of $S(1/8)$, $\frac{1}{8}S'(1/8)$, and $\frac{1}{8}S'(1/8) + \frac{1}{64}S''(1/8)$.

$$\therefore \frac{\pi\sqrt{2}}{128}\sum_{n=0}^{\infty}\binom{2n}{n}\left(\frac{1}{8}\right)^{n}(4n^2+20n+21) = 21S\left(\frac{1}{8}\right)+3S'\left(\frac{1}{8}\right)+\frac{1}{16}S''\left(\frac{1}{8}\right)$$

$$= \frac{\pi\sqrt{2}}{128}\big(21\sqrt{2}+6\sqrt{2}+3\sqrt{2}\big)$$

$$= \frac{72\pi}{128}$$

$$= \frac{9\pi}{16}$$

And thus we are done.

$$\therefore \int_0^{\infty}\frac{1}{(x^4-x^2+1)^3}\,dx = \frac{9\pi}{16}$$

17.2 Ramanujan Product

$$\prod_{n=1}^{\infty}\left(1-\frac{1}{(2n+1)^2}\right)^{(-1)^n(2n+1)}$$
$$\approx\left(1-\frac{1}{3^2}\right)^{-3}\left(1-\frac{1}{5^2}\right)^{5}\left(1-\frac{1}{7^2}\right)^{-7}\left(1-\frac{1}{9^2}\right)^{9}\cdots$$

This is a very cool product! Now we will show that it has a crazy closed form solution too. Solution method guided via online from [13]. To do this, first define the following function.

$$P(x)=\prod_{n=1}^{\infty}\left(1-\frac{x^2}{(2n+1)^2}\right)^{(-1)^n(2n+1)}$$

We are thus solving for $P(1)$. Now, define the function

$$S(x)=\ln(P(x))=\sum_{n=1}^{\infty}(-1)^n(2n+1)\ln\left(1-\frac{x^2}{(2n+1)^2}\right)$$
$$\therefore S'(x)=\sum_{n=1}^{\infty}(-1)^n(2n+1)\frac{-\frac{2x}{(2n+1)^2}}{1-\frac{x^2}{(2n+1)^2}}$$
$$=-2x\sum_{n=1}^{\infty}\frac{(-1)^n(2n+1)}{(2n+1)^2-x^2}$$

We wish to find this sum in terms of a function we know. Via research I found out that we can! Consider the summation expression for $\pi\sec(\pi x)$ using Mittag-Leffler's theorem, which we will morph into the form for our $S'(x)$.

$$\pi\sec(\pi x)=2\sum_{n=0}^{\infty}\frac{(-1)^{n-1}(n+1/2)}{x^2-(n+1/2)^2}$$
$$\therefore -\frac{\pi x}{2}\sec\left(\frac{\pi x}{2}\right)=-x\sum_{n=0}^{\infty}\frac{(-1)^{n-1}(n+1/2)}{\frac{1}{4}x^2-(n+1/2)^2}$$
$$=-2x\sum_{n=0}^{\infty}\frac{(-1)^{n-1}(2n+1)}{x^2-(2n+1)^2}$$
$$=-2x\sum_{n=0}^{\infty}\frac{(-1)^{n}(2n+1)}{(2n+1)^2-x^2}$$
$$=-2x\sum_{n=1}^{\infty}\frac{(-1)^{n}(2n+1)}{(2n+1)^2-x^2}-\frac{2x}{1-x^2}$$
$$=S'(x)-\frac{2x}{1-x^2}$$

Thus,

$$\therefore S'(x)=-\frac{\pi x}{2}\sec\left(\frac{\pi x}{2}\right)+\frac{2x}{1-x^2}$$
$$\therefore S(x)=\int_0^x-\frac{\pi t}{2}\sec\left(\frac{\pi t}{2}\right)dt+\int_0^x\frac{2t}{1-t^2}\,dt$$

Before integrating, I wish to make adjustments to the left integrand in order to get it into a form that yield's a nice u-sub.

$$-\frac{\pi t}{2}\sec\left(\frac{\pi t}{2}\right)=-\frac{\pi t}{2\cos(\frac{\pi t}{2})}$$

$$= -\frac{\pi t}{2\sin\left(\frac{\pi}{2}(1-t)\right)}$$

$$= \frac{\frac{\pi}{2}(1-t) - \frac{\pi}{2}}{\sin\left(\frac{\pi}{2}(1-t)\right)}$$

We can now put this back into the integral.

$$S(x) = \int_0^x \frac{\frac{\pi}{2}(1-t) - \frac{\pi}{2}}{\sin\left(\frac{\pi}{2}(1-t)\right)}\,dt + \int_0^x \frac{2t}{1-t^2}\,dt$$

$$= \int_0^x \frac{\frac{\pi}{2}(1-t)}{\sin\left(\frac{\pi}{2}(1-t)\right)}\,dt - \frac{\pi}{2}\int_0^x \frac{1}{\sin\left(\frac{\pi}{2}(1-t)\right)}\,dt + \int_0^x \frac{2t}{1-t^2}\,dt$$

On the middle and left integrals,

$$u = \frac{\pi}{2}(1-t), du = -\frac{\pi}{2}\,dt$$

And on the right integral,

$$u = t^2, du = 2t\,dt$$

Thus our integrals become

$$= -\frac{2}{\pi}\int_{\frac{\pi}{2}}^{\frac{\pi}{2}(1-x)} \frac{u}{\sin u}\,du + \int_{\frac{\pi}{2}}^{\frac{\pi}{2}(1-x)} \frac{1}{\sin u}\,du + \int_0^{x^2} \frac{1}{1-u}\,du$$

Now, recall that we are looking for $P(1)$, which is $e^{S(1)}$. We can thus substitute in $x = 1$ into the bounds, but for now will only do so to the left integral.

$$= -\frac{2}{\pi}\int_{\frac{\pi}{2}}^{0} \frac{u}{\sin u}\,du + \int_{\frac{\pi}{2}}^{\frac{\pi}{2}(1-x)} \frac{1}{\sin u}\,du + \int_0^{x^2} \frac{1}{1-u}\,du$$

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \frac{u}{\sin u}\,du + \ln\left(\left|\tan\left(\frac{\pi}{4}(1-x)\right)\right|\right) - \ln(|1-x^2|)$$

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \frac{u}{\sin u}\,du + \ln\left(\left|\frac{\tan\left(\frac{\pi}{4}(1-x)\right)}{1-x^2}\right|\right)$$

Now we may substitute $x = 1$ into this expression, but must take a limit as we'd get a $0/0$ case.

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \frac{u}{\sin u}\,du + \lim_{N\to 1}\ln\left(\frac{\tan\left(\frac{\pi}{4}(1-N)\right)}{1-N^2}\right)$$

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \frac{u}{\sin u}\,du + \ln\left(\lim_{N\to 1}\frac{\tan\left(\frac{\pi}{4}(1-N)\right)}{1-N^2}\right)$$

Performing L'hopital's,

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \frac{u}{\sin u}\,du + \ln\left(\lim_{N\to 1}\frac{-\frac{\pi}{4}\sec^2\left(\frac{\pi}{4}(1-N)\right)}{-2N}\right)$$

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \frac{u}{\sin u}\,du + \ln\left(\frac{\pi}{8}\right)$$

For the remaining integral, we do

$$w = \tan\left(\frac{u}{2}\right), du = \frac{2}{1+w^2}\,dw$$

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \frac{\arctan(w)(1+w^2)}{w}\cdot\frac{2}{1+w^2}\,dw + \ln\left(\frac{\pi}{8}\right)$$

$$= \frac{4}{\pi}\int_0^1 \frac{\arctan w}{w}\,dw + \ln\left(\frac{\pi}{8}\right)$$

Using the Taylor series for $\arctan x$,

$$= \frac{4}{\pi}\int_0^1 \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n}}{2n+1}\,dw + \ln\left(\frac{\pi}{8}\right)$$

$$= \frac{4}{\pi}\sum_{n=0}^{\infty}\frac{(-1)^n}{2n+1}\int_0^1 x^{2n}\,dw + \ln\left(\frac{\pi}{8}\right)$$

$$= \frac{4}{\pi}\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2} + \ln\left(\frac{\pi}{8}\right)$$

This sum is just Catalan's constant, G.

$$= \frac{4G}{\pi} + \ln\left(\frac{\pi}{8}\right)$$

We have finally solved for $S(1)$. Then, as $P(1) = e^{S(1)}$, which is the original product that we were solving for, we obtain

$$\therefore P(1) = \frac{\pi}{8}e^{\frac{4G}{\pi}}$$

This is our answer!

$$\therefore \prod_{n=1}^{\infty}\left(1 - \frac{1}{(2n+1)^2}\right)^{(-1)^n(2n+1)} = \frac{\pi}{8}e^{\frac{4G}{\pi}}$$

Further work:

I tried reverse engineering the same method to try to get to other Mittag-Leffler expansions, then integrating to get a nice result. Here are just a couple of the cooler ones that I got, absolutely not an exhaustive list of what's possible.

$$\prod_{n=2}^{\infty}\left(1 - \frac{1}{n^2}\right)^{(-1)^n} = \frac{8}{\pi^2}$$

$$\prod_{n=1}^{\infty}\left(1 - \frac{1}{1+n^2}\right)^{(-1)^n} = \frac{\pi}{2}\coth\left(\frac{\pi}{2}\right)$$

And lastly,

$$\prod_{n=1}^{\infty}\left(1 + \frac{k^2}{(2n+1)^2}\right)^{(-1)^n(2n+1)} = \frac{1}{k^2+1}\exp\left(\frac{8}{\pi}\int_0^{\tanh(\frac{k\pi}{4})}\frac{\operatorname{arctanh} t}{1+t^2}dt\right) \approx P(k) = \frac{e^{\frac{4G}{\pi}}}{k^2+1}$$

This starts off as a bad approximation, but becomes very good for high k values (roughly $k \geq 5$), improving further as k continues to increase. An even better approximation is

$$\prod_{n=1}^{\infty}\left(1 + \frac{k^2}{(2n+1)^2}\right)^{(-1)^n(2n+1)} \approx Q(k) = \frac{\exp\left(\frac{4G}{\pi} - \left(\frac{4}{\pi} + 2k\right)e^{-\frac{\pi k}{2}}\right)}{k^2+1}$$

Which is really good even for k values as low as $\sim 3/2$. For instance, when $k = 7$, the first approximation gives $P(7) \approx 0.0641982383...$, the second approximation gives $Q(7) \approx 0.0641817915...$, and the true value is $\sim 0.0641817992...$ Thus, while the first approximation is correct to 4dp, the second approximation is correct to 8dp! This is better than 100 million terms of the product! Meanwhile, the first approximation 'only' beats about 45,000 terms.

17.3 The answer is real?!

$$\int_0^{\pi} i^{\tan x}\,dx$$

Let the integral above be I. The integral itself has i in it, so surely the answer has a non-zero imaginary component. Right?

$$\int_0^{\pi} e^{\frac{i\pi}{2}\tan x}\,dx$$

$$\int_0^{\pi} \cos\left(\frac{\pi}{2}\tan x\right)dx + i\int_0^{\pi} \sin\left(\frac{\pi}{2}\tan x\right)dx$$

$$I_1 + iI_2$$

Let us first examine I_2

$$I_2 = \int_0^{\pi} \sin\left(\frac{\pi}{2}\tan x\right)dx$$

Using king's rule, we use the fact that $\tan(\pi - x) = \tan x$. Thus,

$$I_2 \int_0^{\pi} \sin\left(\frac{\pi}{2}\tan x\right)dx = -\int_0^{\pi} \sin\left(\frac{\pi}{2}\tan x\right)dx$$

$$\therefore I_2 = 0$$

This shows that the original integral is purely real, and is equal to I_1. It is now thus time to solve I_1.

$$\therefore I = I_1 = \int_0^{\pi} \cos\left(\frac{\pi}{2}\tan x\right)dx$$

First, we notice that the integrand is symmetric and even about $x = \pi/2$. This is because $f(\pi/2 - x) = f(\pi/2 + x)$, where $f(x)$ is the integrand. Thus, we can write our integral as

$$= 2\int_0^{\frac{\pi}{2}} \cos\left(\frac{\pi}{2}\tan x\right)dx$$

$$t = \tan x, dx = \frac{1}{1+x^2}\,dt$$

$$= 2\int_0^{\infty} \frac{\cos\left(\frac{\pi t}{2}\right)}{1+t^2}\,dt$$

As the integrand is even, this can be rewritten as

$$= \int_{-\infty}^{\infty} \frac{\cos\left(\frac{\pi t}{2}\right)}{1+t^2}\,dt$$

Using $I(k)$ from page 23 (so long ago i know!), where $k = \pi/2$, we can give an answer to this integral.

$$= \pi e^{-\frac{\pi}{2}}$$

And we're done.

$$\therefore \int_0^{\pi} i^{\tan x}\,dx = \pi e^{-\frac{\pi}{2}}$$

17.4 A lil integral using recursion

I have done similar integrals to this one before back on pages 40-49, but not this one exactly.

$$\int_{-\infty}^{\infty} e^{-x^2} \cos x \, dx$$

Call this integral I. Using the Maclaurin expansion for $\cos x$, this integral becomes the following

$$\begin{aligned} &= \int_{-\infty}^{\infty} e^{-x^2} \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} \, dx \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \int_{-\infty}^{\infty} e^{-x^2} x^{2n} \, dx \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} J_n \end{aligned}$$

We now want to solve J_n, and sub it back into this expression above for I.

$$J_n = \int_{-\infty}^{\infty} e^{-x^2} x^{2n} \, dx$$

$$\text{IBP with } u = e^{-x^2}, dv = x^{2n}$$

$$\begin{aligned} &= \left[\frac{x^{2n+1}}{2n+1} e^{-x^2} \right]_{-\infty}^{\infty} + \frac{2}{2n+1} \int_{-\infty}^{\infty} e^{-x^2} x^{2(n+1)} \, dx \\ &= \frac{2}{2n+1} \int_{-\infty}^{\infty} e^{-x^2} x^{2(n+1)} \, dx \\ &= \frac{2}{2n+1} J_{n+1} \end{aligned}$$

$$\therefore J_{n+1} = \frac{2n+1}{2} J_n$$

Switching n for $n-1$,

$$\therefore J_n = \frac{2n-1}{2} J_{n-1}$$

Then, after multiplying the top and bottom of the fraction by $2n$, we arrive at

$$J_n = \frac{2n(2n-1)}{4n} J_{n-1}$$

Via recursion, we can then say that

$$\therefore J_n = \frac{2n(2n-1)}{4n} \cdot \frac{(2n-2)(2n-3)}{4(n-1)} J_{n-2}$$

Continuing this down to J_0, we arrive at

$$\therefore J_n = \frac{(2n)!}{4^n n!} J_0$$

J_0 can be found using page 40 (or page 42), and is famously

$$J_0 = \int_{-\infty}^{\infty} e^{-x^2} \, dx = \sqrt{\pi}$$

Putting this into our result for J_n, we get

$$\therefore J_n = \frac{(2n)! \sqrt{\pi}}{4^n n!}$$

We can now sub this into our expression for I

$$\therefore I = \sqrt{\pi}\sum_{n=0}^{\infty}\frac{(-1)^n(2n)!}{(2n)!4^n n!}$$

$$= \sqrt{\pi}\sum_{n=0}^{\infty}\frac{1}{n!}\left(-\frac{1}{4}\right)^n$$

Notice how this sum is the Maclaurin series for e^x at $x = -1/4$.

$$= \sqrt{\pi}e^{-\frac{1}{4}}$$

$$= \sqrt{\frac{\pi}{\sqrt{e}}}$$

Very nice solution!

$$\therefore \int_{-\infty}^{\infty} e^{-x^2}\cos x\,dx = \sqrt{\frac{\pi}{\sqrt{e}}}$$

Also, using the same method we can find that

$$\int_{-\infty}^{\infty} e^{-x^2}\cosh x\,dx = \sqrt{\pi\sqrt{e}}$$

NOTE: I have later noticed that after a u-sub and integration by parts, these two integrals/integrands can be turned into the two respective ones found on page 80. Notice that the only difference between the respective solutions is a factor of 2, which is due to the bounds of these integrals being from -infinity to infinity instead of zero to infinity. Despite this, I have still chosen to include these two integrals in section 1 (List of integrals) of the document.

17.5 Very nice sum solution

$$\int_1^\infty \frac{1}{(x-\ln x)^2}\,dx$$

$$= \int_1^\infty \frac{1}{x^2\left(1-\frac{\ln x}{x}\right)^2}\,dx$$

For the $1/(1-\ln(x)/x)^2$ part, we can use the derivative of the geometric sum.

$$\sum_{n=0}^\infty x^n = \frac{1}{1-x}$$

$$\therefore \sum_{n=0}^\infty nx^{n-1} = \frac{1}{(1-x)^2}$$

The $n=0$ term is just zero. So,

$$\therefore \sum_{n=1}^\infty nx^{n-1} = \frac{1}{(1-x)^2}$$

In our case, x is $\ln(x)/x$. Substituting this into our integral, we get

$$= \int_1^\infty \frac{1}{x^2}\sum_{n=1}^\infty n\left(\frac{\ln x}{x}\right)^{n-1}\,dx$$

$$= \sum_{n=1}^\infty n \int_1^\infty x^{-(n+1)}(\ln x)^{n-1}\,dx$$

$$u = \ln x, du = \frac{1}{x}\,dx$$

$$= \sum_{n=1}^\infty n \int_0^\infty e^{-nu}u^{n-1}\,du$$

$$t = nu, dt = n\,du$$

$$= \sum_{n=1}^\infty \int_0^\infty e^{-t}\left(\frac{t}{n}\right)^{n-1}\,dt$$

$$= \sum_{n=1}^\infty \frac{1}{n^{n-1}}\int_0^\infty e^{-t}t^{n-1}\,dt$$

$$= \sum_{n=1}^\infty \frac{(n-1)!}{n^{n-1}}$$

$$= \sum_{n=1}^\infty \frac{n!/n}{n^n/n}$$

$$= \sum_{n=1}^\infty \frac{n!}{n^n}$$

The answer being the sum of $n!$ over n^n is wild. A pretty short solution too!

$$\therefore \int_1^\infty \frac{1}{(x-\ln x)^2}\,dx = \sum_{n=1}^\infty \frac{n!}{n^n}$$

17.6 Some constants that should have a name

Firstly, I think the $\mathrm{Sphd}(\alpha; x)$ function should be recognised more often.

$$\mathrm{Sphd}(\alpha; x) = \int_0^x t^{\alpha t}\, dt$$

This is because we can get nice results like

$$\int_0^1 x^x\, dx = \sum_{n=1}^{\infty} -(-n)^{-n} = \mathrm{Sphd}(1; 1)$$

$$\int_0^1 x^{-x}\, dx = \sum_{n=1}^{\infty} n^{-n} = \mathrm{Sphd}(-1; 1)$$

$$\int_0^{\infty} x^{-x}\, dx = \mathrm{Sphd}(-1; \infty)$$

Here are some more sums that I think should have a defined constant as its result. First, using page 144, we know that

$$\sum_{n=0}^{\infty} \frac{(xn)^n}{n!} = \frac{1}{1 + W(-x)}$$

For $|x| < 1/e$. I think we could also define a function

$$\mathfrak{W}(x) = \sum_{n=0}^{\infty} \frac{n! x^n}{n^n}$$

Which converges for $|x| < e$. In particular, define the constant $\mathfrak{W} = \mathfrak{W}(1)$.

$$\mathfrak{W} = \sum_{n=0}^{\infty} \frac{n!}{n^n} \approx 2.87985386218\ldots$$

The next function we could define is

$$\sum_{n=0}^{\infty} \frac{x^n}{!n} = \mathfrak{S}(x)$$

Where $!n$ is the sub-factorial. We can give the example constant $\mathfrak{S} = \mathfrak{S}(1)$

$$\mathfrak{S} = \sum_{n=0}^{\infty} \frac{1}{!n} \approx 2.6382270751$$

Next up is the hyper-factorial and the super-factorial.

$$\mathcal{S}(x) = \sum_{n=0}^{\infty} \prod_{k=0}^{n} \frac{1}{k!} = \sum_{n=0}^{\infty} \frac{x^n}{sf(n)}$$

$$\mathcal{H}(x) = \sum_{n=0}^{\infty} \prod_{k=0}^{n} k^k = \sum_{n=0}^{\infty} \frac{x^n}{H(n)}$$

They may have the special values $\mathcal{S}(1) = \mathcal{S}$, and $\mathcal{H}(1) = \mathcal{H}$.

$$\mathcal{S} = \sum_{n=0}^{\infty} \frac{1}{sf(n)} \approx 2.5868345309$$

$$\mathcal{H} = \sum_{n=0}^{\infty} \frac{1}{H(n)} \approx 2.2592954398$$

Like how $n!$ can be generalised to the reals by $\Gamma(n+1)$, $sf(n)$ can be generalised via the Barnes-G function $G(n+2)$, and $H(n)$ can be generalised by the K-function $K(n+1)$. There is also this nice link between all three functions:

$$G(x)K(x) = \Gamma^{x-1}(x)$$

18 June 26

18.1 Gamma reflection integral

The following integral converges for $|a| < 1$ (I believe).

$$\int_0^1 \Gamma\left(1+\frac{x}{a}\right)\Gamma\left(1-\frac{x}{a}\right)dx$$

Using the gamma reflection formula, this is equal to

$$= \int_0^1 \frac{\pi x/a}{\sin(\pi x/a)}\,dx$$

$$t = \frac{\pi x}{a}, dt = \frac{\pi}{a}\,du$$

$$= \frac{a}{\pi}\int_0^{\frac{\pi}{a}} \frac{t}{\sin t}\,dt$$

$$\text{IBP with } u = t, dv = \csc t$$

$$= \frac{at}{\pi}\ln\left|\tan\left(\frac{\pi}{2a}\right)\right| - \frac{a}{\pi}\int_0^{\frac{\pi}{a}} \ln\left|\tan\left(\frac{t}{2}\right)\right|dt$$

$$= \ln\left|\tan\left(\frac{\pi}{2a}\right)\right| - \frac{a}{\pi}\int_0^{\frac{\pi}{a}} \ln\left|\tan\left(\frac{t}{2}\right)\right|dt$$

$$= \ln\left|\tan\left(\frac{\pi}{2a}\right)\right| - \frac{a}{\pi}\int_0^{\frac{\pi}{a}} \ln\left|\sin\left(\frac{t}{2}\right)\right|dt + \frac{a}{\pi}\int_0^{\frac{\pi}{a}} \ln\left|\cos\left(\frac{t}{2}\right)\right|dt$$

$$= \ln\left|\tan\left(\frac{\pi}{2a}\right)\right| - \frac{a}{\pi}\int_0^{\frac{\pi}{a}} \ln\left|2\sin\left(\frac{t}{2}\right)\right|dt + \frac{a}{\pi}\int_0^{\frac{\pi}{a}} \ln\left|2\cos\left(\frac{t}{2}\right)\right|dt$$

One of the definitions of the clausen function is

$$\mathrm{Cl}_2(\theta) = -\int_0^{\theta} \ln\left|2\sin\left(\frac{x}{2}\right)\right|\,dx$$

We can turn our integrals into this

$$= \ln\left|\tan\left(\frac{\pi}{2a}\right)\right| + \frac{a}{\pi}\mathrm{Cl}\left(\frac{\pi}{a}\right) + \frac{a}{\pi}\int_0^{\frac{\pi}{a}} \ln\left|2\sin\left(\frac{\pi-t}{2}\right)\right|dt$$

$$u = \pi - t, du = -\,dt$$

$$= \ln\left|\tan\left(\frac{\pi}{2a}\right)\right| + \frac{a}{\pi}\mathrm{Cl}\left(\frac{\pi}{a}\right) - \frac{a}{\pi}\int_0^{\pi-\frac{\pi}{a}} \ln\left|2\sin\left(\frac{u}{2}\right)\right|du$$

$$= \ln\left|\tan\left(\frac{\pi}{2a}\right)\right| + \frac{a}{\pi}\mathrm{Cl}\left(\frac{\pi}{a}\right) + \frac{a}{\pi}\mathrm{Cl}\left(\pi - \frac{\pi}{a}\right)$$

Done!

$$\therefore \int_0^1 \Gamma\left(1+\frac{x}{a}\right)\Gamma\left(1-\frac{x}{a}\right)dx = \ln\left|\tan\left(\frac{\pi}{2a}\right)\right| + \frac{a}{\pi}\mathrm{Cl}\left(\frac{\pi}{a}\right) + \frac{a}{\pi}\mathrm{Cl}\left(\pi - \frac{\pi}{a}\right)$$

Cool how the Clausen function makes an appearance. Before showing the special values for this integral, also note another form of the solution can be written in terms of the inverse tangent integral, which is

$$\mathrm{Ti}_2(x) = \int_0^x \frac{\arctan t}{t}\,dt = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)^2}$$

Using this, we can have the solution

$$\int_0^1 \Gamma\left(1+\frac{x}{a}\right)\Gamma\left(1-\frac{x}{a}\right)dx = \frac{2a}{\pi}\mathrm{Ti}_2\left(\tan\left(\frac{\pi}{2a}\right)\right) = \frac{2a}{\pi}\sum_{n=0}^{\infty}\frac{(-1)^n\tan^{2n+1}\left(\frac{\pi}{2a}\right)}{(2n+1)^2}$$

This solution comes from applying the Weierstrass substitution on the $t/\sin t$ integral on the previous page. Okay, here are some special values, using either solution representation above.

$$\int_0^1 \Gamma\left(1+\frac{x}{2}\right)\Gamma\left(1-\frac{x}{2}\right)dx = \frac{4G}{\pi}$$

$$\int_0^1 \Gamma\left(1+\frac{x}{3}\right)\Gamma\left(1-\frac{x}{3}\right)dx = \frac{5V}{\pi} - \frac{1}{2}\ln 3$$

$$\int_0^1 \Gamma\left(1+\frac{2x}{3}\right)\Gamma\left(1-\frac{2x}{3}\right)dx = \frac{5V}{2\pi} + \frac{1}{2}\ln 3$$

18.2 Even Zeta formula using Fourier series (2 Methods)

9-10/6. METHOD 1: In this section, we will be deriving the formula for the zeta function of even integers, $\zeta(2k), k \in \mathbb{Z}^+$. This will be done using the Fourier series for $B_{2k}(x)$, which is the Bernoulli polynomial specifically for even integers. We note before starting that the Bernoulli polynomial for even integers is symmetric around $x = 1/2$. As such, if we were to shift the function by 1/2, then it is even and thus all sine terms vanish. Also note that, as $B_{2k}(x)$ has a period of one from $[0, 1]$, $L = 1/2$. We can thus construct the Fourier series accordingly for $[0, 1]$.

$$B_{2k}(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(2\pi n x)$$

Starting with a_0,

$$a_0 = 2\int_0^1 B_{2k}(x)\,dx = \frac{2(B_{2k+1}(1) - B_{2k+1}(0))}{2k+1}$$

Note that

$$B_k(x+1) = B_k(x) + kx^{k-1}$$

As $k \in \mathbb{Z}^+$,

$$\therefore B_{2k+1}(1) - B_{2k+1}(0) = (2k+1)\cdot 0^{2k} = 0$$

$$\therefore a_0 = 0$$

Now it is time to solve a_n. Letting the integral be I_{2k},

$$a_n = 2\int_0^1 B_{2k}(x)\cos(2\pi n x)\,dx = 2I_{2k}$$

We will be solving I_{2k}, then substituting that back into a_n.

$$\text{IBP with } u = B_{2k}(x), dv = \cos(2\pi n x)$$

$$\therefore I_{2k} = \frac{B_{2k}(1)\sin(2\pi n)}{2\pi n} - \frac{B_{2k}(0)\sin(0)}{2\pi n} - \frac{2k}{2\pi n}\int_0^1 B_{2k-1}(x)\sin(2\pi n x)\,dx$$

For integer n, $\sin(2\pi n) = \sin 0 = 0$.

$$= -\frac{2k}{2\pi n}\int_0^1 B_{2k-1}(x)\sin(2\pi n x)\,dx$$

$$\text{IBP with } u = B_{2k-1}(x), dv = \sin(2\pi n x)$$

$$= -\frac{2k}{2\pi n}\left(-\frac{B_{2k-1}(1)\cos(2\pi n)}{2\pi n} + \frac{B_{2k-1}(0)\cos(0)}{2\pi n} + \frac{2k-1}{2\pi n}\int_0^1 B_{2k-2}(x)\cos(2\pi n x)\,dx\right)$$

Note that the remaining integral is I_{2k-2}, and that $\cos(2\pi n) = \cos 0 = 1$ for integer n. We can also take out a factor of $-1/2\pi n$

$$= \frac{2k\big(B_{2k-1}(1) - B_{2k-1}(0) - (2k-1)I_{2k-2}\big)}{(2\pi n)^2}$$

Also, using the same $B_k(x+1) = B_k(x) + kx^{k-1}$ formula we used earlier, we get

$$B_{2k-1}(1) - B_{2k-1}(0) = (2k-1)0^{2k-2} = \begin{cases} 1 & \text{if } k = 1, \\ 0 & \text{otherwise.} \end{cases}$$

We thus consider the $k = 1$ case, and the general $2k$ case for $k \in \mathbb{Z}^+$.

$$\therefore I_2 = \frac{2(1 - I_0)}{(2\pi n)^2}$$

Although k cannot equal zero in our final solution due to divergence of the zeta function, we can find I_0 here in order to kickstart the recurrence.

$$I_0 = \int_0^1 B_0(x)\cos(2\pi nx)\,dx = \int_0^1 \cos(2\pi nx)\,dx = 0$$

$$\therefore I_2 = \frac{2}{(2\pi n)^2}$$

Our recurrence for I_{2k}, for $k \geq 2$, is then

$$I_{2k} = -\frac{2k(2k-1)I_{2k-2}}{(2\pi n)^2}$$

This allows us to solve for I_{2k}.

$$I_4 = -\frac{4 \cdot 3}{(2\pi n)^2} I_2 = -\frac{4 \cdot 3 \cdot 2}{(2\pi n)^4}$$

$$I_6 = -\frac{6 \cdot 5}{(2\pi n)^2} I_4 = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2}{(2\pi n)^6}$$

By now, it becomes clear that

$$I_{2k} = \frac{(-1)^{k+1}(2k)!}{(2\pi n)^{2k}}$$

Thus, as $a_n = 2I_{2k}$

$$a_n = \frac{2(-1)^{k+1}(2k)!}{(2\pi n)^{2k}}$$

As we now have both a_0 and a_n, we can put these into the Fourier series for $B_{2k}(x)$, $0 \leq x \leq 1$.

$$\therefore B_{2k}(x) = \sum_{n=1}^{\infty} \frac{2(-1)^{k+1}(2k)!}{(2\pi n)^{2k}} \cos(2\pi nx)$$

Letting $x = 0$, $B_{2k}(0)$ is just B_{2k} the Bernoulli number(s). We can now move everything not got to do with n in the sum to the other side, leaving behind what is the zeta function of $2k$ for integer $k \in \mathbb{Z}^+$.

$$\therefore B_{2k} = \sum_{n=1}^{\infty} \frac{2(-1)^{k+1}(2k)!}{(2\pi n)^{2k}}$$

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n^{2k}} = \zeta(2k) = \frac{(-1)^{k+1}(2\pi)^{2k}B_{2k}}{2(2k)!}$$

Finally, as B_{2k} alternates in sign (negative for even k), $(-1)^{k+1}B_{2k} = |B_{2k}|$, and we get our answer.

$$\therefore \zeta(2k) = \frac{(2\pi)^{2k}|B_{2k}|}{2(2k)!}$$

This is a very cool solution, showing how the π part increases by two in power, with the coefficient of π being a difficult to recognise pattern involving a power of two, a factorial, AND Bernoulli numbers.

METHOD 2: Interestingly, there is another cool way of deriving this result. The other way is by finding the Fourier series of x^{2m} on $[-\pi, \pi]$, then substituting in $x = \pi$. This doesn't give us the solution to $\zeta(2k)$ directly, but can be equated with a rearrangement of one of the definitions of the Bernoulli polynomial, to which it then drops out. Let's do it.

$$x^{2m} = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx)$$

Lets start with a_0.

$$a_0 = \frac{2}{\pi} \int_0^\pi x^{2m}\, dx = \frac{2\pi^{2m+1}}{\pi(2m+1)}$$

Now we solve a_n. Letting the integral be I_{2m},

$$a_n = \frac{2}{\pi} \int_0^\pi x^{2m} \cos(nx)\, dx = \frac{2}{\pi} I_{2m}$$

$$\text{IBP with } u = x^{2m}, dv = \cos(nx)$$

$$\therefore I_{2m} = \int_0^\pi x^{2m} \cos(nx)\, dx = \frac{\pi^{2m} \sin(n\pi)}{n} - \frac{2m}{n} \int_0^\pi x^{2m-1} \sin(nx)\, dx$$

$$= -\frac{2m}{n} \int_0^\pi x^{2m-1} \sin(nx)\, dx$$

$$\text{IBP with } u = x^{2m-1}, dv = \sin(nx)$$

$$= -\frac{2m}{n} \left(-\frac{\pi^{2m-1} \cos(n\pi)}{n} + \frac{2m-1}{n} \int_0^\pi x^{2m-2} \cos(nx)\, dx \right)$$

For integer n, $\cos(n\pi) = (-1)^n$.

$$= -\frac{2m}{n} \left(-\frac{\pi^{2m-1}(-1)^n}{n} + \frac{2m-1}{n} I_{2m-2} \right)$$

$$= \frac{2m\big(\pi^{2m-1}(-1)^n - (2m-1) I_{2m-2}\big)}{n^2}$$

Solving the base case $m = 1$ gives us our recurrence that we can use to solve for I_{2m}.

$$I_0 = \int_0^\pi 0\, dx = 0$$

$$\therefore I_2 = \frac{2\pi(-1)^n}{n^2}$$

$$I_4 = \frac{4(-1)^n(\pi^3 - 3I_2)}{n^2} = \frac{4\pi^3(-1)^n}{n^2} - \frac{4 \cdot 3 \cdot 2\pi(-1)^n}{n^4}$$

$$I_6 = \frac{6(-1)^n(\pi^5 - 5I_4)}{n^2} = \frac{6\pi^5(-1)^n}{n^2} - \frac{6 \cdot 5 \cdot 4\pi^3(-1)^n}{n^4} + \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2\pi(-1)^n}{n^6}$$

By now, we are able to notice that

$$I_{2m} = \sum_{k=1}^{m} \frac{(-1)^n(-1)^{k+1}(2m)!\pi^{2m-2k+1}}{n^{2k}(2m+1-2k)!}$$

Finally, as $a_n = \frac{2}{\pi} I_{2m}$,

$$a_n = \frac{2}{\pi} \sum_{k=1}^{m} \frac{(-1)^n(-1)^{k+1}(2m)!\pi^{2m-2k+1}}{n^{2k}(2m+1-2k)!}$$

We can now substitute this, alongside a_0, into our fourier series for x^{2m} from $-\pi$ to π.

$$\therefore x^{2m} = \frac{\pi^{2m+1}}{\pi(2m+1)} + \frac{2}{\pi}\sum_{n=1}^{\infty}\sum_{k=1}^{m} \frac{(-1)^n(-1)^{k+1}(2m)!\pi^{2m-2k+1}}{n^{2k}(2m+1-2k)!}\cos(nx)$$

We are now going to make some adjustments to both sides in order to get the sum how we want it. First, we let $x = \pi$, and move some of the terms over to the other side that are not connected to the sum.

$$\therefore \sum_{n=1}^{\infty}\sum_{k=1}^{m} \frac{(-1)^n(-1)^{k+1}(2m)!\pi^{2m-2k+1}}{n^{2k}(2m+1-2k)!}\cos(n\pi) = \frac{\pi^{2m+1}}{2} - \frac{\pi^{2m+1}}{2(2m+1)}$$

$$\therefore \sum_{k=1}^{m} \frac{(-1)^{k+1}(2m)!\pi^{2m-2k+1}\zeta(2k)}{(2m+1-2k)!} = \frac{m\pi^{2m+1}}{(2m+1)}$$

$$\therefore \sum_{k=1}^{m} \frac{(-1)^{k+1}\zeta(2k)}{\pi^{2k}(2m+1-2k)!} = \frac{m}{(2m+1)!} \tag{8}$$

We now wish to equate this to a sum from $n = 1$ to m involving Bernoulli numbers. To do this, we first use the definition that

$$B_m(x) = \sum_{k=0}^{m} \frac{m!}{k!(m-k)!} B_k x^{m-k}$$

$$\therefore B_{2m+1}(x) = \sum_{k=0}^{2m+1} \frac{(2m+1)!}{k!(2m+1-k)!} B_k x^{2m+1-k}$$

Odd indexed Bernoulli polynomials at $x = 1/2$ are equal to zero.

$$\therefore \sum_{k=0}^{2m+1} \frac{(2m+1)!}{k!(2m+1-k)!2^{2m+1-k}} B_k = 0$$

$$\therefore \sum_{k=0}^{2m+1} \frac{(2m+1)!2^k}{k!(2m+1-k)!} B_k = 0$$

We will now break up this sum into its $k = 0$ term, $k = 1$, term, all following even terms up to m, and all following odd terms up to m. Recall that $B_1 = -\frac{1}{2}$. Thus, our sum from before becomes

$$1 - (2m+1) + \sum_{k=1}^{m} \frac{(2m+1)!2^{2k}}{(2k)!(2m+1-2k)!} B_{2k} + \sum_{k=1}^{m} \frac{(2m+1)!2^{2k+1}}{(2k+1)!(2m-2k)!} B_{2k+1} = 0$$

$B_{2k+1} = 0$ for all non-negative integer k.

$$\therefore \sum_{k=0}^{m} \frac{(2m+1)!2^{2k}}{(2k)!(2m+1-2k)!} B_{2k} = 2m$$

$$\therefore \sum_{k=1}^{m} \frac{2^{2k-1}}{(2k)!(2m+1-2k)!} B_{2k} = \frac{m}{(2m+1)!}$$

This now has the exact same answer as the other sum (equation 8) from $k = 1$ to m. Thus, as both summands have the same $1/(2m-2k+1)!$ structure, we hypothesise that perhaps the summands are equal. I'll omit the proof here, but by using induction, this is indeed the case. Thus, we can equate them in order to get our solution for $\zeta(2k), k \in \mathbb{Z}^+$.

$$\therefore \frac{(-1)^{k+1}\zeta(2k)}{\pi^{2k}(2m+1-2k)!} = \frac{2^{2k-1}}{(2k)!(2m+1-2k)!} B_{2k}$$

$$\therefore \zeta(2k) = \frac{(-1)^{k+1}(2\pi)^{2k}B_{2k}}{2(2k)!}$$

$$= \frac{(2\pi)^{2k}|B_{2k}|}{2(2k)!}$$

And we're done, nicely arriving at the same answer as via the first method.

18.3 Odd Dirichlet Beta formula using Fourier series

As a reminder, the Dirichlet Beta function is

$$\beta(k) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^k}$$

Like how there is a formula for the Zeta function at even integers $k \in \mathbb{Z}^+$, there is a formula for the Dirichlet Beta function at odd integers, $k \in \mathbb{N}$. To find this, we wish to find the Fourier series for the Even indexed Euler Polynomials, $E_{2k}(x)$. Like $B_{2k}(x)$, these polynomials are symmetric around $x = 1/2$, However, while $B_{2k}(x)$ has a period of one from $[0,1]$, $E_{2k}(x)$ has a period of two. Thus it is all the cosine terms that vanish, and its Fourier series from $[0,1]$ is

$$E_{2k}(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \sin(\pi n x)$$

First solving a_0,

$$a_0 = 2\int_0^1 E_{2k}(x)\,dx = \frac{2(E_{2k+1}(1) - E_{2k+1}(0))}{2k+1}$$

Note that

$$E_k(x+1) = E_k(x) + 2x^k$$

As $k \in \mathbb{N}$,

$$\therefore E_{2k+1}(1) - E_{2k+1}(0) = 2 \cdot 0^{2k+1} = 0$$

$$\therefore a_0 = 0$$

Now we solve for a_n, letting the following integral be I_{2k}.

$$a_n = 2\int_0^1 E_{2k}(x)\sin(\pi n x)\,dx = 2I_{2k}$$

Before we solve for I_{2k}, we should first solve I_0 as $k = 0$ is a special case that we want to ignore when trying to solve for I_{2k} in general. Note that $n \in \mathbb{Z}^+$, and that $E_0(x) = 1$.

$$I_0 = \int_0^1 E_0(x)\sin(\pi n x)\,dx = \int_0^1 \sin(\pi n x)\,dx = \frac{1-(-1)^n}{\pi n}$$

Alright, now that we only need to consider all $k \in \mathbb{Z}^+$, we perform integration by parts twice.

$$\text{IBP with } u = E_{2k}(x), dv = \sin(\pi n x)$$

$$\therefore I_{2k} = -\frac{E_{2k}(1)\cos(\pi n)}{\pi n} + \frac{E_{2k}(0)\cos(0)}{\pi n} + \frac{2k}{\pi n}\int_0^1 E_{2k-1}(x)\cos(\pi n x)\,dx$$

$$= -\frac{1}{\pi n}\left(E_{2k}(1)(-1)^n - E_{2k}(0) - 2k\int_0^1 E_{2k-1}(x)\cos(\pi n x)\,dx\right)$$

As $E_k(x+1) = E_k(x) + 2x^k$,

$$E_{2k}(1)(-1)^n = (-1)^n E_{2k}(0) + 2(-1)^n \cdot 0^{2k}$$

Then, as $E_{2k}(0) = 0$ for all $k \in \mathbb{Z}^+$, and $0^{2k} = 0$ (as k cannot be zero).

$$E_{2k}(1)(-1)^n = 2(-1)^n \cdot 0^{2k} = 0$$

$$\therefore I_{2k} = \frac{2k}{\pi n}\int_0^1 E_{2k-1}(x)\cos(\pi n x)\,dx$$

$$\text{IBP with } u = E_{2k-1}(x), dv = \cos(\pi n x)$$

$$= \frac{2k}{\pi n}\left(\frac{E_{2k-1}(1)\sin(\pi n)}{\pi n} - \frac{E_{2k-1}(0)\sin 0}{\pi n} - \frac{2k-1}{\pi n}\int_0^1 E_{2k-2}(x)\sin(\pi n x)\,dx\right)$$

$$= -\frac{2k}{\pi n}\left(\frac{2k-1}{\pi n}\int_0^1 E_{2k-2}(x)\sin(\pi n x)\,dx\right)$$

Notice that the remaining integral is I_{2k-2}.

$$\therefore I_{2k} = -\frac{2k(2k-1)}{(\pi n)^2} I_{2k-2}$$

Using I_0 from above, we get the recurrence

$$I_2 = -\frac{2}{(\pi n)^2} I_0 = -\frac{2(1-(-1)^n)}{(\pi n)^3}$$

$$I_4 = -\frac{4\cdot 3}{(\pi n)^2} I_2 = \frac{4\cdot 3\cdot 2(1-(-1)^n)}{(\pi n)^5}$$

$$I_6 = -\frac{6\cdot 5}{(\pi n)^2} I_4 = -\frac{6\cdot 5\cdot 4\cdot 3\cdot 2(1-(-1)^n)}{(\pi n)^7}$$

By now, we can notice that

$$I_{2k} = \frac{(-1)^{k+1}(2k)!(1-(-1)^n)}{(\pi n)^{2k+1}}$$

Notice how this is zero for all even n. We can therefore disregard all even n terms, and only consider odd n.

$$\therefore I_{2k} = \frac{2(-1)^k(2k)!}{(\pi(2n-1))^{2k+1}}$$

Then, as $a_n = 2I_{2k}$ (for odd n since we cut all the even terms)

$$a_n = \frac{4(-1)^k(2k)!}{(\pi(2n-1))^{2k+1}}$$

As we now have both a_0 and a_n, we can put these into the Fourier series for $E_{2k}(x), 0 \le x \le 1$. Note that the $\sin(\pi n x)$ term becomes $\sin(\pi(2n-1)x)$ as well, due to the even terms being cut.

$$\therefore E_{2k}(x) = \sum_{n=1}^{\infty} \frac{4(-1)^k(2k)!}{(\pi(2n-1))^{2k+1}} \sin(\pi(2n-1)x)$$

$$= \sum_{n=0}^{\infty} \frac{4(-1)^k(2k)!}{(\pi(2n+1))^{2k+1}} \sin(\pi(2n+1)x)$$

Finally, we use the fact that the connection between the Euler polynomials and Euler numbers is $E_k = 2^k E_k(1/2)$. As such, let $x = 1/2$.

$$\therefore \sum_{n=0}^{\infty} \frac{4(-1)^k(2k)!}{(\pi(2n+1))^{2k+1}} \sin\left(\frac{\pi(2n+1)}{2}\right) = \frac{E_{2k}}{2^{2k}}$$

Now, as $\sin\left(\frac{\pi(2n+1)}{2}\right) = (-1)^n$ for integer n, we move terms where possible to the other side to obtain

$$\therefore \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^{2k+1}} = \beta(2k+1) = \frac{(-1)^k \pi^{2k+1} E_{2k}}{2^{2k+2}(2k)!}$$

$\beta(2k+1)$ appears! Finally, as E_{2k} alternates in sign (negative for odd k), $(-1)^k E_{2k} = |E_{2k}|$.

$$\therefore \beta(2k+1) = \frac{\pi^{2k+1}|E_{2k}|}{2^{2k+2}(2k)!}$$

Amazing! So similar to $\zeta(2k)$ and the method there, but different.

18.4 log Barnes-G and K-function

On page 60 we solved Raabe's integral, and found that

$$\int_0^1 \ln\Gamma(x)\,dx = \frac{1}{2}\ln(2\pi)$$

In this section we will use known series expansions and formulas previously found here in order to solve

$$\int_0^1 \ln G(x)\,dx$$

Where $G(x)$ is the Barnes-G function. To start, we first use the relation between the Barnes-G function and the integral of $\ln\Gamma(t)$ from zero to x, which was found on page 90.

$$\int_0^x \ln\Gamma(t)\,dt = \frac{x}{2}\ln(2\pi) + x\ln\Gamma(x) - \ln G(1+x) + \frac{x(1-x)}{2}$$

Rearranging and using the fact that $G(1+x) = \Gamma(x)G(x)$, we get

$$\ln G(x) = \frac{x}{2}\ln(2\pi) + (x-1)\ln\Gamma(x) + \frac{x(1-x)}{2} - \int_0^x \ln\Gamma(t)\,dt$$

Now we integrate both sides from zero to one.

$$\therefore \int_0^1 \ln G(x)\,dx = \frac{\ln(2\pi)}{2}\int_0^1 x\,dx + \int_0^1 (x-1)\ln\Gamma(x)\,dx + \frac{1}{2}\int_0^1 x(1-x)\,dx - \int_0^1\int_0^x \ln\Gamma(t)\,dt\,dx$$

$$= \frac{1}{4}\ln(2\pi) + \frac{1}{2}\int_0^1 x - x^2\,dx + \int_0^1 x\ln\Gamma(x)\,dx - \int_0^1 \ln\Gamma(x)\,dx - \int_0^1\int_0^x \ln\Gamma(t)\,dt\,dx$$

Using page 61,

$$= \frac{1}{4}\ln(2\pi) + \frac{1}{4} - \frac{1}{6} + \int_0^1 x\ln\Gamma(x)\,dx - \frac{1}{2}\ln(2\pi) - \int_0^1\int_0^x \ln\Gamma(t)\,dt\,dx$$

$$= \frac{1}{12} - \frac{1}{4}\ln(2\pi) + \int_0^1 x\ln\Gamma(x)\,dx - \int_0^1\int_0^x \ln\Gamma(t)\,dt\,dx$$

We now wish to reverse the order of the double integral on the right. To do this. We know that as x ranges from zero to one, t ranges from 0 to the line $t = x$. When swapping order, this is equivalent to t ranging from zero to one, and then x ranges from $x = t$ to one. Thus, our double integral on the right becomes

$$= \frac{1}{12} - \frac{1}{4}\ln(2\pi) + \int_0^1 x\ln\Gamma(x)\,dx - \int_0^1\int_t^1 \ln\Gamma(t)\,dx\,dt$$

$$= \frac{1}{12} - \frac{1}{4}\ln(2\pi) + \int_0^1 x\ln\Gamma(x)\,dx - \int_0^1 (1-t)\ln\Gamma(t)\,dt$$

$$= \frac{1}{12} - \frac{3}{4}\ln(2\pi) + 2\int_0^1 x\ln\Gamma(x)\,dx \tag{9}$$

Awesome. Let the remaining integral be I_1. Now, in order to solve this integral, we are able to use the Fourier series of $\ln\Gamma(x)$ from zero to one. I might derive this Fourier series myself another time, but for today we will just use its result

$$\ln\Gamma(x) = \left(\frac{1}{2} - x\right)(\gamma + \ln 2) + (1-x)\ln\pi - \frac{1}{2}\ln(\sin(\pi x)) + \frac{1}{\pi}\sum_{n=2}^{\infty}\frac{\ln n}{n}\sin(2\pi n x)$$

$$\therefore I_1 = \int_0^1 x\ln\Gamma(x)\,dx =$$

$$(\gamma+\ln 2)\int_0^1 x\left(\frac{1}{2} - x\right)dx + \ln(\pi)\int_0^1 x(1-x)\,dx - \frac{1}{2}\int_0^1 x\ln(\sin(\pi x))\,dx + \frac{1}{\pi}\int_0^1 x\sum_{n=2}^{\infty}\frac{\ln n}{n}\sin(2\pi n x)\,dx$$

$$= \frac{-\gamma - \ln 2}{12} + \frac{1}{6}\ln\pi - \frac{1}{2}\int_0^1 x\ln(\sin(\pi x))\,dx + \frac{1}{\pi}\sum_{n=2}^{\infty}\frac{\ln n}{n}\int_0^1 x\sin(2\pi nx)\,dx$$

$$= \frac{-\gamma - \ln 2}{12} + \frac{1}{6}\ln\pi - \frac{1}{2}I_2 + \frac{1}{\pi}\sum_{n=2}^{\infty}\frac{\ln n}{n}I_3$$

Solving I_2, we use King's rule,

$$I_2 = \int_0^1 x\ln(\sin(\pi x))\,dx = \int_0^1 (1-x)\ln(\sin(\pi(1-x)))\,dx$$

$$= \int_0^1 (1-x)\ln(\sin(\pi x))\,dx$$

Rearranging,

$$\therefore 2I_2 = 2\int_0^1 x\ln(\sin(\pi x))\,dx = \int_0^1 \ln(\sin(\pi x))\,dx$$

$$u = \pi x, du = \pi\,dx$$

$$= \frac{1}{\pi}\int_0^{\pi}\ln(\sin u)\,du$$

$$= \frac{1}{\pi}\int_0^{\pi}\ln(\sin u)\,du$$

Using page 60,

$$= -\ln 2$$

$$\therefore I_2 = -\frac{1}{2}\ln 2$$

Now we solve for I_3.

$$I_3 = \int_0^1 x\sin(2\pi nx)\,dx$$

$$\text{IBP with } u = x, dv = \sin(2\pi nx)$$

$$= -\frac{1}{2\pi n} + \frac{1}{2\pi n}\int_0^1 \cos(2\pi nx)\,dx$$

$$= -\frac{1}{2\pi n}$$

Substituting these back into our expression for I_1,

$$\therefore I_1 = \frac{-\gamma - \ln 2}{12} + \frac{1}{6}\ln\pi + \frac{1}{4}\ln 2 - \frac{1}{2\pi^2}\sum_{n=2}^{\infty}\frac{\ln n}{n^2}$$

The remaining sum is known to equal the following

$$\sum_{n=1}^{\infty}\frac{\ln n}{n^2} = \frac{\pi^2}{6}(12\ln A - \gamma - \ln(2\pi))$$

$$\therefore I_1 = \int_0^1 x\ln\Gamma(x)\,dx = \frac{-\gamma - \ln 2}{12} + \frac{1}{6}\ln\pi + \frac{1}{4}\ln 2 - \frac{1}{12}(12\ln A - \gamma - \ln(2\pi))$$

$$= \frac{1}{4}\ln(2\pi) - \ln A$$

A nice result in of itself! And finally, we can substitute this result back into equation 9, to obtain our solution to the original problem.

$$\therefore \frac{1}{12} - \frac{3}{4}\ln(2\pi) + 2\int_0^1 x\ln\Gamma(x)\,dx = \frac{1}{12} - \frac{3\ln(2\pi)}{4} + \frac{\ln(2\pi)}{2} - 2\ln A$$

$$= \frac{1}{12} - \frac{1}{4}\ln(2\pi) - 2\ln A$$

$$\boxed{\therefore \int_0^1 \ln G(x)\,dx = \frac{1}{12} - \frac{1}{4}\ln(2\pi) - 2\ln A}$$

And we're done! However... I also noticed that there is a relationship between the Barnes-G function, the Gamma function, and the K-function. Recall that the Gamma function generalises the factorial to real/complex numbers, that the Barnes-G function generalises the superfactorials to the real/complex numbers, and the K-function generalises the hyperfactorials to the real/complex numbers. The relationship between the three is as so.

$$K(x)G(x) = \Gamma^{x-1}(x)$$

This means we can now solve for the integral from zero to one of the natural log of the K-function too!

$$\int_0^1 \ln K(x)\,dx$$

$$= \int_0^1 \ln\left(\frac{\Gamma^{x-1}(x)}{G(x)}\right)dx$$

$$= \int_0^1 (x-1)\ln\Gamma(x)\,dx - \int_0^1 \ln G(x)\,dx$$

$$= \int_0^1 x\ln\Gamma(x)\,dx - \int_0^1 \ln\Gamma(x)\,dx - \int_0^1 \ln G(x)\,dx$$

Using our results from page 61, the previous page, and this page, this is equal to

$$= \frac{1}{4}\ln(2\pi) - \ln A - \frac{1}{2}\ln(2\pi) - \frac{1}{12} + \frac{1}{4}\ln(2\pi) + 2\ln A$$

$$= \ln A - \frac{1}{12}$$

$$\boxed{\therefore \int_0^1 \ln K(x)\,dx = \ln A - \frac{1}{12}}$$

Nice! Note that it is also possible to solve for this one using the direct definition of the K-function. After a few steps, it also boils down to I_1 with some added terms.

All in all, we have now found the three following integrals, all with very cool solutions.

$$\int_0^1 \ln\Gamma(x)\,dx = \frac{1}{2}\ln(2\pi)$$

$$\int_0^1 \ln G(x)\,dx = \frac{1}{12} - \frac{1}{4}\ln(2\pi) - 2\ln A$$

$$\int_0^1 \ln K(x)\,dx = \ln A - \frac{1}{12}$$

18.5 Generalisation for double gamma sums

From page 128-129, we derived the following Gamma sum formula.

$$\sum_{n=0}^{\infty} \frac{\alpha^n}{n!}\Gamma^2\left(\frac{2n+1}{4}\right) = 4\sqrt{\pi}K\left(\frac{\sqrt{2+\alpha}}{2}\right)$$

In this section, we will solve for a general formula for the following sum, which uses the same method.

$$\sum_{n=0}^{\infty} \frac{(-2)^n}{n!}\Gamma\left(\frac{n}{2}+\alpha\right)\Gamma\left(\frac{n}{2}+\beta\right)$$

As we are performing the same method, we'll be explaining most of the steps a lot less than last time. Using the formula for the gamma function,

$$= \sum_{n=0}^{\infty} \frac{(-2)^n}{n!} \int_0^{\infty}\int_0^{\infty} e^{-(x+y)} x^{\frac{n}{2}+\alpha-1} y^{\frac{n}{2}+\beta-1}\, dx\, dy$$

$$= \int_0^{\infty}\int_0^{\infty} e^{-(x+y)} x^{\alpha-1} y^{\beta-1} \sum_{n=0}^{\infty} \frac{(-2\sqrt{xy})^n}{n!}\, dx\, dy$$

$$= \int_0^{\infty}\int_0^{\infty} e^{-(x+y+2\sqrt{xy})} x^{\alpha-1} y^{\beta-1}\, dx\, dy$$

$$x = u^2, y = v^2, dx = 2u\, du, dy = 2v\, dv$$

$$= 4\int_0^{\infty}\int_0^{\infty} e^{-(u^2+v^2+2uv)} u^{2\alpha-1} v^{2\beta-1}\, du\, dv$$

$$= 4\int_0^{\infty}\int_0^{\infty} e^{-(u+v)^2} u^{2\alpha-1} v^{2\beta-1}\, du\, dv$$

Now is where we can use polar coordinates

$$u = r\cos\theta, v = r\sin\theta, du\, dv = r\, dr\, d\theta$$

$$= 4\int_0^{\frac{\pi}{2}}\int_0^{\infty} e^{-r^2(\cos\theta+\sin\theta)^2} r^{2\alpha+2\beta-1} (\cos\theta)^{2\alpha-1} (\sin\theta)^{2\beta-1}\, dr\, d\theta$$

$$= 4\int_0^{\frac{\pi}{2}}\int_0^{\infty} e^{-r^2(1+\sin(2\theta))} r^{2\alpha+2\beta-1} (\cos\theta)^{2\alpha-1} (\sin\theta)^{2\beta-1}\, dr\, d\theta$$

$$w = r^2(1+\sin(2\theta)), dw = 2r(1+\sin(2\theta))\, dr$$

$$= 2\int_0^{\frac{\pi}{2}}\int_0^{\infty} \frac{1}{1+\sin(2\theta)} e^{-w} \left(\frac{w}{1+\sin(2\theta)}\right)^{\alpha+\beta-1} (\cos\theta)^{2\alpha-1} (\sin\theta)^{2\beta-1}\, dw\, d\theta$$

$$= 2\int_0^{\frac{\pi}{2}}\int_0^{\infty} e^{-w} w^{\alpha+\beta-1} (1+\sin(2\theta))^{-(\alpha+\beta)} (\cos\theta)^{2\alpha-1} (\sin\theta)^{2\beta-1}\, dw\, d\theta$$

We can now solve the w integral in terms of the gamma function.

$$= 2\Gamma(\alpha+\beta) \int_0^{\frac{\pi}{2}} (1+\sin(2\theta))^{-(\alpha+\beta)} (\cos\theta)^{2\alpha-1} (\sin\theta)^{2\beta-1}\, d\theta$$

$$= 2\Gamma(\alpha+\beta) \int_0^{\frac{\pi}{2}} (\cos\theta+\sin\theta)^{-(2\alpha+2\beta)} (\cos\theta)^{2\alpha-1} (\sin\theta)^{2\beta-1}\, d\theta$$

The remaining integral is now in the form of the beta function.

$$= 2\Gamma(\alpha+\beta)B(2\alpha, 2\beta)$$

Which, when written in terms of the gamma function, is

$$= 2\Gamma(\alpha+\beta)\frac{\Gamma(2\alpha)\Gamma(2\beta)}{\Gamma(2\alpha+2\beta)}$$

And we're done.

$$\therefore \sum_{n=0}^{\infty}\frac{(-2)^n}{n!}\Gamma\left(\frac{n}{2}+\alpha\right)\Gamma\left(\frac{n}{2}+\beta\right) = 2\Gamma(\alpha+\beta)\frac{\Gamma(2\alpha)\Gamma(2\beta)}{\Gamma(2\alpha+2\beta)}$$

This works i believe for $\Re(\alpha+\beta) < \frac{3}{2}$, but it may be stricter than that depending on the values of α and β. Notice that $2\alpha+2\beta \neq -\mathbb{N}$ for instance. From here, we can get the following cool identity. When $\alpha+\beta = 1/2$, we can use the reflection formula to obtain

$$\sum_{n=0}^{\infty}\frac{(-2)^n}{n!}\Gamma\left(\frac{n}{2}+\alpha\right)\Gamma\left(\frac{n+1}{2}-\alpha\right) = 2\pi^{\frac{3}{2}}\csc(2\pi\alpha)$$

Which works for $\alpha \neq \frac{m}{2}, m \in \mathbb{Z}$. This gives results like

$$\sum_{n=0}^{\infty}\frac{(-2)^n}{n!}\Gamma^2\left(\frac{2n+1}{4}\right) = 2\pi^{\frac{3}{2}}$$

$$\sum_{n=0}^{\infty}\frac{(-2)^n}{n!}\Gamma\left(\frac{3n+1}{6}\right)\Gamma\left(\frac{3n+2}{6}\right) = \frac{4}{\sqrt{3}}\pi^{\frac{3}{2}}$$

$$\sum_{n=0}^{\infty}\frac{(-2)^n}{n!}\Gamma\left(\frac{4n+1}{8}\right)\Gamma\left(\frac{4n+3}{8}\right) = 2\sqrt{2}\pi^{\frac{3}{2}}$$

$$\sum_{n=0}^{\infty}\frac{(-2)^n}{n!}\Gamma\left(\frac{5n+2}{10}\right)\Gamma\left(\frac{5n+3}{10}\right) = \frac{8\pi^{\frac{3}{2}}}{\sqrt{10+2\sqrt{5}}}$$

$$\sum_{n=0}^{\infty}\frac{(-2)^n}{n!}\Gamma\left(\frac{6n+1}{12}\right)\Gamma\left(\frac{6n+5}{12}\right) = 4\pi^{\frac{3}{2}}$$

Lastly, when we differentiate both sides of our formula with respect to α, we get the result

$$\sum_{n=0}^{\infty}\frac{(-2)^n}{n!}\Gamma\left(\frac{n}{2}+\alpha\right)\Gamma\left(\frac{n}{2}+\beta\right)\psi\left(\frac{n}{2}+\alpha\right) = 2\Gamma(\alpha+\beta)\frac{\Gamma(2\alpha)\Gamma(2\beta)}{\Gamma(2\alpha+2\beta)}(\psi(\alpha+\beta)+2\psi(2\alpha)-2\psi(2\alpha+2\beta))$$

This similarly has many cool results. For cases when $\alpha=\beta$, we obtain

$$\sum_{n=0}^{\infty}\frac{(-2)^n}{n!}\Gamma^2\left(\frac{n}{2}+\alpha\right)\psi\left(\frac{n}{2}+\alpha\right) = 2\frac{\Gamma^3(2\alpha)}{\Gamma(4\alpha)}(3\psi(2\alpha)-2\psi(4\alpha))$$

Here are a couple results using the above formula.

$$\sum_{n=0}^{\infty}\frac{(-2)^n}{n!}\Gamma^2\left(\frac{2n+1}{4}\right)\psi\left(\frac{2n+1}{4}\right) = -2\pi^{\frac{3}{2}}(6\ln 2+\gamma)$$

$$\sum_{n=0}^{\infty}\frac{(-2)^n}{n!}\Gamma^2\left(\frac{4n+1}{8}\right)\psi\left(\frac{4n+1}{8}\right) = -\frac{2}{\sqrt{\pi}}\Gamma^3\left(\frac{1}{4}\right)\left(\frac{3\pi}{2}+5\ln 2+\gamma\right)$$

Out of all of the results on this page, a few will be included in the list of results at the top.

18.6 Return to contour integration (2 integrals)

Integral 1: Haven't done much contour integration so here is one.

$$\int_{-\infty}^{\infty} \frac{\sin x}{x(x^2+1)}\, dx$$

To solve via complex methods, we use the fact that $\sin x = \Im(e^{ix})$.

$$= \Im \int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2+1)}\, dx$$

$$= \Im \int_{-\infty}^{\infty} \frac{e^{ix}}{x(x-i)(x+i)}\, dx$$

Let the integrand above be $f(x)$. We now wish to evaluate this via contour integration. Namely, we can choose to use a semicircle contour with a second small semicircle at the origin. This contour consists of a real axis segment from $-R$ to $-\epsilon$, a semicircle C_ϵ of radius ϵ centred at the origin indented into the upper half plane traversed clockwise, a real axis segment from ϵ to R, and a large semicircle C_R of radius R centred at the origin in the upper half plane traversed anti-clockwise. As we are not using the lower-half plane, our two residues are at $x = i$, and the boundary residue $x = 0$.

$$\text{Res}(f, i) = \lim_{x\to i}(x-i)\frac{e^{ix}}{x(x-i)(x+i)} = \frac{e^{-1}}{i(i+i)} = -\frac{1}{2e}$$

$$\text{Res}(f, 0) = \lim_{x\to 0} \frac{xe^{ix}}{x(x-i)(x+i)} = \frac{e^0}{(-i)i} = 1$$

Now, using the Cauchy residue theorem, the integral over the whole loop is equal to $2\pi i$ times the residues strictly inside of the contour.

$$\therefore \oint_f f(z)\, dz = \int_{-R}^{-\epsilon} f(x)\, dx + \int_{C_\epsilon} f(z)\, dz + \int_{\epsilon}^{R} f(x)\, dx + \int_{C_R} f(z)\, dz = 2\pi i\left(-\frac{1}{2e}\right)$$

As $R \to \infty$, the ML-lemma clearly shows that

$$\lim_{R\to\infty} \int_{C_R} f(z)\, dz = 0$$

Then, for the $x = 0$ pole on the boundary, we use the fractional residue theorem. As we are going around only half a circle, and as that semicircle is being integrated clockwise, the contribution is $-\pi i$ times the residue. Thus, as $\epsilon \to 0$,

$$\lim_{\epsilon\to 0} \int_{C_\epsilon} f(z)\, dz = -\pi i \text{Res}(f, 0) = -\pi i$$

Putting everything together, and letting $R \to \infty$ and $\epsilon \to 0$, we obtain

$$\oint f(z)\, dz = \int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2+1)}\, dx - \pi i = 2\pi i\left(-\frac{1}{2e}\right)$$

$$\therefore \int_{-\infty}^{\infty} \frac{e^{ix}}{x(x^2+1)}\, dx = 2\pi i\left(-\frac{1}{2e}\right) + \pi i = \pi i\left(1 - \frac{1}{e}\right)$$

And since our original integral is the imaginary part of the above,

$$\therefore \int_{-\infty}^{\infty} \frac{\sin x}{x(x^2+1)}\, dx = \pi\left(1 - \frac{1}{e}\right)$$

Awesome! Using similar method, we can also find that

$$\int_{-\infty}^{\infty} \frac{\sin x}{x(x^4+5x^2+4)}\, dx = \frac{\pi}{3e}\left(\frac{1}{4e} - 1\right) + \frac{\pi}{4}$$

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^4+5x^2+4}\, dx = \frac{\pi}{3e}\left(1 - \frac{1}{2e}\right)$$

and more similar integrals of the like.

Integral 2: This one will also be using a similarsemicircle contour in the upper-half plane C_R, with a second semicircle C_ϵ around $x = 0$.

$$\int_0^\infty \frac{\ln x}{x^4 - 1}\,dx$$

$$= \int_0^\infty \frac{\ln x}{(x-1)(x+1)(x-i)(x+i)}\,dx$$

Let the integrand above be $f(x)$. We define the branch cut for $\ln z$ to be along the negative imaginary axis. This means that for $x > 0, \ln z = \ln x$, and for $x < 0, \ln z = \ln|x| + i\pi$. Now, we note that the denominator of the integrand has four poles, at $-1, 1, -i$, and i. However, the $x = -i$ pole is not in the upper-half plane so it can be ignored. $x = -1$ and $x = 1$ are poles on the boundary and are integrated clockwise, meaning that that their residues only have a half contribution and are multiplied by -1. This also means we need to add two more semicircle indentations over $x = -1$ and $x = 1$, with radius ρ and σ respectively

$$\text{Res}(f, -1) = \lim_{x\to-1}(x+1)\frac{\ln x}{(x-1)(x+1)(x-i)(x+i)} = \frac{\ln(-1)}{-2(-1-i)(-1+i)} = -\frac{\pi i}{4}$$

$$\text{Res}(f, 1) = \lim_{x\to1}(x-1)\frac{\ln x}{(x-1)(x+1)(x-i)(x+i)} = \frac{\ln 1}{2(1-i)(1+i)} = 0$$

Since we get a 0/0 case when plugging in $x = 1$ into $f(x)$, the pole is removable and we can ignore this semicircle indentation entirely.

$$\text{Res}(f, i) = \lim_{x\to i}(x-i)\frac{\ln x}{(x-1)(x+1)(x-i)(x+i)}$$

$$= \frac{\ln i}{2i(i-1)(i+1)} = -\frac{\ln i}{4i} = -\frac{\pi}{8}$$

Thus, as $x = i$ is the only enclosed pole,

$$\oint f(z)\,dz = \int_{-R}^{-1+\rho} f(z)\,dz + \int_{C_\rho} f(z)\,dz + \int_{-1-\rho}^{\epsilon} f(z)\,dz + \int_{C_\epsilon} f(z)\,dz + \int_\epsilon^R f(x)\,dx + \int_{C_R} f(z)\,dz = 2\pi i\left(-\frac{\pi}{8}\right)$$

As $R \to \infty$, the ML-lemma clearly shows that

$$\lim_{R\to\infty}\int_{C_R} f(z)\,dz = 0$$

Then, as $\epsilon \to 0$,

$$\lim_{\epsilon\to0}\int_{C_\epsilon} f(z)\,dz = \lim_{\epsilon\to0} zf(z) = 0$$

And lastly, as $x = -1$ is a boundary pole integrated clockwise,

$$\lim_{\rho\to0}\int_{C_\rho} f(z)\,dz = -i\pi\text{Res}(f, -1) = -\frac{\pi^2}{4}$$

Putting everything together, as $R \to \infty$, $\epsilon \to 0$, and $\rho \to 0$, we obtain

$$\oint f(z)\,dz = \int_{-\infty}^{-1} \frac{\ln z}{z^4-1}\,dz + \int_{-1}^{0} \frac{\ln z}{z^4-1}\,dz + \int_0^\infty \frac{\ln x}{x^4-1}\,dx - \frac{\pi^2}{4} = -\frac{\pi^2 i}{4}$$

$$\therefore \int_{-\infty}^{0} \frac{\ln z}{z^4-1}\,dz + \int_0^\infty \frac{\ln x}{x^4-1}\,dx = \frac{\pi^2}{4} - \frac{\pi^2 i}{4}$$

For the integral on the left, we are on the negative x-axis and thus $\ln z = \ln|x| + i\pi$

$$\therefore \int_{-\infty}^{0} \frac{\ln|x| + i\pi}{x^4-1}\,dx + \int_0^\infty \frac{\ln x}{x^4-1}\,dx = \frac{\pi^2}{4} - \frac{\pi^2 i}{4}$$

Then, on left integral only,

$$t = -x, dt = -\,dx$$

$$\therefore \int_0^\infty \frac{\ln t + i\pi}{(-t)^4 - 1}\,dt + \int_0^\infty \frac{\ln x}{x^4 - 1}\,dx = \frac{\pi^2}{4} - \frac{\pi^2 i}{4}$$

$$\therefore 2\int_0^\infty \frac{\ln x}{x^4 - 1}\,dx + i\pi \cdot \mathrm{PV}\int_0^\infty \frac{1}{t^4 - 1}\,dt = \frac{\pi^2}{4} - \frac{\pi^2 i}{4}$$

And finally, since we know our original integral is real, we can equate the real and imaginary components of both sides.

$$\therefore \int_0^\infty \frac{\ln x}{x^4 - 1}\,dx = \frac{\pi^2}{8}$$

Done! This was a very nice use of contour integration to solve an otherwise very difficult integral. In future I will find further integrals to solve using contour integration with other contours other than semicircles. We did have that nice problem on pages 145-149 that we solved first by a dog-bone contour then via a rectangular box contour, but we have not yet had any integrals using a keyhole contour for instance.

18.7 Filler problem using geometric sum

(26/6) As said in the subsection title, this one is very simple but it has a very cool answer so i must include it.

$$\int_0^\infty \frac{x^n}{e^x+1}\,dx$$

$$= \int_0^\infty \frac{x^n e^{-x}}{1+e^{-x}}\,dx$$

$$= \int_0^\infty \frac{x^n e^{-x}}{1-(-e^{-x})}\,dx$$

Using a geometric series (since $|-e^{-x}| < 1$ from zero to infinity),

$$= \int_0^\infty \sum_{k=1}^\infty x^n e^{-x}(-e^{-x})^{k-1}\,dx$$

$$= \sum_{k=1}^\infty (-1)^{k-1} \int_0^\infty x^n e^{-xk}\,dx$$

$$t = xk, dt = k\,dx$$

$$= \sum_{k=1}^\infty \frac{(-1)^{k-1}}{k^{n+1}} \int_0^\infty x^n e^{-t}\,dx$$

$$= \Gamma(n+1)\sum_{k=1}^\infty \frac{(-1)^{k-1}}{k^{n+1}}$$

$$= \Gamma(n+1)\sum_{k=1}^\infty \frac{(-1)^{k-1}}{k^{n+1}}$$

The sum part is even for odd k, and odd for even k. Thus, we can break up the sum like so:

$$= \Gamma(n+1)\left(\sum_{k=1}^\infty \frac{1}{(2k-1)^{n+1}} - \sum_{k=1}^\infty \frac{1}{(2k)^{n+1}}\right)$$

$$= \Gamma(n+1)\left(\sum_{k=1}^\infty \frac{1}{k^{n+1}} - 2\sum_{k=1}^\infty \frac{1}{(2k)^{n+1}}\right)$$

$$= \Gamma(n+1)\zeta(n+1)(1-2^{-n})$$

Done already!

$$\therefore \int_0^\infty \frac{x^n}{e^x+1}\,dx = \Gamma(n+1)\zeta(n+1)(1-2^{-n})$$

Using the same method, we can also obtain

$$\int_0^\infty \frac{x^n}{e^x-1}\,dx = \Gamma(n+1)\zeta(n+1)$$

18.8 Generalisation of integral via contour integration

Okay I may've lied when I said the next integral ill do via contour integration will be via a new contour I haven't used before. This is because I first want to show this generalised version of our integral from page 199.

$$\int_0^\infty \frac{\ln x}{x^2-1}\prod_{n=1}^{\alpha}\frac{1}{x^2+n^2}\,dx$$

This will be solved via the same contour as last time, but now we have α poles enclosed, all at a general $x = ki$ for integer $1 \leq k \leq \alpha$. We thus again the use the same branch cut for $\ln x$ too.

$$= \int_0^\infty \frac{\ln x}{(x-1)(x+1)}\prod_{n=1}^{\alpha}\frac{1}{(x-ni)(x+ni)}\,dx$$

Recall that the $x = 1$ pole is removable. This is trivially true for this integral due to how $\ln 1 = 0$. Let the integrand be $f(x)$. Finding our boundary residue at $x = -1$,

$$\text{Res}(f,-1) = \lim_{x\to -1}(x+1)\frac{\ln x}{(x-1)(x+1)}\prod_{n=1}^{\alpha}\frac{1}{(x-ni)(x+ni)}$$

$$= -\frac{\ln(-1)}{2}\prod_{n=1}^{\alpha}\frac{1}{(-1-ni)(-1+ni)} = -\frac{\pi i}{2}\prod_{n=1}^{\alpha}\frac{1}{n^2+1}$$

Now we look at one of the α enclosed pole. Choose a general pole to be $x = ki$ for integer $1 \leq k \leq \alpha$.

$$\text{Res}(f,ki) = \lim_{x\to ki}(x-ki)\frac{\ln x}{(x-1)(x+1)}\prod_{n=1}^{\alpha}\frac{1}{(x-ni)(x+ni)}$$

$$= \frac{\ln(ki)}{2ki(ki-1)(ki+1)}\prod_{\substack{n=1\\n\neq k}}^{\alpha}\frac{1}{(ki-ni)(ki+ni)} = -\frac{\ln(ki)}{2ki(k^2+1)}\prod_{\substack{n=1\\n\neq k}}^{\alpha}\frac{1}{(ki-ni)(ki+ni)}$$

Thus the sum of these residues is the sum from $k = 1$ to α of this expression above. However, we first need to solve that product.

$$P = \prod_{\substack{n=1\\n\neq k}}^{\alpha}\frac{1}{(ki-ni)(ki+ni)} = \left(\prod_{\substack{n=1\\n\neq k}}^{\alpha}(ki-ni)(ki+ni)\right)^{-1}$$

$$= \left(\prod_{\substack{n=1\\n\neq k}}^{\alpha}(n^2-k^2)\right)^{-1} = P_1^{-1}$$

Let's focus on the new product, and take the reciprocal at the end.

$$P_1 = \prod_{\substack{n=1\\n\neq k}}^{\alpha}(n^2-k^2)$$

$$= \prod_{\substack{n=1\\n\neq k}}^{\alpha}(n-k)\cdot\prod_{\substack{n=1\\n\neq k}}^{\alpha}(n+k) = P_2\cdot P_3$$

Solving P_2,

$$P_2 = \prod_{\substack{n=1\\n\neq k}}^{\alpha}(n-k) = (1-k)(2-k)\cdots(-1)\times(1)(2)\cdots(\alpha-1-k)(\alpha-k)$$

$$= (-1)^{k-1}(k-1)(k-2)\cdots(1)\times(\alpha-k)!$$

$$= (-1)^{k-1}(k-1)!(\alpha - k)!$$

And solving P_3,

$$P_3 = \prod_{\substack{n=1 \\ n\neq k}}^{\alpha} (n+k) = (1+k)(2+k)\cdots(2k-1) \times (2k+1)(2k+2)\cdots(\alpha-1+k)(\alpha+k)$$

$$= \frac{(\alpha+k)!}{(2k)k!}$$

Thus, as $P_1 = P_2 \cdot P_3$

$$\therefore P_1 = \prod_{\substack{n=1 \\ n\neq k}}^{\alpha} (n^2 - k^2) = \frac{(-1)^{k-1}(k-1)!(\alpha-k)!(\alpha+k)!}{(2k)k!}$$

$$= \frac{(-1)^{k-1}(\alpha-k)!(\alpha+k)!}{2k^2}$$

And lastly, as $P = P_1^{-1}$

$$P = \prod_{\substack{n=1 \\ n\neq k}}^{\alpha} \frac{1}{(ki - ni)(ki + ni)} = \frac{2k^2(-1)^{k-1}}{(\alpha-k)!(\alpha+k)!}$$

We can now substituting this back into our expression for $\text{Res}(f, ki)$, and find that

$$\therefore \text{Res}(f, ki) = -\frac{(-1)^{k-1} k \ln(ki)}{i(k^2+1)(\alpha-k)!(\alpha+k)!}$$

$$= -\frac{(-1)^{k-1} k (\ln k + \pi i/2)}{i(k^2+1)(\alpha-k)!(\alpha+k)!}$$

Like the integral on page 199,

$$\lim_{R\to\infty} \int_{C_R} f(z)\, dz = 0$$

$$\lim_{\epsilon\to 0} \int_{C_\epsilon} f(z)\, dz = \lim_{\epsilon\to 0} z f(z) = 0$$

Putting everything together, and summing all these enclosed residues from $k = 1$ to α, we obtain

$$\oint f(z)\, dz = \int_{-R}^{-1+\rho} f(z)\, dz + \int_{C_\rho} f(z)\, dz + \int_{-1-\rho}^{\epsilon} f(z)\, dz + \int_{\epsilon}^{R} f(x)\, dx = -2\pi \sum_{k=1}^{\alpha} \frac{(-1)^{k-1} k(\ln k + \pi i/2)}{(k^2+1)(\alpha-k)!(\alpha+k)!}$$

Then, since $x = -1$ is a boundary residue integrated clockwise, its contribution to the LHS of the expression above is $-\pi i$ times its residue. Specifically,

$$\int_{C_\rho} f(z)\, dz = -\pi i \text{Res}(f, -1) = -\frac{\pi^2}{2} \prod_{n=1}^{\alpha} \frac{1}{n^2+1}$$

And, again like the integral on page 199,

$$\int_{-R}^{-1+\rho} f(z)\, dz + \int_{-1-\rho}^{\epsilon} f(z)\, dz + \int_{\epsilon}^{R} f(x)\, dx$$

$$= 2\int_0^\infty \frac{\ln x}{x^2-1} \prod_{n=1}^{\alpha} \frac{1}{x^2+n^2}\, dx + i\pi \cdot PV \int_0^\infty \frac{1}{x^2-1} \prod_{n=1}^{\alpha} \frac{1}{x^2+n^2}\, dx$$

Due to our chosen branch cut such that $z = xe^{i\pi} = -x$ for the negative real axis, meaning $\ln z = \ln x + i\pi$. We can now put it all together with our results, obtaining

$$\therefore \oint f(z)\,dz = 2\int_0^\infty \frac{\ln x}{x^2-1}\prod_{n=1}^{\alpha}\frac{1}{x^2+n^2}\,dx + i\pi\cdot PV\int_0^\infty \frac{1}{x^2-1}\prod_{n=1}^{\alpha}\frac{1}{x^2+n^2}\,dx - \frac{\pi^2}{2}\prod_{n=1}^{\alpha}\frac{1}{n^2+1}$$

$$= -2\pi\sum_{k=1}^{\alpha}\frac{(-1)^{k-1}k(\ln k + \pi i/2)}{(k^2+1)(\alpha-k)!(\alpha+k)!}$$

And finally, equating real and imaginary components, and dividing both sides by two, we obtain our answer

$$\boxed{\therefore \int_0^\infty \frac{\ln x}{x^2-1}\prod_{n=1}^{\alpha}\frac{1}{x^2+n^2}\,dx = -\pi\sum_{k=1}^{\alpha}\frac{(-1)^{k-1}k\ln k}{(k^2+1)(\alpha-k)!(\alpha+k)!} + \frac{\pi^2}{4}\prod_{n=1}^{\alpha}\frac{1}{n^2+1}}$$

Wow! As an added bit, I also applied the same method to the following very similar integral, with the attached solution.

$$\int_0^\infty \ln(x)\prod_{n=1}^{\alpha}\frac{1}{x^2+n^2}\,dx = \pi\sum_{k=1}^{\alpha}\frac{(-1)^{k-1}k\ln k}{(\alpha-k)!(\alpha+k)!}$$

Which as an aside, when equating the real and imaginary components at the end, yielded

$$\int_0^\infty \prod_{n=1}^{\alpha}\frac{1}{x^2+n^2}\,dx = \pi\sum_{k=1}^{\alpha}\frac{(-1)^{k-1}k}{(\alpha-k)!(\alpha+k)!}$$

Which can be cleaned up and solved to become

$$\int_{-\infty}^\infty \prod_{n=1}^{\alpha}\frac{1}{x^2+n^2}\,dx = \pi\frac{(\alpha-1)!}{(2\alpha-1)!}$$

Each of these three results will be shown at the top in reverse order (since that more naturally goes from less to more complex in result).

18.9 An arctan-log problem via a known Fourier series

Another short-ish method, yielding a very cool solution

$$\int_0^\infty \frac{\arctan x}{x(x^2+1)} \ln\left(\frac{1+x^2}{x^2}\right) dx$$

Start off with a u-sub

$$x = \tan t, dx = \sec^2 t \, dt$$

$$= \int_0^{\frac{\pi}{2}} \frac{t\sec^2 t}{\tan(t)(\tan^2 t+1)} \ln\left(\frac{1+\tan^2 t}{\tan^2 t}\right) dt$$

$$= -2\int_0^{\frac{\pi}{2}} \frac{t}{\tan t} \ln(\sin t) \, dt$$

$$\text{IBP with } u = t, dv = \frac{\ln(\sin t)}{\tan t}$$

$$v = \frac{1}{2}\ln^2(\sin t)$$

$$\therefore -2\int_0^{\frac{\pi}{2}} \frac{t}{\tan t} \ln(\sin t) \, dt = \int_0^{\frac{\pi}{2}} \ln^2(\sin t) \, dt$$

Okay now we have a very manageable integrand. This integrand reminds me of the well known fourier series

$$\ln(2\sin x) = -\sum_{n=1}^\infty \frac{\cos(2nx)}{n}$$

Thus,

$$\ln(\sin x) = -\sum_{n=1}^\infty \frac{\cos(2nx)}{n} - \ln 2$$

$$\therefore \ln^2(\sin x) = \sum_{n=1}^\infty\sum_{m=1}^\infty \frac{\cos(2nx)\cos(2mx)}{nm} + 2\ln(2)\sum_{n=1}^\infty \frac{\cos(2nx)}{n} + \ln^2 2$$

$$\therefore \int_0^{\frac{\pi}{2}} \ln^2(\sin t) \, dt = \int_0^{\frac{\pi}{2}} \sum_{n=1}^\infty\sum_{m=1}^\infty \frac{\cos(2nt)\cos(2mt)}{nm} + 2\ln(2)\sum_{n=1}^\infty \frac{\cos(2nt)}{n} + \ln^2 2 \, dt$$

$$= \int_0^{\frac{\pi}{2}} \sum_{n=1}^\infty\sum_{m=1}^\infty \frac{\cos(2nt)\cos(2mt)}{nm} + 2\ln(2)\sum_{n=1}^\infty \frac{\cos(2nt)}{n} \, dt + \frac{\pi}{2}\ln^2 2$$

$$= \int_0^{\frac{\pi}{2}} \sum_{n=1}^\infty\sum_{m=1}^\infty \frac{\cos(2nt)\cos(2mt)}{nm} \, dt + 2\ln(2)\sum_{n=1}^\infty \frac{1}{n}\int_0^{\frac{\pi}{2}} \cos(2nt) \, dt + \frac{\pi}{2}\ln^2 2$$

$$= \sum_{n=1}^\infty\sum_{m=1}^\infty \frac{1}{nm}\int_0^{\frac{\pi}{2}} \cos(2nt)\cos(2mt) \, dt + \frac{\pi}{2}\ln^2 2$$

Let's focus on this integral.

$$\int_0^{\frac{\pi}{2}} \cos(2nt)\cos(2mt) \, dt$$

We note that unless $n = m$, this integral is simply equal to zero as we're just going to get some sine terms of 2 times something involving n and m, that equate to zero under the current bounds. Thus, consider when $n = m$

$$
\begin{aligned}
\int_0^{\frac{\pi}{2}} \cos(2nt)\cos(2mt)\,dt &= \int_0^{\frac{\pi}{2}} \cos^2(2nt)\,dt \\
&= \int_0^{\frac{\pi}{2}} \frac{1+\cos(4nt)}{2}\,dt \\
&= \frac{\pi}{4} + \frac{1}{2}\int_0^{\frac{\pi}{2}} \cos(4nt)\,dt \\
&= \frac{\pi}{4}
\end{aligned}
$$

Thus, substituting this into our expression for the original integral, we get (recalling that $n = m$)

$$
\begin{aligned}
\therefore \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} \frac{1}{nm}\int_0^{\frac{\pi}{2}} \cos(2nt)\cos(2mt)\,dt + \frac{\pi}{2}\ln^2 2 &= \frac{\pi}{4}\sum_{n=1}^{\infty}\frac{1}{n^2} + \frac{\pi}{2}\ln^2 2 \\
&= \frac{\pi^3}{24} + \frac{\pi}{2}\ln^2 2
\end{aligned}
$$

Done!

$$
\therefore \int_0^{\infty} \frac{\arctan x}{x(x^2+1)} \ln\left(\frac{1+x^2}{x^2}\right) dx = \frac{\pi^3}{24} + \frac{\pi}{2}\ln^2 2
$$

19 July 26

19.1 Omega right-half plane semicircle contour

This one also has the omega constant in the answer which is very cool, the omega constant Ω being the number such that $\Omega e^{\Omega} = 1$.

$$\int_0^\infty \frac{x \sin x + 2\cos x + 1}{2x \sin x + x^2 + 1}\, dx$$

In order to solve this one via contour integration, we first need to manipulate the integrand into an easier to work with way. One way this can be done is by writing the integrand as the real or imaginary part of some simpler expression. But first, as the function is even, we'll write

$$= \frac{1}{2}\int_{-\infty}^\infty \frac{x \sin x + 2\cos x + 1}{2x \sin x + x^2 + 1}\, dx$$

Okay, we're going to show that the integrand can be written as

$$\Re \frac{1 + 2e^{ix}}{1 - ixe^{ix}}$$

Expanding e^{ix}, and simplifying,

$$= \Re \frac{1 + 2\cos x + 2i \sin x}{1 + x \sin x - ix \cos x}$$

$$= \Re \frac{(1 + 2\cos x + 2i \sin x)(1 + x \sin x + ix \cos x)}{(1 + x \sin x - ix \cos x)(1 + x \sin x + ix \cos x)}$$

$$= \Re \frac{1 + x \sin x + 2\cos x + i(x \cos x + 2x \cos^2 x + 2\sin x + 2x \sin x)}{1 + 2x \sin x + x^2}$$

$$= \frac{1 + x \sin x + 2\cos x}{1 + 2x \sin x + x^2}$$

Which is our integrand! Thus, we can write our integral as so.

$$\therefore \frac{1}{2}\int_{-\infty}^\infty \frac{x \sin x + 2\cos x + 1}{2x \sin x + x^2 + 1}\, dx = \frac{1}{2}\Re \int_{-\infty}^\infty \frac{1 + 2e^{ix}}{1 - ixe^{ix}}\, dx$$

$$t = ix, dt = i\, dx$$

$$= \Re \frac{1}{2i}\int_{-\infty i}^{\infty i} \frac{1 + 2e^t}{1 - te^t}\, dt \tag{10}$$

Let the integrand be $f(t)$. We will be integrating this one using a semicircle of radius R on the right-half plane ($\theta \in (-\pi/2, \pi/2)$), with base from $-Ri$ to Ri. We now need to find the poles of this integrand, which will be the complex points where

$$1 - te^t = 0$$

This occurs only when

$$t = W_0(1) = \Omega$$

As all other $W_k(1)$ have $\Re(t) < 0$ and thus lie outside the contour. We now wish to find the residue at this t value.

$$\operatorname{Res}(f, \Omega) = \lim_{t\to\Omega} \frac{1 + 2e^t}{\frac{d}{dt}(1 - te^t)} = \frac{1 + 2e^{\Omega}}{-(e^{\Omega} + \Omega e^{\Omega})}$$

$$= -\frac{1 + 2e^{\Omega}}{e^{\Omega} + 1} = -\frac{e^{\Omega} + 1 + e^{\Omega}}{e^{\Omega} + 1} = -1 - \frac{e^{\Omega}}{e^{\Omega} + 1}$$

$$= -1 - \frac{e^{\Omega}}{e^{\Omega}+1}$$

Then as $e^{\Omega} = 1/\Omega$,

$$= -1 - \frac{1}{\Omega+1}$$

When going from $-Ri$ to Ri we are effectively integrating clockwise, and so we introduce a factor of -1 on the RHS. Letting the semicircle arc be C_R, therefore

$$\oint f(z)\,dz = \int_{-Ri}^{Ri} f(t)\,dt + \int_{C_R} f(z)\,dz = -2\pi i\left(-1 - \frac{1}{\Omega+1}\right)$$

We thus now need to find the added integral over the C_R arc, which unfortunately this time is not zero. To do this, we note that over C_R, θ ranges from $\pi/2$ to $-\pi/2$ (since integrating clockwise). Then, as the arc lies on the right-half plane, and we evaluate the integral as $R \to \infty$, we thus let $t = Re^{i\theta}$.

$$\therefore \frac{1+2e^t}{1-te^t} = \frac{1+2e^{Re^{i\theta}}}{1-Re^{Re^{i\theta}+i\theta}}$$

We now look at how this behaves as $R \to \infty$.

$$\frac{1+2e^{Re^{i\theta}}}{1-Re^{Re^{i\theta}+i\theta}} = \frac{\frac{1}{e^{Re^{i\theta}}}+2}{\frac{1}{e^{Re^{i\theta}}}-Re^{i\theta}} \approx \frac{2}{-Re^{i\theta}} = -\frac{2}{t}$$

Thus, asymptotically, the function behaves like $-2/t$. Then, as $t = Re^{i\theta}$, we get that $dt = iRe^{i\theta}\,d\theta$. The integral over the arc is thus

$$\lim_{R\to\infty}\int_{C_R} f(z)\,dz = \lim_{R\to\infty}\int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} -\frac{2}{Re^{i\theta}} iRe^{i\theta}\,d\theta$$

$$= 2i\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta$$

$$= 2\pi i$$

Cool! Okay, putting everything together and letting $R \to \infty$, we have found that

$$\oint f(z)\,dz = \int_{-\infty i}^{\infty i} \frac{1+2e^t}{1-te^t}\,dt + 2\pi i = -2\pi i\left(-1 - \frac{1}{\Omega+1}\right)$$

$$\therefore \int_{-\infty i}^{\infty i} \frac{1+2e^t}{1-te^t}\,dt = \frac{2\pi i}{\Omega+1}$$

Finally, referring to equation 10, recall that we are finding the real part of this integral divided by $2i$.

$$\therefore \Re\frac{1}{2i}\int_{-\infty i}^{\infty i} \frac{1+2e^t}{1-te^t}\,dt = \frac{\pi}{\Omega+1}$$

And since this was exactly our expression for the original integral, we are now done!

$$\therefore \int_0^{\infty} \frac{x\sin x + 2\cos x + 1}{2x\sin x + x^2 + 1}\,dx = \frac{\pi}{\Omega+1}$$

Three very similar results of note using the same method would be

$$\int_0^{\infty} \frac{x\sin x + 1}{2x\sin x + x^2 + 1}\,dx = \pi - \frac{\pi}{\Omega+1}$$

$$\int_0^{\infty} \frac{\cos x}{2x\sin x + x^2 + 1}\,dx = \frac{\pi}{\Omega+1} - \frac{\pi}{2}$$

$$\int_0^{\infty} \frac{x\sin x - x^2\cos x + 1}{(x^2+1)(2x\sin x + x^2 + 1)}\,dx = \frac{\pi\Omega}{1-\Omega^2} - \frac{\pi}{e-1}$$

19.2 Product via odd & even separation

In this section we will be solving the following product

$$\prod_{n=2}^{\infty}\left(1+\frac{1}{1-n^2}\right)^{(-1)^n}$$

Due to the power of $(-1)^n$, we will be solving this by breaking the product into its even and odd terms.

$$\prod_{n=2}^{\infty}\left(1+\frac{1}{1-n^2}\right)^{(-1)^n}=\prod_{n=1}^{\infty}\left(1+\frac{1}{1-4n^2}\right)\prod_{k=1}^{\infty}\left(1+\frac{1}{1-(2k+1)^2}\right)^{-1}=P_E\cdot P_O$$

First solving P_E,

$$\begin{aligned}P_E&=\prod_{n=1}^{\infty}\left(1+\frac{1}{1-4n^2}\right)\\&=\prod_{n=1}^{\infty}\frac{2-4n^2}{1-4n^2}\\&=\prod_{n=1}^{\infty}\frac{n^2-\left(1/\sqrt{2}\right)^2}{n^2-(1/2)^2}\\&=\frac{\prod_{n=1}^{\infty}\left(1-\left(\frac{1/\sqrt{2}}{n}\right)^2\right)}{\prod_{n=1}^{\infty}\left(1-\left(\frac{1/2}{n}\right)^2\right)}\end{aligned}$$

Now, using the fact that

$$\prod_{n=1}^{\infty}\left(1-\frac{x^2}{n^2}\right)=\frac{\sin(\pi x)}{\pi x}$$

We can solve the products with the following results

$$\begin{aligned}&=\frac{\sqrt{2}\sin\left(\pi/\sqrt{2}\right)/\pi}{2\sin(\pi/2)/\pi}\\&=\frac{1}{\sqrt{2}}\sin\left(\frac{\pi}{\sqrt{2}}\right)\end{aligned}$$

Now that P_E has been solved, it is time to solve P_O.

$$\begin{aligned}P_O&=\prod_{k=1}^{\infty}\left(1+\frac{1}{1-(2k+1)^2}\right)^{-1}\\&=\prod_{k=1}^{\infty}\left(\frac{2-(2k+1)^2}{1-(2k+1)^2}\right)^{-1}\\&=\prod_{k=1}^{\infty}\frac{1-(2k+1)^2}{2-(2k+1)^2}\\&=\prod_{k=1}^{\infty}\frac{(2k+1)^2-1}{(2k+1)^2-2}\\&=\frac{\prod_{k=1}^{\infty}\left(1-\frac{1}{(2k+1)^2}\right)}{\prod_{k=1}^{\infty}\left(1-\frac{2}{(2k+1)^2}\right)}\end{aligned}$$

Now, using the fact that

$$\cos(\pi x)=\prod_{k=0}^{\infty}\left(1-\frac{4x^2}{(2k+1)^2}\right)$$

$$= (1-4x^2)\prod_{k=1}^{\infty}\left(1-\frac{4x^2}{(2k+1)^2}\right)$$

$$\therefore \prod_{k=1}^{\infty}\left(1-\frac{4x^2}{(2k+1)^2}\right) = \frac{\cos(\pi x)}{(1-4x^2)}$$

We can solve these products, with the following results

$$= \lim_{x\to 1/2} -\frac{\cos(\pi x)}{\cos\left(\pi/\sqrt{2}\right)(1-4x^2)}$$

$$= \lim_{x\to 1/2} \frac{\pi\sin(\pi x)}{-8\cos\left(\pi/\sqrt{2}\right)x}$$

$$-\frac{\pi}{4\cos\left(\pi/\sqrt{2}\right)}$$

Having finally solved both P_E and P_O, we can multiply them together to obtain the result to the original product.

$$\therefore P_E \cdot P_O = \frac{1}{\sqrt{2}}\sin\left(\frac{\pi}{\sqrt{2}}\right)\cdot -\frac{\pi}{4\cos\left(\pi/\sqrt{2}\right)}$$

$$= -\frac{\pi}{4\sqrt{2}}\tan\left(\frac{\pi}{\sqrt{2}}\right)$$

And we're done!

$$\therefore \prod_{n=2}^{\infty}\left(1+\frac{1}{1-n^2}\right)^{(-1)^n} = -\frac{\pi}{4\sqrt{2}}\tan\left(\frac{\pi}{\sqrt{2}}\right)$$

Using a very similar method, we can also find that

$$\prod_{n=1}^{\infty}\left(1+\frac{1}{1+n^2}\right)^{(-1)^n} = \frac{\tanh\left(\frac{\pi}{\sqrt{2}}\right)}{\sqrt{2}\tanh\left(\frac{\pi}{2}\right)}$$

19.3 Simples filler integral via contour methods

I keep lying! This one too uses a simple contour but I want to include it anyway.

$$\int_{-\infty}^{\infty} \frac{e^{\cos x}\cos(\sin x)}{x^2+1}\,dx$$

I realised that I never had written out here the famous contour integration example of

$$\int_{-\infty}^{\infty} \frac{\cos x}{x^2+1}\,dx$$

I did do it all the back in February last year, but that was via real methods only. Although I don't currently intend on showing the contour integration method of that integral, I will do this one instead due to the method being pretty much identical. While the integral above uses the fact that $\cos x = \Re(e^{ix})$, the numerator in our target integral today can be written as $e^{\cos x}\cos(\sin x) = \Re(e^{e^{ix}})$. Thus, our target integral can be written as

$$= \Re \int_{-\infty}^{\infty} \frac{e^{e^{ix}}}{x^2+1}\,dx$$

$$= \Re \int_{-\infty}^{\infty} \frac{e^{e^{ix}}}{(x-i)(x+i)}\,dx$$

We use a regular semicircle contour in the upper half plane. Trivially, the only pole in this contour is at $x = i$. Thus, the residue is as follows.

$$\operatorname{Res}(f,i) = \lim_{x\to i}(x-i)\frac{e^{e^{ix}}}{(x-i)(x+i)}$$

$$= \lim_{x\to i}\frac{e^{e^{ix}}}{(x+i)} = \frac{e^{\frac{1}{e}}}{2i}$$

Thus, where $f(z)$ is our integrand, and C_R is the semi circle arc part of the contour (integrated anti-clockwise),

$$\oint f(z)\,dz = \int_{-R}^{R} \frac{e^{e^{ix}}}{(x-i)(x+i)}\,dx + \int_{C_R} f(z)\,dz = 2\pi i\left(\frac{e^{\frac{1}{e}}}{2i}\right) = \pi e^{\frac{1}{e}}$$

Since we wish to take the limit as $R \to \infty$, it just remains then to solve

$$\lim_{R\to\infty}\int_{C_R} f(z)\,dz = \int_{C_R} \frac{e^{e^{iz}}}{z^2+1}\,dz$$

Going around this arc, we let $z = Re^{i\theta}$, where $\theta \in [0,\pi]$. Thus, $dz = iRe^{i\theta}\,d\theta$, and the integral becomes

$$= \int_0^{\pi} \frac{e^{e^{iRe^{i\theta}}} iRe^{i\theta}}{R^2e^{2i\theta}+1}\,d\theta = i\int_0^{\pi} \frac{e^{e^{iRe^{i\theta}}}}{Re^{i\theta}+\frac{1}{Re^{i\theta}}}\,d\theta$$

$$= i\int_0^{\pi} \frac{e^{e^{-R\sin\theta+iR\cos\theta}}}{Re^{i\theta}+\frac{1}{Re^{i\theta}}}\,d\theta$$

Through inspection of the numerator, as R approaches infinity and $\theta \in [0,\pi]$, it is bounded. This can be seen below.

$$\lim_{R\to\infty}\left|e^{e^{-R\sin\theta+iR\cos\theta}}\right| = \lim_{R\to\infty} e^{\Re(e^{-R\sin\theta+iR\cos\theta})}$$

$$= \lim_{R\to\infty} e^{e^{-R\sin\theta}}$$

$$= \lim_{R\to\infty} e^{\frac{1}{e^{R\sin\theta}}} = \begin{cases} 1 & \text{if } \sin\theta > 0, \\ e & \text{if } \sin\theta = 0. \end{cases}$$

Awesome. Then, as it is trivial that the denominator grows to infinity as $R \to \infty$, we have found that

$$\therefore \lim_{R\to\infty} \int_{C_R} f(z)\,dz = i\int_0^\pi 0\,d\theta = 0$$

$$\therefore \oint f(z)\,dz = \int_{-\infty}^{\infty} \frac{e^{e^{ix}}}{(x-i)(x+i)}\,dx = \pi e^{\frac{1}{e}}$$

And since our original integral is just the real part of the integral above, we obtain our answer.

$$\therefore \int_{-\infty}^{\infty} \frac{e^{\cos x}\cos(\sin x)}{x^2+1}\,dx = \pi e^{\frac{1}{e}}$$

Acknowledgements

As mentioned in the abstract, many of the problems were solved with help, including but not limited to: internet sources for intermediate step integrals, prior knowledge (hints) from sources for the general method direction, and/or the first step or so of the solution itself. For these problems, the amount/type of help used will be mentioned at the start of the solution. Many other problems here were simply fully copied from the source in which I found it. Citations have been provided where I have felt necessary (particularly for completely copied solutions), but are likely incomplete. If no mention of any of the above is present, then I likely did the problem completely myself. Citations have been omitted for very well known problems, where I have instead simply said that the solution is obviously not my own.

Overall, take this document to be more of a collection of solutions to very difficult integrals/sums/products, than anything else :)

References

[1] Papa Flemmy (Youtube). A beautiful result in calculus: Solution using feynman integration (integral $\cos(\mathrm{x})/(\mathrm{x}^2+1)$), 2018.

[2] Wikipedia. Lambert w function, other formulas.

[3] Maths505 (Youtube). Raabe's integral, 2025.

[4] Maths505 (Youtube). Yes this is (also) your favourite integral, 2025.

[5] Maths505 (Youtube). Let's solve a monster integral, 2025.

[6] Tyma Gaidash (https://math.stackexchange.com/users/905886/tyma gaidash). Mathematics Stack Exchange. https://math.stackexchange.com/q/5083488 (version: 2025-07-20).

[7] Maths505 (Youtube). A beautiful integral and stunning result!, 2025.

[8] xpaul (https://math.stackexchange.com/users/66420/xpaul). Mathematics Stack Exchange. https://math.stackexchange.com/q/5000278 (version: 2024-11-18).

[9] Maths505 (Youtube). An exponentially fascinating integral, 2025.

[10] Maths505 (Youtube). A monster integral, 2025.

[11] Sungjin Kim (https://math.stackexchange.com/users/67070/sungjin kim). Mathematics Stack Exchange. URL:https://math.stackexchange.com/q/959546 (version: 2014-10-05).

[12] Shreyash Pachore (https://math.stackexchange.com/users/1642314/shreyash pachore). Mathematics Stack Exchange. URL:https://math.stackexchange.com/q/5134643 (version: 2026-05-02).

[13] Michael Penn. A nice product from ramanujan, 2023.